Next-Generation Agent Technology Route Survey

Jev-class fast-thinking models (System One) and fast / slow cooperation frameworks

Survey report · analysis and survey only, no training plan

Generated: 2026-10-29

Contents

Next-Generation Agent Technology Route Survey: Jev-Class Fast-Thinking Models and Fast/Slow Cooperation Frameworks . . . 3
Abstract . . . 3
1. Core logic (the whole report is organized around these six statements; full text in § 2.2.5) . . . 3
2. Report structure . . . 4
4. Method . . . 8
5. Evidence credibility and caveats . . . 8
6. Conventions . . . 8
1 • Object and definitions: what Jev-class models are, what the backbone constraints are, and what each paper is saying . . . 9
1.1 Jev / System One: definition, interface primitives, architecture reconstruction and independent evaluation . . . 9
1.2 Open-source decoder reproductions and the recipe matrix (with small-model evidence) . . . 12
1.3 Backbone models and hybrid-architecture constraints (survey only, no selection) . . . 18
1.4 Cross-paper comparison of problem definitions: the same word, different definitions . . . 21
2 • Propositions and core logic: what this report is organized around . . . 28
2.1 The four propositions and their tests . . . 28
2.2 The locked discussion draft: what we are doing, why, where existing work supports it and where it does not . . . 37
2.3 Adversarial review of this discussion draft: omissions and blind spots . . . 44
3 • Semantics: did the model actually read what the business side wrote . . . 49
3.1 D1 in-context learning: can it "learn in the prompt", and which levers can business staff use . . . 49
3.2 D3 logic primitives and rule language: letting business experts write rules in text . . . 53
4 • Consistency: the primary metric, its denominator, and failure under symbolic interventions . . . 58
4.1 D2 symbolic instruction following: rules, negation, arithmetic/time, fallback, hierarchy and adversarial context . . . 58
4.2 D4 consistency: split into 30 definable, measurable sub-questions . . . 62
5 • System: business goal, property list, and the positions of System1+System2 . . . 69
5.1 Business goal and properties: which class of cost is eliminated, which position is supported . . . 69
5.2 D5 Agent / knowledge systems: fast/slow cooperation, memory, routing and guards . . . 75
6 • Learning side: algorithms, data, backbone selection, output shape and chain of thought . . . 81
6.1 Learning algorithms, dataset availability and base-model selection patterns . . . 81
6.2 Hypothesis review: can SFT bring generalization in the general sense . . . 92
6.3 RL or SFT: under the constraints of "keep ICL" and "inject skill semantics" . . . 96
6.4 Does task-agnostic RL have vitality; can business expectations be made differentiable . . . 99
6.5 Training objective: what the output should look like, and whether to introduce chain of thought . . . 102
7 • Evaluation: the existing checklist, the coverage, and what cannot be decided . . . 110
7.1 "How evaluation should be done" already has an answer: the 14-item checklist and its compliance rate . . . 110
7.2 Existing evaluation sets classified by "what is tested" . . . 111
7.3 Coverage matrix: twelve business properties × existing evaluation sets . . . 113
7.4 Numbers across evaluation sets are incomparable, and this is structural . . . 114
7.5 Working backward from business goals: which capabilities no existing evaluation set can currently decide . . . 114
7.6 Chapter summary and blind spots . . . 115
8 • Review and adversarial check: working hypotheses, the report's own constructs, methodological defects and attacks . . . 116
8.1 The four working hypotheses, the report's own constructs and my methodological defects . . . 116
8.2 Seven adversarial attacks on the § 2.2 framework . . . 121
8.3 Work-problem matrix and blind-spot summary . . . 125
8.4 List of open questions (without solutions) . . . 128
9 • References . . . 131
9.1 arXiv papers (283 in total) . . . 131
9.2 TypeSafe official documentation and pages . . . 140

9.3 Open-source reproductions, tools and model cards . . . 141
9.4 Other web pages . . . 141

Next-Generation Agent Technology Route Survey: Jev-Class Fast-Thinking Models and Fast/Slow Cooperation Frameworks

> Survey report • 2026-10-28 • **Analysis and survey only; no training plan and no model-selection conclusion**
>
> Printable version: nga-survey.pdf (A4, generated by `tools/build_pdf.py` from the Markdown in this repository; the README and chapters 1–9 are merged in order, with a table of contents and page numbers)
>
> Plan documents derived from this report (**not part of the report**): proposal/00-research-proposal.md: task decomposition and pre-registered conclusions for nine rounds of experiments, with three stage gates; experiments/: the pre-experiment task set and build scripts. The report itself stays "analysis only, no plan".

Abstract

This report surveys **Jev-class (System One) single-pass decision models** and the **fast/slow cooperation agent frameworks** they enable. The purpose is to understand the problem before we train such a model ourselves: which works solve which class of problem, and which blind spots we have identified that existing work does not cover.

The evidence base is **283 arXiv papers** (about 70 of them read in full), 17 pages of TypeSafe official documentation, 25 GitHub READMEs, 6 model cards, and 14 blog and leaderboard pages, with a time window concentrated on September–October 2026. **Every number in the report was checked against the cached source text; anything whose source could not be found was deleted.**

The organizing axis of the report was fixed in the 2026-10-28 discussion as "**generalization of semantic understanding**", not the original "business cost". The synthesis of six rounds of discussion is locked as the **discussion draft** (§ 2.2: what we want to do, the business goal, support from existing work, the defect registry, the final framework); § 2.3 and § 8.2 are its adversarial review. **P0″–P5″ in § 2.2.5 are the final framework statement of this report.**

1. Core logic (the whole report is organized around these six statements; full text in § 2.2.5)

#	Statement	What it replaces
P0″	**Semantics to the model, arithmetic to the code, questions are first-class objects.** The model maps the state into a **per-question** scaled typed observation; the code does thresholding, sufficiency checks and the union bound, and its parameters may only be fitted on held-out ground truth. Equivalence classes are **induced by the questions**, not declared separately.	"Move the decision authority into code" (whose optimum is a lookup table); "business-declared equivalence classes" (themselves a non-idempotent judgment)
P1″	**The primary metric is a pair**: on pre-registered, held-out (input, question set) pairs, **within-question invariance** (including swapping in a set of level names that carry no semantics, and adding or removing irrelevant questions) and **cross-question re-partition** (the same input must move under questions that partition differently). A system-level metric; the model-level proxy is ordinal stability of the scores and rank change across questions.	"Accuracy is the primary metric"; a "consistency rate" with no fixed denominator
P2″	**Accuracy is the third leg**: a declared floor, measured under the decision-level paired protocol. Without it, both P1″ items can be passed by a degenerate model that "reads only the question wording and ignores the state".	"Correctness comes second" (which under lexicographic order equals unbounded)
P3″	**Idempotence lives in thresholds and contracts; the operator is fixed**: the code thresholds the level distribution by **cumulative mass**, and the residual that clears neither side goes to C (unknown / deny); exclusive candidates use margin + explicit abstain; no argmax, no expectation. On the model side: monotone, thresholdable scores whose margin exceeds the within-class variance, native unknown (currently 7%), within-question invariance.	"Jev turns non-idempotence into idempotence"
P4″	**Every invariance is paired with a discrimination**, made concrete as the "within-question / cross-question" pair. The credit lies only in this fixing.	Four formulations that named only the invariance

#	Statement	What it replaces
P5″	**Unit price is an enabling condition; the trade-off has been decided by a design choice**: the fixed order (task definition → input → candidate branches) places the cache on the "same rule set" axis; we do not have Jev's "extra questions nearly free"; what we keep is the one to two orders of magnitude from not decoding; A4 chose the expensive, rule-sensitive side.	"The business goal is to be cheap"

One structural finding runs throughout: in eleven rounds of discussion, the propositions of seven rounds (idempotence is the goal / the core is turning it into code / consistency rate is the core metric / force it to read the definitions / train "capability" / independent scoring is comparable / expert-defined levels, the fourth being my own) have degenerate optima under no constraints (lookup table, lookup table, constant classifier, ignore the names, learn the generator grammar, separately calibrated and mutually incomparable, priors by level position); every time it was "the invariance was named, the discrimination was left blank". When the next proposition is raised, first check whether its unconstrained optimum is a degenerate model (§ 2.2.6).

Known boundaries of the discussion draft and the fixed structure (end of § 2.2.1, § 2.3): the nine omissions have each been discussed to completion. The questioner fixed a structure different from Jev's: **task definition → input → candidate branches** (cache along the rule axis); the candidate set is declared exclusive or stackable; **levels are defined by business experts** and declared with a type; the model outputs a level distribution and the harness thresholds by cumulative mass; labels are given by the business at construction time and by outcome retrospection in production; the input is a mix of semantic specification + dialogue record + historical context; **the deliverable is a template standard released for business staff to write in, plus a model that can read it**. Three things remain open: (a) denominator data and rule-diversity data for "cross-question re-partition" (zero after local sampling and checking, § 6.1.4.3); (b) the hybrid vs attention-only controlled experiment (defect 11); (c) direct evaluation by business-staff authors (defect 13, required before the template is released).

2. Report structure

Chapter	Content	What it answers
1 Object and definitions	Definitions, interface primitives, the vendor's zigzag, architecture reconstruction, independent evaluations; a recipe matrix of 15+ open-source reproductions; hybrid-backbone constraints; cross-paper definitional disagreements	What it is, who has reproduced it, whether our backbone can carry it, whether two numbers can be compared side by side
2 Propositions and core logic	The four propositions tested one by one; the **locked discussion draft** (what to do and the fixed structure / business goal / existing work against the fixed design / the 18-row defect registry / final framework / the trajectory of eleven rounds of discussion); the nine omissions discussed one by one, with results	**What we want to do, why, where it holds up, where it does not**
3 Semantics	D1 in-context learning and the prompt lever for business staff; D3 logic primitives and rule language	Whether the model actually reads the definitions, criteria, examples and rules the business side writes down
4 Consistency	D2 symbolic instruction following; D4 the 30 sub-dimensions of consistency	All the evidence for the primary metric, and how its denominator should be fixed
5 System	Cost structure, twelve business properties, the five positions of System1+System2; D5 fast/slow cooperation	What it is worth, what it can promise, where in the system it belongs
6 Learning side	Algorithm lineage and data availability; SFT generalization; RL vs SFT; task-agnostic RL and differentiability; output shape and chain of thought	What to train, how to train it, what cannot be trained
7 Evaluation	The 14-item checklist and compliance rate, coverage matrix, incomparability, undecidable items	Which business properties existing evaluation sets cover and which they miss
8 Review and adversarial check	Verdicts on the four working hypotheses, this report's own eight constructs, my nine methodological defects, seven adversarial attacks, blind spots and open questions	**Where this report may be wrong**
9 References	283 arXiv papers + documentation + repositories + web pages, all with links	—

Mapping of the **five requirement dimensions** to chapters (definitions follow the latest discussion):

Dimension	Definition	Where
D1 In-context learning	Not a cost level. Whether the single-pass model after decision fine-tuning still has ICL and can use it; **whether business staff can use it directly through prompt engineering**; effect size and failure modes of each lever	§ 3.1

Dimension	Definition	Where
D2 Symbolic instruction following	Decision consistency under rules, negation, arithmetic/time, fallback options, instruction hierarchy, and adversarial context	§ 4.1
D3 Templating / skill injection	Define a set of **logic primitives** so that non-technical business experts write business rules in text and the model executes them faithfully	§ 3.2
D4 Consistency	More than the four items of "numerical consistency". Enumerate **30 sub-dimensions** (six groups A–F) as an exploration of the problem definition	§ 4.2
D5 Agent / knowledge system	Fast/slow cooperation framework, memory, compression, tool routing, guards, and "does faster decision equal a better agent"	§ 5.2
## 3. Core conclusions		

3.1 Object: what it is and what it is not

1. **Jev's value lies in interface, cost, latency and calibration, not accuracy**. The early audit conclusion from 28 papers was that "typed readout by itself shows no accuracy advantage independent of same-class label-probability readout" (2609.32160); an off-the-shelf chat model with its logits read directly is already a Jev-style decision model (2610.02076). The sharper formulation is not "the value lies in the interface contract" but that **the contract hands the bookkeeping of the decision to code and leaves the semantics in the model**: the step of mapping an open-world situation into a scaled observation has no code form at all (§ 2.1.7).
2. **The architecture has been reconstructed and reproduced by more than 15 open-source recipes**: shared state prefix + isolated question branches + list-style probability readout. Decision fine-tuned models from 0.6B to 9B are available, and **a domain fine-tuned small model beats a general-purpose large one** (SecJev-0.8B is 20.51 pp above Kev-9B, 2610.03073; AgentDecision-4B scores 91.5% on out-of-distribution tasks against Jev's 82.0%, 2610.21412).
3. **Backbone constraint**: Qwen3.5 / 3.8 is a hybrid of Gated DeltaNet + attention; the recurrent layers do not recognize attention masks, so a tree mask cannot isolate sibling branches. Existing practice is one row per question (Kev) or prefix forking (8–19× faster inference, but gradients through the cache are not yet implemented). Literature on "training-time branch isolation" for hybrid architectures: **zero hits** (§ 1.3).
4. **The same word has different definitions across papers**: "decision model" is defined by interface, contract or purpose; "confidence" is four quantities that cannot be merged; ECE has at least five algorithms; "flip rate" has five denominators; "RLCD" has four meanings. This is the root of why cross-paper numbers cannot be compared directly (§ 1.4.10 lists the six points of disagreement that must be fixed before the problem can be defined).

3.2 Semantics: this is the weakest layer

1. **Label names override definitions, while the type is always correct**. A name-definition swap takes mean AUC from **93.8% → 23.2%** (below random), while the **type error rate stays at 0% throughout**; a correct type ≠ having read the definition (2609.26758); **deleting all definitions leaves accuracy unchanged** (0.8487), while definitions alone can sustain 0.7971 and accuracy drops to 0.2281 when labels and definitions conflict, with the cause localized in prompt rendering (2610.02586); for 1–12B small models in the few-shot setting, **semantic coverage is exactly zero** (2511.21038). **No paper directly measures "whether decision fine-tuning preserves or destroys ICL"** (§ 3.1).
2. **Rules can be read out, but are executed only 22%–72% of the time** (2610.18579); on matched pairs containing arithmetic/time, accuracy goes 87.7% → 57.4% from difficulty 1 to 3; when the correct answer is absent the refusal rate is only **7%**, and one added threshold brings it to **79%**; a fallback option drops Intern from 97.9% to 23.1% (§ 4.1).
3. **D3 already has primitive prototypes to borrow from** (three primitives, fan-out, classification trees, composite scoring, confidence routing; SemTree's Yes/No/Unknown node tree; the "probability-aware, deterministic adjudication" union bound), but **all work on compiling natural-language business rules into verifiable logic targets generative LLMs; there is no rule language for single-pass decision models yet**; and rule artifacts must be **compiled per judge** (changing the judge: 67.70% → 47.87%, 2610.13999) (§ 3.2).

3.3 Consistency: the primary metric, but the denominator is not fixed

1. **"Consistency rate" is not a metric until the denominator is fixed**. For the same model: **99.83%** under the byte-level rerun protocol, 0.11%–8.30% under order permutation, **53.7%** under the answer-preserving adversarial-variant protocol (attacks flip 46.3% of correct decisions, and among decisions still executed at a 0.8 threshold

23.5% are wrong, 2609.31142): **a gap of 46 percentage points between the two ends**. Of five candidate denominators, two are already achieved or can be bought by construction (so their value as a training objective is zero or very low), and the only one with business meaning, the "**business-declared equivalence class**", **has never been defined by anyone** (§ 2.1.8, § 7.5 row 13).

2. **Aggregate accuracy is blind to the changes we care about**: across 8 option orders, accuracy changes by **<2 pp** while the cumulative failure set grows by **27–29%** (2610.35826); when corrections and regressions cancel out, accuracy is unchanged while the decisions have changed (2609.27678). Repeatability itself is solved and **is misleading**: repeated wrong answers account for most of the observed errors (§ 4.2.2).
3. **D4 is far more than four items**: the report enumerates **30 sub-dimensions** (A same-input stability / B option-representation dependence / C logical coherence / D uncertainty semantics / E input-content sensitivity / F interface and action), each with a quantifiable definition and existing evidence; for the same fact and the same rule, the probability varies with **25 factors unrelated to the business** (§ 4.2).

3.4 System: cost is an enabling condition, not the goal

1. **What is eliminated is the decoding + round-trip + parsing cost**: rubric grading is 16–357× more expensive and 5–350× slower; moderation is about 0.000025 USD per call and about 100 ms, with AUROC indistinguishable from 13 of 14 frontier systems; a local 2B at 30.9 ms with zero output tokens against 8,758 ms for a frontier API. **Only two things can be promised to the business** (the output is always within the option set; one forward pass answers the whole request); calibration, invariance to equivalent phrasings, fallback, unknown, self-knowledge, probabilistic self-consistency, auditability and risk budget: **all eight have counterexamples measured by independent audits** (§ 5.1.2).
2. **There are five engineering-distinct positions for supporting System1+System2**: the first four cells (entry gate, process monitor, evidence/memory surface, readout of the slow model's output) have positive evidence, while **the fifth cell, "closed-loop executor", fails across the board**. And "faster decision ≠ better agent": the model most accurate on isolated questions solves only 69.5% of agent tasks; delegating tool selection without fitting a threshold costs 19.0 points; a Jev controller raises cost by +19% with an unchanged pass rate (§ 5.1.4, § 5.2.1).
3. **The proposition carries its own gradient**: push determinism all the way into the harness and the model becomes redundant (one deterministic rule over typed policy fields gets everything right; a controller without a model solves **79%** of τ^2-bench telecom). So the upper bound of the usable range is how structurable the business fields are, and the lower bound is the repetition of the decisions (§ 5.1.5).

3.5 Learning side: what should change is the shape, not the objective function

1. **The learning algorithms almost never "change the algorithm"**: what changes is the readout position, a proper scoring term added to the objective, and the data composition. What can be confirmed effective is LoRA, data composition, structural isolation and temperature; **reinforcement learning on single-pass models is negative or ineffective** (the published RLCD recipe is 2.5–3.0 points below cross-entropy; REINFORCE −10.2 F1; GRPO no gain), and all positive evidence is tied to reasoning chains; but the evidence base for this conclusion is only three heterogeneous data points (§ 6.1, § 8.1.4).
2. **"SFT cannot bring generalization in the general sense": the strong form fails, the weak form holds**. Transfer is reliable on new samples / new sources within the same task family; but for new schemas (SFT gain about zero at 27B), new symbolic operations, prior shift (the same SFT gives +10.0 or −12.0 on financial sentiment), calibration transfer and reasoning depth, the four layers have **no positive evidence at all**. What moves results across layers is base-model scale, the instruction-tuned starting point, the supervision distribution and structural constraints, **not the supervision objective** (§ 6.2).
3. **"Whether to use RL" is not the variable that decides ICL and skill injection**. The only two methods that improve the ability to "define the task from context" are both SFT (MetaICL task-level meta-training, symbol tuning); what decides whether ICL survives is **whether the task definition varies between training samples**. Skill injection is a question of **storage location**, and a single-pass readout **cannot execute a procedure** (§ 6.3).
4. **Task-agnostic RL is indeed alive, but non-differentiability and RL's payoff surface barely intersect**. What RL generalizes is **second-order properties** (confidence, calibration, constraint satisfaction), and wherever a verifier is the reward it is systematically arbitraged; the only thing that is both non-differentiable and uncertifiable is "agent downstream outcomes and decision → action", and that is exactly where RL's measured gain is zero. So "the objective is non-differentiable, therefore use RL" **does not hold** (§ 6.4).
5. **The shape of the output deserves changing more than the objective function**. Proper scoring rules have hit the ceiling (with known labels they share the optimum with RLCD-style rewards; under annotator disagreement they provably cannot beat cross-entropy). Three of the four output shapes are wrong: boolean **lacks the third truth**

value (soft accuracy 0.771→0.978 after restoring P(U)); the choice readout **is ordinal, not probabilistic** (4.2 points of error at the ordinal layer, 21.0 at the probability layer); ordinals are **systematically compressed** (40 datasets × 9 models; scale does not recover it). Numbers should not use scalar regression (an interval decision tree takes MAE from 5.18% to 1.84%), and arithmetic must be outsourced (§ 6.5).

6. **Chain of thought should be introduced, but at inference time it should not be a token stream**. Training-time distillation is optimal (27B matches Jev on JEVal at about 0.1 s); layer looping is the only zero-token route with depth-extrapolation evidence, but it requires a looped pretrained backbone and is **incompatible with our hybrid base model, with no bridging work**; **reading others' chains is the most underrated** (+4.1 pp under a 1,500-token budget, +8.6 pp on truncated items, saving 71.3% of cost). System 1's own unrolled form is the **question graph**, not a token stream. The hard limit is that Jev cannot make the decisions that "an LLM can only make by writing out intermediate steps" (97.5% solvable in one forward pass, **74%** of the rest), and **it does not know its own limit** (§ 6.5).
7. **Reading this chapter through the core logic yields one extra conclusion**: untuned readout already reaches **76–97%** of post-tuning accuracy, the tuning gain shrinks with scale from 15.8 points to **2.3 points**, "pretraining provides the representation, tuning only adjusts how it is read" (2610.15501). So the question is not "how to train the readout to be more accurate" but "**how to keep the readout from losing the semantics already present in the representation**"; and the only ready-made means aimed at this are rendering fixes (+0.1511) and symbol tuning, **neither of which has been tried on a decision model** (§ 2.1.7).

3.6 Endgame size: the 27B / 35B tier changes how the evidence is read

1. **The endgame target is fixed at the Qwen 27B / 35B tier**, and the evidence at this tier **points in a different direction** from the small-model tier. First separate the architectures: **27B is dense, and Qwen has no dense 35B; every 35B is the `35B-A3B` MoE (about 3B active)**; if the endgame lands on 35B-A3B, then beyond the three structural assumptions of § 1.3 there is one more, "whether per-token expert routing introduces new coupling between isolated question branches", on which **this report has zero hits**. At 27B: the decision fine-tuning gain has shrunk to **2.3 points** (untuned readout reaches 76–97% of post-tuning, "pretraining provides the representation, tuning only adjusts how it is read"); **the gain on new schemas is about zero** (87.8–88.1 against 89.1 untuned); **letting it reason is still clearly better than readout** (96.9 against 89.1; on numerical schemas readout gets only 56–66% at every scale, reasoning 98–100%); the strongest positive result at this tier is **distillation**, not decision SFT (Reasoning-to-Readout self-distillation brings 27B level with Jev at about 0.1 s). The counter-evidence must also be kept in mind: **a domain fine-tuned small model can beat a general-purpose large one** (SecJev-0.8B is 20.51 pp above Kev-9B), so "the endgame must be large" needs an independent justification. **Taken together**: what is worth investing in at this tier is not "training the readout to be more accurate" (only 2.3 points left) but "keeping the readout from losing the semantics already in the representation" and "distilling reasoning into the readout"; accordingly, the role of the small-size ladder changes from "a scaled-down endgame" to **a testbed for structural changes** (§ 6.1.5).

3.7 Evaluation and self-review

1. **The "how to evaluate" literature has given its answer, and the compliance rate of existing papers is low**: on the 14-item checklist, "same-backbone label log-probability baseline" is met by only 5 of 26 applicable papers, "option-name invariance test" by only 1 of 27, "two latency figures, with and without cache" by only 3 of 20. **That is, in the published numbers "the credit of the interface" and "the credit of the model" have basically never been separated**. By business property, **the two cells "configurable by business staff" and "auditable" are zero**; the largest gap is the business goal itself: **end-to-end cost per correct result has no reproducible protocol at all**, and one self-audit corrected its own "23.9% saving" to **4.3%** (§ 7).
2. **Three of the four working hypotheses are "conclusion right, mechanism wrong"**. The reason is that the questioner's judgments come from deployment observation, which is reliable about the **phenomenon**; attributing a mechanism requires controlled experiments, and the literature lacks controlled experiments exactly at these points. **Where there is disagreement, keep the phenomenon and redo the mechanism** (§ 8.1.1).
3. **Eight frameworks in the report are my own constructs**, not literature conclusions (seven generalization levels, five kinds of differentiability, three layers of idempotence, five System1+System2 positions, the decision repetition axis, the question graph, twelve business properties, D4's 30 sub-dimensions), and the last three carry the highest overfitting risk. In the body, "under this classification cell X is zero" should be read as "**under this report's classification**" (§ 8.1.2).
4. **Reaching agreement is not passing a test**. Of the seven adversarial attacks, **two statements of the framework were overturned** (the equivalence class is itself a newly added source of non-idempotence; "the model handles

semantics, the code handles bookkeeping" holds only when the code's parameters are not fitted on the model's candidates), **three must be qualified** (pre-registration, system level / model level kept separate, credit only in the fixing), **one was undecided and is now decided by a design choice** (the operation that makes it cheap and the operation that destroys rule sensitivity **are the same one**: cacheable independent encoding is about 5× cheaper and rule sensitivity goes 1.00 → 0.24, 2609.37832; the order fixed in the eighth round chose the expensive, rule-sensitive side, and the residual is recurrent compression on the hybrid backbone, §8.2.1 A4 postscript two), **one is a permanent discount** (each of the four load-bearing points rests on a single preprint from within six weeks). §8.2.4 lists five observations that would overturn the framework, and the measurements needed for three of them can be made now (§8.2). After the sixth round of discussion A1 was dissolved (equivalence classes are induced by the questions) and A4 was upgraded to a backbone-specific version (the 48/64 recurrent layers pass on only a question-independent summary); see §2.2.4 rows 11–12.

4. Method

- **Scope**: anchored on TypeSafe's Jev / System One, covering the first-party documentation, third-party architecture reconstruction, independent evaluations and audits, open-source reproductions (mainly the decoder route), and the upstream academic literature of each of the five dimensions.
- **Sources**: 357 arXiv abstracts (two rounds of keyword scanning + targeted additions), about 70 of them read in full, **283 actually cited in the body**; 17 pages of TypeSafe official documentation; 25 GitHub READMEs; 6 Hugging Face model cards; 14 blog / leaderboard / specification pages.
- **Time**: Jev was released on 2026-09-15, and the great majority of Jev-related literature falls between 2026-09-19 and 2026-10-26. Upstream classic literature (ICL, calibration, tree attention) goes back to 2021.
- **Number verification**: every number was checked against the cached abstract / full text / documentation. Anything whose source could not be found was deleted or marked "unverified".

5. Evidence credibility and caveats

- The great majority of the evidence is **arXiv preprints** (not peer reviewed), with an extremely narrow time window (within 6 weeks of release). Several describe themselves as "early evidence".
- Jev's training details (RLCD, data, backbone) are **not public**; all architecture descriptions come from black-box probes and inference from open-source reproductions.
- The benchmark protocols differ between reproductions; **numbers across papers cannot be compared directly** (§1.4).
- This report **does no model selection and writes no training plan**. Chapter 8 is dedicated to review and attack: it lists my own methodological defects, this report's own constructs, and the observations that would overturn the core framework. **Before reading the conclusions, it is recommended to read §8.1.2 and §8.1.3 first.**
- The conclusions of chapter 2 come from **discussion** rather than from single papers, the only chapter in the report for which this is so; §2.2 is the locked discussion draft, and the final framework is P0″–P5″ in §2.2.5 (P0′–P5′ in §8.2.3 is an intermediate version).

6. Conventions

- The parenthetical after a number in the text is its source: `[arXiv ID](link)` or a page name.
- "Hosted model / hosted" means TypeSafe's Jev API; "decoder reproduction" means an open-source decision model built on a causal language model such as Qwen / Gemma; the "encoder family" (Laya, Verdict, Jeva, NDM, RubriX, etc.) is cited only as comparative evidence and **is not our technical direction**.
- Section numbers have three to four segments (e.g. §4.2.2, §6.5.2): the first segment is the chapter, the second is the component of that chapter (originally the individual dimensions / follow-up questions), followed by section and subsection.

1 • Object and definitions: what Jev-class models are, what the backbone constraints are, and what each paper is saying

> This chapter establishes the factual basis for the whole report and contains no judgments. § 1.1 is the object itself (interface primitives, the vendor's acknowledged zigzag, black-box architecture reconstruction, independent evaluations and audits); § 1.2 is the recipe matrix given by more than fifteen open-source reproductions, together with the small-model evidence; § 1.3 is where hybrid backbones (Gated DeltaNet class) break the structural assumptions that the Jev-style interface depends on, **survey only, no selection**; § 1.4 is the cross-paper definitional disagreement, which explains why the numbers in later chapters cannot be compared with each other directly.
>
> How to read: § 1.1– § 1.2 answer "what it is and who has reproduced it", § 1.3 answers "whether our backbone can carry it", § 1.4 answers "whether two numbers should be placed side by side when you read them".

1.1 Jev / System One: definition, interface primitives, architecture reconstruction and independent evaluation

> This section answers: what exactly a "Jev-class fast-thinking model" is, what interface it exposes, what architecture black-box probes infer, and what conclusions independent evaluations reach. All numbers were checked against the cached original texts.

1.1.1 Basic facts

Item	Content	Source
Vendor / release	TypeSafe AI (San Francisco, founded 2024); Jev released 2026-09-15; Jev 1.14 released 2026-10-14	Wikipedia, docs/models
Pricing	Input $0.042 / Mtok, output not billed (the output is not text)	docs/models
Context	64k per request, 32k per question; can be relaxed with the `TypeSafe-Danger-Zone-Long-Context: true` header; the number of questions per request is unlimited, constrained only by the two caps	docs/models
Training	"RLCD" (Reinforcement Learning for Calibrated Decisions), synthetic data only, details not disclosed; no fine-tuning on customer data, one set of weights serves all accounts	ML primer, docs/models, Wikipedia
Positioning	"For building AI-driven software rather than agents": code owns the control flow, deterministic rules and side effects, the model only answers narrow, structured judgments; "ask independent questions together, then combine the answers in code"	How to build

1.1.2 Interface primitives (the starting point of our later "logic primitives" discussion)

TypeSafe's System One API has only three question types, plus `state`:

Primitive	Returns	Constraints	Notes
Noul	Probability P(true) that a proposition is true	Optional `criteria`: when the yes/no boundary is subtle, use structured `true` / `false` descriptions, each with definitions and examples	primitives/advanced
Choice	A probability distribution over the options + a confidence	Options carry `criteria` (descriptions) and `examples`; JSON structure may be used	Same as above
Score	A probability distribution over ordered levels + a score + a confidence	`criteria` is an ordered description of levels from low to high, ⩾2 levels, API maximum 10 levels	primitives/score
State	The material the questions are about: text, a JSON object or an array, "may contain relevant context, examples and other information that helps answer"; the metaphor is "material handed to a panel of experts"	The documentation explicitly allows examples inside state	concepts/state

Confidence is defined as `confidence = (p_max − 1/n) / (1 − 1/n)`: it looks only at the highest probability, is 0 for the uniform distribution and 1 for certainty (docs/confidence). This means "confidence" is not entropy and not the top-2 gap, but only the height of the top-1 relative to the uniform distribution. Several papers point out that this definition is not the same thing as "the ability to detect errors" (see D § 4.1).

The official patterns form the primitive library for "composing small judgments in code":

- **Speculative fan-out**: ask several independent questions in parallel within one request (patterns/fan-out).
- **Composite scoring**: the results of several Noul / Score calls are combined in code as a weighted sum (patterns/composite-scoring).
- **Confidence routing**: one global lower bound (0.6 in the example) + a separate threshold for each action class; below the threshold, hand over to a human (patterns/confidence-routing).
- **Walking a taxonomy**: one Choice per layer, walking down the tree layer by layer in code (primitives/advanced).
- Engineering advice from the **Agent skill document**: "put the questions and the threshold constants in one file for easy review; agents are not good at writing questions, edit them collaboratively"; "if you only need the best option, take the highest confidence and do not set thresholds everywhere" (agent-skill).

The use-case map (use-case map) divides decision shapes into ten classes: classification, detection, scoring, routing, search, retrieval, ranking, verification, ML feature extraction, structured extraction. These ten classes are the reference taxonomy when we define the D3 primitive set.

1.1.3 The vendor's acknowledged "zigzag" (jev-1.13 model jaggedness)

The official documentation lists 9 failure classes and gives "use this instead" advice (model-jaggedness/jev-1.13):

#	Failure mode	Official advice
1	Literal reading	Write out precise conditions and criteria for each option
2	Math and numbers: cannot count reliably (character counts, word frequencies, list items), "recognizes the shape of the answer rather than tallying one by one"	Put arithmetic in code
3	Date and time comparison: reads dates as text; before/after, interval and window judgments are unreliable	Extract the components and compare them in code
4	Indirection: double negation, attributes of attributes, multi-hop	Reduce the number of hops; point directly at the relevant state
5	State with a large amount of irrelevant detail	Filter first; send only what the question needs
6	Adversarial content	Write precise prompts; test the boundaries before going live
7	Instructions contradicting the criteria	Align the criteria with the instructions
8	Choice option order: biased toward options placed earlier	Reorder the options and check that the answers agree
9	Generation	Use a generative model

This table is almost the "official problem list" for D2 / D4; third-party papers (below and in the per-dimension chapters) have quantified every one of these items and found more failures that the vendor does not list.

1.1.4 Architecture reconstruction (black-box probes)

Archer Hume's "Jev's Architecture Unmasked" (2026-09-17, link) concludes from API probes:

- **A causal Transformer** (the author guesses it contains sparse MoE but explicitly says "this is speculation"), with **a shared state prefix + mutually isolated question branches + direct probability readout**.
- **Evidence for branch isolation**: placing a secret code in a sibling question, the queried question gives probability 0.00; deleting that sibling question leaves the result unchanged; placing the same statement in state raises the probability.
- **Evidence for list-style readout**: with the target option placed first, 12/16 correct; placed last, 16/16 correct, which shows the options are not scored independently.
- **An irrelevant option changes the odds ratio between existing options**: after adding an irrelevant option, the mean log-odds between the two original options drops from +0.38 to +0.11, and all ten random blocks drop (mean −0.28, 95% interval −0.36 to −0.19). This rules out the mechanism "fixed independent logits + an unchanged softmax".
- **Serving characteristics**: server time is almost flat up to about 100 questions and rises linearly after that; measured in tokens, question text costs about twice as much as state.
- **The tokenizer** matched none of 192 public tokenizers across 415 probes; each digit is tokenized separately.

This reconstruction directly determines the structure of the open-source reproductions (see § 1.2) and the branch-isolation difficulty under hybrid architectures (see § 1.3).

1.1.5 Independent evaluations and audits: what Jev is good at and what it is not

1.1.5.1 Broad capability

Source	Setup	Main conclusions
2609.37647	37 datasets zero-shot, 346,009 requests costing under 10 dollars; compared against the exact next-token probabilities of Qwen3.8-27B and Gemma-4-E4B	IMDB / SST-2 / HellaSwag / ARC 95–99%, Belebele 86.7% across 122 languages; beats Qwen on 27 of 37 datasets and Gemma on all; all three degrade on low-resource languages, fine-grained/noisy labels, and rubric-based quality judgments. Choice probabilities are well calibrated; **binary probabilities rank well but are placed inaccurately relative to a fixed 0.5 threshold**; on UNFAIR-ToS, tuning the threshold on the training set raises micro-F1 from 0.50 to 0.75
2610.11978	13 multiple-choice benchmarks, compared against 19 LLMs	On knowledge and common sense, on par with frontier models; best on MMLU-Redux and ARC-Challenge; **MathQA 17.7 points below the frontier median, below all 19 LLMs**
2609.24574	Computational social science, 18 tasks, 7,977 items, compared against 19 LLMs	Behind the best LLM for the task on 14 of 15 tasks, median gap 11.6 macro-F1, but median cost 44 times lower; confidence better calibrated than the verbalized confidence of 16 of the 19 LLMs; on the "empathy in peer-support conversations" task, **highly confident but close to random**
2610.18529	Medicine, 15 datasets, 8,742 items	On text-driven tasks on par with leading models, **clearly weaker on number-driven tasks**; strong consistency under changes of question format; most models cannot recognize "the correct answer is missing" and do not use the optional abstain; **normalizing per-option binary scores retains higher accuracy than Choice scoring**
2610.33258	Content moderation on 142,658 human-annotated comments, compared against 14 frontier reasoning models	Zero-shot AUROC 0.83, statistically indistinguishable from 13 of the 14 frontier systems, at a cost 1–3 orders of magnitude lower; "confidence" mixes two different kinds of uncertainty, **human disagreement** and **decision risk**
2609.26550	JEV-as-a-Judge, compared against 16 generative and reward-model judges	On every task where the verdict can be read directly from the text, within 3 points of GPT-6 at 0.36% of its cost, median latency 0.15 s; **behind on math / code / logic tasks where the verdict must be derived**; a confidence-gated cascade beats GPT-6 by 0.9 points on 1,610 pairs at 41% of the cost
2610.35721	Clinical trial eligibility, 2,048 pairs	In batch mode, on par with Qwen Flash on TrialGPT / MIMIC-IV, significantly better on MIMIC-III (0.768 vs 0.534), with the advantage coming mainly from recall on the "insufficient information" class (56.3% vs 9.9%); latency 1.18–1.40 s vs 2.80–8.90 s, cost 53%–74% lower
2609.24052	Screening 499,500 Texas crash narratives, 195,857 coded with a 27-question schema	F1 0.908 against human labels; cost is determined by schema size, not narrative length; **calibration varies by model, not by paradigm, so every model must be audited**; recalibration reduces calibration error by 3.3 times
2609.23136	6G Open RAN intent interpretation	Jev meets the 1 s near-real-time budget on 99.8% of calls; two hosted LLMs reach 17.9% and 0%

1.1.5.2 Meta-audits and ecosystem

- **Early evidence audit** (2609.32160): reviews the 28 papers published within 9 days of release and concludes that "**typed readout itself has not yet shown an accuracy advantage independent of comparable label-probability readout**; Jev's clearest gains are in latency and cost, and an accuracy gap remains on hard tasks; in practice confidence is mostly used to decide when to hand off to a stronger model or a human". It gives a 14-item evaluation checklist.
- **Ecosystem** (2609.30216): 2,170 public GitHub projects as of 2026-09-22; attribute judgment and scoring are the most common uses; action selection, content filtering and model/tool selection vary widely across domains.
- **The practitioner guide** (2610.33775) proposes a five-family taxonomy: encoder classifiers, reward models, non-autoregressive decision engines, contrastive decision models, extraction models. Our direction falls in the **decoder-backbone** subclass of "non-autoregressive decision engines".
- **Leaderboards**: Decision Index 0.4 changed its main ranking to "Full score = 60% private tests on the same skills + 40% private decision tasks in new domains", with skill scores corrected for chance level (apolinario/decision-index); Jev leads version 0.2 (systemonemodels.org).

1.1.5.3 Stability and bias (overview; details in D § 4.1)

- Extremely high repeatability: on a 24-question cultural values survey with 288,000 responses, ICC 0.997 (2609.36399).
- But "same score, different verdicts": on ContractNLI, after changing the visible hypothesis, the requested answer and the output order, corrections and regressions cancel out, accuracy is unchanged while the decisions have changed (2609.27678).
- Broadly good on cognitive biases, but sensitive to irrelevant numerical information and option order (2610.19619).

1.1.6 Section summary: what Jev is and what it is not

1. **What it is**: a causal Transformer whose only interface is "state + typed question → probability", with a shared prefix, branch isolation and list-style readout, winning on calibration training and very low cost.
2. **What it is not**: not a more accurate classifier, not a model that can do arithmetic / compare dates / count, not a scorer immune to option order and irrelevant options, and not a model for which "confidence = knowing what it does not know".
3. **What it means for us**: Jev's value lies in the **interface contract** (probabilities that can be thresholded, composed and audited), not in the weights. As long as this contract is reproduced, a small model + domain data can beat it on specific tasks (see § 1.2). The truly hard part is making this contract **stay consistent** across the dimensions D1–D5 (see § 4.2).
4. **This item was revised in 17 and § 8.1**: saying "the value lies in the interface contract" is not enough, because it does not say what the contract is for. The sharper version is "the contract hands the **bookkeeping of the decision** to code and leaves the **semantics** in the model". Note that this is not "turning the decision into code": the step that maps an open-world unstructured situation onto a scaled observation record has no code form at all; it is semantic generalization, and it is the only irreplaceable part of the whole construct (§ 2.1.7). This version predicts which kinds of deployment succeed (the decision authority is in code) and which fail (the decision authority goes back to the model), and it agrees with the five-cell table of § 5.1.4. But it **cannot** be read further as "so the system is idempotent": the host Jev's own repeat jitter, the effect of menu composition on the metrics, the effect of harness planning and queuing, and "judgments that cannot be made without writing out intermediate steps" are four items that together restrict it to "shrinking the non-idempotent surface into a few boundable, priceable fields" (§ 2.1 § 2.1, § 8.1.1.2).

1.2 Open-source decoder reproductions and the recipe matrix (with small-model evidence)

> This section answers: which works reproduce the Jev interface with open-source causal language models (Qwen / Gemma, etc.), what backbone, readout head, loss, data, calibration and "branch isolation" method each uses, and how far small models (0.6B–9B) actually get. The encoder family (Laya, Verdict, Jeva, NDM, RubriX, Bongard, etc.) is outside our technical direction and is cited only for comparison.

1.2.1 A shared skeleton

All decoder reproductions follow the three-part structure reconstructed by Archer Hume:

1. **Shared state prefix**: state is encoded only once (KV cache / prefix forking / separate rows).
2. **Mutually isolated question branches**: a tree-shaped attention mask (attention-only backbones), or one row per question, or prefix forking (hybrid backbones, see § 1.3).
3. **Readout**: read the candidate-letter logits of the LM head at the last position (letter readout), or place a "pointer head" at the end position of each option / mean-span pooling, then softmax over the legal options, and finally multiply by a fitted temperature.

One important piece of prior evidence: **readout works without training**. The LDM paper finds that the readout of an untuned 27B model already reaches 97% of its post-tuning accuracy (2610.15501); FunDM uses 24 of the 36 layers of Qwen3-8B and a small head trained without labels to recover about 99% of multiple-choice accuracy, with 1.56–1.57 times throughput and about one third less peak memory (2610.17175); the conclusion of LLM-as-Jev is written straight into its title: "LLMs are already Jev-style decision models" (2610.02076).

1.2.2 Recipe matrix

Name	Backbone	Readout / head	Training objective	Data	Calibration	Branch isolation	Key numbers	Source
Kev 1.0	Qwen3.5-0.8B/4B/9B, Qwen3.8-27B	rank-16 LoRA + pointer head	Cross-entropy (details in the README)	`decision-v7`: 10,000 public-dataset samples + 896 generated policy samples + 1,680 samples over 60 rule structures	One temperature stored per checkpoint (0.8B 2.35, 4B 2.41, 9B 2.19, 27B 1.32; 27B/9B fitted on a never-trained held-out set)	Attention-only backbones use a tree mask + position reset; **Qwen3.5/3.8 ignore the mask because of DeltaNet, so each question gets its own row**	Kev-9B puts ⩾0.9 probability on a wrong answer in 2.4% of questions from new sources; date-fact supplementation (`KEV_DATE_FACTS`) raises deadline-policy questions from 0.80 to 0.90 (Jev 0.93); provides `/v1/systemone/permute` for multi-order testing	jaredpalmer/kev
Tev1	Qwen3.5-4B	Outputs a single answer letter (letter)	SFT	37,840 unique training samples, about 17 dollars, 25 minutes	Not stated	One question per request	880/1,000 on main decisions, 300/300 on policy transfer	togethercomputer/tev1, Together blog
Bespoke Nimble 9B	Qwen3.5-9B + LoRA	Enum field (1–255 choices)	SFT	**Contrastive data curation**: "change one fact so the answer flips"; 2,676 public training samples + 324 evaluation	Not stated	8,192-token context	Macro-F1 −12.0 under popularity shift in the finance domain (see D4)	bespokelabsai/nimble, model card
pplx-decider v1.2	27B (inherits v1.1-27B)	Multiple readouts per request	Details not disclosed	Not disclosed	—	Joint within a request, independent across requests	Decision Index 68.09 (joint request) vs 67.75 (separate) vs 62.30 for v1.1; **asking several facets of the same message jointly helps (PhishNChips +5.1), asking independent candidates jointly hurts (−0.2 to −0.4) → "separate requests for independent candidates"**	model card
OneJev 1.6	Qwen3.5-0.8B/4B/9B, Qwen3.8-27B (multimodal, vision tower frozen)	Answer position	Cross-entropy + Brier	99,193 questions (94,707 public), lr 5e-6, 16,384 tokens, 1 epoch	Brier term	—	One 720p screenshot on an H200: 0.8B 1 question 31 ms / 10 questions 51 ms; 4B 64 / 104 ms; 9B 81 / 131 ms	OmniJev/OneJev
Open-Jev (Zefan-Cai)	Qwen 2B/9B/27B	LoRA (27B rank 8) + scalar decision head	—	Frozen 148,639-row training mixture	Saved temperature	—	The 127,787-row internal evaluation of 27B v1.1 passed an independent audit	Zefan-Cai/Open-Jev
open-jev / TOD (daseinlabs)	Gemma 4 12B + LoRA; or a small head on frozen Gemma 3 4B	Head	—	User-supplied data	—	—	cookbook 01: 84.2% / 66.3% zero-shot → 97.0% / 100% (team / priority)	daseinlabs/open-jev

Name	Backbone	Readout / head	Training objective	Data	Calibration	Branch isolation	Key numbers	Source
qwen-rlcd	Qwen3.5-0.8B-Base (18 DeltaNet + 6 attention)	letter / likelihood / PMI / `<|read|>` head	Inference only (training shelved)	—	—	**Prefix forking** replaces the tree mask; forking gives 8–19 times speedup over the full forward; on 0.8B, forking vs sequential error 4.8e-5; the gradient check through the cache has not been done	shamazharikh/qwen-rlcd	
Mapika decider	Stock Gemma-4-31B-it / Qwen3.6-27B (weights unchanged)	Readout	No training	—	**Temperature fitted by option count** $T(n)=\max(0.05, 10.124 - 1.633 \cdot \ln n)$	—	Gemma-4-31B-it Decision Index 57.33, 2nd of 70; Qwen3.6-27B 51.35, 8th; decider-31b v2 (merged fine-tune) 63.90	Mapika/decider
StartLux-Decision v1.1	4B / 9B / 27B (two variants, Instant and AutoThink)	Single-pass (Instant uses bidirectional attention in the full-attention layers); AutoThink generates reasoning when the top probability is < 0.9	Three stages: shared decision ability → differentiation → reasoning supervision + RL for AutoThink	After adding the scoring / safety / preference / logic families in stage 2, +8.6 points on the 18 corresponding DecisionPivot tasks, only +0.3 on out-of-family tasks	**Policy-aware calibration**: confidence-accuracy gap ⩽2.7 pp across 184 settings	—	AutoThink over Instant: +18.4/+12.4/+11.5 points of paired accuracy at 4B/9B/27B; 27B AutoThink 86.9%, Instant 75.4%; 27B Instant ranks 1st among 174 Decision Index entries (2026-10-16)	2610.17309, StartLuxLabs
Releasing the Inner Jev	Gemma 4 31B / 12B, Qwen3.6-27B, Qwen3.5-4B	LoRA + a decision head initialized from a slice of the LM head + a learned correction term	Teacher labels (Qwen3.5-122B-A10B, Qwen3.6-27B); **soft labels "may" not help**	OpenDecisions	—	—	Gemma 4 31B without fine-tuning 79.6 vs Jev 80.1, 84.4 after fine-tuning; **LoRA is the source of the improvement, insensitive to rank**	2610.17205
LLM-as-Jev	4B frozen (with 0.6B experiments)	Reads logits directly; list-wise loss + KL regularization	—	—	—	—	JevBench 94.8% with native thinking under a token budget enabled; 0.6B on Banking77 (77 classes) from 22.2% → 60.5%; letter readout cannot handle 77 classes	2610.02076
LDM	Three base models (including Gemma-4-31B)	One prefill; at the last position the LM head normalizes only the legal-option logits	DDS (diverse decision supervision) + SPRD (self-questioning reasoning distillation)	DecisionVista-Synth / Human	—	—	+1.8 / +1.5 points over step-matched baselines; 182 times median latency speedup over reasoning rollouts; routing recovers 92% of the reasoning advantage with 27–58% of the reasoning tokens	2610.15501
SII-Decision	4B / 27B (one full-rank, one LoRA)	—	**Reasoning-to-Readout self-distillation**	—	—	—	27B ties Jev on JEVal; a typical query takes about 0.1 s	2610.03935

Name	Backbone	Readout / head	Training objective	Data	Calibration	Branch isolation	Key numbers	Source
Chinese Decision	Qwen3.5-0.8B/2B/4B	Candidate-letter logits at the last position, normalized only over legal options	1 epoch, 13.048M decisions	Same as left	Temperature fitted on the validation set; Q8_0 quantization	Each question is compiled into a shared instruction + candidate prompt	CJ-Bench 307,900 decisions: 4B overall 85.97%, general 75.69%, medical/legal/finance average 84.00%	2610.23170
Intern-Decision	0.8B / 2B / 4B multimodal	One decision placeholder per field inside a complete JSON skeleton, read out with the native LM head	Autoregressive masked LM	—	Temperature scaling: Brier 0.628→0.550, ECE 0.213→0.089	All fields evaluated in a single causal forward	4B seven-suite average 90.02% vs 88.74% for Jev 1.13; a three-field 289-token request takes 44.16 ms on an RTX 4090; confidence handoff takes JevBench-Hard from 73.87%→87.39%	2610.22131
AnyJev	Frozen backbone	**Question-agnostic head**: affine map into the final-layer space, reusing the original output head	—	—	—	—	Across 11 models, reversing the option list changes 33% of answers on average; rotation lowers the order flip rate from 0.33 to 0.14 / 0.18	2610.00831, nokia-applied-research/AnyJev
OpenJev-RLCD	Qwen3-1.7B	Explicit probabilities	"Calibrate first, then reinforce": the same proper scoring rule as the reward	Two real reasoning tasks	Intrinsic	—	Compared with SFT / RFT / GRPO (each with temperature scaling), accuracy equal or better, selective prediction better in all cases; on GSM8K answer verification a single query can decide 81% of items at ⩽5% error, GRPO 19%	2609.38850
PACT	—	Single-token typed decision	Pairwise anchored calibration fine-tuning	—	—	—	Accuracy 84.6 vs 85.2 for the published recipe (no significant difference); lowest position bias, relabeling 9.8 vs 13.8%; variance across seeds halved, NLL −26%; "the value is in robustness, not accuracy"	2609.35865
this-that-model 1.0	2B	Reads directly from the hidden state at a designated position, restricted to the declared option set	—	Internal suite of 42 families	Brier 0.042	All questions in a request answered in the same forward	30.9 ms on a notebook GPU; third-party 68-question queue 0.941 vs Jev 0.765	2609.23886, FLock-io
System X	Qwen3-0.6B → v4 Qwen3-1.7B LoRA	LLM-as-Jev readout + a "text reply" option; when it wins, generation continues from the KV cache	—	Reason-9k + Bitext	ECE 0.292→0.169	Scores on a prompt-cache copy; differs from full-sequence scoring by 1.4e-6	v4 JevBench 55.0%→61.0%; cache reuse saves 15–39% latency	2610.25456
Our Own Jev	Frozen backbone + rank-16 LoRA + pointer head	Answers three questions in one forward	—	148K agent coding sessions	—	—	Replaces a frontier judge costing about 1,000 dollars per pass; 1,000 sessions in about 3 minutes on one GPU for about 0.06 dollars	2610.29052

Name	Backbone	Readout / head	Training objective	Data	Calibration	Branch isolation	Key numbers	Source
Decision Model Factory	Selectable backbone	Computes probabilities only for the allowed answers, reusing the shared context	Short recipe: train→calibrate→validate→evaluate	—	—	A dedicated runner gives more than 2 times speedup on multi-question loads	On Japanese tasks all five models beat their backbones, three beat the public best	2610.34956
SecJev	Kev-based 0.8B–9B	Kev single-pass candidate scoring	Scenario-weighted training	SecJev-Corpus: 14 tasks, 8 sources	—	—	**SecJev-0.8B beats the general-purpose Kev-9B by 20.51 pp of task macro-accuracy**; 0.8B/4B are close in accuracy and latency to answer-only generative fine-tuning, with lower peak memory	2610.03073
AgentDecision-4B	4B	—	Trained on the states the agent reaches after its own mistakes, with Jev's answers as labels	DecisionGym	Thresholds fitted per choice	—	Out-of-distribution tasks 91.5% vs Jev 82.0%; in-distribution makes the agent 81.1–89.0% cheaper and up to 2.5 times faster	2610.21412
SanSi / LEV (recurrent)	LMs pretrained to be recurrent (LEV 2.6B, etc.)	Reads the options after each recurrent step; all steps trained jointly	Proper scoring rule	—	A small stopper fits thresholds on unlabeled requests	—	SanSi 72.0% on 10,027 decisions from 59 sources, +13.5 over a same-shape non-recurrent model; LEV 2.6B JevBench from 63.2%→76.2% going from depth 1→4; on-demand depth saves 15% compute	2610.07730, 2610.20951
Block-causal	—	Candidate-agnostic block-causal attention + position reset	—	—	—	Isolation + position reset	Isolation alone takes the test flip rate from 8.30%→3.82%, OOD 7.54%→3.71%; resetting positions alone is worse	2610.01601
Prism-1	Qwen3-8B / Gemma 4 12B	Position- and opponent-agnostic scoring	—	—	—	—	One shuffle under letter readout changes 20.0% / 11.9% of answers; Prism-1 has zero flips	2610.22066
Runoff	Any (training-free)	First scores each option alone to keep a few candidates, then lets each candidate occupy each position for a list-wise comparison	None	—	—	Two forwards	Order-independent by construction; zero flips for two open-source models on five datasets, versus 9.3% average for cyclic shifts; 3–4 times faster at 20 options	2610.18233
ufakzeka-karar	Turkish	Option-blind head (order-invariant)	—	—	—	—	—	2610.06744
JEV entailment reranking	NeoHorse-1-4B	**mean-span pooling** replaces the option-end pointer	**Selective adaptation**: LoRA kept on the attention and Gated DeltaNet projections, FFN adapters removed	MC11	—	—	Trainable parameters more than halved with an accuracy gain	2610.35654

Supplementary training theory: an exact risk calculation under a Gaussian surrogate shows that **the student inherits the teacher's bias; more student data cannot remove it, and agreement with the teacher cannot expose it**; a fixed representation lowers risk only when the sample size is below some threshold; temperature scaling removes all excess risk only when the direction of the teacher's coefficients is correct (2610.26797). This agrees with the empirical

finding in "Releasing the Inner Jev" that "teacher labels help, soft labels may not", and it is the origin of the "teacher bias inheritance" sub-dimension in D4.

1.2.3 Can small models actually do it: evidence summary

Evidence	Conclusion
SecJev-0.8B > Kev-9B (+20.51 pp) (2610.03073)	**Domain data > parameter count**
AgentDecision-4B out-of-distribution 91.5% > Jev 82.0% (2610.21412)	**Training on the states of one's own trajectories > a general-purpose model**
0.6B Banking77 22.2% → 60.5% (2610.02076)	Small models can learn multi-class list-style readout
Chinese Decision 4B medical/legal/finance 84.00% (2610.23170)	1 epoch and 13M decisions suffice for a usable small Chinese decision model
this-that 2B third-party 68 questions 0.941 > Jev 0.765 (2609.23886)	Single source, small sample, but the direction agrees
JevCascade: 150-class full menu 69.0% → curated 5-candidate menu 95.4%, **a span of 26 points, larger than the gap between 0.8B and 4B**; menu interventions are predictable to within 4.2 points (2610.07716)	**Candidate-set design is more first-order than model size**
Microtask eligibility gap: off-the-shelf Qwen3 0.6/1.7/4/8B pass the practitioner threshold on 0/16 of 4 agent microtasks, 4-bit quantization changes nothing, Llama-3.x fails 12/12; recommends "put the SLM behind a cheap baseline that already meets the bar, and use it only where the baseline fails" (2610.00025)	**Untrained small models are not enough**: either train them or place them behind a baseline
Measure Before You Train: on 739 real emails, logistic regression with about 200 labels is the most accurate on every field, and few-shot prompting cannot close the gap; but when "the most accurate system" is composed into actions it most often ignores emails that need handling (18% vs 4% for the best LLM); a cost-chosen threshold on the noisy probabilities halves this and makes it the cheapest system (2610.32955)	**Measure before training; accuracy is not the product metric; the threshold is part of the product**
Jev-like (Qwen3-1.7B + LoRA + 37 open-source datasets): in-domain confidence tracks accuracy almost exactly; out of domain, a 0.99 gate validated in-domain at 99.85% fails (mustafasemi-ai/jevlike)	Small-model calibration **does not transfer** to new distributions
Continued post-training of small models: SDFT depends on strong ICL and instruction-following ability and fails on small models; forgetting and insufficient acquisition correlate linearly with distribution shift (2610.13113)	Continued training of small decision models must control distribution shift, which relates directly to D1

1.2.4 Consensus and disagreement read from the recipe matrix

Consensus

1. LoRA + a small readout head + a temperature suffice to reproduce the interface; full rank is not required.
2. Hard labels from a teacher (a larger model) work; contrastive / counterfactual data (Nimble, Certo, the DecisionPivot training set) works for rule sensitivity.
3. Calibration must be fitted on a **held-out distribution** (as Kev-27B/9B do), and fitting by **option count** (Mapika) or **deployment policy** (StartLux) is more stable.
4. Order bias must be solved **structurally** (block-causal, Prism-1, Runoff, AnyJev rotation), not by shuffling the training data (Prism-1 shows that published decision fine-tunes still flip).

Disagreement / undecided

1. Whether soft labels help (Releasing the Inner Jev says "may not help", OneJev uses Brier, OpenJev-RLCD uses a proper scoring rule as the reward).
2. Single-pass vs recurrent (SanSi / LEV) vs on-demand reasoning (StartLux AutoThink, Selective Reasoning, LDM routing): all three paths claim large gains on "hard problems", but there is no controlled comparison on the same benchmark and the same backbone.
3. Branch isolation **at training time** under hybrid backbones (Gated DeltaNet): Kev sidesteps it with separate rows, qwen-rlcd's prefix forking is inference only, and no work reports results for "training with a tree mask on a hybrid backbone" (see § 1.3).

1.2.5 What it means for us (analysis only)

- Reproducing the interface itself is not the hard part; **domain data, candidate-set design, per-choice fitted thresholds and held-out calibration** are where the scores come from.
- Small models (0.6B–4B) can beat general-purpose 9B–27B models and Jev in narrow domains, but **not without training**, and calibration does not transfer.

- Almost every work in the recipe matrix treats "examples / definitions / rules" as **training data** rather than **inference-time input**; this is exactly where the gap for D1 and D3 lies.

1.3 Backbone models and hybrid-architecture constraints (survey only, no selection)

> Protocol: **survey only, no selection**. The user asked that "anything with mamba has this class of problem; survey first, select later". This section organizes: the three structural assumptions the Jev-style interface depends on; at which step hybrid backbones break them; how each open-source reproduction works around this; what can be borrowed from the inference-systems literature; and which questions nobody has studied.

1.3.1 The Three Structural Assumptions the Jev-Style Interface Depends On

Archer Hume's probes of the hosted Jev (architecture) and all open-source reproductions (§ 1.2) share the same skeleton:

1. **Shared state prefix**: the state is encoded once and all questions reuse its cache. "Causal attention describes which positions may use which information, not the order in which input tokens must be executed", so the prefill phase can be parallel.
2. **Question branch isolation**: each question (with its options) can see only the state and itself, never its sibling branches. Isolation gives the interface its semantics: "asking whether the customer is angry should not change which queue the ticket goes into". Probe evidence: a secret placed in a sibling branch → 0.00; placed in the state → the probability rises.
3. **Position reset / readout**: each branch's position IDs restart from the end of the state; the readout is taken at the branch's last token.

On an **attention-only** model, 1–3 can be implemented in one forward pass with a tree-shaped attention mask + position-ID reset (Kev README: "on attention-only models, rows and the mask give identical probabilities"; Kev). The block-causal paper shows that isolation alone has value: isolation by itself takes the test flip rate from 8.30%→3.82% and OOD from 7.54%→3.71%, while resetting positions alone is worse (2610.01601).

1.3.2 At Which Step Hybrid Backbones Break the Assumptions

Fact	Source
Qwen3.5-9B-Base: 32 layers = **24 Gated DeltaNet (linear attention) + 8 full attention**, hidden 4,096	Kev-9B model card
Qwen3.5-0.8B-Base: 24 layers = **18 Gated DeltaNet + 6 gated full attention**	qwen-rlcd
"Qwen3.5 and Qwen3.8 interleave attention layers with Gated DeltaNet layers; the latter are recurrent and **ignore the attention mask**"	Kev README
"A tree-shaped attention mask cannot isolate sibling branches inside the recurrent layers"	qwen-rlcd
The recurrent layers of hybrid models **update state in place**; cache entries with partial prefix overlap cannot be rolled back and can only be hit by exact match	Marconi 2411.19379
Cache state is **history-dependent**: the same token sequence yields different cache states under different sequences of cache operations, so prefix matching is not sufficient to guarantee correct reuse	Cache Engineering 2610.28577

Verified layer configurations (read directly from each repository's `config.json` on Hugging Face on 2026-10-28; all are `Qwen3_5ForConditionalGeneration`, `full_attention_interval = 4`, i.e. 1 full-attention layer and 3 Gated DeltaNet layers in every 4 layers; vocabulary 248,320, context 262,144):

Model	Layers	Full attention / linear attention	hidden	Attention heads / KV heads	FFN
Qwen3.5-0.8B (Base / Instruct identical)	24	6 / 18	1,024	8 / 2	3,584
Qwen3.5-2B (same)	24	6 / 18	2,048	8 / 2	6,144
Qwen3.5-4B (same)	32	8 / 24	2,560	16 / 4	9,216
Qwen3.5-9B (same)	32	8 / 24	4,096	16 / 4	12,288
Qwen3.5-27B, Qwen3.6-27B, **Qwen3.8-27B**	64	16 / 48	5,120	24 / 4	17,408

The three generations of 27B have exactly the same shape; they differ only in weights. The Qwen3.8 series **has no size below 27B** (Hugging Face only has `Qwen3.8-27B`, `Qwen3.8-2.4T-A95B` and `Qwen3.8-Flash-Next`, the latter two being MoE), and Qwen3.6 only has 27B and 35B-A3B. So the "small to large" same-architecture ladder can only be Qwen3.5-0.8B / 2B / 4B / 9B → 27B (any of 3.5 / 3.6 / 3.8), which matches the four levels actually used by reproductions

such as Kev and OneJev (§ 6.1.5). **The endgame size is fixed at the 27B / 35B level** (2026-10-28 terms), and there is a fact that must be separated first: **27B is dense, and Qwen has no dense 35B: every 35B is the `35B-A3B` MoE (about 3B active)**. If the endgame lands on 35B-A3B, then beyond the three structural assumptions below there is one more item, "whether per-token expert routing introduces new coupling between isolated branches", on which this report has **zero hits** (§ 6.1.5). The local `/mnt/cephfs/models/` already holds complete, unquantized checkpoints of all three 27B models, Qwen3.5-27B, Qwen3.6-27B and Qwen3.8-27B, so a "same shape, cross-generation weights" comparison can be run directly. This turns the variable "small end and large end differ in generation" into a measurable quantity: the difference of the same decision evaluation between 3.5-27B and 3.8-27B is the difference brought by cross-generation pretraining, independent of size.

Implication: the recurrent layers mean assumption 2 (isolation) cannot be achieved by a mask: tokens of branch A "flow into" branch B through the recurrent state. Assumption 1 (shared prefix) still holds, but what is reused is a **recurrent-state snapshot**, not KV.

1.3.3 The Three Workarounds Used by Open-Source Reproductions

Method	What it does	Cost / evidence	Source
Row-by-row execution (Kev)	One row per question: "state + that question", with the same positions as the tree-mask scheme; the state cache is computed once and reused by every row; the server batches by a token budget (one row of at most 16,384 tokens per forward pass), "memory does not grow with the number of questions, and answers do not depend on batching"	Isolation is exact; every row has to re-read the state's recurrent state; `forward()` (the source of all published numbers) and the server agree within fp32 rounding	Kev
Prefix forking (qwen-rlcd)	Prefill the state and keep the cache (KV for attention layers + conv and recurrent state for DeltaNet layers); copy along the batch dimension, one row per branch; right-pad all "question + answer" branches and run them as one batch at the same position offset; read the hidden state at each branch's last real token	8–19× faster than one forward pass per branch; chunking branches bounds memory; on real 0.8B weights the fork vs sequential error is 4.8e-5 (tolerance 1e-3), and the negative control with the recurrent state zeroed gives 6.3; **"the gradient check through the cache is deferred to training time"**: training is on hold	qwen-rlcd
Shared-context anchoring + candidate prefix sharing + decision-guided pruning (FastJEV)	Reuse the recurrent initial state and skip the unused final recurrent cache; keep the intermediate states that later branches need; prune depth by relative score change on a small unlabeled set	Candidate depth −43.75–45.83%, retaining 93.66–97.52% of task score; **"candidate prefix sharing reduces repeated computation but increases latency"**; "sharing granularity and execution scheduling must be designed together"	2610.11379

Two further architecture-related choices on the training side:

- **Selective adaptation**: keep LoRA on the attention and Gated DeltaNet projections and drop the FFN adapters; on MC11 this beats the all-module baseline with trainable parameters cut by more than half (2610.35654); Kev-9B's LoRA is also placed on the attention, MLP and DeltaNet projections (45.4M parameters).
- **StartLux Instant**: bidirectional attention over the prompt inside the **full-attention layers** (llama.cpp non-causal flag, vLLM plugin); only the attention layers are changed, the recurrent layers are untouched (StartLux). This shows that on a hybrid backbone the freedom to "change how the model reads" is concentrated in the few attention layers.

1.3.4 What Can Be Borrowed from the Inference-Systems Literature

Work	Applicable layer	Key point	Source
Hydragen	Attention	Compute attention separately for the shared prefix and the unique suffixes	2402.05099
DeFT	Attention	IO reuse and load balancing for tree-shaped KV	2404.00242
Block-Attention	Attention	Documents are chunked and their KV computed independently; the last chunk sees everything	2409.15355
IPPD	Attention	Virtual position IDs + mask manipulation to answer multiple same-context questions in parallel within one prompt, without changing the architecture	2609.05707
Visual Jev	Both (implemented by batching isolated suffixes)	Image and shared context encoded once, isolated question suffixes batched: at N=32, warm state is 8.9× faster than serial and 3.4× faster than batching with prefix recomputation, with higher peak memory; typed heads show no consistent advantage	2609.25845
Marconi	Recurrent	Admission / eviction policies for prefix caching in hybrid models: token hit rate up to 34.4×, TTFT −71.1%	2411.19379

Work	Applicable layer	Key point	Source
Cache Engineering	Both	The agent directly drops / repositions the cache; serving challenges from history dependence	2610.28577
FastJEV	Both	See 8.3	2610.11379

The attention-layer works (the first four) solve different faces of **the same problem**: IO for the shared prefix on a tree. They all assume the mask is effective, so they only cover the attention layers of a hybrid backbone; for the recurrent layers there are only two papers, Marconi and FastJEV.

1.3.5 Attention-Only Backbones as the Control

Attention-only backbones used in reproduction work: the Qwen3 series (Qwen3-1.7B in OpenJev-RLCD; Qwen3-8B in FunDM; Qwen3-0.6B–8B in System X / SHIFT-JEV / Microtask), Gemma 3 (daseinlabs' decision head on a frozen Gemma 3 4B), Qwen3-8B / Gemma 4 12B (Prism-1), Gemma 4 31B (Releasing Inner Jev). On these backbones the "rows and mask are equivalent" statement in the Kev README holds, and the tree-mask scheme can be trained directly (gradients through the shared prefix have no recurrent-state problem).

The gap in cost comparison: no work compares "hybrid backbone + row-by-row / forking" against "attention-only + tree mask" at equal parameter count on decision accuracy, calibration and latency. The existing numbers are scattered: OneJev (Qwen3.5 hybrid) 0.8B 31/51 ms, 4B 64/104, 9B 81/131 (OneJev); Intern-Decision 4B 44.16 ms (2610.22131); this-that 2B 30.9 ms (2609.23886); the measurement conditions all differ.

1.3.6 Matrix for This Section

Question	Attention-only	Hybrid (DeltaNet)	Evidence status
Shared state prefix (inference)	KV reuse, mature	Recurrent-state snapshot reuse (Kev, qwen-rlcd, FastJEV)	Both feasible
Branch isolation (inference)	Tree mask, one forward pass	Row-by-row / prefix forking (multi-row batch)	Feasible on hybrid; memory and scheduling costs have numbers (8–19×, FastJEV's latency warning)
Branch isolation (**training**)	Tree mask, gradients flow naturally	Forked cache needs gradients through the copied recurrent state	**qwen-rlcd explicitly defers; no literature at all**
Prefix-cache serving	vLLM etc., mature	Marconi-style admission / eviction	One paper for hybrid
Changes to the reading mode (bidirectional etc.)	All layers changeable	Only attention layers changeable (StartLux)	Existing practice
Adapter placement	Anywhere	Attention + DeltaNet projections, drop FFN (2610.35654)	One paper
Decision quality of hybrid vs attention-only at equal parameter count	—	—	**None**

1.3.7 Blind Spots (To Be Filled Before Selection)

1. **Branch isolation at training time** has no literature on hybrid backbones: Kev trains "row-by-row" (one row per question), which is equivalent but costs repeated encoding of the state; qwen-rlcd's forking scheme is "deferred" for training. Either accept the row-by-row training cost or prove that gradients through the copied recurrent state are correct; neither has public numbers.
2. **Whether recurrent layers are more sensitive than attention to "irrelevant detail in a large state"**: the D2/D4 evidence (Jev retrieves without error within 20k tokens, but multiple updates interfere with each other) all comes from the hosted Jev, whose architecture is unknown; there is no open-source hybrid vs attention-only comparison.
3. **Structural mitigation of order bias** (block-causal, Prism-1, Runoff) and how to implement it on hybrid backbones: block-causal depends on the mask; Runoff is two forward passes (architecture-independent); Prism-1's implementation details are unverified.
4. **Quantization and fused kernels**: Kev uses fused DeltaNet kernels + CUDA graphs on CUDA, with bf16 deviating from fp32 by at most 0.03/0.05; Chinese Decision's Q8_0 changes top-1 by 0.4–1.1%. These are all single-point numbers, with no breakdown by layer type.

5. ~~Qwen3.6 / 3.8 layer configurations have no verifiable numbers~~: verified from `config.json` on 2026-10-28 and listed in 8.2; the remaining gap is that Qwen3.8 has no small size, so the small→large ladder must cross generations (Qwen3.5 small models → 3.8-27B), and the cross-generation pretraining / post-training difference is itself a variable that no literature controls for.

1.4 Cross-paper comparison of problem definitions: the same word, different definitions

> Protocol: the user asked to "summarize the problem definitions in detail; the academic definitions in different papers are very likely to differ". This section only organizes and compares at the level of definitions and does not propose our own definition (that is the next stage's work), but § 1.4.10 lists "the points of disagreement we must answer before defining the problem". All quotations were checked against the cached full text / abstract / documentation; definitions that could not be checked are marked "unverified".

1.4.1 Why Definitions Diverge

Jev was released only six weeks ago, TypeSafe has published no training details or formal definition, and the community is doing three things at once: reproduction (which needs operational definitions), auditing (which needs testable definitions) and theorizing (which needs axiomatic definitions). The three motivations give mutually incompatible definitions of "decision model", and **the same metric name (ECE, flip rate, confidence, RLCD) refers to different quantities in different papers**. Definitions must be aligned before numbers can be compared across papers; this is also the root of README § 4 "numbers across papers are not directly comparable".

1.4.2 What the "Decision Model" Is Called and What It Refers To

Name	Key points of the definition	Source
System One model (TypeSafe)	"trained for calibrated decisions: their probabilities are optimized against outcomes to reflect uncertainty. Calibration is measured across groups of predictions; it does not guarantee that an individual answer is correct"; "designed to return stable answers across repeated evaluations" (Self-consistent); writes no replies, produces no code, generates no explanations; classified under "Machine Native Intelligence"; the training path is RLCD (the "third post-training path" alongside RLHF and RLVR)	system-one, how-to-build, ML primer
Typed decision model (TDM)	The generic name in the audit papers; emphasizes the "typed" interface rather than the training method: "Type-Safe Is Not Error-Free", "Labels Override Definitions", "TDMs under Prevalence Shift", "How Reliably Do TDMs Use Fallback", "open TDMs"	2609.32160, 2609.26758, 2610.02586, 2610.26325, 2610.19856, 2609.31142
Typed decision models, also called System One models	"return such probabilities without generating text"; the object of interest is "decision gates": "Software that hands branching decisions to a model needs a declared option and a probability it can threshold"	2610.00346
Large Decision Model (LDM)	"processes the task in one prefill, projects the last input hidden state through the LM head, and normalises only the allowed option logits"; the Choice 2–8 and Score 3–6 limits "are implementation choices, not part of the LDM definition"; calls the training **decision tuning** and the output **readout**	2610.15501
General decision model	A decision instance is a quadruple (state, instruction, criteria, gold): "state provides the information needed for the decision, instruction specifies the decision goal, criteria defines the legal decision space"; the readout is called **First-Token Decision Readout** (the distribution at the answer position restricted to the legal option labels; choice / noul / score obtained in one forward pass)	2610.03935

Name	Key points of the definition	Source
Narrow Decision Model (NDM)	Eight desiderata: D1 typed output (a distribution over a closed set, never free text); D2 calibration ("among answers given probability p, about p are right", measured, adjusted and reported separately per question type); D3 single-pass (all questions and options of one request answered in one forward pass, the state encoded only once); D4 option-permutation invariance (reordering options only permutes the output distribution, nothing else changes); D5 question isolation (an answer does not depend on other questions in the same request or their options); D6 abstention (each Choice may include an explicit "none of these", whose probability is calibrated separately); D7 read, not memorize (options are recognized by their descriptions; unseen options and unseen organizational structures can be handled); D8 (a structural item, not excerpted). Claims to give "a formal statement of the System One decision contract", including **declared / undeclared abstention mechanisms** and a **set-valued training objective**, and requires that "the evaluation protocol for the accept-or-reject mechanism be completed before the held-out data is opened" (contract tests)	2610.25553
Prefill-only decision model	"score every candidate in a menu in a single forward pass and never decode"; the core property is **readout stability** (Definition 1: the distribution after restricting the menu to a subset equals the original distribution renormalized on that subset)	2610.07716
System 1 decision models	"architectures that map an input directly to a typed, structured decision in a single forward pass"; five-family taxonomy: encoder classifiers, reward models, non-autoregressive decision engines, contrastive decision models, extraction models	2610.33775
System 1 / can it be left to System 1	The question is "can it be left to a fast model that answers with calibrated probabilities in a single forward pass, or does it still need a generative LLM"	2610.21244
Probability-only model	"return calibrated probabilities for fixed choices in milliseconds and generate no text"	2610.09188
Candidate-scoring decision model	"return a probability distribution over a presented set of answers"; three training ingredients (soft targets, frozen backbone, temperature) determine "what these probabilities estimate"	2610.26797
Decision models (FunDM)	"score candidate answers or actions": action selection and answer selection are treated as the same class	2610.17175
Direct-decision model	"JEV-like direct-decision models", emphasizing skipping reasoning and emitting the distribution directly; used in the ordinal-bias study	2609.38827
One-pass decision model	A conclusion-style definition: "best used as fast selectors over a state that a reasoning model has computed and checked", i.e. the decision model is a **selector**, not a **calculator**	2610.24929
System 1.5	A looped model: each loop step reads the options once; "one model serves every budget from 1 to 8 loops"	2610.07730
Atomic action	"fast, single-step decisions such as classification, semantic endpointing, Boolean checks, and scoring, formulated as constrained generation with single-token outputs" (still generative decoding, just a single token)	2609.36115
Native typed decision-making	An action encoder embeds each candidate action description in parallel and scores it against the LLM's internal state to obtain a typed action distribution	2609.36116
Machine intuition	"recognising patterns and judging situations without explicitly unfolding every intermediate step", as an independently trainable capability	2609.39111
Typed probabilistic assessor	Laya's self-description: answers choice / noul / score questions over workflow state	2609.33843
System One with a natural-language interface	Opposes "structured schema objects encoding state, intent and candidates"; argues that both the request and the candidates should be expressed in natural language, with a candidate space that can be an open set in the millions	2610.11316
"LLMs are already Jev-style decision models"	A frozen chat model with logits read directly already satisfies the definition; "decision model" is a readout mode, not a model category	2610.02076

Points of disagreement: (1) whether the subject of the definition is the **interface** (TDM, LDM, JEVal), the **properties** (the eight NDM items, readout stability), the **training method** (TypeSafe's RLCD, "optimized against outcomes") or the **use** (decision gate, fast selector, can it be left to System 1); (2) whether "generates no text" is part of the definition: Koa-decision and SALSA generate a single token, System X continues from the KV cache, StartLux AutoThink reasons first and then reads out, and all call themselves decision models; (3) whether the option limit (255 / 2–8 / millions) belongs to the definition; (4) "calibration" is a description of the training objective in TypeSafe, a contract clause in NDM, and a hypothesis to be tested in the audit papers.

1.4.3 Formalization of the Request / Contract

Formalization	Content	Source
TypeSafe's three primitives	Noul (a single probability), Choice (a distribution over ⩽255 options), Score (a distribution over 2–10 ordered levels); one request = one state + multiple questions; state and questions are separated, and examples may go in the state or in the question's examples field	§ 1.1, state docs
Kev's question schema	`{"type": "choice", "instructions": …, "criteria": {name: definition, …}}`; options are given as "name + definition" pairs	jaredpalmer/kev
JEVal quadruple	(state, instruction, criteria, gold); criteria defines the legal decision space	2610.03935
LDM	Choice with 2–8 named options, yes/no, Score with 3–6 levels; the limits are implementation choices	2610.15501
NDM contract	Closed-set distribution + calibration per question type + single-pass + permutation invariance + question isolation + explicit abstention + reading option descriptions; declared / undeclared abstention; set-valued objective; a synthetic-data engine with "validity guards"; "specification first"	2610.25553
Probability Contracts	Treats the Event (Noul) and Choice interfaces as two possibly inconsistent "probability contracts": ask the same fact through both interfaces, then compare "canonical posterior error", "complement residual" and "coarsening residual"; Jev's Event and Choice interfaces **change the binary action in 32.8% of valid pairs** at a 10% latency cost	2609.37470
Multiple questions over shared state	Verdict: "answering many small typed questions about one piece of state"; a structured mask makes the state encoding independent of the questions, the questions independent of each other, and the option order irrelevant "by construction"; Intern-Decision places one decision placeholder per field inside a complete JSON skeleton	2610.20778, 2610.22131
Single-token constrained generation	Koa-decision's atomic label tokens; SALSA's class→token mapping	2609.36115, 2510.22691
Natural-language request + natural-language candidates	MetaEncoder: no schema, candidates can number in the millions	2610.11316
Action space	FunDM / Dyad / AgentDecision: candidates are action descriptions; DecisionGym's three levels (L1 isolated questions, L2 step-level over recorded trajectories, L3 controlling a running agent)	2610.17175, 2609.36116, 2610.21412

Points of disagreement: whether Event and Choice are the same contract (2609.37470 shows they are not on Jev); when "option = name + definition", which one the model follows (2609.26758: the name; 2610.02586: labels override definitions); whether multiple questions should be "isolated" (NDM D5, Verdict) or "jointly beneficial" (pplx-decider: asking multiple facets of the same message jointly +5.1, asking independent candidates jointly −0.2 to −0.4, see § 1.2).

1.4.4 What "Confidence" Means

Definition	Content	Source
TypeSafe `confidence`	Compresses the distribution shape into one number in 0–1: "1 when all the probability is on one outcome and 0 when the probability is spread evenly"; Noul has no such field; § 1.1 gives its formula, which depends only on the top probability ((p_max − 1/n)/(1 − 1/n))	confidence docs, § 1.1
Top probability	"confidence to be the probability of its verdict"; also notes that Jev's native field "is a statistic of the whole label distribution whose formula is not published", with Spearman 0.976 between the two; SanSi likewise "Confidence is the top probability"	2609.26550, 2610.07730
Expected error	ufakzeka-karar returns temperature-scaled probabilities + "an expected error that serves as a 'not sure' signal"	2610.06744
Learned uncertainty score	SHIFT-JEV learns one risk score from three signal classes (uncertainty, format, option consistency) and compares it with the top-probability score (at a 10% target risk, 3–5 vs 5–15 / 27 exceedances)	2610.34329
Two kinds of uncertainty (training view)	OpenJev-RLCD: the outcome is "one-hot when there is a right answer (epistemic uncertainty) and spread when humans disagree (aleatoric uncertainty)"; "Aleatoric uncertainty: when RL cannot help"	2609.38850
Two kinds of uncertainty (deployment view)	Human disagreement vs decision risk; under asymmetric loss the two require opposite escalation signals, "no single confidence number serves both"; splitting a judgment into parallel sub-questions and reading the dispersion recovers the disagreement signal	2610.33258
Calibration vs sensitivity	"calibration and sensitivity (how well confidence separates right from wrong answers) are distinct"; the most sensitive model's confidence tracks "which kind of situation will fail" rather than "which answer is wrong"	2610.09683
Answer confidence vs self-knowledge	"Do you know the answer" style Nouls are separated from answer confidence; surface cues can masquerade as self-knowledge	2610.01006
The decision model "does not know its own limits"	"its confidence flags the errors it makes itself but not" errors inherited from input errors	2610.24929

Definition	Content	Source
Answer quality (answer mass)	AnyJev uses "total probability mass on the legal options" as the readout's normalizer and as a validity metric	2610.00831

Points of disagreement: confidence is (a) a statistic of the distribution shape (TypeSafe), (b) the top-1 probability (most audits), (c) a learned risk score (SHIFT-JEV), or (d) two quantities that cannot be merged (2610.33258). (a) and (b) are highly correlated in ranking but **not equivalent at the threshold level** (the same 0.6 threshold means different things with 2 options and with 10 options), and neither is an "error detector" (§ 4.2 D1).

1.4.5 How "Calibration" Is Defined and Measured

Definition / metric	Content	Source
TypeSafe	"measured across groups of predictions; it does not guarantee that an individual answer is correct"	system-one
NDM D2	"among answers given probability p, about p are right", measured, adjusted and reported separately **per question type**	2610.25553
ECE, 15 equal-width bins	Choice uses the top option's probability, Noul uses its probability; "top-label ECE, 15 equal-width bins, the last bin closed"	2609.37647, 2610.26325
ECE, 10 equal-width bins	JEVal: "ten equal-width confidence bins"; SanSi: ten equal-width bins, only on answerable items	2610.03935, 2610.07730
ECE, 15 equal-mass bins	AnyJev	2610.00831
Equal-mass bins + bias-aware estimation	Adopted by the benchmark paper on the grounds that "the estimate depends on the binning scheme"	2610.00346
ECE relative to a finite-sample null	TypedBench: "calibration relative to a matched finite-sample null ECE", also reporting smooth ECE, Brier, NLL, coverage at 5% risk, flip rate, JSD, RPS, QWK and decision value under asymmetric costs; evaluated across "friendly / adversarial" phrasings	2610.11392
smooth ECE	ufakzeka-karar (temperature scaling lowers it on the dev set and raises it on held-out support questions, 0.027→0.045)	2610.06744
Signed confidence-accuracy gap	Laya audit: −0.214, accuracy above confidence in every occupied bin, "sign agreement collapses every binned ECE variant to the same value"; the direction determines which way the cascade fails	2609.33843
Proper scoring rules	Brier, log-loss, RPS as training objectives and evaluation (jevlike, Open-Jev, NDM, SanSi, OpenJev-RLCD, UniJev CORA)	§ 6.1 family C
Selective prediction / AURC	Risk-coverage curve after sorting by confidence (OpenJev-RLCD, PACT, 2609.37647)	2609.38850, 2609.35865
Policy-aware calibration	Measures the confidence-accuracy gap under the deployment policy (selection, routing), ⩽2.7 pp across 184 settings; "downstream selection distorts the confidence-accuracy relation even when the model is unchanged"	2610.17309
Risk budget / false-omission risk	Default to a conservative fallback when uncertain, and control the false-omission risk within the subset "assigned the utility-maximizing action"; naive thresholds exceed the budget in about half the settings, Learn Then Test 7.8%	2610.19811
Calibration under prior shift	Candidate restriction, EM prior correction, 35 pre-registered paired tests	2610.26325
Probability axioms / coherence	Deviation of negation-pair sums from 1: Jev 0.064, Qwen3.8-27B first token 0.293, verbalized probability 0.122; total variation between hierarchical marginalization and direct asking: Jev 0.219–0.349, Laya 0.424–0.689	2609.33209, 2609.33971
Cross-interface consistency	Canonical posterior error, complement residual, coarsening residual	2609.37470

Points of disagreement: (1) the binning scheme (10/15, equal-width/equal-mass, closed last bin) directly changes the ECE value, so it is not comparable across papers; (2) whether "calibration" includes coherence (2609.33209 / 2609.33971 say it does not and must be listed separately); (3) whether calibration is measured on "isolated questions" or under the "deployment policy" (StartLux, 2610.19811 and 2609.37470 hold that only the latter is the product metric); (4) whether Noul calibration is measured separately for the positive and negative classes (2609.37647 uses the Noul probability itself, TypedBench uses friendly / adversarial phrasings).

1.4.6 How "Consistency / Flip" Is Defined

Definition	Content	Source
Repeatability	ICC of repeated calls on the same request (0.997); the test-retest floor ("Jev is not deterministic, so … the flip rate between two identical runs"); repeats in a fixed order as a control; same-text rerun flips 0.17%	2609.36399, 2609.26758, 2610.35826, 2609.31142

Definition	Content	Source
Single-shuffle flip	One shuffle under letter readout changes 12–20% (Qwen3-8B 20.0% / Gemma 4 12B 11.9%, the published fine-tune 9.1%)	2610.22066
Cumulative failure set	Tracks individual items across eight presentations: "accuracy varies by less than two percentage points, yet its cumulative failure set expands by 27–29% relative to the original order"; i.e. **stable accuracy ≠ stable items**	2610.35826
List reversal	Averaged over 11 models, reversing the option list changes 33% of answers	2610.00831
Order invariance (definition)	Runoff Definition 1: the decision rule gives the same distribution for any option order; NDM D4: reordering only permutes the distribution	2610.18233, 2610.25553
Luce's two axioms	Order independence + adding an option does not change the relative odds of the other options; theorem: only a softmax in which "each option's score depends on itself alone" satisfies both; "a well-chosen added option steers 4–34% of decisions"	2610.22066
IIA / clones / decoys	Pre-registered tests of Jev's option-set dependence (decoys, fillers)	2610.16655
Readout stability	Distribution after menu restriction = original distribution renormalized (Definition 1); when it holds, the restricted predictor is exact	2610.07716
Definition-level flip	"track changes in the selected definition rather than in the returned name": after swapping the name-definition binding, a model that follows definitions should return a different name; AUC 93.8% → 23.2%	2609.26758
Labels override definitions	Three interventions: deleted / contradictory / reversed definitions	2610.02586
Answer-preserving variants	Keeps only the 5,586 variants that "should not change the answer" and attacks only correct decisions: 46.3% flips vs 0.17% for same-text reruns	2609.31142
Equivalent wording vs identical-text repeats	"changes answers more across equivalent wordings than across identical-text repeats"	2610.30045
Ordinal compression	"candidate-space compression": correlated with the number of options, direction not unique, large models still biased	2609.38827
Position bias (training metric)	PACT: "answer flips under relabelling" 9.8 vs 13.8%	2609.35865
Question isolation	NDM D5; Verdict's structural mask; pplx-decider's "joint / separate request" experiment	2610.25553, 2610.20778

Points of disagreement: (1) "flip rate" has at least five denominators (single shuffle, cumulative over eight presentations, list reversal, relabelling, answer-preserving variants); (2) whether order invariance is a **guarantee by construction** (Prism-1, block-causal, Verdict, NDM) or an **empirical measurement** (AnyJev, 2610.35826); (3) whether the object of "consistency" is the name, the definition or the distribution (the core distinction of 2609.26758); the 30 sub-dimensions of § 4.2 are an enumeration on top of these definitions.

1.4.7 "RLCD" Has at Least Four Meanings

Usage	Meaning	Source
TypeSafe	The "third post-training path" alongside RLHF and RLVR, "probabilities are optimized against outcomes"; **algorithm, reward and data are all unpublished**	ML primer
Proper-scoring reward with reasoning chains	OpenJev-RLCD: after each reasoning chain, read out the committed distribution and score it with a strictly proper scoring rule; a variance identity distinguishes "per-chain scoring" from "mixture scoring" (the latter = RLVR + a diversity reward); "calibrate first, then reinforce"	2609.38850
$RLCD^2$ (differential reward for reasoning quality)	StartLux AutoThink: the difference in proper score between the readouts before and after reasoning + a soft length penalty; only for models that generate reasoning	2610.17309
Umbrella term for five "RLCD algorithms"	UniJev compares Shaped RLCD, GRPO, CtR, RLCD-lite and Rewarding Doubt; its Table 1 lists the CE + Brier of Von / OneJev / Open-Jev and the CE of Kev / NanoJev as "direct gradient" controls	2610.25784
"The RLCD recipe released by open-source System One models"	SBERT2S1 refers to a score-function estimator with reward normalization (REINFORCE-style), 2.5–3.0 points below cross-entropy	2610.02486
"Reinforcement learning against proper scoring rules"	The benchmark paper's description of Laya: "trained with reinforcement learning against proper scoring rules, a recipe its developers call RLCD"	2610.00346
Stand-in	jevlike: "our open-source stand-in for TypeSafe's unpublished RLCD. We claim no equivalence"; in practice CE + 0.5 • Brier	mustafasemi-ai/jevlike
Equivalence theorem	NDM: "proper scoring rules applied directly when labels are known … reach the same optimum as the policy gradient of an RLCD-style reward without its variance"	2610.25553

Usage	Meaning	Source
Community naming	Model names such as `SFT + RLCD` (S^2-MAS), `qwen3-0.6b-rlcd-decision`, `rlcd-modernbert-151m`, `Qwen-2.5-1B-RLCD`, with varying training details	2610.27746, HF probe records

Points of disagreement: when reading "RLCD" one must first ask three things: is there a reasoning chain, is the reward correctness or a proper score, and is it a policy gradient or a direct gradient. The conclusion of §6.1.3, "no positive evidence for RL on single-pass models", holds only for the one meaning "no reasoning chain + policy gradient".

1.4.8 How "Abstention / Fallback / Don't Know" Is Defined

Definition	Content	Source
Explicit option + separate calibration	NDM D6: the probability of "none of these" is calibrated separately; two mechanisms, declared (present among the options) and undeclared (absent from the options, relying on probability)	2610.25553
Probability band	TypeSafe cookbook: 0.30–0.70 maps to `uncertain` and goes to a human, with the explicit statement "The band is illustrative; it is neither a calibrated guarantee nor an optimized threshold"; confidence routing uses a global floor (0.6 in the example) + per-action-class thresholds	consistency cookbook, confidence-routing
Abstention in the evaluation protocol	JevBench counts yes/no answers whose probability falls inside the band as "abstention = error", so Releasing the Inner Jev moves such answers to 0.801 / 0.199	2610.17205
Execution of fallback rules	PFR test: whether the switch between the regular assignment and the fallback assignment is correct; Intern drops from 97.9% to 23.1% on TREC50, "even when the correct class is given explicitly"	2610.19856
Correct answer absent	Wrong substitutes are still accepted even when an explicit reject option exists (the arithmetic-dependent rejection bottleneck, 7% → 79% after thresholding, see §4.1); legal domain: after the correct verdict is replaced, generative LLMs still pick one in 38–86% of items, the most cautious decision model 30%	2609.39496, 2610.29961
Out-of-scope items	TypedBench: "out-of-scope items, whose true option is withheld"; abstention rate measured separately with / without "none of the above"	2610.11392
Answerable / unanswerable	SanSi computes ECE only on "answerable items", plus a stopper whose threshold is fitted on unlabeled requests	2610.07730
Silent hallucination	RSAJEV: "assigning higher confidence to unseen than to seen classes (AUROC 0.38)"; prototype retrieval restores "a calibratable abstention channel"	2610.15742
Certified abstention	SHIFT-JEV certifies the abstention threshold with Learn-then-Test; 2610.19811 replaces abstention with a conservative fallback action when there is no human hand-off and controls the false-omission risk	2610.34329, 2610.19811
Hand-off option ineffective	"neither an explicit nor a trained hand-off option closes this gap" (inherited input errors)	2610.24929
Ternary contract	true / false / unknown admission (§3.2)	2610.10978

Points of disagreement: abstention is an **option** (NDM, TypedBench's NOTA condition), a **probability band** (TypeSafe, JevBench), a **separate risk score** (SHIFT-JEV, ufakzeka), or a **fallback action** (2610.19811, PFR). The four definitions correspond to four failures: the option is not chosen (legal 30%), the band is not calibrated (TypeSafe's own statement), the score does not transfer (jevlike out of domain), the fallback is not executed (Intern 23.1%).

1.4.9 Definitions at the Agent Layer

Definition	Content	Source
Record-state mismatch / claim laundering	The decision layer "follows the record instead [and] turns the agent's claim into the decision"; the record dominates in all 12 model-domain configurations	2610.25947
Three-level delegation	L1 isolated questions, L2 step-level over recorded trajectories, L3 controlling a running agent; "confidence thresholds fitted on trajectories"; a non-inferiority test as the acceptance criterion	2610.21412
Controller	The decision model as agent controller shows "good judgment, bad advice" (ranking AUROC 0.78, but mean "done" probability 0.50 while 87% actually pass, see §4.2)	2610.24649
System Switch	When the fast model should stop and hand off to the reasoning model: defer the most uncertain 30% by confidence	2610.09683
Fast selector	The decision model "over a state that a reasoning model has computed and checked"	2610.24929
Can it be left to System 1	The same typed request sent to 30 models, four task groups + 138 closed-loop environments	2610.21244

Definition	Content	Source
Quadruple / joint reliability	Controlled tests of rule transfer, joint reliability and information use; "equivalent rewrites have opposite task-level effects"	2610.30045
Two time scales	A slow teacher writes instructions in a paused environment, a fast student executes while running; under a "live clock" the faster model overtakes	2610.20665
Decision-level continual learning	Typed feedback (accept / revoke) becomes bounded causal state	2610.31521

Points of disagreement: whether "correct" at the agent layer means **isolated accuracy** (L1), **step-level agreement** (L2) or **task success / cost / clock** (L3, live clock); the conclusion of § 5.2, "faster decisions ≠ a better agent", is exactly the manifestation of these three definitions not agreeing.

1.4.10 Summary Table of Disagreements and "What We Must Answer Before Defining the Problem"

Dimension	Definition A	Definition B	Definition C	Consequence
Decision model	Interface (TDM / LDM)	Property contract (eight NDM items, readout stability)	Use (gate / selector / can it be left to System 1)	Decides whether the evaluated object is "the model" or "model + threshold + code"
Request	Three primitives + limits	Quadruple (state, instruction, criteria, gold)	Natural-language candidates / action space	Decides the input form of the D3 rule language
Confidence	Distribution-shape statistic	top-1 probability	Learned risk score / two quantities	Thresholds do not transfer across different option counts
Calibration	Population frequency matching (10/15 equal-width/ equal-mass)	Coherence + axioms	Error under deployment policy / risk budget	ECE numbers not comparable across papers
Consistency	Single-shuffle flip	Cumulative failure set / answer-preserving variants	Definition-level flip	"Unchanged accuracy" hides per-item instability
Order / IIA	Empirical flip rate	Guarantee by construction (Luce)	Readout stability	Whether hybrid backbones can satisfy the constructive guarantee (§ 1.3)
RLCD	Unpublished	Proper-scoring reward with reasoning chains	Direct-gradient stand-in	The conclusion on "whether RL helps" depends on the meaning
Abstention	Explicit option (calibrated separately)	Probability band	Risk score / fallback action	Four different failure modes
Multiple questions	Isolation (NDM D5)	Jointly beneficial (pplx)	Read dispersion after decomposition (2610.33258)	Semantics of fan-out and decomposition
Agent correctness	Isolated accuracy	Step-level agreement	Task success / cost / clock	"Faster ≠ better"

Before we write our own problem definition, at least six of these disagreements must be fixed first: (1) whether "decision model" is defined by interface or by contract; (2) whether confidence uses top-1 or a distribution statistic, and whether it should be separated from "self-knowledge"; (3) which binning calibration uses, whether it includes coherence, and whether it is measured in isolation or under the deployment policy; (4) which denominator the flip rate uses, and whether order invariance should be a guarantee by construction or an empirical measurement; (5) whether abstention is an option, a band, a score or an action; (6) which meaning the word "RLCD" has in our documents. This section does not fix any of these; it only points out the evidence and failure cases already in the literature for each choice.

2 • Propositions and core logic: what this report is organized around

This chapter is the logical centre of the whole report, and the only chapter whose conclusions come from **discussion rather than a single paper**. § 2.1 tests four propositions put forward in sequence; § 2.2 is the **locked discussion draft** (what we are doing, the business goal, where existing work supports and falls short, the final framework); § 2.3 is an adversarial review of the discussion draft itself.

The four propositions: (1) Jev converts generative judgments that do not guarantee idempotence into "indicator output + deterministic execution in the harness"; (2) functional idempotence is the design purpose; (3) the core logic is the generalization of semantic understanding, not turning into code; (4) the decision consistency rate is the core metric, and right or wrong comes second.

Test results: proposition 1 has a **premise that does not hold, but a diagnosis that can be adopted** (six pieces of counter-evidence, § 2.1.3); proposition 2, as a purpose, **must be rewritten**, otherwise its optimum is a lookup table (§ 2.1.6); proposition 3 **holds**, and it holds by this chapter's own reductio (§ 2.1.7); proposition 4 **holds but the reason should be stronger**: consistency is the precondition for accuracy to be promisable (§ 2.1.8). The synthesis of the six rounds of discussion and the final framework P0″–P5″ are in § 2.2, which replaces P0′–P5′ of § 8.2.3.

One structural finding runs through all of it: propositions 1, 3 and 4 each name only "**invariance**" and leave "**discrimination**" blank, so their unconstrained optima are, respectively, a lookup table, a lookup table and a constant classifier.

2.1 The four propositions and their tests

Protocol: still analysis and survey only. This section answers the first follow-up question: **what is Jev's business goal, what are its properties, which current business problem does it solve, and how does it ultimately support a new kind of System1+System2 agent**.

Conclusions first: (1) **The starting proposition of this section is "the migration of idempotence", but it does not survive testing**: the proposition says Jev adds no new business-processing capability and merely replaces generative judgments that do not guarantee idempotence with "typed indicators + deterministic execution inside the harness". § 2.1 tests it with six pieces of counter-evidence, and the conclusion is that **both the premise and the closing statement must change**: the host Jev is itself not idempotent (0.78%–1.33% of decisions change under identical repeated calls, and four papers report its non-determinism), the format guarantee is not something it brings (the existing lineage of grammar-constrained decoding) and it carries a "constraint tax", the harness's own execution plan changes the answer, and capability is not conserved (it is lost outright in multi-step/stateful places, and the harness receives no signal). The version that can be defended is: **it does not eliminate non-idempotence; it shrinks the surface of non-idempotence to a countable, boundable, priceable set of fields, at the cost of failures turning from explicit into silent** (§ 2.1.4). **Treating idempotence as a "design purpose" rather than an "accomplished fact" is a separate proposition and must be tested separately**: taken literally it has already been achieved (same-request ICC 0.997, same-text rerun flips 0.17%) yet has bought nothing ("consistent answers also preserve errors"; repeated errors account for most of the observed errors), and it can be bought without a model (a lookup table is strictly idempotent; a deterministic rule is correct on all six policies; a model-free controller solves 79%). **An objective function whose optimum is "no model" cannot be handed to a model team.** The testable version is a triple: same decision within an equivalence class + still able to produce a decision outside the equivalence class + a unit price one to two orders of magnitude lower. Translated into model-side requirements this means **monotone thresholdable scores, native unknown, and invariance within business equivalence classes**, while idempotence itself lives in the thresholds and the deterministic contract (§ 2.1.6). (2) The business goal that follows is **not to make decisions more accurate, but to push the unit price of "a decision" down to the point where it can be called on every branch of a program**, so that a large class of judgments that "were never worth handing to a model" become worth it. (3) In its list of properties, **only two are structurally guaranteed** (the output is necessarily within the declared option set; one forward pass answers the whole request); all the rest (calibration, invariance, fallback, self-knowledge, probabilistic self-consistency) are **empirical properties that vary with the workload**, and several independent audits have

> already measured counterexamples. (4) The way it supports System1+System2 is not "one fast model plus one slow model" but **five positions that are entirely different in engineering terms**; the strongest evidence is for "outsourcing every external judgment of the slow model", and the weakest evidence is precisely for "letting the fast model act as the Agent in a closed loop by itself": the failure of that cell is directly predicted by "the decision being moved back into the model".
>
> **Re-check note (§8.1)**: this section §2.1 has already tested the starting proposition in place; §8.1.1.2 further separates **the diagnostic part that should be adopted** from **the descriptive part that should be replaced**, and marks this section's "three layers of idempotence" (high overfitting risk), "five positions A–E" (medium), "decision repetition axis" (high) and "twelve business properties" (known to be incomplete) as the report's own constructs rather than literature conclusions.
>
> This section was rewritten during the 2026-10-28 re-check. The first draft treated the proposition below as the foundation of this section, and that was a mistake: the proposition's **premise had not been tested**, while the evidence that overturns the premise **had already been cited by this report itself**, scattered across 04, 05, 06, 09, 10 and §1.4. This section now handles it in the order "prior work → the parts that hold → counter-evidence → correction".

2.1.1 The Proposition and Its Prior Work

Starting proposition (put forward by the questioner): Jev-class models add no new business-processing capability; they split business processing, which used to be handled end to end by a generative judgment that was "generalizing, configuration-driven, but not guaranteed idempotent", into two halves: **the model produces only one type-constrained indicator**, and **the harness (the caller's code) is responsible for turning the indicator into deterministic, idempotent decision execution**.

This is not a new proposition; two papers in the literature have already written it as a title and a theorem, and they must be recorded here first:

- **"Probabilistic perception, deterministic authority"**: the model's output **may only be admitted as a scored observation record**; an admission policy (threshold fitted on a held-out set to a declared false-positive-rate cap) maps the score to true / false / **unknown**, where **unknown means deny**; a **deterministic contract whose sufficiency is checked** then makes the decision. The paper gives the quantitative form of the proposition: **"the probability that perception changes the verdict is bounded above by the sum, over all perceived fields of that contract, of 'wrong-admission rate + unknown rate'"** (a union bound indexed by the contract). A registered study, two constructed domains, two sensor families, 36,000 model calls, and **not a single cell overturns the bound**; perception-caused "deny→allow" flips occurred in 13 of 21,000 test verdicts, all from 3 self-contradictory records, and every flip fell below its registered bound. Note one key detail: **the scores of both sensor families are "informative but uncalibrated"**, which means the bound **does not depend on calibration**; it depends on the construction "unknown means deny" (2610.10978).
- **FlashScript** gives the most precise wording: compile one successful Agent run into a reusable neuro-symbolic program, where actions and data transformations become code, instance values become parameters, and **decisions that need model judgment become LLM calls**; and "**compilation does not eliminate non-determinism; it localizes it, so that the deterministic program and its neural calls can be reviewed and tested separately**". On 120 WorkBench analysis tasks these programs are correct on every run and issue exactly the same tool calls, with about **70 times** fewer input tokens and about **1/3** the runtime of a "skill-guided Agent", whose repeated runs are inconsistent with one another; on compound versions of the tasks the skill-guided Agent's accuracy falls to **23%** and its output inconsistency rises, while the compiled programs stay correct and identical run after run, and generalize to all 108 unseen instances of the same template (2610.33876).

"Localize", not "eliminate", is the keyword for every correction in the rest of this section.

2.1.2 The Parts of the Proposition That Hold

Three parts, and all three hold.

(1) Decision authority does indeed land in the harness, and this is measured, not a design preference. Delegating tool selection without fitting a confidence threshold lowers the success rate by **19.0 points**; the effective practice is to **fit a threshold separately for each choice** on the Agent's own trajectories and hand anything below the threshold back to the LLM (2610.21412). Naive threshold selection exceeds the target risk in about half of the settings with a defined test risk, Learn-Then-Test in only **7.8%**, and the authors conclude that "relying on a class-probability interface

is not enough for deployment decisions" (2610.19811). Under asymmetric costs, **using the model's probabilities can be worse than taking its top-1 directly** (2610.11392): the indicator itself contains no cost structure, and the cost structure can only live in the harness.

(2) Every system that actually works does have this shape. JEVDB treats typed decisions as SQL semantic operators, with deterministic Yannakakis-style semi-join reduction and semantic Bloom filtering in front and the model only scoring (2610.02046); NIOCIS folds execution logs into a loop-folded action graph, where **restricting the candidate set per node** contributes +3.5 points while graph traversal contributes zero within the interval (2610.25985); Jev-Guard first routes on an editable policy tree and then does conditional classification (2610.18326); SemTree's tree of Yes/No/Unknown nodes can be repaired node by node (2610.13999); Mnemon is explicitly "**rules with an explicit budget** turning a judgment into a small View" (2609.36059); CARE keeps the deterministic fast path for common commands and calls the neural verdict only for boundary commands (2607.21642); field prediction for enterprise sales orders goes further still: the LLM is **used only in the offline rule-discovery stage**, and inference time is a transparent key lookup that needs no model serving (2610.34258).

(3) The real gain is not "determinism" but "boundability". This is the contribution of the first paper in § 2.1.1: once non-idempotence is compressed into "the wrong-admission rate and unknown rate of each of a handful of perceived fields", the verdict-flip probability of the whole system has an upper bound that is **computable, minimizable, and independent of calibration**, and "which of several sufficient contracts to choose" becomes a valid cost model (2610.10978). **This is a far weaker but far more real promise than "idempotence".**

2.1.3 Six Pieces of Counter-Evidence

Counter-evidence 1: the premise does not hold. The Jev side is not idempotent either, and the generative side was never measured for comparison.

Measurement	Number
Host Jev under **exactly identical** repeated calls	**4 of 510 labels change** (0.78%); in another set, **up to 1.33%** of decisions on 300 questions change between two identical runs
Number of papers reporting "host model is non-deterministic"	**4** ([53, 61, 70, 79] as relayed by 2609.32160); checklist item R2 therefore requires reporting repeated-run stability
Repeat noise of the open-source reproduction	0.17% (2609.31142), **lower than the host model**
The generative side	Koa-decision uses atomic label tokens + SFT to **reduce classification to a deterministic one-step decoding problem**, reaching 85.5% on production intent routing, on par with the strongest frontier models and several times faster at the median; against Jev, "comparable accuracy, faster median", plus support for multimodal and multi-label (2609.36115)

Conclusion: **the dichotomy "generative is non-idempotent / Jev is idempotent" does not hold under existing measurements**. Both sides sit in the same band below 2%, and the measured jitter of the host Jev is larger than that of the open-source single-pass model; an ordinary decoder with atomic label tokens is already "deterministic one-step decoding". More importantly: **no paper has measured Jev's repeat jitter side by side with a temperature-0 generative baseline**. So the premise is neither confirmed nor falsified; nobody has measured it.

Counter-evidence 2: L1 (format idempotence) is not brought by Jev, and buying it has a measured price.

The early audit lists **constrained decoding as an independent prior lineage**, and in its comparison table credits the row "guaranteed valid output" directly to "grammar-constrained decoding": **"finite-state and grammar-constrained decoders already guarantee that the output satisfies the schema"** (2609.32160).

And the "constraint tax" measures the price: over 15,000 generations (Qwen2.5-0.5B / 1.5B, SmolLM2-1.7B), hard answer-schema decoding raises schema validity from 61.5% to **100.0%** but pushes answer accuracy from 19.7% **down to 11.0%**, and lifts the share of "schema-valid but wrong" from 49.5% to **88.9%**; on the deterministic calendar tool-calling task closest to an industrial setting, Qwen2.5-1.5B reaches 91.5% executable accuracy with prompted JSON but only **48.0%** under the same hard schema, with both modes 100% schema-valid. In the authors' words, **"the errors are semantic, not structural"**, and they recommend reporting four numbers separately: schema validity, answer accuracy, executable accuracy and the **"valid-but-wrong rate"** (2605.26128).

This is a direct blow to the proposition: **typing trades "visible breakage" for "invisible error"**. For the harness this is not a net gain: a parse failure is a loud failure, while "valid but wrong" is a silent one. The corresponding measurement on Jev already exists: swapping the no/yes names across rubrics pushes AUC from .815 to .581 while **type errors are zero** (2609.32160 T3).

(An honest qualification: the constraint tax was measured with grammar decoding on ⩽3B generative models, not on Jev's trained readout; the failure mode it names matches what T3 measured on Jev, but it is not itself a measurement of Jev.)

Counter-evidence 3: the "indicator" is not a measurement of the world, so deterministic rules built on it are not portable.

A pre-registered IIA test of 299,463 calls: **every** added option shifted the log-odds between the gold label and the strongest wrong label, with a mean-square systematic shift of **0.20–0.52 nats**, **1.2–3 times the repeated-run jitter**, even when the added option received no more than 0.1% of the probability mass; the shift is directional; the distortion grows steadily with the logarithm of the number of options, reaching about **1 nat** at 255 options; out-of-domain padding pushes the odds away from the gold label. And **scoring options one by one with independent yes/no questions removes the option-set dependence, at the cost of 8.7 points of accuracy**. The authors' conclusion: **"its probabilities are best read as relative to the option table they were computed against"** (2610.16655).

Consequence for the proposition: one **business-irrelevant** menu change moves the indicator systematically by **1.2–3 times the model's own noise**. So the "deterministic" threshold rule in the harness is valid only for the option table, the wording and the set of option names it was fitted against. This is the exact opposite of a "portable deterministic rule", and the only structural fix costs 8.7 points. Checklist item C3 independently records the same phenomenon: thresholds fitted on a choice set **do not hold on held-out items for every fallback model**, and the fitted temperature differs across workloads (2609.32160).

Counter-evidence 4: the harness is not a protected zone of determinism. Its own execution plan changes the answer.

- In semantic query engines, **"the choice of query plan affects not only the cost of a query but also its result"**; accuracy is plan-independent **only under pointwise deterministic policies**; and **when the escalation band is calibrated on the plan's own candidates, selection pushdown is not quality-safe**; "which tuples to drop" is NP-hard (2610.08089).
- A systematic review of 139 paper families: **"a semantic decision engine like Jev can return a valid answer and still miss a network deadline, pick an infeasible action, or leave a service in an unverified state."** 50 families claim their engine fits a control loop or a time budget, and **only 4 back the claim with matching measurements**; of the 139 families, **only 4 report deadline attainment**. And the gap concentrates where "the decision has no deterministic computation step": those 72 families make 22 claims, **none of them supported**, and only 2 name an owner for coverage. The sharpest sentence: **"a decision that meets a 10-second budget for every isolated request meets it for no request once decisions queue ahead of replayed execution time."** (2610.06425)

So L3 execution idempotence holds only under a **fixed plan and fixed load**. Once the plan is selectable, once there is queuing, it no longer holds. The three-layer table in the first draft was too lenient toward the harness.

Counter-evidence 5: the conversion is lossy, and the loss falls exactly on the class of business logic that most needs to be kept.

The proposition says "business-processing capability is unchanged; only the carrier changes". This is falsifiable, and it has already been falsified: Jev **cannot produce the decisions that "an LLM can only make by writing out intermediate steps"**: on items the reference LLM can solve in one forward pass it is right 97.5% of the time, on the rest only **74%**; once the answer depends on **two or more updates** from the same user, accuracy falls, reaching 53%–80% at sixteen updates; given a stale state summary, it answers according to the stale summary. Most fatal of all, **it does not know its own limits**: confidence can flag its own mistakes, but not the mistakes it inherits from stale input, and **neither the explicit nor the trained hand-off option catches the latter** (2610.24929).

So the idea that "the harness catches the rest" fails on this class of failures: **the harness receives no signal at all.** Multi-step, stateful business processing is not "moved to a different carrier"; it is dropped.

Counter-evidence 6: determinism is sometimes precisely the wrong target.

- Collapsing the distribution to one action throws away something measurably valuable: averaged over respondents, **the probability vector is closer to the human item marginals than the rounded answer**, and for ranking respondents it is roughly equal to a persona regression fitted on 163 real people (2610.14912); confidence conflates two kinds of uncertainty, human disagreement and decision risk, which under asymmetric loss demand **opposite escalation signals**, so **no single confidence number can serve both** (2610.33258).
- But handing the full vector downstream also has a price: full output lowers average downstream cost in **27/27** model-task comparisons, yet pushes the secret-recovery rate on the held-out set to 62.85%–75.35% (51.04%–

57.64% using only the chosen class; random is 50%); **confidence fields increase leakage while lowering cost inconsistently** (2610.23668).

So "indicator → deterministic action" is a **lossy compression with an information cost**, not a free architectural victory; how much to compress and which fields to expose is a bill to be computed workflow by workflow.

2.1.4 The Corrected Proposition

Folding the six pieces of counter-evidence back in, the version that can be defended is:

> **Jev-class models do not turn "non-idempotent" into "idempotent". They do three things: shrink the surface of non-idempotence from "any position in free-text output" to "the values of a handful of typed fields"; thereby make that surface countable, boundable and priceable (the union bound does not depend on calibration; it depends on the construction "unknown means deny"); and the cost is that failures turn from explicit into silent, and that capability is lost outright on the class of judgments that "require writing out intermediate steps".**

Differences from the first draft:

	First draft (proposition copied as is)	Corrected
Key verb	convert / eliminate	**localize, shrink, price** (taken from 2610.33876, 2610.10978)
L1 format idempotence	Jev's contribution	**The existing lineage of constrained decoding**, and it carries a "constraint tax" (2605.26128)
L3 execution idempotence	Holds by construction	**Holds only under a fixed plan and fixed load** (2610.08089, 2610.06425)
Conservation of capability	Implicitly holds	**Does not hold**; the loss concentrates in multi-step/stateful places and the harness receives no signal (2610.24929)
Semantics of the "indicator"	A measured value the harness can build rules on	**A relative quantity with respect to that option table**; business-irrelevant menu changes cause shifts of 1.2–3 times its own noise (2610.16655)
What to pursue	Determinism	**A boundable error surface**; determinism itself is sometimes the wrong target (2610.33258, 2610.14912, 2610.23668)

The three-layer idempotence table is still useful, but it must be rewritten (this table is the report's own construct, not the conclusion of any paper):

Layer	Promise	Carrier	Status after correction
L1 format	The output is always a valid typed value	Interface structure	Holds by construction, **but is not unique to Jev** (grammar-constrained decoding already guarantees it), and it trades "parse failure" for "valid but wrong"
L2 semantic	Equivalent business situations give equivalent indicators	Attention mask / readout structure	**The weakest layer**. Equivalences that can be written into the mask are solvable (flip rate 8.30%→0.11%, Prism-1 zero flips by construction); those that cannot (wording, tone, label names, menu composition, irrelevant logs) all fall, and **the effect of menu composition has been measured at 1.2–3 times its own noise**
L3 execution	The same indicator gives the same action	Harness code	Holds **only under a fixed plan and fixed load**; plan selection changes the result, and queuing voids every latency guarantee

2.1.5 Corollary: First Separate the "Business Goal" from the "Capability Goal"

Most of the disappointment with Jev in the existing literature comes from conflating two things.

	Capability goal	Business goal
Proposition	Single-pass typed readout is **more accurate** than a generative readout of the same size	Single-pass typed readout lowers **the marginal cost of the same judgment by one to two orders of magnitude**
Existing evidence	**Does not hold**. The audit of 28 early papers concludes that "typed readout itself shows no accuracy advantage independent of a comparable label-probability readout" (2609.32160); reading logits directly from an off-the-shelf chat model is already a Jev-style decision model (2610.02076)	**Holds, and is the most stable finding in the whole literature** (see the table in § 5.1.1)
Corollary	Expecting a gain from "switching to single-pass readout" has no basis	Expecting "making judgments cheap" to change system design has a basis

This distinction is not rhetorical. It determines what the **optimization objective** (§ 6.5) and the **evaluation protocol** (§ 5.2) should be organized around: if the business goal is unit price, then "accuracy on par" is not a failure but a **precondition**; what actually has to be defended is "accuracy does not drop significantly while unit price falls by one to two orders of magnitude".

2.1.6 Treating Idempotence as a "Purpose" Rather Than a "Fact": Only Written This Way Is It Testable

What § 2.1.3 overturns is the **descriptive** proposition ("Jev turned non-idempotent into idempotent"). But there is also a **normative** version: functional idempotence is the purpose for which we design Jev-class models; if it cannot be idempotent, on what grounds does it become enterprise decision intelligence. These are two different propositions, and overturning the former does not overturn the latter, so it must be tested separately. The result of the test: **as a purpose it must be rewritten, otherwise its optimum is "no model".**

Objection one: taken literally, this purpose has already been achieved, and it has bought nothing.

Evidence	Number
Intraclass correlation of repeated calls on the same request	**ICC 0.997** (24 questions, 288,000 answers, 2609.36399)
Magnitude of repeat noise	Negation inconsistency is about **5 times** the repeat noise, i.e. the repeat noise itself is small (2609.33209)
Same-text rerun flip rate of the open-source reproduction	**0.17%** (2610.35826), lower than the host Jev's 0.78%–1.33%

And § 4.2 § D4's verdict on this sub-dimension (A1 repeatability) is "**solved and misleading**", based on two findings from the same legal-document study: **"consistent answers also preserve errors": repeated wrong answers account for most of the observed errors**; and after changing the visible assumptions and the order, **corrections and regressions cancel each other out, so accuracy is unchanged while the decisions have changed** (2609.27678). So the property "same input, same output" is one we already hold at roughly the 99.8% level, and it has not made the system trustworthy, because **an idempotent error is a systematically repeated error**: the same customer is denied every single time, and aggregate accuracy cannot see it.

Objection two: taken literally, this purpose can be bought without a model, so it cannot be the design goal.

The supremum of functional idempotence is a **lookup table**: memoize by input hash and any model attains strict 1.0 idempotence. Pushing further in this direction, the report already has three measured endpoints: after parsing policy fields into typed values, **one deterministic rule is correct on all six policies**; the decision space of bounded Agent workflows saturates, and **a model-free controller solves 79% on τ^2-bench telecom** (2610.24871); in enterprise field prediction the LLM is used only in the offline rule-discovery stage, and **inference time is a transparent key lookup**. If an objective function's optimum is "no model", it is not an objective function that can be handed to a model team.

Objection three: what the enterprise side actually mandates is not bit-level idempotence either. Audit and compliance require four things: **same case, same verdict** (= L2, stability within an equivalence class), **a reproducible decision record** (one can re-derive "why it was decided this way"), **a declared and bounded error rate**, and **appealability**. Bit-level idempotence is merely the **cheapest route** to buying the first, which is why it is easily mistaken for the requirement itself. But it is neither sufficient nor necessary:

- **Not sufficient**: hard schema decoding raises validity from 61.5% to **100%** while pushing answer accuracy from 19.7% to **11.0%** and lifting "valid but wrong" from 49.5% to **88.9%**; the authors conclude that "**the errors are semantic, not structural**" (2605.26128). A system that is 100% idempotent and silently wrong 11% of the time is worse than one that is 99.8% stable but flags the uncertain batch.
- **Not necessary**: the class of deployed systems this argument appeals to (scorecards, credit scoring) is indeed idempotent, but their licence to operate comes from being **monotone, thresholdable at a declared FPR, and auditable**; idempotence is merely a by-product of "a fitted scoring function", not the reason they are accepted.

So the purpose should be rewritten as a triple, not a single property. Each end has a degenerate solution: the optimum of idempotence is a lookup table (zero generalization: unseen situations get no answer), and the optimum of generalization is a free-text LLM (zero stability). Jev-class models occupy the segment in between, so the testable purpose must state all three items at once:

> **(a) Give the same decision within a business-declared equivalence class (L2); (b) still produce a decision outside the equivalence class (generalization); (c) a unit price one to two orders of magnitude lower than generative.** Remove any one item and the remaining two have trivial optima.

This rewrite does not soften the goal; it translates it into three model-side requirements, and note that none of them is "be idempotent". The construction that can be copied directly is 2610.10978: the model output is admitted only as a **scored observation record**, the threshold is fitted on a held-out set to a **declared FPR cap**, the score maps to true / false / **unknown**, **unknown means deny**, and a **deterministic contract** that performs a sufficiency check makes the decision; the probability that "perception changes the verdict" is then bounded by Σ(admitted-and-wrong + unknown), **and this bound needs no calibration** (pre-registered study, 36,000 calls, no cell overturns it). Here idempotence lives in **the thresholds and the contract**, not in the model. What should be designed on the model side is:

Requirement	Why	Current state
Monotone, thresholdable scores (rather than self-reported probabilities)	The union bound needs only an admit/deny region, not calibrated values	Doable; and the readout is in fact **ordinal, not probabilistic** (ordinal-layer error 4.2 points vs probability-layer 21.0 points, § 6.5.2.2), which fits exactly
A native unknown / abstain class	"Unknown means deny" is the construction that frees the bound from calibration	**The worst cell in the whole report**: the correct-refusal rate when the answer is absent is **7%**; fitting a threshold gets to **79%** (2609.39496)
Invariance within business-declared equivalence classes (L2)	This is what "same case, same verdict" is	**No paper has defined a "business equivalence class" and measured stability on it** (§ 7.5 row 13); this is a vacant position that can be occupied

The last item must be remembered on its own, because it is where this design purpose is most likely to break in an enterprise, and it has nothing to do with model noise: the indicator is a **menu-relative quantity**. The pre-registered 299,463 calls show that every option added to the option table moves the log-odds between the gold label and the strongest wrong label, with a systematic shift of **1.2–3 times** the same model's own rerun fluctuation (2610.16655). In other words: **business staff add one new category to the drop-down, no business fact changes, and yet every threshold moves**. Under the "same input, same output" protocol this change does not even count as an input change, which is exactly why "functional idempotence" must be defined by **equivalence class** and not by "identical input bytes".

2.1.7 The Third Proposition: "The Core Logic Is the Generalization of Semantic Understanding, Not Turning Into Code"

§ 2.1.4 summarizes the adoptable part of H4 as "moving decision authority from the model to the code", and § 2.1.6 uses "the optimum of idempotence is a lookup table" to rebut treating idempotence as the purpose. Placed side by side, these two passages **contradict each other**: if "moving to code" is the right direction, its optimum is likewise a lookup table; **the reductio I used against idempotence applies unchanged to my own wording**. So this objection holds, and it holds by this section's own mode of argument: what we are building is an AI system; the core logic is the ability to generalize semantic understanding, not turning judgment into code.

First make precise "what the code gets". In the construction of 2610.10978, what the deterministic contract gets is **not the judgment but the arithmetic on top of the judgment**: threshold comparisons, sufficiency checks, summing the union bound. Mapping an open-world unstructured situation to a scaled observation record is a step with **no way to write it as code**; it is semantic generalization, and it is the only irreplaceable part of the whole construction. So the correct wording is not "decision authority moves to code" but: **the model is responsible for semantics, the code for bookkeeping**. The report's original wording would make the model look like a replaceable component; that was an error of expression, already corrected in § 1.1.6 and § 8.1.1.2.

The strongest single piece of evidence for this proposition (and the strongest "the model is irreplaceable" evidence in the whole report): RLCDAlignBench uses **one generic question**, zero-shot, across ten classes of alignment failure, 44 benchmarks and 5 target models, with a median AUROC of **0.886**, **beating supervised baselines** on most benchmarks, at **1/63** the cost of an LLM judge (2609.29429). A rule engine cannot do this, and a lookup table even less: it has no "table". On the business-cost axis this evidence would be recorded merely as "63 times cheaper"; only on the semantic-generalization axis does what it really proves become visible.

But this proposition must accept the same test as "idempotence", and it likewise has a degenerate solution. A model that gives the same answer for every input "generalizes" perfectly. Generalization is meaningful only under a

discrimination requirement, and sub-dimension D1 of §4.2 §D4 has already separated these three things: calibration, discrimination and sensitivity are not one thing. So the testable version has three parts, not one sentence:

> **(a)** Invariance within **business-declared equivalence classes** (this is "semantics"); **(b)** discrimination on **business-meaningful distinctions** (this is "understanding"); **(c)** on **new situations outside these**, either be right or abstain (this is "generalization").

It is worth noting that these three parts are **structurally isomorphic** to the triple that §2.1.6 derived from the cost axis (same decision within the equivalence class / a decision outside the equivalence class / a declarable error bound). Two mutually independent axes deriving the same three-part structure is a sign that it is fairly stable; it also shows that **the dispute is not about what the goal is, but about which item is the primary one**.

The most valuable corollary of this proposition: L2 is the measurable definition of "semantic understanding". This report has been treating "equivalent business situations give equivalent indicators" as a consistency property (L2), but under the semantic-generalization reading, **two business-equivalent situations are the same semantic situation**; so L2 is not a nicety added on top of consistency, it **is** the operational definition of "having understood the semantics". The nature of the vacant position in §7.5 row 13 (**no paper has defined a "business equivalence class" and measured stability on it**) therefore changes too: it is not an unmeasured consistency sub-item, it is **the core metric of the whole direction, which nobody has measured**.

And the conclusion under this reading is harsher than under the cost reading, not kinder. The measurements say: single-pass typed readout is weakest precisely at the semantic layer.

Measurement	Number	What it shows
Name-definition swap	Mean AUC **93.8%** → **23.2%** (below random), type error rate 0% throughout (2609.26758)	Type-correct ≠ read the definition. The model uses the **label string**, not the semantics
Deleting all definitions	Accuracy **unchanged** (0.8487), while definitions alone can sustain 0.7971; when labels and definitions conflict, 0.8511 → **0.2281** (2610.02586)	The only place where developers can write rules **is not read**
1–12B few-shot	**Semantic coverage is exactly zero**; ICL "adjusts how the input is projected onto stable semantic directions" rather than remapping label meaning (2511.21038)	Context cannot move the semantics
Cross-judge transfer	SemTree's tree compiled for judge A moved to judge B: 67.70% → **47.87%** (2610.13999)	Semantics do not transfer across models
Judgments that need intermediate steps	97.5% on items solvable in one forward pass, **74%** on the rest; 53%–80% at sixteen updates; **it does not know its own limits** (2610.24929)	Semantic generalization fails at multi-step, and **without self-knowledge**

There is also one direct conflict, and it is the most important finding of this section: the operation that makes it cheap and the operation that destroys semantic sensitivity are the same operation. Certo's joint scorer (state + rules + candidates **read together**) has a rule-sensitivity recall@1 of **1.00**; after switching to cacheable independent encoding (candidates encoded once and reused, about **5 times cheaper** with 77 candidates), rule sensitivity drops to **0.24** (2609.37832, §6.2.3). That is, **the cost axis and the semantic axis are in direct engineering opposition**, while this report had previously recorded the two separately in §5.1.1 (cost structure) and §6.2 (generalization levels) without juxtaposing them. With semantic generalization as the primary item, this trade-off must be priced explicitly; the cheap side cannot be chosen by default.

For our own continued training, what this proposition changes is the direction of model selection. If the core logic is semantic generalization, then decision fine-tuning is **not** where semantics gets added: on Qwen3.5 0.8B→27B the untuned readout already reaches **76–97%** of post-tuning accuracy, the tuning gain shrinks from 15.8 points to **2.3 points**, 1/16 of the data recovers most of it, and training only the letter head "recovers almost none of the gain"; the authors' explanation is "**pretraining provides the representation, tuning only adjusts how it is read**" (2610.15501). The same paper also gives a discriminating control: the same untuned 27B scores 96.9 when reasoning is allowed versus 89.1, and on the three date-arithmetic and numerical-comparison schemas the readout is only 56–66% at **every scale** while reasoning is 98–100%. So under this reading the question to ask is not "how to train the readout to be more accurate" but **"how to keep the readout from losing the semantics already present in the representation"**; and the literature offers only two ready-made means aimed at this, neither tried on decision models: changing the rendering layer to write only definitions, with meaningless identifiers as labels (accuracy **+0.1511**, 2610.02586), and symbol tuning (using meaningless symbols as labels to force reading the definitions, which directly improves the ability to **flip in-context label semantics**, 2305.08298, §6.3.2; §3.1 blind spot 2 already records "nobody has tried it on decision models").

2.1.8 The Fourth Proposition: "The Decision Consistency Rate Is the Core Metric, and Right or Wrong Comes Second"

This proposition is on the same line as § 2.1.7: if invariance within an equivalence class **is** the operational definition of "having understood the semantics", then the consistency rate is naturally the primary metric. This section has three parts: the reason for it that is stronger than the one the questioner gave, the item it must add, and the exchange rate it must price explicitly.

(i) It holds, and with a reason stronger than "more important": consistency comes before accuracy.

Accuracy is a property that can be measured and accepted; consistency is the property **that makes the measurement itself well-defined**. The whole calibration-free union-bound construction depends on "the threshold is fitted on a held-out set to a declared FPR" (2610.10978), and a threshold is valid only while the score distribution is stable. Once scores within an equivalence class can drift, **the threshold fails and the bound no longer holds**; at that point the question "how accurate is it" does not even have a definite answer. The measured price: in one paper the threshold was fitted in-sample, **targeting 5% and reaching up to 17% on the held-out set** (2610.02267); SHIFT-JEV, with thresholds certified by Learn-then-Test, still exceeds the target risk of 10% in **3–5** of 27 environment-seed pairs (5–15 for max-probability scores, 9–14 for conformal sets, 2610.34329). So the correct statement is not "consistency matters more than accuracy" but **consistency is the precondition for accuracy to be promisable**.

There is also one independent piece of support: aggregate accuracy is blind to the changes we actually care about. Across 8 option orderings for Jev, **accuracy changes by less than 2 percentage points while the cumulative failure set grows by 27–29%** (2610.35826); after changing the visible assumptions and the order, **corrections and regressions cancel each other out, so accuracy is unchanged while the decisions have changed** (2609.27678). Putting right or wrong second comes with a condition: **right or wrong can no longer be measured with aggregate accuracy either**; it must use the decision-level paired protocol (cumulative failure set + equivalence-class flip rate). § 8.4 Q4.6 has already listed this as "needs a standard, not a new method".

(ii) The item that must be added: the unconstrained optimum of the consistency rate is a constant classifier.

This is the third time this section meets the same structure. The optimum of idempotence is a lookup table (§ 2.1.6); the optimum of "moving to code" is also a lookup table (§ 2.1.7); the optimum of the consistency rate is **giving the same answer for every input**. Each of the three propositions names the "invariance" half and leaves the "discrimination" half blank, so the unconstrained optimum of each is a degenerate model.

And this is not a hypothetical failure: the report has its measured form. **Ordinal scales are systematically compressed** (40 datasets × 9 models, worse the finer the scale, and model scale does not recover it, 2609.38827): this is exactly the model buying stability with "not using the scale"; **homogeneous demonstrations collapse the accuracy of 0.8B–8B models to below 12%** (2605.08295): this is collapse to the majority class. So the usable form of this proposition must have two terms:

> **Consistency rate as the objective function, accuracy as a constraint** (declare an accuracy floor that must not be breached), rather than the lexicographic order "consistency first, accuracy second". Under lexicographic ordering, "second" is, in engineering terms, "unbounded".

(iii) The thing that must be fixed first: the denominator of the consistency rate. § 1.4 has already pointed out that "flip rate" has **five different denominators** in the literature. Seen by denominator, the current state of this metric is entirely different:

Denominator	Current state	Value as a training objective
Repeated calls on the same request	**Already at ICC 0.997 / flips 0.17%**	**Zero**. We already have it (§ 2.1.6)
Option order permutation	Block-causal takes flips from 8.30%→0.11%, Prism-1 is **zero by construction** (2610.22066)	**Low**. Can be bought by construction, no need to train
Wording / tone / label names	Partially measured, all fall; name-definition swap AUC 93.8%→23.2%	**Medium-high**, but it is the same problem as "reading the definition" (§ 2.1.7)
Answer-preserving adversarial variants	**Attacks flip 46.3% of Jev's correct decisions**, fallacious arguments without commands flip **77.2%**; among decisions still executed at the 0.8 threshold, **23.5% are wrong** (2609.31142)	**High**

Denominator	Current state	Value as a training objective
Business-declared equivalence classes	**No paper has defined it and measured on it** (§ 7.5 row 13)	**This is the vacant position**

This table yields a sharp conclusion: **the "consistency rate" of the same model is 99.83% under the "byte-identical rerun" protocol and 53.7% under the "answer-preserving adversarial variants" protocol, a gap of 46 percentage points**. So the sentence "the consistency rate is the core metric" **is not a metric** until the denominator is fixed, and fixing the denominator is itself the unoccupied vacant position this report has found.

(iv) The exchange rate is measured, not assumed. Buying consistency back costs accuracy, and four independent measurements give the price list:

Consistency bought	Price paid
Hard schema decoding: validity 61.5% → **100%**	Answer accuracy 19.7% → **11.0%**, "valid but wrong" 49.5% → **88.9%** (2605.26128)
Independent yes/no scoring: **removes option-set dependence**	**−8.7 points** (2610.16655)
Cacheable independent encoding: about **5 times cheaper**	Rule-sensitivity recall@1 1.00 → **0.24** (2609.37832)
PACT's pairwise anchoring + ordinal transport	The authors themselves state "**no single component improves raw accuracy; the value is in robustness**" (2609.35865)

But note that rows two and three point in different directions: **independent scoring pays in accuracy, cacheable encoding pays in rule sensitivity (that is, semantics), while by-construction methods of the block-causal and Prism-1 kind have so far reported no accuracy cost**. So "consistency costs money" is not a uniform conclusion: **buying by construction (changing the mask, the positions, the branch isolation) is currently the free side, while buying through a loss term or through serving optimization is the paying side**. This is the same conclusion as § 6.1 "structural changes are the only algorithmic changes with large effect sizes", holding once more on the question of metric ordering.

2.2 The locked discussion draft: what we are doing, why, where existing work supports it and where it does not

> This section is the synthesis of the six rounds of discussion on 2026-10-28, **locked as the discussion draft**. It answers four questions: what we are doing (§ 2.2.1), what the business goal and purpose are (§ 2.2.2), where existing work supports it (§ 2.2.3), and where it falls short (§ 2.2.4, the defect registry); it then gives the final framework (§ 2.2.5) and the trajectory of the discussion (§ 2.2.6). § 2.3 is an adversarial review of this discussion draft itself. **P0″–P5″ in § 2.2.5 replace P0′–P5′ in § 8.2.3 and are the final statement of the framework for the whole report.**

2.2.1 What we are doing

A **single-pass, typed-readout** decision model (endgame at the Qwen 27B / 35B tier, with the 0.8B–9B ladder as the testbed for structural changes), whose interface is:

> **The state is encoded once; every question that business staff write down in words is projected once onto this state, yielding a scaled typed observation (P(true), a distribution over options, an ordinal grade); the output levels (the set of names and its type: boolean / nominal / ordinal) are defined by the business and injected with the question; the code does arithmetic on top of the observations (thresholds, sufficiency checks, union bound) and makes a definite decision.**

The core requirement of this interface is fully characterized by one example (given by the questioner):

state	Question Q1: "did I eat a meal?"	Question Q2: "did I consume liquid?"
I eat rice	1.0	0
I eat milk	1.0	1.0

That is: **the business rule (written in the question) determines the basis of the decision**. For the same state, the two sentences are of the same class under Q1 and of different classes under Q2. This has two corollaries, which respectively fix one earlier difficulty of this report and fill one earlier gap:

1. **Equivalence classes are induced by the question; no separate artifact is needed.** § 8.2.1 A1 pointed out that a "business-declared equivalence class" is itself a non-idempotent semantic judgment; this example shows that the equivalence class is not declared separately, it follows from the question, and questions are already first-class objects of the interface.
2. **The discrimination metric now has a concrete form: cross-question re-partition.** The same state **must move** under questions that partition differently ("I eat rice" must get 1.0 under Q1 and 0 under Q2). This is exactly the discrimination counterpart that § 8.2.1 A6 requires for every invariance.

But to state this example precisely, three things it exposes about itself must be recorded:

- **"Independent of semantics" holds only in a narrow sense**. It does not need open-world understanding, but predicate evaluation is still entangled with two things: **world knowledge** (milk is a liquid, rice is a solid: this comes from pretraining) and **contestable business conventions** ("does drinking only milk count as a meal?" is answered differently by different business sides; "rice" can be congee). The 1.0 in the table is a **convention**, not a fact read off the text; and a convention can only come from the question text or from examples.
- **It is a one-step predicate**. Real business rules compose (thresholds, exceptions, nested conditions), and the evidence shows that performance decays fast after composition (§ 2.2.4 rows 2–4).
- **It has a degenerate solution**: look only at the surface words of the question and ignore the state entirely (see "liquid", go look for liquid words). This solution passes both the "within-question invariance" check and the "cross-question re-partition" check at the same time. So the metric needs a **third leg**: the accuracy floor (§ 2.2.5 P2″).

The fixed structure (rounds eight to eleven, afternoon of 2026-10-28). After the nine omissions were discussed one by one, the interface of § 2.2.1 changed from "Jev's architecture" into **an architecture we fixed ourselves**, which differs from Jev in three places. First the terminology mapping, because the questioner's vocabulary is the reverse of the Jev literature: where this report cites the literature, "state" means the **input** (Jev's usage); the questioner's "state" means the **output business state** (the level). From § 2.2 onward the report consistently writes "input" and "state (business state)".

Component	Fixed as	Difference from Jev	Direct consequence
Input	Supplied by the Agent's context engineering: a mixture of three things, a **semantic specification** (similar to a skill given to an agentic model), the **dialogue record**, and **historical context**	Jev defines only one opaque state string	In the mixed input, "record vs observation" coexist (defect 14); the specification part is reused across samples, the dialogue and history are per-sample
Order	Task definition (system prompt) → input → one branch each for the candidate rules / states	In Jev the input is the shared prefix and the question branches come after	The encoding of the input is conditioned on the **task definition** (the question-aware-encoding side, the direction of Bongard's third stage, 2609.39111) but not on the specific candidates; cross-sample caching runs along the axis of "the same set of rules", aligned with business usage (rules change weekly, cases arrive by the second); Jev's advantage that "additional questions are nearly free" is one **we do not have**
Type of the candidate set	Declared by whoever writes the question: **exclusive** (pick one state) or **stackable** (several rules hold at once). Both exist	Jev has only Choice (exclusive) and Noul (single item)	Exclusive uses candidate isolation + joint softmax (IIA is free at the ranking layer, not guaranteed at the probability layer, 2610.01601); stackable uses one independent Noul per candidate (IIA holds at both the ranking and the probability layer, but **−8.7 points**, 2610.16655). The rice / milk table is stackable
Levels (the set of states)	**Defined by business experts**: the name, the definition of each level, and the **type** (boolean / nominal / ordinal); none of the three may be missing; the number of levels is capped by the harness	In Jev the caller supplies the candidates, but nominal and ordinal are not distinguished	When the names carry no semantics the harness renders them as neutral ids + definitions and maps back (the fix for defect 1 goes from load-bearing to the only path); "ordered" is the expert's declaration, checkable by "adjacent-level confusion higher than distant-level confusion"; ordinal compression worsens as the scale gets finer (2609.38827)
Model output	One **level distribution** per question (or, when stackable, one P(true) per candidate)	Same as Jev's Score / Noul	The **ranking** of the distribution is trustworthy, its **mass** is not (ordinal-layer error 4.2 points, probability-layer error 21.0 points, § 6.5.2)

Component	Fixed as	Difference from Jev	Direct consequence
Harness contract	The code decides on the distribution: **cumulative-mass thresholding** ($P(>k) \geqslant \tau$ goes to A, $P(<k) \geqslant \tau$ goes to B, the **residual** goes to C, where C is the unknown / deny branch); exclusive candidates use the **margin** between the top score and the runner-up plus an explicit abstain candidate; τ is fitted on held-out labels per (question, level-set version, rule version)	Jev leaves the threshold to the caller and prescribes no operator	No argmax (when adjacent levels have similar mass any perturbation flips it), no expectation (a bimodal distribution gets hidden as the middle level); decision idempotence now has a concrete measure: the ratio of the distance from cumulative mass to τ over the within-class variance
Labels	Given by the business when the dataset is constructed; once in production, **business-outcome retrospection** decides whether a decision was right or wrong	—	Retrospective labels are **decision-level** (one outcome for N questions, credit assignment has no solution), **selective** (no counterfactual on the rejected side), and **late** (when the label for rule version k arrives, production is already on version k+2); they can refit thresholds, they cannot directly supervise each readout
Deliverable	Not the model itself but **a template standard** (how to write questions, levels, types, candidate-set types, criteria and examples), released for business staff to write, plus a model that can read this template	Jev's deliverable is an API	Defect 13 (zero evaluation of business-staff authors; 91.8% of SKILL.md files are defective) changes from an "incidental risk" into **a risk of the product itself**; the template must be compilable and checkable, and compiled per judge (changing the judge: 67.70% → 47.87%)
Size	Lock the dataset and training plan at the smallest size (the smallest available locally are Qwen3-0.6B attention-only and Qwen3.5-0.8B hybrid; **there is no 0.3B**), then climb the ladder; the endgame may be either 27B or 35B-A3B	—	0.6B versus 0.8B is exactly the "attention-only vs hybrid" comparison of defect 11

Each of the three differences from Jev in this table (order, candidate type, expert levels) swaps one of Jev's guarantees for a property we have to measure ourselves: Jev's "additional questions are nearly free" becomes "caching along the rule axis"; Jev's Choice ranking invariance becomes "IIA separately for exclusive / stackable"; Jev's fixed candidates become "expert-defined levels must be read". They correspond to rows 11, 16 and 17 of the defect registry respectively.

2.2.2 Business goal and purpose

The business goal is not a more accurate classifier (typed readout itself has no accuracy advantage independent of same-backbone label log-probability readout, 2609.32160); it is to make "**judgments made according to business rules**" satisfy three things at once, which neither of the two existing paradigms can deliver together:

Requirement	Why it is a business requirement	Generative LLM	Lookup table / deterministic rules
(a) Within-question consistency: under the same question, states of the same class get the same decision, so that thresholds and the contract are effective and one can **declare** an upper bound on verdict flips that does not depend on calibration (2610.10978)	This is the technical form of "like cases decided alike, declarable error rate, auditable", the precondition for an enterprise to accept automated judgment	Not satisfied (the compliance rate for the same requirement changes substantially under a rewording, 2610.05278)	Satisfied
(b) Cross-question re-partition: when the business changes a rule, the basis of the decision changes with it, and by **reading the definition** rather than the label name	This is the technical form of "business staff adjust the basis of decisions through text"	Partly satisfied (it can execute part of a rule by writing out intermediate steps)	Not satisfied (no answer for unseen situations)
(c) Unit price one to two orders of magnitude lower, answering all the questions in a request in one forward pass	This is the precondition for "callable on every branch of the program", **an enabling condition, not a goal**	Not satisfied	Satisfied

Purpose: **to make System1 + System2 cooperation possible**. Per § 5.1.4, S1 has positive evidence in the first four positions (entry gate, process monitoring, evidence/memory surface, readout of the slow model's output), while the fifth cell, "closed-loop executor", fails across the board. This discussion draft narrows S1's responsibility to: **giving boundable typed observations for the questions posed by S2 or by business code**; the decision authority lives in the contract, the scheduling authority lives in S2.

2.2.3 Existing work against the fixed design

Component by component of the structure fixed in § 2.2.1: which literature directly supports the component, which is its cost or counter-evidence, and which cell is covered by no work at all. All entries are expanded in chapters 1 and 3–7; this is only an index.

Component	Direct support	Cost / counter-evidence	Covered by no one
Order: task definition → input → candidate branches	Candidate-isolated attention: each candidate block sees only the shared prefix and itself, flips 8.30% → 0.11% (2610.01601); Prism-1 has zero flips by construction (2610.22066); Bongard moved from "encode only the input" to "question-aware encoding", stating that the former "cannot focus on what the question cares about" and that the latter "has a cost: answers change when other instructions are added, and the cache only hits for the same instruction set" (2609.39111)	Cacheable independent encoding loses rule sensitivity 1.00 → 0.24 (2609.37832); our candidate blocks can see the input, so this is not that kind of independent encoding, but on a hybrid backbone the channel through which candidates see the input is 16 attention layers token by token + 48 recurrent layers of compression (§ 1.3.2)	**Zero hits for the hybrid vs attention-only comparison** (defect 11); "readout unchanged when unrelated questions are added or removed" measured by no one
Candidate scoring: isolation + softmax for exclusive, independent Noul for stackable	Calibration-free union bound: score → true / false / unknown, unknown meaning deny, not one cell overturned in 36,000 calls (2610.10978); independent scoring removes menu relativity (2610.16655)	Independent scoring **−8.7 points** (same source); the menu-relativity shift is 1.2–3 times its own noise; confidence is four quantities that cannot be merged (§ 1.4)	**Cross-candidate comparability** as a sixth denominator of consistency defined by no one (defect 16); the instability of entity-linking-style NIL thresholds across mentions is an analogy, not a measurement on a decision model
Levels defined by experts, type declared with the levels	Rendering only the definition with neutral ids for the names **+0.1511** (2610.02586); symbol tuning (2305.08298); interval decision trees turn numbers into ordered levels, MAE 5.18% → 1.84% (2609.28587)	Deleting the definitions leaves accuracy unchanged, definitions alone 0.7971 (2610.02586); name-definition swap AUC 93.8% → 23.2% (2609.26758); ordinal compression worsens with finer scales and model scale does not recover it (2609.38827); the score primitive is the worst (Jev 1.13 on akhilaaa3: score 80.3% vs choice 95.7%, § 6.1.4.3)	Symbol tuning never tried on a decision model; whether "the expert declares ordered" agrees with model behavior checked by no one (defect 17)
Harness contract: cumulative-mass thresholding, C as residual, margin + abstain	Choice readout ordinal-layer error 4.2 points vs probability-layer 21.0 points (§ 6.5.2.2): the ranking is trustworthy; 23.5% of decisions still executed at a 0.8 threshold are wrong (2609.31142): absolute thresholds are not trustworthy; when the answer is absent a held-out-set threshold takes the rejection rate 7% → 79% (2609.39496)	The Score primitive returns "distribution + expectation", and the expectation hides a bimodal distribution as the middle level (Jev docs, 2609.39111); thresholds shift with the menu and the rule version (defect 7)	The margin distribution (distance from cumulative mass to τ against within-class variance) reported by no one; the amount by which the C branch systematically inflates because of ordinal compression measured by no one
Labels: given by the business at construction, outcome retrospection in production	Outcome retrospection is the only label source that bypasses "conventions are contestable"; 35% annotator disagreement under the same criteria (§ 6.1.4.3)	RL gain from downstream outcomes to decisions is zero (§ 6.4.1.3); verifiers used as rewards get arbitraged (§ 6.4)	Credit assignment from a decision-level outcome to N readouts done by no one; the missing counterfactual on the rejected side (reject inference) handled by no one on a decision model
Deliverable: template standard + a model that reads it	Rule artifacts compiled per judge (2610.13999); SemTree's Yes/No/Unknown node tree and the prototype of the three primitives (§ 3.2)	**91.8%** of 138,133 SKILL.md files are defective (2608.08453); changing the judge 67.70% → 47.87%	**Zero evaluation of business-staff authors** (defect 13); zero hits for a rule language aimed at single-pass decision models

Component	Direct support	Cost / counter-evidence	Covered by no one
Input: a mixture of specification + dialogue + history	One generic question zero-shot across 44 benchmarks, median AUROC 0.886 (2609.29429): semantic generalization is alive on the cheap architecture	Record vs observation: the model trusts the record more, 53%–100% (2610.25947); on typed fields one deterministic rule gets everything right and a controller without a model solves 79% (§ 5.1.5): the higher the structured share, the more the model is squeezed out	How much of the specification part of the mixed input gets compressed by the recurrent layers measured by no one
Size and ladder	Domain-fine-tuned small models beat general large ones: SecJev-0.8B is 20.51 pp above Kev-9B (2610.03073); at 27B distillation ties Jev at about 0.1 s (2610.03935)	At 27B the decision-tuning gain is down to 2.3 points and about zero on new schemas (2610.15501); every 35B is MoE (defect 15)	Zero hits for expert routing × candidate isolation
Data and evaluation protocol	24 local datasets, 40 models (§ 6.1.4); decision-level paired evaluation: cumulative failure set, flip rate, axiom violations (2610.35826, 2609.33209); a 14-item checklist (2609.32160)	Single-pass readout is enough on one-step predicates: within 3 points of GPT-6 at 0.36% of the cost (2609.26550); moderation AUROC 0.83 indistinguishable from 13/14 frontier systems (2610.33258); but on business-criteria tasks Jev 1.13 is 8–9 points below the generative frontier (§ 6.1.4.3)	Denominator and rule diversity for "cross-question re-partition": **zero**; Chinese decision data aimed at business rules: zero

2.2.4 Shortcomings of existing work: the defect registry

Each row: defect / which denominator it is measured by / measured effect size / known fix / type of fix (interface · structure · SFT · RL · none) / residual / whether it is blocked by measurement. **The type column is the most important information in this table**: it answers "which defects should not get RL spend".

#	Defect	Measure	Measured effect size	Known fix	Type	Residual / blocker
1	**Label names override definitions** (the business rule is written in the definition but is not read)	Name-definition swap; deleting definitions	AUC 93.8% → **23.2%** with a 0% type-error rate (2609.26758); deleting definitions leaves accuracy **unchanged** (2610.02586)	Render only the definition and use meaningless identifiers for labels **+0.1511**; symbol tuning (2305.08298)	**interface → SFT**	Symbol tuning never tried on a decision model; both have a degenerate solution (ignore the names entirely)
2	**Rule read out but not executed**	Execution rate of policy modifications	Readout correct, execution **22%–72%** (2610.18579)	Split the rule into a question graph and ask node by node (§ 6.5.5.1)	structure / interface	Composed calibration after decomposition **measured by no one**
3	**Silent scope omission**: applies the general rule but drops the nested exception	NormBench, 2,290 items	Generative baselines already behave this way (2606.08932)	None	none	Harder for a single-pass model; no fix
4	**Arithmetic / time**	Difficulty-graded paired comparisons	87.7% → **57.4%** (2610.28522); on numeric schemas readout 56–66% at every scale, reasoning 98–100% (2610.15501)	Arithmetic outsourcing; numbers go through ordered bins (MAE 5.18% → 1.84%, 2609.28587)	interface	Requires first judging "this item needs arithmetic", and the model **does not know its own limits**
5	**Fallback / unknown** (the worst cell in the whole report)	Correct rejection rate when the answer is absent	**7%**, a held-out-set threshold can reach **79%** (2609.39496); a fallback option takes Intern 97.9% → 23.1% (2610.19856)	Held-out-set threshold; native unknown class	interface → SFT	Training of native unknown never reported on a decision model
6	Option order / position	Shuffle flip rate	8.30% → **0.11%**; Prism-1 zero flips	Block-causal, position reset, branch isolation	**structure**	**Solved and free**; should get no further training budget
7	**Menu relativity**: add one option and every threshold shifts	Pre-registered IIA test	Shift is **1.2–3 times** its own noise; independent scoring removes it but **−8.7 points** (2610.16655)	Independent Noul scoring	interface	Pay 8.7 points; or accept "thresholds versioned with the menu"
8	**Ordinal compression**	40 datasets × 9 models	Systematic, worse the finer the scale, model scale does not recover it (2609.38827)	Ordinal loss (Kev has it but it is not enabled); coarse scale	SFT / interface	Effect size unknown
9	**Answer-preserving adversarial variants**	JevAdvBench	Flip **46.3%** of correct decisions; **23.5%** of decisions still executed at a 0.8 threshold are wrong (2609.31142)	None	none	This is the worst of the five denominators of the "consistency rate"

#	Defect	Measure	Measured effect size	Known fix	Type	Residual / blocker
10	**Multi-step / stateful judgment**	Solvable in one forward pass vs the rest	97.5% vs **74%**; 53–80% after 16 updates; **does not know its own limits** (2610.24929)	Layer looping (needs a loop-pretrained backbone, incompatible with the hybrid base model); reading others' chains +4.1 / +8.6 pp (2610.25768); distillation	structure / interface / SFT	The handoff option cannot catch inherited errors
11	**Candidate branches see the input through the recurrent layers** (identified in round six as "question-conditioned state representation", rewritten in round eight under the fixed order)	Hybrid vs attention-only, same input, N candidates, graded by how salient the required aspect is	**Unmeasured**. Under the fixed order the encoding of the input is conditioned on the task definition, not on the specific candidates; the channel through which candidate blocks see the input is 16 attention layers token by token + 48 recurrent layers of compression (§ 1.3.2); aspects of the input that a candidate needs but that are not salient may be compressed away and fail silently	Controlled experiment (local Qwen3-0.6B attention-only vs Qwen3.5-0.8B hybrid); Bongard-style question-aware encoding (paying with the cache)	**structure**	**Zero hits, blocked by measurement**; the comparison can be run now
12	**Consistency rate on equivalence classes** (the main metric itself)	Pre-registered, held-out (state, question set)	**Defined by no one, measured by no one**	—	—	**The core metric of the whole direction is blocked by measurement**; until it exists no optimization targeting consistency can be done
13	**Business-staff authors**	Questions / rules written by non-technical authors	Zero evaluation; analogous evidence: **91.8%** of 138,133 public SKILL.md files are defective (2608.08453); rule artifacts must be compiled per judge, changing the judge 67.70% → 47.87% (2610.13999)	None	none	The premise "configurable by business staff" is itself unverified
14	**Record vs observation**: the model trusts the record more	Paired claim/ reality	53%–100% (2610.25947)	S1 training data containing post-error states	SFT	Only one paper
15	**Expert routing × branch isolation** (only if the endgame lands on 35B-A3B)	—	Unmeasured. Qwen has no dense 35B; every 35B is MoE (about 3B active)	—	—	**Zero hits**; must be filled before choosing 35B
16	**Cross-candidate comparability** (newly identified in round nine: the premise for taking the top score among exclusive candidates)	IIA of ranking / probability after adding or removing candidates; margin between the top score and the runner-up	Isolation + joint softmax: IIA free at the ranking layer, probability layer changes with the menu (2610.01601); independent Noul: both layers hold but **−8.7 points** (2610.16655)	Divide by candidate-set type (exclusive / stackable); unknown via margin + explicit abstain candidate	interface	Margin distribution reported by no one; the sixth denominator of consistency defined by no one
17	**Expert-defined levels must be read** (round eleven: level names, definitions and types are set by business experts)	Whether the distribution moves after swapping adjacent-level definitions or reordering the levels; whether adjacent-level confusion is higher than distant-level confusion	With the names removed only the definitions remain, and definitions alone give 0.7971 vs 0.8487 with names (2610.02586); the unconstrained optimum is a prior by level position (which shows up as ordinal compression + middle bias)	Neutral ids + definition rendering with map-back; symbol tuning; the level set (K, names, definitions) must vary across training samples; the harness caps the number of levels	interface → SFT	Can be tested locally by re-rendering directly (no new labels needed); zero data with Chinese definitions
18	**The three properties of retrospective labels** (round ten: production labels come from business-outcome retrospection)	Label latency; coverage of the rejected side; credit assignment from decision level to readout level	Decision-level (one outcome for N questions), selective (no counterfactual on the rejected side), late (when the label for rule version k arrives, production is already on version k+2)	Used only to refit thresholds; the rejected side supplemented by exploration or manually reviewed samples	interface (contract side)	When used to supervise readouts or as an RL reward, the degenerate solution is to learn an outcome proxy; credit assignment done by no one

Three things read directly off the table:

1. **RL is the first-choice fix for no row of this table.** Among the fixable defects, the first choice is interface (1, 4, 5, 7) or structure (6, 11); SFT is the second step (1, 5, 8, 14). This agrees with the conclusions of § 6.3 and § 6.4: for pure readout, a proper scoring rule and an RLCD-style reward share the same optimum (2610.25553), and RL has independent value only when the reward depends on a sampled intermediate object; none of the defects in this table involves an intermediate object. Where RL really belongs is **decisions about the computation itself** (which questions to ask, how many layers to loop, when to hand off to the slow model; LEV's on-demand depth saves 15–24%, 2610.20951); that is scheduling, not readout.
2. **Two rows are blocked by measurement (11, 12), and they are precisely the two most important rows.** Until "the consistency rate on a pre-registered (state, question set)" is defined, any "training that targets consistency" is in-sample fitting (target 5% while the held-out set reaches at most 17%, 2610.02267).
3. **The fix for row 1 and requirement (b) of § 2.2.1 are the same thing.** "Business rules adjust the basis of decisions" is technically "read the definition, not the name". So the rendering fix and symbol tuning are promoted from "optional" to **load-bearing**.

2.2.5 The final framework (replaces § 8.2.3)

> **P0″ Semantics to the model, arithmetic to the code, questions are first-class objects.** The model maps the state into a scaled typed observation **for each question separately**; the code does thresholds, sufficiency checks and the union bound on top of the observations. The code's parameters may only be fitted on held-out ground truth, never on the model's own candidates (§ 8.2.1 A3). Equivalence classes are **induced by the questions** and are not declared separately (§ 2.2.1).
>
> **P1″ The main metric is a pair, not a single item.** On pre-registered, held-out (input, question set): **within-question invariance** (same decision for inputs of the same class under the same question; "the same question" includes the same question with a different set of level names that carry no semantics, and adding, removing or reordering **unrelated** questions in the prefix; rounds seven and eight) and **cross-question re-partition** (the same input must move under questions that partition differently). This is a system-level metric; the trainable model-level proxies are the ordinal stability of scores and the rank change across questions (§ 8.2.1 A5).
>
> **P2″ Accuracy is the third leg, not second priority.** It is the declared floor, measured with the decision-level paired protocol; without it, both items of P1″ can be passed by a degenerate model that "reads only the question wording and ignores the state".
>
> **P3″ Idempotence lives in the thresholds and the contract, and the operator is fixed.** The code does **cumulative-mass thresholding** on the level distribution ($P(>k) \geqslant \tau$, $P(<k) \geqslant \tau$, the rest is C, i.e. unknown / deny); exclusive candidates use the margin between the top score and the runner-up plus an explicit abstain; no argmax, no expectation. Three requirements on the model side: monotone thresholdable scores (and, within pre-registered equivalence classes, the distance from cumulative mass to τ exceeds the within-class variance), native unknown (currently 7%), within-question invariance. τ is fitted on held-out labels per (question, level-set version, rule version), and refitted in production by outcome retrospection (defect 18).
>
> **P4″ Every invariance gets a discrimination counterpart**, now made concrete as the "within-question / cross-question" pair. The credit belongs to this fixed choice only, not to the principle (§ 8.2.1 A6).
>
> **P5″ Unit price is an enabling condition; the trade-off has been settled by a design choice.** The fixed order (task definition → input → candidate branches) puts cross-sample caching on the axis of "the same set of rules", aligned with business usage; the cost is that Jev's "additional questions are nearly free" is one **we do not have**: the encoding of the input is conditioned on the whole task definition, and adding a new question changes existing readouts (defect 11, the new item in P1″). What is kept is "one to two orders of magnitude cheaper than generative", which comes from not decoding and has nothing to do with prefix sharing. A4 is therefore no longer open; instead **the expensive, rule-sensitive side has been chosen** (§ 8.2.1 A4, second postscript).

2.2.6 How the discussion progressed

Round	Proposition (the questioner)	Test result	Unconstrained optimum
1	Jev turns non-idempotent judgment into "metric + deterministic execution"	Premise does not hold (the host Jev is itself non-idempotent); the diagnosis can be adopted	—

Round	Proposition (the questioner)	Test result	Unconstrained optimum
2	Functional idempotence is the design purpose	Must be rewritten as a purpose	**Lookup table**
3	The core logic is semantic generalization, not turning into code	Holds, and holds by this report's own reductio	(the optimum of "turning into code" is a **lookup table**)
4	Decision consistency rate is the core metric, correctness is second priority	Holds, the reason should be stronger (consistency is the precondition for accuracy to be committable)	**Constant classifier**
5	Semantic sensitivity is normal in generative models too; the identified problems are finally strengthened with RL	First half right but not an exemption; second half: RL is the first choice for no row of this table	(my own "force reading the definition" also has a degenerate solution: **ignore the names**)
6	Business rules adjust the basis of decisions; same state, multiple questions	Holds; fixes A1 and supplies the discrimination metric; "independent of semantics" holds only in a narrow sense	**Read only the question, ignore the state**
7	Survey the data first; discuss only decision idempotence; train "ability" not tasks; levels injected through context	The triplet form exists, the re-partition denominator is zero; idempotence written as margin; the testable form of ability is no degradation on held-out rule families; injection lands on option-name invariance	(the optimum of "training ability" on synthetic rules is to **learn the generator grammar**)
8	Definition first, cases after, what is shared is the system prompt	Holds, and is the side Bongard tested; P5″ rewritten, A4 settled, defect 11 rewritten	—
9	Score candidates one by one and take the top score, premised on comparable scores	The premise is not free: comparability at the ranking layer is free, the probability layer costs 8.7 points; rice / milk is stackable, so the top score cannot be taken	(the degenerate solution of independent scoring is separately calibrated, mutually incomparable scores)
10	Output a level distribution, the harness routes by >k / <k / =k	Holds, the operator is fixed as cumulative mass, C is the residual; expectation and argmax both excluded	(the expectation hides a bimodal distribution as the middle level)
11	Levels are defined by business experts themselves; labels given at construction, retrospective in production	Holds, but the definition becomes the only information and definitions are read worst; "ordered" is a declaration to be checked; retrospective labels are decision-level, selective, late	**A prior by level position** (ordinal compression + middle bias)

Pattern: in seven of the eleven rounds the proposition (including one of my own) has a degenerate optimum under no constraints, and every time it was "named the invariance, left the discrimination blank". This is the one methodological lesson this report should remember most.

2.3 Adversarial review of this discussion draft: omissions and blind spots

§8.2 attacks the statement of the framework. This section attacks the **discussion itself**: what the six rounds never talked about. Ordered by impact on subsequent work.

Omission 1: the six rounds never talked about data, and the data form required by the interface of §2.2.1 exists only in half. "Same state, multiple questions, different partitions" requires training and evaluation data to be triplets of **(state, question set, label per question)**. The first draft of this section wrote "all existing decision datasets are (state, single fixed question, label)"; after sampling each of the 24 local datasets (§6.1.4.3) this turned out to be **wrong**: `typed-decisions` (1,200 states × 5 questions), `akhilaaa3/decision-bench` (586 questions), the synthetic part of Open-Jev v1.1 (48 visible rule templates, three-valued logic, with abstain) and `bev-decision` (56,091 multi-question rows) are all in triplet form. What is missing is two more specific things: (a) **cross-question re-partition**, i.e. competing conventions over the same state (the second column of the rice / milk table), is zero locally; every multi-question set is different facets of the same state or different views of the same rule; (b) **rule diversity**: the full Open-Jev mix has 147,139 rows but only 4,101 distinct question strings, while §6.3.2 says that what decides whether the ability to "define the task through context" lives or dies is precisely whether the task definition varies between training samples. The label-source problem was also measured: among the 6,000 question labels of `typed-decisions`, **35% have inconsistent annotator argmax**; business conventions remain contestable under the same criteria, and this is not an annotation-budget problem. So for requirement (b) there is currently **evaluation data with an accuracy floor, but no denominator for "cross-question re-partition" and no rule diversity for training "ability"**; the former can be generated by pairing existing components (rule text + code evaluation), and of the latter two, convention labels on natural-language states can only come from the business side.

Omission 2: A1 is fixed only for questions written by people; in the Agent loop questions are machine-written, and A1 comes right back. §2.2.1 says equivalence classes are induced by the question, so no separate artifact is

needed; this holds for questions that business staff write **offline**. But the purpose in § 2.2.2 is S1+S2 cooperation, and in the first four positions a substantial share of questions are **generated at runtime by S2 (a generative model)**. Then the question itself is a non-idempotent generation, the equivalence class is non-idempotent with it, and the denominator of P1″ changes on every call. The discussion did not distinguish the variable "who writes the question", and it decides whether P0″ holds. The defensible version is: **runtime-generated questions must also pass through the contract** (versioned question templates + slot filling, not free generation), otherwise S1's consistency guarantee cannot propagate upward to the Agent.

Omission 3: the example assumes everything is decidable; unknown never appears. All four cells of the table have definite answers. But real rules do not cover every situation (the silent scope omission of defect 3 is exactly this), and P3″ requires native unknown, currently only 7%. The discussion never stated **what the correct output is when the question does not cover the state**. "I eat rice" giving 0 under "did I consume liquid?" implies the convention "rice = cooked dry rice"; if the business side did not write that convention, the correct output should be unknown, not 0. This means the "within-question invariance" of P1″ must also hold on unknown (states that are equally uncovered must equally give unknown), and this item **does not even have a denominator yet**.

Omission 4: consistency is defined only on the state side; paraphrase consistency on the question side was not discussed. Two questions that are paraphrases of each other ("did I consume liquid?" / "did I take in fluids?") must induce the same partition, otherwise every word business staff change changes the equivalence class. The evidence is already in the report but did not enter the discussion: changing the criteria changes 28% of predictions with no improvement (§ 3.1.3); the compliance rate for the same requirement changes substantially under a rewording (2610.05278). So P1″ is actually **three items**: within-question invariance, cross-question re-partition, and **partition invariance under question paraphrase**. The third item is symmetric to the first and has the same degenerate solution.

Omission 5: the cost account for N questions was never done; after round eight, changed to "dissolved by a design choice". This item originally assumed Jev's architecture (input as shared prefix, question branches after), and so worried that question-conditioned encoding would pay N prefills. The order the questioner fixed is task definition first, input in the middle, candidate branches last (the fixed structure of § 2.2.1), with cross-sample caching along the axis of "the same set of rules"; the N questions never shared the encoding of the input, so nothing was "lost". The two remaining things are recorded elsewhere: Jev's "additional questions are nearly free" is one we do not have (already removed from P5″); the encoding of the input is conditioned on the whole task definition, and adding a new question changes existing readouts, which is the invariance newly added to P1″. The typical value and the upper bound of N are externalized to the harness; the questioner considers them not the focus of this stage.

Omission 6: rules change, contract parameters must be refitted, where do the labels come from; round ten gave the answer, and the answer carries three properties. The questioner: labels are given when the business dataset is constructed, and once in production, **business-outcome retrospection** decides whether a decision was right or wrong. Labels at construction time carry no new information (the 35% annotator disagreement under the same criteria remains as is). Retrospective labels are new; what they can do is refit the thresholds in the contract (one decision against one outcome), and what they cannot do is determined by three properties (defect 18): **decision-level**, one outcome for N questions, which readout was wrong is unknown; **selective**, only executed decisions have outcomes, the rejected side or the side handed to humans never has a counterfactual, and refitting on it pushes the threshold in one direction (the reject-inference problem in credit scoring); **late**, when the label for rule version k arrives, production may already be on version k+2, so after every rule change there is a window of running the new rule with old thresholds. The positive item: it is the only label source that bypasses "conventions are contestable", because it turns "does milk count as a meal" from a convention into "which way of judging gives better outcomes"; but this trains the downstream decision, not predicate evaluation, and the two must be accounted for separately, otherwise what the model learns is an outcome proxy. The frequency of rule updates differs by business; the questioner externalizes it to the harness; the real focus is that **the template must be released as a standard for business staff to write** (the deliverable row of § 2.2.1, defect 13).

Omission 7: ordinals and numbers did not enter the "question-induced partition" framework; rounds ten and eleven fixed the form. The questioner: do not have the model output a number, have it output a **distribution** over states (business states), e.g. a distribution over five levels 1–5, and let the harness decide how to use it (>3 goes to A, <3 goes to B, =3 goes to C), not mechanical classification; and 1–5 are not set by us but **defined by business experts themselves**. The version after testing: (a) thresholds stay in the harness, compatible with P3″; but ">3" applied to a distribution has three readings: argmax flips most easily, the expectation hides a bimodal distribution as the middle level, and **cumulative mass** $P(>3) \geqslant \tau$ is the monotone thresholdable score that P3″ wants; take the third, with the C branch defined as the **residual** that passes neither threshold, i.e. the unknown / deny branch (2610.10978). (b) "A state

represented by a single token may be fine" disagrees with the evidence: ordinal compression measures exactly the readout of discrete grades (2609.38827), and the score primitive is the worst of the three (Jev 1.13 on akhilaaa3: 80.3% vs choice 95.7%); the ranking of the distribution is trustworthy, its mass is not (4.2 vs 21.0 points), so τ must be fitted on held-out labels. (c) What makes it "not mechanical" is not the harness but a loss that recognizes order: the "mass above 3" from a nominal five-way softmax has no ordinal meaning; an ordinal / cumulative-link loss (Kev has it but it is not enabled, defect 8) is what gives it meaning; K−1 cumulative boolean readouts are the way to write order into the structure. (d) Expert-defined levels move three things into the expert's hands: the definition becomes the only information and definitions are read worst (deletion unchanged, definitions alone 0.7971, 2610.02586); "ordered" is the expert's declaration, must declare a type with the levels and be checkable by "adjacent-level confusion higher than distant-level confusion"; the artifacts experts write have themselves zero evaluation (defect 13). The training implication: the level set (K, names, definitions) must vary between training samples (§ 6.3.2), and the harness caps the number of levels (the finer the scale, the heavier the compression). (e) The phrase "confidence state" should be avoided: levels should be business states (severity, urgency), confidence is the mass of the distribution; writing confidence into a level definition stacks two layers of uncertainty and is one of the sources of the 35% annotator disagreement.

Omission 8: the order I gave (interface → SFT → RL only for scheduling) has been run by no paper; withdrawn in round ten. The questioner does not follow this pattern; the next stage is "business discussion + a model reinforcement plan under business constraints". The order was a prediction I derived from three independent groups of evidence, not a measurement, and withdrawing it costs nothing; but the evidence that produced it is not withdrawn with it: in the defect registry RL is the first-choice fix for no row, at the 27B tier the SFT gain on new schemas is about zero, and verifiers used as rewards get arbitraged. So whatever form the reinforcement plan takes, every measure must answer two questions: **which row of the registry it corresponds to**, and **whether its unconstrained optimum is a degenerate model**. The "type" column of the registry is the version indexed by business constraints. Read together with omission 6: if "reinforcement" means RL from the business-outcome retrospection signal, that signal is exactly the decision-level, selective, late kind, credit assignment across N questions has no solution, and the degenerate solution is learning an outcome proxy (defect 18). The scope of this report remains "analysis only, no plan"; the reinforcement plan is a separate document.

Omission 9: no round defined what the input is; filled in round eleven. The questioner: the input is supplied by the Agent's context engineering and can be a **semantic specification** (similar to a skill given to an agentic model), a **dialogue record**, and may also include **historical context**. This decides three things: (1) the input is mixed, "record vs observation" coexist (defect 14, the model trusts the record more, 53%–100%), the structured share varies by business, and the gradient of § 5.1.5 (the more structured, the more the model is squeezed out) has to be examined business by business; (2) the specification part is reused across samples, the dialogue and history are per-sample, consistent with the caching axis of the fixed order, but the longer the specification, the more heavily it is compressed through the 48 recurrent layers (defect 11); (3) among the local datasets the ones most like this input are typed-decisions (structured account fields + dialogue threads) and akhilaaa3 (multilingual dialogue streams + ticket records), and the synthetic part of Open-Jev (pure fields) is the least like it; the data selection for the pre-experiments is set accordingly. Terminology: the questioner's "state" means the output business state, the reverse of the Jev literature; from § 2.2 onward "input" and "state (business state)" are distinguished.

Summary (after the nine items were discussed): of the nine, **5 is dissolved by a design choice**; **1, 6, 9 received the questioner's input** (the triplet form exists, the re-partition denominator is zero; the three properties of retrospective labels; the input is a mixture of specification + dialogue + history); **2, 3, 4, 7 changed the metrics and the form** (runtime questions must pass through the contract; unknown enters the denominator, C is the residual; the question-side paraphrase item and the level-name item as symmetric counterparts; cumulative-mass thresholding, expert-defined levels with declared types); **8 withdrawn**. Beyond the locked draft only three things remain open: (a) denominator data for "cross-question re-partition" and rule-diversity data (to be built); (b) the hybrid vs attention-only comparison (to be run); (c) direct evaluation of business-staff authors (required before the template standard is released).

Round seven: item-by-item discussion of omissions 1–4 (2026-10-28; the questioner gave four inputs, the tested versions follow).

Input 1 (data): survey the local datasets first and judge whether experiments can be done with existing data. The result is in § 6.1.4.3, and omission 1 has been rewritten according to it. The gist is three sentences: the triplet form **exists** (four sets), the denominator for "cross-question re-partition" is **zero**, and the rule diversity needed to train "ability" is **zero** (147,139 rows, 4,101 questions). The former is enough to measure the P2″ floor and several items of defects 5, 6, 13; the latter two have to be built, but what is built is a new pairing of existing components (rule text + code

evaluation; Open-Jev's generator already has three-valued logic and per-family held-out splits), not a new task type. What really cannot be built is convention labels on natural-language states: `typed-decisions` measures 35% annotator disagreement on question labels.

Input 2 (idempotence): this time only decision idempotence is discussed; business functional idempotence is out of scope. This narrowing is compatible with P3″: decisions exist only after thresholding, so "decision idempotence" is a property of the contract in the first place, not of the model. But after narrowing there is still one denominator that must be fixed, otherwise this item falls into the trap of §2.1.8 again: **same-byte state gives the same decision** is already achieved by deterministic readout (99.83%) and has zero value as a goal; **same business situation gives the same decision** is exactly the "within-question invariance" of P1″, and its denominator is still defect 12. One more thing hidden by the narrowing: **decision idempotence after thresholding is a function of the score margin**. For a state whose score is near the threshold, any unrelated perturbation flips its decision (a substantial share of the 46.3% flips in 2609.31142 are of this kind). So what can be measured at the model layer is not a "decision flip rate" but the **margin distribution**: what fraction of decisions have a score-to-threshold distance greater than the score variance within the same equivalence class. This quantity needs no new labels, only equivalence classes; and the unknown interval (§2.2.5 P3″) is exactly the institutionalized treatment of low-margin states. Conclusion: the narrowing is accepted, but "decision idempotence" must be written as "the fraction of decisions, on pre-registered equivalence classes, whose margin exceeds the within-class score variance", otherwise it degenerates into the item that is already free.

Input 3 (what to train): do not train the model to understand a specific task; train its ability to "understand logic and business requirements" and to give a pre-designed output accordingly. This agrees with §6.3.2: the two methods that improve the ability to "define the task through context" (MetaICL, symbol tuning) are both SFT, and the decisive variable is whether the task definition varies between training samples; this is exactly the rule-diversity-is-zero item of omission 1, so the data implication of input 3 is **build rule diversity first**. But three qualifications apply. First, "ability" is not a training objective that can be written down; what can be written down is a distribution over (rule, state, label), and "ability" is a generalization claim about **held-out rule families**; the `family_disjoint … 1 ood` split of Open-Jev v1.1 is exactly this test form, and 5,120 ood rows are available locally to test directly. Second, the degenerate-optimum check: on synthetic rule data the unconstrained optimum is to **learn the generator's rule grammar** (a parser), which generalizes to new rules in the same grammar and gives no guarantee at all on the "milk is a liquid" half; on natural-language rules the evidence is that definitions are not read (AUC 93.8% → 23.2%, deletion unchanged) and rules are read out but executed only 22%–72%. So by the evidence "understanding business requirements" must be split into **reading the rule** (semantics; an interface and rendering problem) and **evaluating the rule** (arithmetic / time / multi-step, 87.7% → 57.4%; beyond what is solvable in one forward pass only 74%); the latter half has a structural upper bound for single-pass models (2610.24929) and is **out of reach of "training ability"**; it can only be handled by question-graph decomposition (defect 2) and arithmetic outsourcing (defect 4). Third, at the 27B tier the SFT gain on new schemas is about zero while untuned readout already reaches 76%–97% of the tuned level (§6.1.5); this means the ability to "understand business requirements" is either already in the pretrained representation or SFT cannot fit it in; for the endgame size the question changes from "how to train this ability" to "**does decision tuning preserve the ability the base model already has**", and this item has zero measurements in the whole literature (§3.1). So input 3 holds, but its testable form is: on held-out rule families, the decision-tuned model does not degrade relative to untuned readout.

Input 4 (output space): the output levels (1–5 or a–e) are defined by the business, injected by the Agent's context engineering, and not trained into the model. Consistent with the interface (a runtime-defined candidate set is exactly Jev's Choice primitive, and Hanno's 23,900 rows are constructed this way). But "inject, do not train" lands exactly on the cell in the literature with the largest effect size and the lowest compliance rate: **option-name invariance**, tested by only 1 of 27 applicable papers; under yes/no names the flip rate is up to 70.4 pp higher than the 0/1 control, while random strings as names are close to neutral (§3.1.2.2). That is, the model is **not invariant** to "injected names"; either train it to invariance (symbol tuning) or have the harness **normalize** the names: the model sees only neutral identifiers plus definitions, and the code maps the result back to business names (the rendering-fix approach, +0.1511). The choice between the two depends on whether the names carry semantics: 1–5 and a–e do not, and normalization is lossless; `billing` / `bug_report` do, and normalization means pushing all the semantics in the name back into the definition, which is exactly the part the model reads worst. The defensible rule: **names that carry no semantics are normalized by the harness, names that carry semantics count as part of the question text**, and both classes enter the within-question invariance test of P1″. Second qualification: 1–5 is **ordinal**, a–e can be either ordinal or nominal; this is not a difference of names but of **primitive**. The same level set rendered as Choice loses the order (Choice readout is ordinal, not probabilistic, §6.5.2), and rendered as Score has systematic compression

(2609.38827); in akhilaaa3's own results Jev 1.13 gets 80.3% on score and 95.7% on choice. So what is injected is not only the set of names; the **type** (nominal / ordinal / boolean) must also be declared, and this item has to be trained (the ordinal loss of defect 8). Third, the questioner uses "state" here for the output levels, while "state" in § 2.2.1 and omission 9 means the input; the suggestion is to call the output side **levels** (output space), otherwise the problem of omission 9 is hidden by this usage. Finally, this is the **only one of the four that can be tested on local data right now**: re-render any choice set into the four name sets {1–5, a–e, business words, random strings} and compare flip rates; no new labels are needed.

Changes to § 2.2.5 after the four-item discussion: the first model-side item of P3″ changes from "monotone thresholdable scores" to "monotone thresholdable scores whose margin within the equivalence class exceeds the within-class variance"; the "within-question invariance" of P1″ explicitly includes replacement of the option-name set; requirement (b) of § 2.2.1 gains the sentence "output levels are injected with a declared type". Omissions 5–9 await discussion.

3 • Semantics: did the model actually read what the business side wrote

> If the core logic is the generalization of semantic understanding (§2.2 P0), then the first question to ask is: **does the model actually read the definitions, criteria, examples and rules that the business side writes down in text**. This chapter is the whole body of evidence on that question.
>
> The protocol of §3.1 (formerly dimension D1) is not a cost level. It is: whether the decision-fine-tuned single-pass model still has, and can use, ICL; **whether business staff can use a Jev-class model directly through prompt engineering**; and the effect size and failure modes of each lever. §3.2 (formerly dimension D3) is the other half: define a set of logic primitives so that non-technical business experts can write business rules in text and have them executed faithfully.
>
> The two sharpest conclusions of this chapter: **label names override definitions** (a name-definition swap takes mean AUC from 93.8% → 23.2%, while the type-error rate stays at 0% throughout: correct type ≠ read the definition), and **deleting all definitions leaves accuracy unchanged**. They strike directly at the premise that "business staff can configure it".

3.1 D1 in-context learning: can it "learn in the prompt", and which levers can business staff use

> Protocol (per the latest discussion): not a cost level. Three questions to answer: 1. Does the decision-fine-tuned single-pass model **still have / can it use** ICL (examples, definitions, criteria, state structure)? 2. How to **make good use** of this ability? 3. Which levers can **business staff use through prompt engineering**, how large are the effects, and in which situations do they fail?

3.1.1 First, Back to the Basic Facts of ICL (Upstream Literature)

A decision model is obtained by fine-tuning a causal LM, so the ICL regularities of LMs are our starting point. The conclusions directly relevant to "business staff writing examples":

Fact	Source
The **true labels of examples are not required**: randomly replacing example labels barely hurts the classification / multiple-choice performance of 12 models; what matters is the label space, the input distribution and the format	2202.12837
Only large models can learn flipped labels: small models ignore flipped labels and rely on pretrained semantic priors; overriding priors is an ability that emerges with scale	2303.03846
1–12B open models have exactly zero semantic override rate under few-shot: inverted examples can only raise "prompt alignment" at the cost of accuracy; ICL is "adjusting how inputs project onto stable semantic directions", not remapping label meanings	2511.21038
Homogeneous labels collapse 0.8B–8B models to ⩽12%: the model treats the tokens in the label slot as an exhaustive answer vocabulary; if examples carry meaningless tokens like {foo,bar,vex,nit,orb}, 42–67% of the probability falls on the example set, P(dog) < 0.2%; under 4-way classification three models score 0%	2605.08295
ICL decomposes into two separable mechanisms, **task Schema and Binding**: Schema transfers 100% through late MLPs, Binding transfers 62% through the residual stream; prior knowledge interferes through attention misrouting (72.7% recency bias); 38% binding failure in high-prior scenarios; the same holds on Mamba	2512.17325
Label words are anchors: shallow layers aggregate semantics into the label-word representation, deep layers use it as the reference for prediction	2305.14160
Symbol tuning: fine-tuning with arbitrary symbols replacing natural-language labels forces the model to learn the input-label mapping; more robust on unseen ICL tasks and underspecified prompts	2305.08298
Calibrate before use / batch calibration: few-shot instability comes from bias toward specific answers (last position, pretraining frequency); estimating the bias with content-free inputs corrects it	2102.09690, 2309.17249
Task vectors / function vectors: ICL compresses the example set into one vector; a few attention heads carry the task representation	2310.15916, 2310.15213
Rules can be amortized, pairings cannot: a static vector is equivalent to one synthetic example and necessarily fails on high-rank mappings (word-level bijections)	2610.00526
ICL amplifies latent symbolic circuits: the abstraction-induction-retrieval three-stage circuit already exists before accuracy rises; examples merely feed inputs to the same circuit	2609.36265

Fact	Source
Many-shot: hundreds to thousands of examples keep improving results, especially for large label spaces; but highly sensitive to the selection strategy, and limited gains on open generation tasks	2404.11018, 2405.00200, 2603.05829
Example **ordering can be attacked**: merely reordering examples gives nearly 80% attack success on LLaMA-3	2502.14628
ICL and instruction following are not the same thing: 3-shot ICL alignment is still worse than instruction tuning	2405.19874
ICL hurts calibration, the source being example-selection bias; a cross-model cross-task universal calibrator can cut the calibration error of frontier models by more than 50%	2610.17096, 2506.12796
Post-training **reduces distribution coverage and in-context steerability**	2510.06084

Hard constraints that follow from these facts: at our target scale of 0.6B–9B, **what ICL can do is "recognize the task pattern + adjust the boundary along existing semantic directions"; what it cannot do is "flip label meanings with examples" or "make the model learn a mapping that conflicts with its prior"**. When business staff write examples, the examples mainly demonstrate the format, expose the label space and pull the input distribution closer; they do not "teach" the model new rules.

3.1.2 ICL Evidence on Decision Models (Jev and Reproductions)

3.1.2.1 Where the Official Interface Puts ICL

- The `state` documentation explicitly allows "relevant context, examples and other information helpful for answering" (concepts/state).
- Choice options carry `criteria` (a description) and `examples`; Noul's `criteria.true` / `criteria.false` can each carry a definition and examples; each Score level is a description (primitives/advanced).
- The fix for item 1 of the vendor's zigzag table, "literal reading", is exactly "write precise conditions and criteria for each option" (jaggedness).

In other words, the vendor designed all three carriers of ICL, **definitions, examples and criteria**, into the interface; the question is whether the model actually uses them.

3.1.2.2 Are Definitions Used? "Label Names Override Definitions"

This is the most important and most counter-intuitive chain of evidence in D1:

Evidence	Number	Source
Change only the "name-definition" pairing, not the question, the state, or the name and definition texts themselves: on 1,200 decision tasks with task-specific definitions, the decision flip rate under yes/no names is up to 70.4 pp higher than the 0/1 control; **reassignment under yes/no names takes mean AUC from 93.8% to a below-chance 23.2%**; with random strings as names the flip rate is close to the neutral control; type-error rate 0% throughout	Correct type ≠ following the definition	2609.26758
When the rule is written only in the definition, the model mostly answers by label: **deleting all definitions leaves accuracy unchanged** (0.8487), although definitions alone can sustain 0.7971; when label and definition contradict, accuracy drops from 0.8511 to 0.2281; **the cause is localized to the prompt rendering `"{label}: {definition}"`**; switching to definition-only (labels replaced by A/B identifiers) gives +0.1511 accuracy; a "two-call test" is given to decide whether a model reads definitions	The definition is the only place a developer can write rules, but the model reads the label	2610.02586
HateDecide: supplying the dataset's hate-speech definition **changes up to 28% of predictions without consistently improving classification**; decomposing the definition into multiple questions is significantly effective in only 20% of comparisons	Definitions change answers but do not improve them	2610.03324
Online moderation: Jev generally benefits from **retrieved precedents** (improves exact policy selection and harmful-content discrimination under a fixed answer space); Laya instead "retrieval often changes its positive rate or no-violation rate without improving discrimination"	Precedents (examples) work for the decoder family, unstable for the encoder family	2610.07953
Structured-output schema descriptions: label definitions in the system prompt beat those in the schema description (GPT-4.1 / 5.4 lower by 11–13 points); but when schema and prompt conflict, wrong schema instructions drop accuracy by 5–45 points; adding an intermediate reasoning field before the label gives schema-only accuracy +15–24	The **position** and **rendering** of definitions are first-order variables	2608.08254

Reading: the decision-fine-tuned model keeps the LM's "label-word anchor" mechanism. It answers from the semantic prior of the option **name**; the definition is read only when the name carries no information (random strings, A/B). This is fully consistent with the §3.1 finding that "small models cannot flip label semantics": a definition is an attempt to rewrite the label meaning, and small models cannot do that.

3.1.2.3 Are Examples Used? Indirect Evidence

No paper runs a "k-shot example curve" experiment **directly** on a decision model. The indirect evidence that can be found:

- **JevKT** (knowledge tracing): zero-data Jev AUC .706, **.722 after adding examples and similar-learner statistics**, ahead of deep KT models up to 16 learners and on average up to 64; this gain is Jev-specific, three other LLMs with byte-identical requests all score below it (2610.11135). This is the closest thing to evidence that "examples in state work".
- **Retrieved precedents** help Jev (table above, 2610.07953).
- **Nimble's contrastive data curation** ("change one fact so the answer flips") uses example pairs at training time, not at inference time (bespokelabsai/nimble).
- **Jev-LDE**: train a small model (System 1) to perform one Keep / Delete / Replace edit on the retrieved example set, then feed it to the target LLM; consistent gains on classification benchmarks and transferable (2609.33609). It shows that "which examples to pick" can itself be a decision task.
- **Many-shot compression**: MemCom compresses many-shot representations layer by layer and beats the surprisingly strong baseline of "just use fewer examples" at 3–8× compression (2510.16092); cheat-sheet ICL distills many examples into one text summary (2509.20820).
- **State length**: in TypedBench the host model's accuracy stays flat up to 30k tokens, decoders slow down, encoders truncate silently (2610.11392); item 5 of the vendor's zigzag table warns that "state with lots of irrelevant detail should be filtered first". Putting examples into state runs straight into this.

Gap: no work answers "how does the k-shot curve of the same backbone change before and after decision fine-tuning". This is the first blind spot of D1.

3.1.2.4 Are Wording and Criteria Used? "Read, but Not Necessarily Acted On"

- Across the nine suites of TypedBench, **the host model is wording-sensitive and systematically under-confident**, and decoders degrade with structural complexity; the "share of variance attributable to wording" is used as a reported metric (2610.11392).
- Rule transfer: **equivalent rewrites have opposite task-level effects**, "gains on unfamiliar rules may still depend on familiar wording"; "a decision service changes its answer more between equivalent wordings than between genuinely different rules" (2610.30045).
- Reading Right, Acting Wrong: policy modifications are **read out correctly** but **acted on** in only 22%–72% of cases (2610.18579, details in D2).
- Definition optimization can be a training target: Goldsmith treats "structured annotation definitions" as trainable text objects and revises them by textual gradients with an executable structural loss on a small gold set, beating OPRO / APE / PromptBreeder (2610.09489); CALICO lets non-technical domain experts turn a codebook into an editable, versionable, optimizable prompt, two coders gaining +13.0 / +7.4 points respectively, and the optimized prompt captured coder-specific interpretations (2609.14726).

3.1.3 Prompt Levers Available to Business Staff: Effect Sizes and Failure Modes

Organizing the evidence above by "knobs business staff can turn":

Lever	Expected effect	Known effect size	Failure mode	Evidence
Option names	The strongest lever	Name-definition reassignment takes AUC 93.8%→23.2%; misleading names push guard fail-open rates to 93–100%	When names have polarity (yes/no, allow/deny, safe/unsafe) they override everything; when name and definition disagree the model follows the name	2609.26758, 2610.12292
Definitions / criteria	Supposed to be the place to write rules	Deletion leaves accuracy unchanged, contradiction 0.85→0.23; definition-only +0.15	Masked by names; changes 28% of predictions without improvement; rendering decides whether it is read	2610.02586, 2610.03324
Examples (in state or in options)	Demonstrate format, expose label space, pull distribution closer	JevKT +0.016 AUC; precedents help Jev	Homogeneous examples collapse; small models cannot flip semantics; longer examples run into "irrelevant state" degradation; example order can be attacked	2610.11135, 2605.08295, 2511.21038
Wording / rewrites	Should have no effect	Equivalent rewrites have opposite effects; the host model is wording-sensitive	This is a variance source, not a lever; take the median over multiple wordings	2610.30045, 2610.11392

Lever	Expected effect	Known effect size	Failure mode	Evidence
Candidate set (menu) design	First-order variable	150 classes→5 curated candidates: 69.0%→95.4%, a span larger than the gap between 0.8B and 4B	Similar candidates, many candidates (Laya 31% with 50 nearest-neighbor tools), missing "other" option makes refusal fail	2610.07716, 2610.02267
Question decomposition (one big question split into several Nouls)	Auditable, composable	Decomposition is significantly effective in only 20% of comparisons; but normalized per-option binary scores after decomposition are more accurate than Choice scoring in medicine	Decomposition introduces probabilistic inconsistency (see D4 hierarchical consistency)	2610.03324, 2610.18529
State structure (JSON vs text, filtering)	Reduce irrelevant information	Vendor recommends filtering; host model flat up to 30k	Decoders slow down with length; irrelevant detail degrades	docs/state, 2610.11392
Language	Should have no effect	Indonesian regional languages with human Indonesian translation +37.2 points; Bengali literal lock-in 1.9–6.4× higher than English	Under low-resource languages all ICL levers weaken	2610.14112, 2610.22526
Writing rules into state rather than into definitions	Bypass name masking	Archer's probe: a statement placed in state raises probability more than one placed in a sibling question	No direct controlled experiment	Archer Hume

3.1.4 Alternatives and Complements to ICL (Not Our Main Line, but Worth Knowing)

- **Turning context into weights**: Text-to-LoRA, Doc-to-LoRA, Internalizer, Cartridges and LatentSkill are all "hypernetwork / self-learning compresses a document or skill into LoRA / KV"; LatentSkill beats in-context skills by +21.4 success rate on ALFWorld (2606.06087, 2602.15902, 2610.11715, 2506.06266). These methods keep "business staff write text" at the front end, but the model side is no longer ICL.
- **Task operators**: the output of each attention head is an affine transform of its context-masked version, so an input-dependent operator can replace the examples (2610.01054).
- **Meta-learned logic**: 1.5B–7B small models learn syllogisms by few-shot meta-learning and beat GPT-4o / o3-mini in low-data regimes (2505.14313); MetaICL (2110.15943). This is the route of "training the model to learn from examples"; its relation to decision fine-tuning has not been studied.
- **Hierarchical demonstrations beat flat logs**: on vaguely described tasks, grouped demonstrations raise pass rate 76.7%→90.7%, with no gain under precise descriptions; only sub-goal grouping works (2606.20978). Directly relevant to "business staff supplying process examples".

3.1.5 Which Works Solve Which Kind of Problem (D1 Mini-Matrix)

Problem	Existing work	Status
Are definitions read	2609.26758, 2610.02586 (incl. diagnostic test)	**Quantified**, cause localized to rendering and name polarity
Do examples work	JevKT, precedent retrieval, Nimble training-time contrast	Indirect evidence, **no k-shot curve**
Wording sensitivity	TypedBench, Rule transfer	Quantified, no mitigation
Optimization of definitions / prompts	Goldsmith, CALICO, Jev-LDE	Tools exist, aimed at generative LLMs or external editors
ICL ceiling of small models	Semantic anchors, In-context fixation, Schema/Binding	Upstream LM conclusions clear, **not re-verified on decision-fine-tuned models**
Effect of ICL on calibration	CALIBFORMER, Surprise calibration	Upstream LM, not measured on decision models
Menu / candidate-set design	JevCascade, Fast Models Slow Evidence	Quantified, the strongest lever

3.1.6 D1 Blind Spots (Identified by Us, Not Covered by Existing Work)

1. **Does decision fine-tuning preserve ICL**: no paper compares the k-shot curve, label-flipping ability or definition-following rate of the same backbone before and after fine-tuning. Spectrum Tuning hints that post-training reduces in-context steerability, but it was not measured on decision models.
2. **A training-side fix for "definitions masked by names"**: the Symbol tuning idea (meaningless symbols as labels to force reading of definitions) has not been tried on decision models; 2610.02586 only gives a rendering-level fix.
3. **Where to put examples**: in state, in the option's `examples`, or in Noul's true/false criteria; the effects of the three positions have no controlled comparison. Archer's probe only shows "state is visible, sibling questions are not".

4. **Interaction of examples × branch isolation**: examples placed in state are shared by all questions and add to the cost of every branch; placed in the question branch, each question carries its own copy. Nobody has measured the accuracy/cost trade-off of these two layouts.
5. **ICL and calibration**: the example-selection bias introduced by ICL breaks calibration (upstream evidence), while calibration is the core selling point of decision models; the interaction of the two is unmeasured.
6. **"Prompt usability" from the business staff's perspective**: CALICO targets generative LLMs; there is no usability study for decision models of "non-technical staff write criteria → model following rate".
7. Whether **median over multiple wordings / batch calibration** can be ported to the single-pass decision interface (which natively supports fan-out at low cost) has not been done.

These blind spots are turned into measurable problem definitions in § 8.4.

3.2 D3 logic primitives and rule language: letting business experts write rules in text

> Protocol (per the latest discussion): define a set of **logic primitives** so that non-technical business experts can make the model understand business rules by **writing text**, just as a Skill uses text to make the model understand a work capability. This section covers: (1) the evidence on the Skill paradigm itself; (2) compiler-style work on "natural-language rules → executable structure"; (3) primitives and composition patterns already present on decision models; (4) the experience of "policy-as-prompt" guards; (5) blind spots.

3.2.1 The Skill Paradigm: How Far Can Text Teach a Model a "Work Capability"

Fact	Source
Agent Skills specification: `SKILL.md` = YAML front matter (`name` ⩽64 characters, `description` ⩽1024 characters) + arbitrary Markdown body; optional `scripts/`, `references/`, `assets/`; **progressive disclosure**: only about 100 tokens of metadata are loaded at startup, the <5000-token body is loaded on activation	agentskills.io/specification
SkillsBench: 87 tasks, 8 domains, 18 model-harness configurations; curated Skills raise mean pass rate 33.9%→50.5% (+16.6 pp), +4.1 to +25.7 per configuration; **focused Skills of ⩽3 modules beat large all-in-one ones**; a small model with Skills can match a large model without	2602.12670
138,133 public SKILL.md files: **91.8% have at least one defect** (88.8–94.6%); mainly weak routing metadata, bloated or non-executable bodies, poor resource organization; Skills with effective routing metadata are retrieved more reliably	2608.08453
SkillJuror: progressive disclosure changes runtime behavior before it changes overall outcomes	2606.11543
Skill coverage: task outcomes do not show which parts of a Skill were executed or whether they were executed as instructed; trajectory-level coverage metrics are needed	2606.20659
SkillSentry: even when an Agent has completed a task under Skill guidance, it may be unstable due to deviating from the procedure or executing steps wrongly; a DSL expresses runtime assertions as guarantees	2608.09253
AFTER: 382 enterprise tasks, 6 roles, 22 procedural skills; one round of refinement +3.7–6.7 points; controlled evaluation of transfer across tasks / roles / models	2606.23127
Auto-Policy, not Auto-Skill: "a Skill describes how an Agent should behave; a Policy decides which behavior is **allowed** to become an action"; existing formats cover only the former and leave the latter to the model; auto-generating more Skills amplifies the gap, not safety	2608.25091
LatentSkill: compresses text skills into LoRA via a hypernetwork, ALFWorld +21.4 success rate, prefill tokens greatly reduced	2606.06087
Decision models screening malicious Skills: SkillScreen scores from SKILL.md in a single pass, flagging only 20–25% of benign Skills at 99% recall	2610.17727

Implications for D3: Skills prove that "text can teach a generative Agent to do things", but their reliability comes from **the model's ability to reason at execution time**; and they only cover "how to do it", not "what is allowed". The "business rule language" we want is closer to Policy than to Skill, and the executor is a **single-pass, non-reasoning** decision model. These two differences mean the Skill approach cannot be copied directly.

3.2.2 Natural-Language Rules → Executable Structure: Compiler-Style Work

All of these works answer "how does text written by a business expert become something a machine executes faithfully". They target generative LLMs, but their **intermediate representations** are exactly the candidates for our logic primitives.

Work	Input → intermediate representation	Key result	Source
Artic	Natural-language workflow → artifact-driven workflow (each step declares the artifacts it reads/writes, constraints gate artifacts, explicit control transfer); the compiler identifies steps that depend on too much state or have hard control logic and optimizes them	488 instances in 11 real domains, task resolution rate +28 pp; cross-model and repeated-execution consistency +32 / +56 pp	2608.21341
FlashScript	Successful Agent runs → neuro-symbolic program (actions and data transforms become code, instance values become parameters, **decisions that need model judgment become LLM calls**); three-tier placement + three techniques for replacing most calls with code	120 WorkBench tasks: the NSP's tool calls are fully identical on every run, about 70× fewer input tokens, 1/3 runtime; on composite tasks the Skill-guided Agent drops to 23% while the NSP stays correct; all 108 unseen instances generalize	2610.33876
TraceCompiler	Clusters of noisy Agent trajectories → mostly deterministic workflow; a dependency is admitted only when a consumer's parameters contain a value uniquely attributable to an earlier producer, each hard edge carrying auditable evidence	—	2608.02680
Formal Skill	Markdown skill → JSON metadata + action schema + Python executor + hook control logic + skill-local state; "executable state machine + hook policies"	Competitive scores on Harness-Bench with far fewer tokens	2605.19604
P2T (Executable Governance)	Policy documents → normalized machine-readable rules; a compact DSL encodes **harm, scope, conditions, exceptions, required evidence**	—	2512.04408
SIRNA	Business rule documents → SMT formulas (LLM translation + verification), consistency-checked against the SMT representation of the implementation; tax cost calculation case	—	2608.00396
Cedar autoformalization	Agent prompts / MCP tool descriptions / policy documents → Cedar policy language (generate-critique loop)	MedAgentBench	2606.26649
ARc (Automated Reasoning checks)	Natural-language policy → formalization (optionally human-guided), statements verified by automated formal reasoning at inference time	A public service aimed at regulated industries such as finance / healthcare	2511.09008
Logic Sketch Prompting	Introduces typed variables, a deterministic condition evaluator and a rule-based validator into the prompt	Highest and reproducible for all three open models on two pharmacological compliance tasks	2512.22258
NormBench / Deontic trees	Statutes → span-anchored deontic trees (general rules, nested exceptions, counter-exceptions)	Diagnoses "silent scope omission"	2606.08932
Business rules recovered as cascaded lookups	Historical transactions → knowledge-graph-guided candidate business keys (column combinations) → cascaded lookup tables, rather than opaque statistical patterns	Sales-order fields (payment terms, shipping point, sales attribution)	2610.34258
AgenTest	System prompt / tool definitions / domain knowledge → mined business rules → tests for the Agent's business-logic defects	Rules are "scattered across the system prompt and tool definitions, or implicit in domain knowledge"	2610.28116
Typed action contracts	The model only interprets intent and proposes actions; typed action contracts, permission-aware capability exposure, scoped context, pre-side-effect validation, consumer-side execution boundaries	"Bounded autonomy" for enterprise AI	2604.14723
Bellman LMs	The decision space in bounded Agent workflows tends to saturate → "discretize, then decide": solve the decision once as an optimization problem and reuse it	Faster, cheaper, more reliable	2610.24871
LLM as interface, not oracle	The LLM only does schema-constrained feature extraction + deterministic plausibility checks; the final decision goes to structured ML	Pediatric appendicitis	2606.19183

Shared intermediate-representation elements (appearing two or more times): the **read/write artifacts** of a step; the hierarchy of **conditions / exceptions / counter-exceptions**; statements that **require evidence**; **points requiring model judgment** explicitly isolated as "calls"; a **deterministic evaluator** responsible for arithmetic / comparison; **typed variables**. This is the minimal candidate set for defining our logic primitives.

3.2.3 Primitives and Composition Patterns Already Present on Decision Models

3.2.3.1 Official Primitives and Patterns (Details in § 1.1)

- Three question primitives: Noul (P(true)), Choice (distribution over ⩽255 options), Score (ordered distribution over ⩽10 levels); all accept JSON-structured `instructions` / `criteria` / `examples`.
- Five composition patterns: fan-out, composite scoring, confidence routing (global floor + per-action threshold), walking a classification tree level by level, "put the constants in one file for easy review".
- Two cookbooks directly demonstrate "business rules → list of Nouls": the **self-consistency cookbook** runs a car-insurance claim through a 14-Noul scoring rubric 15 times to check whether each answer is stable (consistency_noul_cookbook); the **autoresearch cookbook** uses 29 Scores + 9 Nouls to turn free text into 67 columns of numeric features fed to CatBoost (autoresearch_feature_discovery).
- The example rules on the official evaluation page look exactly like business rules: "every reimbursement must have a receipt; if it cannot be read, have the employee resubmit", "meals over 75 dollars need a manager's signature when the description and the receipt disagree" (evals.typesafe.ai).

3.2.3.2 Third-Party Primitives / Composition Structures

Work	Primitive / structure	Key result	Source
SemTree	Rule decision = semantic decision tree: each node is a natural-language question with explicit **Yes/No/Unknown criteria**, answered by a pluggable semantic judge; the branch structure is **compiled against the answers that judge actually gives**; the execution path is the reasoning, errors are fixed at the node where they occur; corrected nodes become training labels for the judge; a tool-using LLM editor revises questions and subtrees while observing execution results	ShARC 8,276 cases: a tree executed by the host judge 67.70% (2.99 questions/case), on par with asking the final label directly 68.23%; a tree compiled for another judge gets only 47.87% when the judge is swapped, a tree recompiled on the new judge's answers 64.39%; node repair takes BANKING77 82.04%→86.16%; one explicit exemption node cuts false positives on a Chinese e-commerce moderation clause 24→5; the trained editor takes dev accuracy 64.47%→73.35%	2610.13999
Probabilistic perception, deterministic adjudication	Model outputs serve only as **scored observation records**; an admission policy maps scores to true / false / **unknown** under a declared false-positive ceiling; unknown means reject; a deterministic, sufficiency-checked contract adjudicates; the probability that perception changes the adjudication has a **union bound** (sum of each perception field's false-admission rate + unknown rate); minimizing the estimated bound is an effective cost model for choosing contracts	36,000 calls over 2 domains and 2 sensor families, not one unit refutes the bound; all 13 "reject→allow" flips among 21,000 test adjudications come from 3 contradictory records	2610.10978
Clinician-led inspectable rules	Two clinicians define, finalize and adjudicate a rule base of 4 safety checks, formally verified for internal consistency, used as a guard on LLM output; **every adjudication can be recomputed from one named rule and 16 facts**	The largest disagreement class against a matched Llama-3.3-70B judge is structural	2610.30690
Jev-Guard	An editable **policy tree**: first route on the tree to the applicable policy by the caller's audit context, then classify safety conditioned on the retrieved policy	93.72% safety accuracy on 7,033 cases at 58 ms; hierarchical routing 93.10%; full pipeline 240 ms	2610.18326
JEVDB	SQL extension: semantic filter, join, classify and sort go to the decision model, uncertain ones escalate to a generative LLM; **a semantic Bloom filter screens candidates with "registered necessary conditions"**; Yannakakis semi-join reduction	Lowest latency on all 21 SemBench queries, lowest cost on 19; Shelob 540K candidate pairs	2610.02046
Cost-accuracy optimization of semantic queries	Expected error from each decision's calibrated confidence, weighted by its contribution to the output (fan-out) to obtain expected output error; the query plan affects both cost and result	Formal problem definition	2610.08089
Code Owns the Simulation, Jev Owns the Evaluation	A clean boundary: what **can be judged from the input description** (evaluation) Jev can do (CRT 99%); what **depends on things beyond the predicted input** (simulation) Jev cannot	Reflection tests, one-shot matrix games, ALFWorld, robot control	2610.01834
NIOCIS	Execution logs → loop-collapsed action graph (forward transitions + recovery back-edges); at each state ask one typed question "which declared option is the next hop + stop"; **node-conditioned option space**	1,615 production security-operations episodes	2610.25985

Work	Primitive / structure	Key result	Source
SharedKV-BT	Each active node of a behavior tree exposes stage-local fields and candidates, scored in parallel over a shared KV; the selected decision goes to a separate execution system	Robot manipulation	2610.07327
NumericJev	A "range selection" primitive: a multi-way decision tree recursively refines a numeric value	See D2	2609.28587
JevForest	Feature acquisition within a budget: path votes of bootstrapped trees decide the next question to ask Jev	AG News four questions 0.729 vs static ordering 0.667, random 0.583	2610.10615
LAVOIR	Places "missing information slots" next to the options; one forward pass returns both the decision distribution and the **value of information** of each slot	Statistically indistinguishable from the Bayes ceiling on seen schemas (encoder family, cited as control)	2609.30706
Reading Right	Fact probes + code composition	Code composition 0.96–1.00 (see D2)	2610.18579

3.2.3.3 Rubric / Checklist Primitives (Upstream)

- TICK: LLM-generated instruction-specific checklists make evaluation interpretable (2410.03608); Rubrics as Rewards uses instance-specific rubrics as RL rewards (2507.17746); ReVeR derives **hard rules and grading principles** from the instruction itself and judges point by point with tool augmentation (2610.18766); rubric adjudication lacks uncertainty and needs criterion-weighted aggregation (2610.21214); decision models as rubric judges are "cheap, fast, and wrong in the same places" (2609.29769).

3.2.4 "Policy-as-Prompt" Guards: Experience of Feeding Business Rule Text Directly to the Model

Work	Conclusion	Source
gpt-oss-safeguard (120b / 20b)	Open-weight **reasoning** models that read the developer's policy and the content together and output a conclusion plus reasoning; "reasoning helps the model generalize to newly written policies"; traditional classifiers are fast but need retraining when the policy changes	OpenAI
DynaGuard	Dynamic guard for user-defined policies, with optional CoT	2509.02563
SafePyramid	1,000 dialogues, 3,000 policies, 61,699 natural-language rules, three levels: L0 single rule, L1 rule dependency, L2 entirely new policy framework defined in context; **the best model, GPT-5.5, exactly identifies all violated rules in only 54.0% / 35.3% / 12.9%**	2606.29887
LPG	Compresses the internal deliberation of policy grounding into 10 latent tokens; a 4B model reaches 84.5% safety accuracy, about 11× faster than Qwen3-4B-Thinking	2605.17329
Policy-as-Prompt (community governance)	Handing moderation rules to ChatGPT is not enough; the impact on community governance must be considered	2607.12149, 2502.18695
PolicyGuard	DLP classification criteria defined entirely by a **plain-text policy file editable by non-engineers**	2608.02687
BARRED / PAM	Train custom policy classifiers from synthesized training data using only a task description + a few unlabeled samples	2604.25203, 2505.19766
Text de-identification	Natural-language de-identification rules + instruction tuning, outputting auditable rationales	2608.14693
Multi-turn constraint tracking	StateWise: the model maintains its own revisable constraint state (with tombstones); persistent failures and recurrences measured with executable checkers	2610.33766
Jev in moderation	Benefits from retrieved precedents; definitions change predictions without stable improvement (see D1)	2610.07953, 2610.03324

Implication: even the strongest reasoning models reach exact execution rates of only 35% / 13% on "rule dependency" (L1) and "new policy framework" (L2). For a single-pass model to be reliable, rules must be **decomposed** to single-rule (L0) granularity, with dependencies and composition left to code. This is exactly the structure of SemTree and of "probabilistic perception, deterministic adjudication".

3.2.5 Which Works Solve Which Kind of Problem (D3 Mini-Matrix)

Problem	Existing work	Status
Can text teach a model to do things	SkillsBench, AFTER	Proven for generative Agents; **none** for single-pass models
Rule text → intermediate representation	Artic, P2T, SIRNA, Cedar, ARc, LSP, Deontic trees	Several IRs exist, aimed at generative LLMs / formal verification

Problem	Existing work	Status
Separation of judgment points from deterministic logic	FlashScript, TraceCompiler, Formal Skill, LLM-as-interface, Reading Right	**Consensus**: "decisions become calls, the rest becomes code"
Trees / contracts on single-pass models	SemTree, probabilistic perception deterministic adjudication, Jev-Guard, JEVDB, NIOCIS, SharedKV-BT	Prototypes exist; SemTree's "compile against the judge's answers" and Unknown nodes are closest to our needs
Editing interface for non-technical staff	CALICO, Goldsmith, PolicyGuard, SemTree's node repair	Exist, but only SemTree targets a decision judge
Ceiling of direct execution of policy text	SafePyramid	Clear: L1/L2 unreliable
Test sufficiency of rules	Skill coverage, SkillSentry, AgenTest	Aimed at generative Agents

3.2.6 D3 Blind Spots

1. **No "rule language" for single-pass decision models**: all IRs (Artic, P2T, Cedar, LSP) assume the executor can reason or can call an evaluator. The subset of rules a single-pass model can execute faithfully (L0 single condition? with one exception?) has not been defined.
2. **The "executability" of primitives has not been graded**: Code Owns the Simulation gives one boundary ("evaluation vs simulation"), Reading Right gives another ("read vs compose"), 2610.24929 gives a third ("solvable in one pass vs needs intermediate steps"). The three boundaries have not been unified into one primitive-level classification.
3. **Unknown / abstain as a first-class primitive**: both SemTree and "probabilistic perception" treat Unknown as a node output, but D2 shows that fallback and refusal are the weakest point of single-pass models. The contradiction between "every primitive must be able to say I don't know" and "the model cannot say I don't know" has not been studied head-on.
4. **Wording invariance of rules**: Lost in the Fallback and 2610.30045 show that equivalent wordings lead to different decisions. A rule language must either prescribe the wording (templates) or train the model to be wording-invariant; no work has taken either path on decision models.
5. **Composition reliability of rules**: SemTree averages 2.99 questions per case; the hierarchical-consistency evidence in D4 (TV 0.219–0.349) shows that probabilities after decomposition are not self-consistent. For "how errors propagate" when several primitives are composed in code, the only formal starting points are semantic query optimization (2610.08089) and the union bound (2610.10978).
6. **Compiling per judge**: SemTree shows that a tree must be compiled for a specific judge (swapping the judge drops 67.70% to 47.87%). This means a compilation step is needed between "the rules the business expert writes" and "the tree the model actually executes", and what that compiler should look like for a small single-pass model has not been studied by anyone.
7. **The separation of Skill and Policy** has not yet appeared on decision models: Auto-Policy points out that the two differ, Jev-Guard's policy tree is a Policy, TypeSafe's agent-skill is a Skill; no work defines both at once and measures their interaction.

4 • Consistency: the primary metric, its denominator, and failure under symbolic interventions

> Per P1 in §2.2, **the consistency rate is the primary metric and accuracy is the declared floor**. This chapter is the full body of evidence for that primary metric, in two halves.
>
> §4.1 (the former D2 dimension) is consistency under **symbolic interventions**: rules, negation, arithmetic/time, fallback options, instruction hierarchy, adversarial context. Its core finding is that the bottleneck is not in "reading" but in "doing": policy modifications are read out correctly but executed in only 22%–72% of cases. §4.2 (the former D4 dimension) splits "numerical consistency" into **30 sub-dimensions** (A same-input stability / B option-representation dependence / C logical coherence / D uncertainty semantics / E input-content sensitivity / F interface and actions), each with a quantifiable definition and the existing evidence.
>
> This must be read together with §2.1.8: **the "consistency rate" of the same model is 99.83% under the byte-level rerun protocol and 53.7% under the answer-preserving adversarial variant protocol, a gap of 46 percentage points**. So before the denominator is fixed, "consistency rate" is not a metric; and the only denominator with business meaning (the equivalence classes the business declares) has never been defined by anyone.

4.1 D2 symbolic instruction following: rules, negation, arithmetic/time, fallback, hierarchy and adversarial context

> Protocol: the model receives a **symbolic** instruction (rule, threshold, condition, hierarchy, fallback rule); can it execute it faithfully in a single forward pass? Difference from D1: D1 cares about "whether the examples / definitions are used"; D2 cares about "given a rule that has been understood, is the decision produced according to the rule". Difference from D3: D3 cares about "how to write rules so that they get executed".

4.1.1 Upstream baseline: rule following in generative LLMs

Fact	Source
RuLES: across 14 simple text scenarios, LLMs' ability to follow "do not do X" rules can be bypassed by jailbreaks; programmatic evaluation is needed	2311.04235
RuleArena: real rules in three domains (airline baggage fees, NBA trades, tax law) require long context, logical reasoning and exact arithmetic, beyond first-order logic	2412.08972
Scaled rule-following failure: none of 126 leading model variants can count accurately above a model-dependent, grammar-sensitive counting capacity threshold; the failure is abrupt and persists as scale and inference time increase	2605.02028
Verifiable instruction following overfits to a small set of constraints and generalizes poorly to unseen constraints (IFBench); RLVR reward-hacks (constraint satisfaction rises, answer quality falls; CALIF corrects this)	2507.02833, 2610.19525
WISE: rephrasing the same operational requirement changes the compliance rate substantially; five "avoid/exclude" phrasings fall below the positive baseline	2610.05278
Instruction hierarchy: non-adversarial benign conflicts are also common; failures decompose into three classes: "identifying the relevant instruction / resolving the conflict / reasoning correctly but outputting a violation"	2604.09075, 2606.07808
Silent scope omission (SSO): models apply the general rule but quietly drop nested exceptions and counter-exceptions; NormBench, 2,290 Chinese and English statutory clauses	2606.08932
The **constraint tax** on small models: <3B models lose answer accuracy and executable accuracy under structured-output constraints; constrained decoding removes structural errors but exposes scale-dependent semantic gaps	2605.26128, 2609.23742
State tracking needs "reasoning written out in the moment": on a lock-state task, most of the violation reduction comes from reasoning at the moment of action; retained earlier reasoning contributes little, and neither filler text nor an accurate state table can substitute for it	2610.25760

Implication: generative LLMs can compensate for part of rule execution by writing out intermediate steps; a single-pass decision model has no such route, so D2 failures will be "harder".

4.1.2 Evidence on decision models: grouped by failure type

4.1.2.1 Reading right, acting wrong (composition failure)

Reading Right, Acting Wrong (2610.18579) is the core diagnosis for D2:

- On the six most accurate systems, **73%–85% of decision errors occur on states where "every probed fact is answered correctly"**, and this share grows as accuracy rises.
- 14 "capable" fast systems all err on **the same 141 states** (random expectation: 0).
- On generated states where "the probes suffice to determine the action", models that can read every fact correctly and can decide perfectly from the facts **drop 20–45 points as soon as they have to do both things from the trajectory at once**; using their own fact answers recovers 15–28 points; **composing the action from the facts in code reaches 0.96–1.00**.
- On real states "the decision itself is often missing": on ALFWorld, "state the next sub-goal" gives JEV +16 points, while any block of facts gives at most +5.
- Deleting history collapses accuracy where "early observations matter"; **policy modifications are read out correctly but executed in only 22%–72% of cases**; calibration error 2.6%–26.6%.
- Decoder fast systems have latency 27–225 ms, likelihood readout from open-source LMs 317–947 ms, reasoning models 2.3 s.

Conclusion: the bottleneck of single-pass models is the **composition** "facts → action", not "reading". This directly supports TypeSafe's own design principle ("ask independent questions together, compose in code") and the direction of D3 (primitives + composition in code).

4.1.2.2 Arithmetic, time and negation

Evidence	Numbers	Source
The vendor's zigzag: unreliable counting, dates read as text	—	jaggedness
MathQA 17.7 points below the frontier median, below all 19 LLMs	—	2610.11978
In medicine, numerically driven tasks are clearly weaker than text-driven tasks	—	2610.18529
DecisionPivot: on 2,400 binary contrast pairs, Perplexity Decider **gives the higher yes probability to the positive item in 92.3% of pairs, yet classifies both sides correctly in only 74.1% of pairs**; pairwise accuracy from difficulty 1→3 goes 87.7%→57.4% (Instant 90.7%→65.3%); "arithmetic and temporal contrasts expose large performance gaps"; API chat models put top probability ⩾0.99 in 63–93% of cases	**Ranked right, classified wrong** is the typical shape of the arithmetic/time class	2610.28522
Kev's date-fact supplement (code computes the day count and writes it into the state) lifts the deadline-policy questions 0.80→0.90	Direct empirical support for "hand arithmetic to code"	jaredpalmer/kev
Negation complementarity: Jev's mean absolute violation of P(A)+P(¬A)=1 is 0.064; Qwen3.8-27B first-token readout 0.293, verbalized readout 0.122; 19.0% of Jev's pairs deviate by >0.1, and the violation is ~5× its repeat-noise floor; **the mean probability of a single-label statement is 0.380, whereas consistency requires 1/3**	Negation is not free	2609.33209
TypedBench has a dedicated negation suite (Suite G) and a severity-ladder suite (F)	—	2610.11392
NumericJev: refines numerical ranges recursively with a multi-way decision tree; a 10-way tree needs only two rounds for a 100-value grid; 2.93 pp higher than "choosing directly from a candidate list that contains the correct answer"; range-normalized MAE 1.84% vs 5.18% for direct selection	Numerical output can be built in code from a "range selection" primitive	2609.28587
Choice2Seq: Jev's selection accuracy on arithmetic / lexical retrieval is 97.1% / 99.5%, but **serial construction retains only 33.0% / 2.1%**; restricting local options to the characters supported by the candidates, and explicit position indices, improve this	"Picking the right answer" and "constructing the answer step by step" are two different capabilities	2610.16191
Same-number reference swaps: cells in financial statements with the same value but different roles; Jev as an evidence verifier lets wrong-role references through and also rejects valid alternative references; explicit column labels help	Numerical matching ≠ role identification	2610.08675
Sensitive to irrelevant numerical information	—	2610.19619

4.1.2.3 Refusal, fallback and "the correct answer is absent"

Evidence	Numbers	Source
The arithmetic-related refusal bottleneck: answer-present accuracy 99%, correct refusal only 7%; this holds across number magnitude, operation depth, context and refusal label; **native Boolean verification reaches 99% exact match on the same answer-absent arithmetic items**, showing "can verify but cannot refuse"; choosing one threshold on a separate dev set takes the refusal rate 7%→79% while retaining 97% answer-present accuracy, with no retraining	Refusal is a readout-layer/threshold problem, not a capability problem	2609.39496
Lost in the Fallback: under non-adversarial conditions, the fallback rule itself makes models fail: on TREC50, Intern drops from 97.9% on regular assignment to 23.1% on fallback assignment, and this persists even when the correct class is given explicitly; **logically equivalent phrasings of the fallback rule cause large divergence in individual decisions while overall accuracy barely changes**	A fallback rule is a kind of "symbolic instruction" that is understood but not executed	2610.19856
Refusal policy transfer: a Jev policy calibrated on DBpedia rejects 69.3% of covered inputs on Emotion, and the reverse policy loses detection entirely; with 10 intent candidates Laya detects 99.0% of out-of-domain queries but rejects 48.8% of covered queries	Refusal thresholds do not transfer across tasks	2610.03387
Law: when the correct precedent is absent, generative LLMs still pick one on 38–86% of items, the most cautious decision model on 30%; on CaseHOLD, decision model 88.1% vs best LLM 88.2%, at 1/55 the cost	Decision models' abstention is **relatively** better but far from reliable	2610.29961
Medicine: most models can neither recognize that the correct answer is missing nor use the optional abstention	—	2610.18529
Memory retirement gate: Jev never retires wrongly on 141 same-session non-changes, but **retires too early**: the binary gate retires on 39.2% of "merely hinted" changes; a three-way decision brings this down to 15.5% but at a 15.3-point detection cost; cascading and simply raising the threshold cannot make the distinction	Distinguishing "established vs merely hinted" with a binary gate needs a third option	2610.15710
Self-knowledge: with no answer-relevant information, Jev still gives a salient option up to 0.80; on news outside its knowledge boundary, confidence exceeds accuracy by 0.21–0.33 and recalibration cannot fix it; but **targeted yes/no questions** reach AUROC 1.00 for "is the outcome already settled" and 0.95 for "is the evidence sufficient" (confidence itself 0.85)	"Asking a question about the evidence" is more reliable than "looking at confidence"	2610.01006

4.1.2.4 Instruction hierarchy and adversarial context

Evidence	Numbers	Source
Decision hijacking (InjecAgent, 510 cases): malicious content moves action probabilities but rarely makes Jev pick the attacker's target; override markers reduce the effect; an adaptive attack raises the success rate of new verification calls from 1.8% to 3.5%, correlated with small initial decision margins	Direct injection has limited effect on Jev	2609.28613
JevOut: within 64 target evaluations, short "background/procedural" context additions steer 61.4%–73.2% of initially correct decisions to a predetermined wrong option; 91.6% are judged by humans as natural and containing no decisive evidence; affects high-confidence decisions and transfers across models	**Natural context** is far more dangerous than direct injection	2609.30243
JevAdvBench: 862 items, 12 black-box attacks, flipping 46.3% of Jev's correct decisions (reruns flip only 0.17%); **flawed arguments containing no commands flip 77.2%**, close to direct injection at 92.0%; eight open-source TDMs 34.0%–51.7%; among decisions still executed under a 0.8 threshold, 23.5% are wrong	Thresholds cannot block attacks	2609.31142
Option-channel attack: seven open-source models as guards, allow/block accuracy 36%–72% (random 50%); six policy-irrelevant lines of server log raise the fail-open rate from 0% to 63%; **giving the permissive option a misleading name (definition and judged text unchanged) makes the four models that place labels in the input fail open 93%–100%**; all defenses are defeated; decisions flipped by the attack are no less confident than the original decisions	Option names are an attack surface (same root as D1 "names override definitions")	2610.12292
JevProbe: 2,273 tests over 228 real Jev applications, **81.5% show decision flips**, 98.6% of which lead to potential downstream safety impact	Reachability at the application layer	2610.18612
DICE: bounded substitutions in each of three adapter outputs (state extraction, question generation, output interpretation) can change the final decision, while Jev and the API response itself are not under the attacker's control	Interface boundaries are an attack surface	2610.14404
Poisoned into Action: an execution-aware protocol, 1,659 Jev-1.13 calls; under natural retrieval the standalone payload arrives in all 60 trials	Poisoned evidence reaches the action through retrieval	2610.15537
Same attack, different judges: 50 attack variants in 13 families, the first controlled comparison across Jev, 5 GPT/GLM judges, Llama Guard 3 and Prompt Guard 2	—	2610.17820

Evidence	Numbers	Source
TrustJev: protected policy scoring + bounded evidence correction (the correction budget computed from the trusted policy and the candidates); a 0.6B implementation achieves the highest Balance against 14 baselines on three datasets (authentication, instruction hierarchy, executable authorization rules); on the authorization benchmark 95.77% of sources answer all 32 attacks correctly (strongest comparison 80.98%)	**Architecturally separating "policy authority" from "untrusted evidence"** is currently the only structural defense	2610.34826
Calibration and deferral for prompt-injection detection: LLM judge AUROC 0.897, Jev 0.871, Prompt Guard 2 86M 0.757; under distribution shift the allow/block/defer policy needs separate calibration	—	2610.20907
Safety decisions: high overall accuracy and low mean calibration error can mask "high-confidence misjudgment as safe within specific attack groups"; under a strict miss limit, automatic allow is rare, and choosing allow/block thresholds separately increases automation mainly through more blocks	Report per attack group	2609.33401

4.1.2.5 Rule transfer and wording

- "Which Differences Matter" (2610.30045) uses **executable arithmetic rules** to test "rule transfer, joint reliability, information use" separately: adaptation gain increases with the answer consistency of the nearest sources; fully conflicting quadruples give small gains whose sign flips across source splits; **equivalent rewrites have opposite task-level effects**; an improvement from an external table does not prove that the model used the table's values and mapping.
- Certo (2609.37832): a Qwen3-4B joint scorer (state + rule + candidates read together) has rule-sensitivity recall@1 of 1.00; switching to cacheable independent encoding (candidates encoded once and reused, about 5× cheaper with 77 candidates) **loses rule sensitivity significantly** (the abstract says 1.00 → 0.24; the number at that point in the full text is masked, so the abstract is taken as authoritative), and shortlist+rerank cannot recover it; targeted counterfactual supervision recovers it on a held-out synthetic rule task, but the authors **did not isolate "whether the prediction depends on the given rule"**.
- Tables: decision models match LLMs on row classification / fact verification / anomaly detection and are cheaper, and three are more robust to question phrasing, **but no model is robust to how the table is organized**; for every model at least one third of row decisions change with the format (2610.19536).
- Graph tasks: accurate adjacency identification does not guarantee broader structural correctness (2610.06354).

4.1.2.6 Decision models as rule judges

- Across nine panels and seven benchmarks, Jev and Decider lag significantly behind flash-tier LLM judges in at most 6 of 27 comparisons, while the latter are 16–357× more expensive and 5–350× slower; **they err in the same places** (2609.29769).
- Agent trajectory safety: on the JEV benchmark, mean positive-class F1 of 77.8 vs 74.1 for the strongest generative model (2609.34862).
- CALM bias tests: all eight TDMs show bias on 3–8 of 12 items; **an emotional tone pulls the typical TDM's accuracy from 99% to 42%**; all eight change their verdict when the answer positions are swapped (2026 LLM judges barely change); good calibration does not prevent bias (2610.18122).
- First-token readout **overestimates** position bias: three Qwen3 judges do not begin with the verdict token on 12%–49% of pairs, and forced readout returns "whichever appears first" (2610.00054). This is a caution for reproductions that "read out from logits".

4.1.3 Which work solves which class of problem (the D2 mini-matrix)

Problem class	Diagnostic work	Mitigation work	Status
Facts→action composition	Reading Right	Composition in code (same paper)	**Diagnosed clearly; the solution is in the interface, not the model**
Arithmetic / time	The vendor's zigzag, DecisionPivot, MathQA, Choice2Seq	Kev date facts, NumericJev range tree, "arithmetic in code"	The solution is an engineering convention; no model-side breakthrough
Negation / complementarity	2609.33209, TypedBench G	none	No mitigation
Refusal / fallback	2609.39496, Lost in the Fallback, 2610.03387, law/medicine/memory gate	Dev-set threshold (7%→79%), three-way decision, yes/no questions about the evidence	Threshold-level mitigation; **fallback semantics not being executed** has no solution
Natural-context attacks	JevOut, JevAdvBench, option channel, JevProbe, DICE, Poisoned	TrustJev (structural), override markers (weak), reporting per attack group	Only TrustJev is a model-side structural solution

Problem class	Diagnostic work	Mitigation work	Status
Rule transfer / wording	2610.30045, Certo, tables, WISE	Counterfactual supervision (Certo, not isolated)	No general solution
Judge bias	2610.18122, 2610.00054	Test before use	No mitigation
Hierarchy / exceptions	NSHA, white-box diagnosis, NormBench (generative)	none	**Zero research on decision models**

4.1.4 D2 blind spots

1. **The execution rate of the three "meta-rules" (fallback / exception / hierarchy) on single-pass models**: Lost in the Fallback tested only fallback; "nested exceptions silently dropped" (SSO) and "conflicting instructions from multiple sources" have no benchmark on decision models.
2. **A training-side solution to "reading right, acting wrong"**: Reading Right's solution is composition in code; nobody has tried to train a model to compose within a single pass (perhaps it should not be done at all), and nobody has quantified the boundary of "which compositions can be learned and which must be handed to code".
3. **A training objective for negation and complementarity**: 2609.33209 gives the measure; no work has used P(A) +P(¬A)=1 as a training constraint.
4. **A general threshold for refusal**: the threshold in 2609.39496 is task-specific; 2610.03387 proves it does not transfer. "Estimating candidate-set coverage" has not been defined as a problem in its own right.
5. **An evaluation protocol for defenses against natural-context attacks**: JevAdvBench and the option-channel attack prove that "confidence does not change"; TrustJev is only on three datasets; there is no result evaluating TrustJev on JevOut / JevAdvBench.
6. **An "executability grading" of rules**: which rules are doable in a single pass (lookup, matching, single-hop comparison) and which are not (counting, two or more state updates, arithmetic); only 2610.24929 gives a proxy measure, "how many thinking tokens a reference LLM needs" (Jev answers 97.5% of single-pass-solvable items correctly, the rest only 74%).

4.2 D4 consistency: split into 30 definable, measurable sub-questions

> Protocol (per the latest discussion): D4 is not just the four items "repetition, order, wording, calibration"; it has to **define and understand the question itself**. This section lists, one by one, the factors that appeared in the literature of the past two months and that make "the same business judgment" return different probabilities; for each item it gives the definition, the way to measure it, the existing evidence and the existing mitigations, and finally gives the matrix and the blind spots. This section is an exploration of the problem definition, not a solution.

4.2.1 Why Split It

Jev's selling point is "probabilities that can be thresholded, composed and audited". But the evidence below shows: **for the same fact and the same rule, the probability changes with 25 factors unrelated to the business**. These changes are often invisible in accuracy: 2610.35826 found that after Jev was given 8 option orders the accuracy changed by less than 2 percentage points, but the cumulative failure set grew by 27–29%; 2609.27678 found that "corrections and regressions cancel each other out, accuracy is unchanged but the decisions changed". So the D4 measurement must be **decision-level, paired, item by item**, not aggregate accuracy.

4.2.2 Sub-Dimension Checklist

Format for each item: **Definition** → **Measurement** → **Evidence** → **Existing mitigation** → **Status**.

A. Stability on the Same Input

A1 Repeatability (same input, same configuration, multiple calls) - Definition: the variance of the output probability for completely identical requests. - Measurement: intraclass correlation coefficient ICC; top-1 flip rate; absolute probability difference. - Evidence: 24 cultural-values questions, 288,000 answers, ICC 0.997 (2609.36399); Jev's negation inconsistency is about 5 times its repetition noise, which shows that the repetition noise itself is small (2609.33209); "consistent answers also preserve errors": repeated wrong answers account for most of the observed errors (2609.27678). - Mitigation: none needed; but high repeatability does not equal correctness, see A2–A4. - Status: **solved and misleading**: high repeatability makes people mistake it for robustness.

A2 Numerical Precision / Quantization / Merge Drift - Definition: the change of top-1 for the same weights under different precision, quantization or after merging. - Measurement: proportion of top-1 changes; maximum probability deviation. - Evidence: Kev's server-side bf16 differs from fp32 probabilities by at most ~0.03 (GPU) / 0.05 (Mac), about 1 top-1 change per 300 questions (Kev README); Chinese Decision's merge and Q8_0 quantization each change 0.4–1.1% of top-1, the aggregate metrics are almost unchanged (2610.23170); Laya-typed's default CUDA path produced 5 paragraph swaps, and the probability shifts disappeared after an FP32 rerun (2610.14631); the damage from 4-bit quantization (RTN/GPTQ/AWQ) varies with model size and did not push any configuration into the passing range (2610.00025). - Status: **measurable, small in magnitude, but flips at the threshold boundary**.

A3 Version / Model Drift - Definition: the change of decisions for the same business rule across model versions or across models. - Evidence: 2610.35826 covers two Jev versions, 12 reproductions, 6 LLMs; the tree SemTree compiled for judge A drops from 67.70% to 47.87% when moved to judge B (2610.13999); the rankings in Decision Index 0.4 fluctuate across versions (Decision Index). - Status: **almost no systematic research**; critical for our "self-trained model + business rule library" (whether rules must be recompiled with the model).

B. Dependence on Option Representation

B1 Option Order / Position - Definition: whether the decision changes after the option order is shuffled. - Measurement: flip rate of one shuffle; expansion of the cumulative failure set across 8 presentations; residual after cyclic-shift averaging. - Evidence: row 8 of the official jaggedness page admits it (jaggedness); Archer Hume's probe: card placed first 12/16, placed last 16/16 (architecture); after reordering, Jev's accuracy changes <2 pp but the cumulative failure set grows +27–29%, the gain from voting is limited, "unanimously agreed answers are also often wrong" (2610.35826); for Qwen3-8B / Gemma 4 12B letter readout, one shuffle changes 20.0% / 11.9%, the published decision fine-tuning still changes 9.1%; changing only the letters without moving positions changes 9.1% / 14.0%, moving only positions without changing letters changes 12.0% / 22.7%; training on shuffled data plateaus (2610.22066); reversing the order changes 30% of Laya's answers (2610.02267); order mainly changes the answer when the model is uncertain about the top few, yet list-style answers look confident (2610.18233). - Mitigation: block-causal attention (2610.01601); Prism-1's position- and opponent-independent scoring (zero flips); Runoff's two stages (zero flips, cyclic-average residual 9.3%); AnyJev rotation + label-prior correction; PACT has the lowest position bias (2609.35865); Kev's `/permute` endpoint. - Status: **clearly defined, mature measurement, structural mitigations exist**; but the mitigations all target Choice, and the position problem of Score (B5) is counted separately.

B2 Option-Set Dependence / Independence of Irrelevant Alternatives (IIA) - Definition: whether the log-odds between the correct option and the strongest wrong option change after adding an option that is neither correct nor the strongest competitor. - Measurement: ratio of the root-mean-square systematic shift of the log-odds (nats) to the between-run variation. - Evidence: Jev 1.13, pre-registered, 299,463 calls: every added option moves the log-odds, 0.20–0.52 nats, 1.2–3 times the between-run variation, even when the new option takes only 0.1% probability; the shift is directional (options that resemble the wrong answer draw probability from it, +0.10 nats; paraphrases of the wrong answer get less than half of its 1/3); it grows with log(number of options), about 1 nat at 255 items; out-of-domain padding pushes the odds away from the correct answer; **multiple questions packed into one call do not affect each other (deviation <0.001)**; independent per-option Noul removes the dependence but accuracy is 8.7 points lower (2610.16655); Archer Hume: an irrelevant option moves the log-odds +0.38→+0.11, in all 10 blocks (architecture). - Mitigation: Prism-1's opponent independence; JevCascade's menu curation (0.8B on a curated 5-candidate menu 95.4% vs 4B on all 150 labels 80.0%, 2610.07716). - Status: **pre-registered evidence and a formal definition exist**; "the probability is only valid relative to its option list" should become part of the interface documentation.

B3 Dependence on Option Names / Definitions - Definition: how much the decision changes when option names are changed or definitions are removed or altered. - Evidence: see D1 § 3.1.2.2 for details: replacing option names with meaningless codes takes AUC 93.8→23.2 (2609.26758); definitions can be removed (0.8487→0.7971), contradictory definitions 0.8511→0.2281 (2610.02586); giving the permissive option a misleading name raises the guard's false-allow rate to 93–100% (2610.12292); putting the literal meaning of an idiom into the option list draws away 65–96% of the remaining errors (2610.22526). - Status: **consistent conclusion: option names are the strongest lever and the largest attack surface**.

B4 Probability Allocation under Multiple Correct Options - Definition: whether probability is distributed fairly among several options when they are all correct. - Evidence: three decision models, even when they rank all correct items above all distractors, still put at least one correct item below 1/n in 28–50% of the lists; five identical correct items differ in probability by 17–79 pp; **a threshold cannot identify the correct set** (2610.29217). - Status: **defined,**

measured, no mitigation. Multi-label business rules ("which of the following clauses are triggered") must be split into multiple Nouls.

B5 Ordinal Scale Compression / Tendency toward the Middle - Definition: in Score-type questions, decisions concentrate on fewer effective levels than the gold standard. - Measurement: effective label utilization (the proportion of the gold standard's effective support covered by the decisions); TVD. - Evidence: JEV 1.13 on 6,400 ANLI questions reaches 74.95% accuracy, yet gives 38.8% of predictions and 51.3% of errors to Neutral, while the gold standard is nearly balanced; on 36 ordinal datasets it uses only 60–76% of the effective gold support, versus 87–102% on 4 nominal tasks; finer scales worsen the compression; random order weakens but does not remove it; widening the level spacing raises accuracy but does not restore utilization; larger models recover inconsistently; targeted training can mitigate it (2609.38827). - Status: **defined, measured, a training mitigation direction is given**. Business scoring rules on a "1–10 scale" must take this into account.

B6 Question-Form Equivalence (Noul / Choice / Score / Packed / Separate) - Definition: whether the probabilities agree when the same judgment is expressed as Noul, Choice, Score, or as several questions packed together / asked separately. - Evidence: for Jev the single-label probabilities of "the label is X" over the three options sum to 1.14 on average (2609.33209); normalized per-option Noul preserves higher accuracy than Choice scoring (medicine, 2610.18529) but is 8.7 points lower on intent classification (2610.16655): the direction is **inconsistent**; pplx-decider asking several facets of the same message jointly gets 68.09 vs 67.75 separately, and joint asking over independent passages drops slightly by 0.2–0.4 (pplx-decider); Kev provides a "packed vs separate" comparison tool. - Status: **the evidence contradicts itself and depends on the task**; "which primitive a business rule should be expressed in" has no general answer and is a quantity that D3 primitive design must measure.

C. Consistency of Logic and Decomposition

C1 Hierarchical / Decompositional Coherence - Definition: whether asking the fine label directly agrees with the distribution reconstructed from "ask the coarse class first, then the fine label". - Measurement: total variation TV at the coarse-class level; change in accuracy and ECE after reconstruction. - Evidence: 72,000 questions: Jev's TV 0.219–0.349, Laya 0.424–0.689; on CLINC150 reconstruction changes Jev's accuracy by −22.9 pp [−24.9, −20.9] and Laya's by +21.3; on MASSIVE Laya's accuracy +9.2 but ECE 0.046→0.124 (2609.33971); SemTree's multi-step tree is on par with asking directly (67.70 vs 68.23, 2610.13999). - Status: **defined, clearly measured, the direction differs by model**; the "rule decomposition" of D3 must answer "are the probabilities still usable after decomposition".

C2 Probability Axioms such as Negation / Complement / Conjunction - Definition: P(X)+P(¬X)=1, exclusive labels sum to 1, a conjunction is not larger than its single terms. - Evidence: Jev's negation pairs deviate by 0.064 on average (CI 0.055–0.072), Qwen3.8-27B first token 0.293, verbalized 0.122; Jev's violation is about 5 times the repetition noise and concentrates where it is uncertain; Qwen rejects both the statement and its negation in 196/480 pairs (2609.33209); Jev gets famous probability puzzles right every time but violates the same rule every time on adapted versions; 54% of 15,000 answers on 30 conjunction tasks violate it (2610.19619). - Mitigation: detectable without labels (axiom tests). - Status: **defined, label-free measurement exists, no training-side mitigation**.

C3 Invariance under Equivalent Wording - Definition: whether a logically equivalent rewording of the rule or question changes the decision. - Evidence: "decision services change their answer across equivalent wordings more often than across repetitions of the identical text" (2610.30045); logically equivalent formulations of fallback rules produce substantive disagreement (2610.19856); 21 rewrites of table tasks: three decision models are robust to the question wording, **none is robust to the table organization**, and for every model at least 1/3 of the row decisions change with the format (2610.19536); on medical tasks Jev keeps strong consistency under changes of question format (2610.18529). - Status: **clearly defined; the evidence shows "question wording" is more stable than "data format"**.

C4 Framing Effects and Cognitive-Bias Manipulation - Definition: whether the probability changes when only the presentation (gain/loss framing, anchoring, irrelevant numbers) changes and the facts do not. - Evidence: at the level of the selected answer Jev avoids most human biases, but it is sensitive to irrelevant numerical information and option order, and several bias manipulations significantly change the returned probability even when the selected answer is unchanged (2610.19619); the effect of gain/loss framing on high-stakes LLM advice (upstream control, 2610.29294). - Status: **selected answer stable, probability unstable**: thresholded systems are affected.

D. The Meaning of Confidence

D1 Calibration vs Discrimination vs Sensitivity (three things, not one) - Definition: calibration = probability matches frequency; discrimination = AUROC of ranking right against wrong; sensitivity = whether confidence can separate right from wrong. - Evidence: models with similar accuracy differ greatly in AUROC, and the confidence of the most sensitive model tracks "which kind of situation will fail" rather than "which answer is wrong"; the gain from deferring the 30% least certain decisions has rank correlation 0.87 with AUROC (2610.09683); Jev ranks completion requests best (AUROC 0.78 vs rules 0.62, small LLM 0.60) yet is poorly calibrated: mean "completed" probability 0.50 while 87% actually pass (2610.24649); in Indonesian the ECE does not increase but the ability to rank its own errors degrades on all three datasets (2610.14112); Intern-Decision's Brier 0.628→0.550 and ECE 0.213→0.089 come from the training objective (2610.22131). - Status: **the concepts are separated; the three must be reported separately**.

D2 Two Kinds of Uncertainty (human disagreement vs decision risk) - Evidence: 142,658 moderation comments: the model's AUROC 0.83 is indistinguishable from 13/14 frontier models; "confidence" mixes annotator disagreement with decision risk, and under asymmetric loss the two demand opposite escalation signals; splitting the judgment into parallel sub-questions and reading their dispersion recovers the disagreement signal, but only for the weakest model and only when the decomposition aligns with the source of disagreement (2610.33258). - Status: **defined; mitigation is conditional**.

D3 Self-Knowledge (is low confidence "random" or "don't know") - Evidence: with no relevant information at all Jev still gives the salient option up to 0.80; on news beyond its knowledge boundary confidence exceeds accuracy by 0.21–0.33 and recalibration on early months cannot fix it; targeted Nouls are sharper: whether the outcome is already settled AUROC 1.00, whether the evidence is sufficient 0.95 (vs answer confidence 0.85); "do you know the answer" seems to flag fictitious entities and post-boundary news (0.91), but after removing dates or using real names it is no better than answer uncertainty (2610.01006); on option lists without semantics the accuracy is random, yet the best-calibrated model still reports mean confidence 0.82; open-source models' confidence is inverted, and gating by confidence is worse than random (2610.22526). - Status: **defined; surface cues disguise themselves as self-knowledge**.

D4 Calibration and Thresholds under Prevalence (Prior) Shift - Evidence: Nimble-9B has only one more LoRA than Qwen3.5-9B, is better calibrated on intent tasks but Macro-F1 −12.0 on financial sentiment; under moderate shift, Jev's ECE and threshold risk are significantly higher than fine-tuned DeBERTa only on financial sentiment (+1.7/+2.5 pp), with an average gap ⩽1 point over 2,000 scenarios; EM prior correction (label-free) lowers ECE by 2.7/1.8 points (2610.26325); SHIFT-JEV: KL anchor + temperature takes the calibration error under shift 0.289→0.087, and a chi-square test detects 95–99% of the class-prior shifts missed by confidence/embedding detectors (2610.34329). - Status: **measurement and cheap mitigation exist**.

D5 Inherited Teacher Bias - Evidence: exact risk computation on a Gaussian proxy: the student inherits the teacher's bias, more student data does not remove it, and agreement with the teacher cannot prove correctness (2610.26797); in the memory-retirement task, LLM majority-vote gold labels share most errors with the upgraded model of the same family (2610.15710); AgentDecision-4B uses Jev answers as labels (2610.21412); Releasing Inner Jev uses Qwen3.5-122B / Qwen3.6-27B as teachers (2610.17205). - Status: **clear in theory, almost never measured empirically on decision models**. A direct risk for our "small model + distillation" route.

E. Dependence on Input Content

E1 Numbers / Dates / Arithmetic (see D2 for details) - Evidence: MathQA is 17.7 points below the best LLM (2610.11978); numerically driven medical tasks are an explicit capability boundary (2610.18529); refusal bottleneck 99%/7% (2609.39496). - Status: **boundary is clear, the vendor recommends "put arithmetic in code"**.

E2 State Length / Irrelevant Detail / Indirection - Evidence: rows 4 and 5 of the official jaggedness page; Jev has no mis-retrieval record within 20k tokens and 64 options, but drops to 53–80% when the answer depends on two or more updates from the same user (16 updates), because similar records interfere with each other (2610.24929); Laya 31% on 50 nearest-neighbor tools, Jev 98% (2610.02267); Archer Hume: serving time is essentially flat up to ~100 questions, question text costs about 2 times the state (architecture). - Status: **"reading" is stable, "aggregating" is not**; boundary = intermediate steps required.

E3 Flips from Natural Background Information (non-instructional adversarial) - Evidence: JevOut: within 64 target evaluations, short background / procedural additions steer 61.4–73.2% of initially correct decisions to the predetermined wrong option, without any explicit answer instruction (2609.30243); six lines of policy-irrelevant server

logs raise the guard's false-allow rate 0%→63% (2610.12292). - Status: **defined, high risk, no effective mitigation** ("escalate the least certain" does not work: decisions reversed by the attack are not less confident).

E4 Record vs Observation (claim vs reality) - Evidence: Claim Laundering: in all 12 configurations the record influences the commitment decision more than the observation; after a claimed success that actually failed, 53–100% still stop / proceed / purchase; a linear probe can tell them apart (AUROC 0.77–1.00); EVIGATE's middle-layer veto +18.7 (ALFWorld) / +5.0 / +5.1, but offline calibration does not transfer to live trajectories (2610.25947); given a stale summary, Jev follows it (2610.24929); in compressed summaries 83% of failures trace back to summary entries with no environment support (2610.20927). - Status: **defined; the most dangerous consistency problem in agent scenarios**.

E5 Language / Culture / Persona - Evidence: Saudi vs US personas reproduce 87% of the human differences in English and 62% in Arabic; the difference comes from the question language, not the persona language, and confidence is lower in Arabic (2609.36399); Indonesian regional languages: human translation into Indonesian gains up to +37.2 points, ECE does not increase but the ability to rank errors degrades (2610.14112); the same medical question from English to Yoruba loses up to −39 pp accuracy (2610.31315); Bengali is captured by the literal meaning 1.9–6.4 times more often than English, "the resistance of commercial models is a property of the test language" (2610.22526). - Status: **defined; the only Chinese evidence is Chinese Decision's CJ-Bench** (2610.23170).

E6 Style / Emotion / Judge Bias - Evidence: CALM's 12 bias tests: all 8 TDMs are biased on 3–8 items, emotional tone has the largest effect, rewriting the better answer in an emotional tone takes a typical TDM's accuracy 99%→42%, and TDMs are more biased than the LLM judges of 2026 (2610.18122); JEV-as-Judge confidence routing fails under style adversaries and on reference-free prose (2609.26550). - Status: **defined, measured, no mitigation**.

F. Interface and Composition

F1 Readout Method (first token / verbalized / typed head / pointer) - Evidence: first-token readout overestimates position bias: for a Qwen3 judge, 12–49% of the pairs do not begin with the verdict token, a forced read returns "who is listed first", and after swapping 89.7% flip vs 47.5% when reading after generation (2610.00054); first-token negation bias 0.293 vs verbalized 0.122 (2609.33209); a typed head has no consistent advantage over LM-head readout (2609.25845, 2609.32160); JevCascade's readout stability is an ordinal property, not a probabilistic one (menu interventions predict with 4.2 points of error, probability-level variants 21.0 points) (2610.07716). - Status: **the readout choice affects bias measurement results**; reproduction work should report the readout type.

F2 Selection vs Sequential Construction (Choice2Seq) - Evidence: on arithmetic / lexical retrieval Jev's whole-selection accuracy is 97.1% / 99.5%, but only 33.0% / 2.1% is retained through local sequential construction (2610.16191). - Status: **defined**; a business rule "judged step by step" and "judged as a whole" can give completely different results.

F3 Request Independence / Packing Side Effects - Evidence: Nimble's request expansion on TREC DL2019/2020 causes 9.1%/8.0% strict binary flips and changes 21.4%/18.9% of the select-or-abstain actions; all 70 flips trace back to schema fields serialized into the scoring prompt; Von's fixed shared separator positions remove the flips (2610.14631); Jev's packed questions do not affect each other, <0.001 (2610.16655). - Status: **the hosted model passes, open-source reproductions not necessarily**; an acceptance metric for reproductions.

F4 Consistency after Decision→Action Composition - Evidence: the most accurate system, once composed into actions, most often ignores emails that need handling (18% vs 4% for the best LLM); a cost threshold on the noisy probability halves this (2610.32955); DICE: the three adapters for state extraction / question generation / output interpretation can each change the final decision while Jev's response is unchanged (2610.14404); union bound and cost-accuracy formalization (2610.10978, 2610.08089). - Status: **defined; product metric ≠ decision accuracy**.

F5 Adversarial Injection (explicit) - Evidence: JevAdvBench 862 questions; option-channel attacks give 36–72% accuracy (random 50%); JPDH program decision hijacking in real applications (2610.18612); injection detection AUROC 0.897 (2610.20907); Jev is most robust on tasks with "an explicit rule to look up" (0 bypasses for 8 forgeries, "why would anyone make such a request" bypasses 9% vs 56% for GPT-6 Astra), while on phishing detection, which needs holistic judgment, the same forgeries bypass 19–29%, and without attack it detects only 62% and mislabels 27% of benign requests (2610.17820); TrustJev separates policy authority from evidence (2610.34826). - Status: **many definitions, few mitigations**; see D2 § 4.1.2.4 for details.

F6 Consistency of Refusal / Fallback / Abstention (see D2 § 4.1.2.3 for details) - Evidence: arithmetic 99%/7%; with missing legal authority LLMs still choose 38–86%, the most cautious decision model 30% (2610.29961); refusal policy transfer 69.3%/48.8% (2610.03387); in memory retirement the binary gate retires prematurely 39.2%, the ternary gate

drops this to 15.5% but detection −15.3 (2610.15710); in medicine most models are weak at both "recognizing a missing correct answer and using the optional abstention" (2610.18529). - Status: **the weakest link**.

4.2.3 D4 Matrix: Which Work Solves Which Kind of Problem

Sub-dimension	Defined	Measured	Mitigated	Representative work
A1 Repeatability	✓	✓	Not needed	2609.36399, 2609.27678
A2 Precision/quantization	✓	✓	—	Kev, 2610.23170, 2610.14631
A3 Version drift	Weak	Weak	—	2610.35826, SemTree
B1 Order	✓	✓	✓ (structural)	2610.22066, 2610.18233, 2610.00831, 2610.01601, 2609.35865
B2 IIA	✓	✓	Partial	2610.16655, Prism-1, 2610.07716
B3 Option names/definitions	✓	✓	—	2609.26758, 2610.02586, 2610.12292, 2610.22526
B4 Multiple correct	✓	✓	—	2610.29217
B5 Ordinal compression	✓	✓	Training direction	2609.38827
B6 Question form	✓	Contradictory	—	2609.33209, 2610.18529, 2610.16655, pplx
C1 Hierarchical coherence	✓	✓	—	2609.33971
C2 Probability axioms	✓	✓ (label-free)	—	2609.33209, 2610.19619
C3 Wording	✓	✓	—	2610.30045, 2610.19856, 2610.19536
C4 Framing	✓	Partial	—	2610.19619
D1 Calibration/discrimination/sensitivity	✓	✓	Training objective	2610.09683, 2610.24649, 2610.22131
D2 Two kinds of uncertainty	✓	✓	Conditional	2610.33258
D3 Self-knowledge	✓	✓	—	2610.01006, 2610.22526
D4 Prevalence shift	✓	✓	✓ (EM/KL anchor)	2610.26325, 2610.34329
D5 Teacher bias	Theory	—	—	2610.26797
E1 Numbers	✓	✓	Code	2610.11978, 2609.39496
E2 Length/irrelevant	✓	✓	Filtering	2610.24929, 2610.02267
E3 Natural background flips	✓	✓	—	2609.30243, 2610.12292
E4 Record vs observation	✓	✓	Partial (probe veto)	2610.25947, 2610.20927
E5 Language/culture	✓	✓	Translation/cascade	2609.36399, 2610.14112, 2610.31315, 2610.22526
E6 Style/emotion	✓	✓	—	2610.18122
F1 Readout	✓	✓	Report readout type	2610.00054, 2610.07716
F2 Selection vs construction	✓	✓	—	2610.16191
F3 Request independence	✓	✓	Format fix	2610.14631
F4 Decision→action	✓	Partial	Starting point for formalization	2610.32955, 2610.14404, 2610.08089
F5 Explicit injection	✓	✓	Weak	2609.31142, 2610.12292, 2610.17820
F6 Refusal/fallback	✓	✓	—	2609.39496, 2610.29961, 2610.03387

4.2.4 D4 Blind Spots

1. **There is no "consistency benchmark" that puts these sub-dimensions together**: DecisionPivot (2610.28522) is the closest (a control group of 48 tasks and 7 roles), but it covers only part of B1/C3/F6; TypedBench (2610.18488) and System1-Arena each cover another part. No work reports A–F at the same time.
2. **There is no consistency data for "Chinese + business rules"**: the E5 evidence is all English / Indonesian / Arabic / Bengali / Yoruba; for Chinese there is only CJ-Bench's accuracy and ECE.
3. **A3 version drift and D5 teacher bias have no empirical evidence**, yet these two items decide "whether the rule library must be recompiled with the model" and "whether a distilled small model inherits Jev's B2/B5/C2 defects".
4. **The evidence on B6 question-form equivalence contradicts itself**, which means every primitive of D3 must be tested separately on the target model for "Noul or Choice".
5. **Mitigations cover only the two items B1 and D4** (structural order independence, prior correction). The mitigations for B2–B5, C1–C3, E3 and E6 all stop at "report it" or "a training direction".

6. **Decision-level paired evaluation has not become the norm**: most reproduction work still reports only accuracy and ECE; 2610.35826 and 2609.27678 show that this systematically overestimates robustness.

5 • System: business goal, property list, and the positions of System1+System2

This chapter answers "**what it is worth, what it can promise, and where it sits in the system**". Note that P5 in §2.2 has already demoted cost from a "business goal" to an "enabling condition", so the cost evidence in §5.1 should be read as a **precondition**, not a goal.

§5.1 gives the class of cost that is eliminated (the one to two orders of magnitude from not decoding), the Jevons-style consequence, which two of the twelve business properties can be promised to the business and which eight have counter-examples measured by independent audits, and the five engineering-distinct positions that support System1+System2 (the first four cells have positive evidence; the fifth cell, the "closed-loop executor", fails across the board). §5.2 (the former dimension D5) is the evidence on fast/slow cooperation Agent frameworks; its core finding is "**faster decisions ≠ a better Agent**".

A built-in gradient: push determinism all the way into the harness and the model becomes redundant (one deterministic rule on typed policy fields gets everything right; a model-free controller solves 79%). So the upper bound of the usable range is how structurable the business fields are, and the lower bound is the repetition of the decisions.

5.1 Business goal and properties: which class of cost is eliminated, which position is supported

Protocol: still analysis and survey only. This section answers the first follow-up question: **what Jev's business goal is, what its properties are, which current business problem it solves, and how it ultimately supports the new System1+System2 agents**.

Conclusions first: (1) **The starting proposition of this section is "the migration of idempotence", but it does not survive testing**: the proposition says Jev adds no new business-processing capability and only replaces generative judgments that do not guarantee idempotence with "typed metrics + deterministic execution in the harness". §2.1 tests it with six pieces of counter-evidence and concludes that **both the premise and the conclusion must change**: the host Jev is itself non-idempotent (0.78%–1.33% of decisions change under identical repeated calls; four papers report its non-determinism), the format guarantee is not something it brings (it belongs to the existing lineage of grammar-constrained decoding) and carries a "constraint tax", the harness's execution plan itself changes the answer, and capability is not conserved (it is lost outright at multi-step/stateful points, and the harness receives no signal). The version that can be defended is: **it does not eliminate non-idempotence; it shrinks the surface of non-idempotence to a countable, boundable, priceable set of fields, at the cost of failures turning from explicit into silent** (§2.1.4). **Treating idempotence as a "design purpose" rather than an "accomplished fact" is a separate proposition and must be tested separately**: taken literally it has already been achieved (same-request ICC 0.997, same-text rerun flip rate 0.17%) yet buys nothing ("a consistent answer also preserves errors"; repeated errors account for most of the observed errors), and it can be bought without a model (a lookup table is strictly idempotent; deterministic rules are correct on all six policies; a model-free controller solves 79%). **An objective function whose optimum is "no model" cannot be handed to a model team**. The testable version is a triple: same judgment within an equivalence class + a decision still produced outside the equivalence class + a unit price one to two orders of magnitude lower; it translates into the model-side requirements of **a monotone thresholdable score, native unknown, and invariance within business equivalence classes**, while idempotence itself lives in the thresholds and the deterministic contract (§2.1.6). (2) The business goal that follows **is not to make decisions more accurate, but to push the unit price of "a decision" down to the point where it can be called on every branch of a program**, so that a large class of judgments that "were not worth having a model make" becomes worth it. (3) In its property list, **only two items are structural guarantees** (the output is necessarily within the declared option set; one forward pass answers the whole request); the rest (calibration, invariance, fallback, self-knowledge, probabilistic self-consistency) are all **empirical properties that vary with the workload**, and several independent audits have already measured counter-examples. (4) The way it supports System1+System2 is not "one fast model plus one slow model" but **five engineering-distinct positions**, among which the strongest evidence is for "outsourcing every external judgment

> of the slow model", and the weakest is precisely "letting the fast model run the Agent loop by itself". The failure of that cell is directly predicted by "the decision being moved back into the model".
>
> **Re-check note (§8.1)**: §2.1 has already tested this section's starting proposition in place; §8.1.1.2 further separates the **diagnostic part that should be adopted** from the **descriptive part that should be replaced**, and marks this section's "three layers of idempotence" (high overfitting risk), "five positions A–E" (medium), "decision repetition axis" (high) and "twelve business properties" (known to be incomplete) as the report's own constructs rather than literature conclusions.

5.1.1 The Business Problem: Which Class of Cost Is Eliminated

5.1.1.1 What Is Eliminated Is the Cost of "Decoding a Passage of Text for Every Judgment"

Evidence	Measured quantity	Protocol
Prefix-only decision model	One call is **one to two orders of magnitude** cheaper than a **same-size** generative language model; under interventions that keep the menu fixed and change only the option set, post-intervention accuracy is already determined by the cached first-pass distribution (2610.07716)	Structural source of the cost: no decoding
As a rubric judge	LLM judges are **16–357× more expensive and 5–350× slower**, while the two decision models are significantly behind in at most 6 of 27 comparisons (2609.29769)	Same human criteria, same panels
As a judge (bias protocol)	TDMs are **37× cheaper** than 2026 LLM judges and more consistent (2610.18122)	CALM bias tests
Social-science text annotation	Median **cost 44× lower**, at the price of a median 11.6 macro-F1 below "the best LLM per task" (behind on 14 of 15) (2609.24574)	18 tasks, 7,977 items, zero-shot, same protocol
Alignment-failure detection	Cost **1/63** of an LLM judge; one generic question, zero-shot median AUROC 0.886 (2609.29429)	10 failure classes, 44 benchmarks
Content moderation	**About 0.000025 USD per call, about 100 ms**; zero-shot AUROC 0.83, **statistically indistinguishable** from 13 of 14 frontier systems (paired bootstrap, n=56; this is "failure to reject equality", not a proof of equivalence), cost 1–3 orders of magnitude lower (2610.33258)	142,658 human-labeled items
Local 2B	**30.9 ms** per decision (single laptop GPU), **zero output tokens**; versus 8,758 ms for a frontier API call, where the host systems that answer well first "think" for 21–212 generated tokens per question, each billed; a single consumer GPU sustains 32 decisions/s (2609.23886)	Local inference, state never leaves the machine
Agent single-hop routing	Typed encoder **10 ms** (typed selective 19.5 ms), versus **2.2 s** per LLM hop (2610.25985)	1,615 production security-operations trajectories
Guard	Safety classification averages **58 ms**, routing + classification end to end 240 ms; the permission gate cuts mean review time per task from 4.80 s for an SFT-LLM judge to 1.33 s (2610.18326)	7,033 evaluation cases
Inference-latency comparison	The largest System-1 model matches or exceeds fast closed-source LLMs on **non-Agent** accuracy, with better probability calibration and **1/4** the latency; the most accurate are reasoning LLMs, at about **70×** the cost of fast LLMs (2610.21244)	92k questions / 105 tasks / 268 sources

In one sentence: what is eliminated is not "model inference cost" but **decoding cost + round-trip cost + parsing cost**. All three are tied to "the answer is a passage of text", not to "the answer is a judgment". 2609.23886 puts it most plainly: what the program needs back is "not prose, but one of n declared options plus a number that a threshold can be set on".

5.1.1.2 So "Judgments That Were Not Worth Making" Become Worth Making

This is exactly what TypeSafe meant when it named its product after Jevons, and it is the starting point of the only paper that treats this as an **organizational** problem rather than a model problem: when the total marginal cost of machine evaluation falls far enough, **organizations consume more machine evaluation wherever latent demand is sufficient and complementary costs do not dominate**; the paper separates prediction, machine evaluation, organizational judgment and delegation into four functional activities whose costs need not fall together, and it states the price explicitly: evaluations will **share evidence, criteria and errors**, will **scale up a mis-specified rubric**, will run on representations from which "the consequential qualifiers have already disappeared", and will **actually change the allocation of decision rights** through thresholds and exception routing (2610.01231).

This "price" must be written next to the "benefit", because it predicts almost every counter-example in §5.1.2: when a mis-specified criterion is run cheaply over a billion records, the errors are not independent. 2609.29769 measured one

concrete form of this: **on the decision model's "most confident errors", a flash-tier LLM judge repeated the same error more than 85% of the time**, whereas independent errors should give about half. So a cascade can only save money; a cascade with a flash-tier fallback beats the strongest single judge by at most 2.7 points even with an oracle threshold.

5.1.1.3 Businesses That Can Be Made Cheap Share One Structure: The Decisions Repeat

Three mutually independent papers point to the same condition.

- **Semantic databases**: using LLMs for relational discrete judgments causes high latency and high cost; switch to typed decision models for semantic filter/join/classify/rank and escalate only the uncertain cases to a generative LLM. On the 21 SemBench queries latency is lowest on all and cost lowest on 19; on Shelob with 540k candidate pairs mean F1 is 95.7%–97.5%, a semantic Bloom filter first prunes **87.4%** of the candidate pairs, and reusable conditional index scoring then **reduces reasoning-model escalations by 55.2%** (2610.02046).
- **Bounded Agent workflows**: the decision space **saturates**: the Agent keeps meeting the same states, and after enough conversations it stops meeting new ones; yet the LLM does not treat these decisions as an optimization problem and re-derives them from context every turn. Solve it once as an MDP and reuse it: on τ^2-bench telecom a model-free controller solves **79%**; Qwen3.8-27B rises from 29% to **96%** with **93%** fewer model calls; for Claude Opus 5.5, which already solves 96%, running it this way cuts per-task calls and cost by about **98%** and solves 2 more tasks (2610.24871).
- **Production tickets / security operations**: the Agent pays per decision, rereading the state and thinking for seconds at every hop, "while its actions are finite and the decisions repeat" (2610.25985).

Corollary: the business value of a Jev-class model is proportional to **decision repetition** and inversely proportional to **decision novelty**. This gives "where to put Jev" a decidable engineering criterion, and it explains why the "closed-loop Agent" in § 5.1.4 is the weakest cell: states in a closed loop hardly ever repeat.

5.1.2 The Property List: What Can Be Promised to the Business and What Cannot

The table below lists, item by item, "the properties on which the business side would place an order"; the right column holds only **counter-examples**, because the positive examples are already in § 5.1.1 and § 1.1.

#	Property	Can it be promised	Measured counter-examples / boundaries
P1	**The output is necessarily within the option set declared by the caller**; a format error is impossible	**Structural guarantee**	No counter-example. This is the only property that "holds by construction" (2609.23886). But it guarantees only the **type**, not the **semantics**: swapping the no/yes names of a rubric can push Jev's AUC from .815 to .581 with **zero type errors** (2609.32160 relaying T3)
P2	**One forward pass answers all questions in a request**; extra questions are nearly free (**this is a property of the Jev architecture; the fixed order of § 2.2.1 does not have it**, caching moves to the rule axis)	**Structural guarantee** (shared prefix + isolated branches, § 1.1.4)	The latency advantage can be erased by caching: one paper's latency advantage disappears once caching is added, and two sister papers report **opposite cost directions** on different workloads; the item "if you report a latency advantage, report both with and without caching" is complied with by only **3** of 20 applicable papers (2609.32160)
P3	**The probabilities are calibrated**; downstream can set thresholds directly	**Varies with workload**	The host model follows the written policy but is **sensitive to wording and systematically under-confident**; under asymmetric costs, **using its probabilities can be worse than taking its top-1 directly** (2610.11392). Decision models "can often point to the most likely outcome, yet substantially overestimate its probability" (2610.03935). Another audit found that three frontier models have a lower median calibration error (0.157 vs 0.066) (2609.24574)
P4	**The same judgment gets the same answer under equivalent phrasings**	**Cannot be promised**	Every TDM is biased on **3–8** of 12 bias tests; rewriting the better answer in an emotional tone can push a typical TDM's accuracy from **99% down to 42%**; **all** eight TDMs change their verdict when answer positions are swapped, whereas 2026 LLM judges almost never do; **good calibration does not protect against bias** (2610.18122)
P5	**Low confidence = should escalate**	**Cannot be promised, and it is two quantities**	Confidence conflates two kinds of uncertainty, **human disagreement** and **decision risk**; under asymmetric moderation losses the two call for **opposite escalation signals**, so **no single confidence number can serve both** (2610.33258). Also measured: escalating the least confident batch of decisions **does not catch the decisions flipped by attacks** (2610.12292)

#	Property	Can it be promised	Measured counter-examples / boundaries
P6	**Falls back / abstains when uncertain**	**Cannot be promised; this is the worst cell**	The correct-rejection rate when the correct answer is absent is only **7%** (accuracy 99% when the answer is present), while native boolean checks reach 99% exact match on the same questions; a threshold selected on an independent dev set lifts rejection accuracy from 7% to **79%** (2609.39496). Paired fallback-robustness tests find that **correctly identifying the category does not transfer into correctly choosing the fallback option**, even when the correct category is given explicitly; **logically equivalent** phrasings of the fallback rule cause significant divergence (2610.19856)
P7	**"I don't know" can be expressed**	**One value missing from the interface**	In Choice answers P(A) and P(¬A) are presented as summing to 1, while the sum is missing a term P(U)≠0; restoring P(U) raises Choice's median soft accuracy from **0.771 to 0.978**. Even in binary decisions the model "wants to answer a third option: I don't know" (2609.35342)
P8	**Knows what it does not know**	**Cannot be promised**	With no answer-relevant information at all, it gives **0.80** to some salient option; beyond the observed knowledge boundary, confidence exceeds accuracy by **0.21–0.33**, and recalibrating with earlier months **does not close the gap**. **Targeted yes/no questions** are sharper instead: asking "has the outcome settled" gives AUROC 1.00, "is the evidence sufficient" 0.95 (confidence on the same questions only 0.85) (2610.01006)
P9	**Probabilities are mutually consistent**	**Cannot be promised**	On 480 negation pairs, the sum of "the label is X" and "the label is not X" deviates from 1 by **0.064** on average; three single-label probabilities sum to **1.14** on average; the violation is about 5× its rerun noise and concentrates where it is **uncertain** (2609.33209)
P10	**Business staff can configure it directly with text**	**Partly holds; this is the core of D1**	It reads examples and definitions, but **label names override definitions**, deleting the definitions leaves accuracy unchanged, and small models cannot flip label semantics in context (§ 3.1). In moderation, Jev **benefits from retrieved precedents** (improving precise policy selection and harmful-content discrimination when the answer space is fixed), whereas retrieval for the open-source control Laya often only changes the positive rate without improving discrimination (2610.07953)
P11	**Decisions are auditable**	**Structurally missing**	It generates no text, so "**there is no textual rationale for human inspection**" (2610.11392); "rationales and chains of thought do not apply" (2610.22526). All existing compensations are outside the model: routing on an editable policy tree (2610.18326), node-by-node repair of a Yes/No/Unknown node tree (2610.13999), writing judgments into an evidence record (2609.27535)
P12	**Risk can be budgeted**	**Only certifiable by post-hoc statistics**	When no human fallback is possible, uncertain items must still have an action; naive threshold selection exceeds the target risk in **about half** of the settings with a defined test risk, whereas Learn-Then-Test exceeds it in only **7.8%** (2610.19811). "Relying on a class-probability interface is insufficient for deployment decisions"

How to read it: P1–P2 are the product; P3–P12 are **empirical facts that must be re-measured per workload**. This table is also the list of objects under test for the evaluation coverage in § 5.2.

5.1.3 What the Business Goal Is Not

1. **Not replacing LLMs**. The most accurate are still reasoning LLMs, at about 70× the cost (2610.21244); on social-science annotation the median lag is 11.6 macro-F1 (2609.24574); on Indonesian regional languages five of six commercial LLMs are 4–14 points above Jev (2610.14112).
2. **Not replacing "task-specific models that can be trained whenever labels exist"**. The item "when enough labels exist, use a task-specific model (encoder, classic classifier, reward model or probe) as a baseline" is complied with by only 7 of 13 applicable papers; what has been measured is that a fine-tuned encoder is not significantly worse than open-source TDMs on ranking, with nominally lower ECE and faster (2609.32160). Under its fixed protocol this report treats the encoder family only as a control.
3. **Not "accuracy on par" counting as a win**. On-par conclusions often come from test sets on which **strong models have already saturated**; the item "report the score gaps between models, do not claim parity on saturated tests" cannot be reliably scored from the text (T1, as above).
4. **Not skipping threshold engineering**. Every effective deployment treats the threshold as **an independent object to be fitted on separate data and frozen** (2609.39496, 2610.19811, 2610.14112, § 5.2).
5. **Not "once judgments are outsourced, semantics no longer need attention"**. Mis-specified rubrics get scaled up, errors become correlated, and decision rights get quietly rewritten by thresholds and exception routing (2610.01231).

5.1.4 How It Supports System1+System2: Five Different Positions

"Fast thinking + slow thinking" is not one structure in engineering terms but **five**. They differ in what they require of the model, in their failure modes, and in the strength of the existing evidence.

Position	What S1 does	Strongest evidence	Which S1 property it requires
A Entry gate / cascade	Decides "does this need to slow down" before the slow model	At equal cost, **a confidence cascade beats any scheme that re-asks the same model**, and **a curated menu beats scaling up the model**: 0.8B at 95.4% on a curated 5-candidate menu, 4B at 80.0% on the full 150-label menu (2610.07716); routing to an LLM can match that LLM while saving **24–77%** of API cost (2610.14112)	P3 calibration, P5 confidence usable for routing → **neither can be directly promised**
B Process monitoring / intervention	Continuously monitors the slow model **while it runs** and takes over when necessary	SOMA: the scenario layer writes goals and planned actions in natural language, the awareness layer has S1 judge whether the behavior has occurred, and the control layer accumulates judgments, warning first and taking over only if the behavior persists. One shared S1 and one controller, **no training needed**, curing two failures: under over-thinking, reasoning tokens drop by up to **36%** with mean accuracy up; on unanswerable questions abstention rises from **70.5% to 87.2%** (2610.26793)	P10 text configuration, P3 calibration; P6 fallback **not required** (the controller accumulates outside)
C Evidence plane / memory plane	Takes over all the "many small judgments" the slow model needs	Mnemon splits memory work into "many independent yes/no judgments" (S1, dozens per second) and "a little slow work" (S2 writes queries, names needs, organizes answers): LoCoMo **91.7%** (highest of 14 systems), LongMemEval-S 83.8%, under 4k tokens of context per question, lowest effective cost index on LoCoMo; with a reasoning model answering, 92.2% / 94.4%; from 100K to 10M tokens of history, per-question cost rises only **1.11×** (2609.36059). Jev-Mem uses S1 as the control plane (memory typing, query routing, retrieval-budget allocation, graph traversal, candidate scoring, adaptive stopping): LoCoMo judge score 0.777 (+11.0% over the strongest baseline), build time 158 s (**6.6×**), mean query latency 0.93 s (**−36.7%**) (2609.23986). In autonomous research, JevGPT has S2 set the evidence goals while S1 locates support and assesses relevance, judging sources independently in parallel (2610.18272)	P1 type, P2 batch nearly free; low accuracy requirement per single judgment (one wrong is not fatal)
D Readout of slow-model output	Reads **someone else's** (possibly truncated) reasoning chain and commits a decision	Seven reasoners × five multiple-choice benchmarks (35 pairs): under 500–16,000 token budgets, Jev's mean accuracy is **above forced answering** (with estimated cost including reasoning and answering); at a 1,500-token budget overall **+4.1 pp**, truncated questions **+8.6 pp**, 17 pairs significant under Holm-corrected bootstrap; a policy selected on the training fold saves **71.3%** of estimated end-to-end cost on the held-out fold while losing only 0.58 pp (2610.25768)	P1, P3; this is the most underrated cell
E Closed-loop executor / RL component	Makes consecutive decisions in the environment by itself	**Weakest**. In the 138 closed-loop environments of System1-Arena, large open-source LLMs have the highest success rate (at most 32.9%), and **small System-1 models rarely succeed** (2610.21244); in 33 closed-loop Doom episodes over three seeds, **no variant reached the exit** (2610.09683); the decision model most accurate on isolated questions solves only 69.5% of Agent tasks (§ 5.2). The only stable sub-form is **a frozen model as a component of RL training**: in three positions (reference policy, exploration judge, replay scoring) it beats standard RL learners on nine MiniGrid tasks and three Atari games (including where the learner alone makes no progress at all), with the model itself untrained; the same paper establishes that Jev can satisfy what each RL object requires of an answer, **except the cardinal use of the value function** (2610.11692)	Requires P5+P6+P9 to hold simultaneously → **none of the three holds**, so failure in this cell is to be expected

5.1.4.1 One Engineering Fact Runs Through These Five Cells

S1's benefit grows linearly with "the number of judgments it replaces", while its risk grows linearly with "the severity of the consequences it decides alone". Cells A/B/C/D all leave the "consequence" with S2 or with code: the cascade hands uncertain items back, the monitor only warns, one wrong judgment on the memory plane is not fatal, and the readout still has a whole reasoning chain in front of it. Cell E hands the consequence to S1 itself, so all the defects of P5/P6/P9 are exposed at once.

This also explains why the saturation phenomenon of § 5.1.1.3 matters: the BLM of 2610.24871 is the limit of this logic. **When decisions repeat enough to be solved once and reused, not even the model needs to be online** (a model-free controller solves 79%). In other words, Jev-class models occupy the segment where "decision repetition is high enough to be worth making cheap, but not yet high enough to be solved offline".

5.1.4.2 One More Cell: Treating "Decisions" as Database Operators or Graph Operators

This is not an "Agent", but it may be the largest in business volume: semantic databases treat typed decisions as semantic filter/join/classify/rank operators (2610.02046); NIOCIS folds execution logs into a "loop-folded action graph" and asks one typed question per state (2610.25985); Jev-Guard routes first and classifies second on an editable policy tree (2610.18326). What this cell has in common is that **the option set is given by an external structure**

(schema / graph / policy tree), so the dependence on P10 in the §5.1.4 table is transferred to the engineering structure.

5.1.4.3 The Business Ceiling May Not Be in the Model

NIOCIS gives the most honest number in this report: a typed encoder matches the logged next hop on 3,875 held-out decisions **32.2%** of the time (34.9% within the declared options); and cross-trajectory leave-one-out puts the ceiling of "already-published states" at **25.6%–44.9%**, varying as the state key gets finer. That is, **recurring states disagree with each other more often than they agree** (2610.25985). The ablation attributes the gain to **restricting the candidate set per node** (+3.5 points over a flat catalog) rather than to graph traversal (zero within the interval).

Corollary: in real business, "the accuracy ceiling of the decision model" is often **the consistency ceiling of the recorded process itself**. What this means for §5.2 is: **evaluating only on public benchmarks cannot measure this ceiling**.

5.1.5 Section Summary and Blind Spots

The starting proposition, as it stands after testing: Jev-class models do **not** turn non-idempotence into idempotence. They shrink the surface of non-idempotence from "any position in a free-text output" to "the values of a few typed fields", which makes that surface countable, boundable and priceable (the union bound on the probability of a verdict flip does not depend on calibration; it depends on the "unknown means reject" construction); the price is that failures turn from explicit into silent, and capability is lost outright on the class of judgments that "need the intermediate steps written out". The most precise verb comes from the literature itself: compilation **does not eliminate** non-determinism, it **localizes** it. The six pieces of counter-evidence are in §2.1.3; three of them overturn the proposition's premise rather than its conclusion: the host Jev changes 0.78%–1.33% of decisions under identical repeated calls; the format guarantee belongs to the existing lineage of grammar-constrained decoding, and a hard schema pushes answer accuracy from 19.7% down to 11.0%; the choice of query plan changes the result, and queueing voids every latency guarantee.

This proposition carries its own gradient: push determinism all the way into the harness and the model becomes redundant (once policy fields are parsed into typed values, one deterministic rule is correct on all six policies; a model-free controller solves 79% on τ^2-bench telecom; in enterprise field prediction the LLM is used only in the offline rule-discovery stage, and inference is a transparent key lookup). So the upper bound of the usable range is "how structurable the business fields are" and the lower bound is "the repetition of the decisions".

The three-layer idempotence table (the report's own construct): L1 format is guaranteed by the interface but is not unique to Jev; L3 execution holds only under a fixed plan and a fixed load; **L2 semantic is the weakest layer**: equivalences that can be written into the attention mask are solvable (flip rate 8.30%→0.11%, Prism-1 zero flips by construction), everything that cannot be written into the mask falls, and the systematic shift caused by menu composition has been measured over 299,463 pre-registered calls at **1.2–3×** its own noise, so a "metric" can only be read as a quantity relative to that option table.

Business goal: push the marginal cost of one judgment down to where it can be written on every program branch, thereby turning "evaluation/judgment" from a scarce resource into a resource that can be consumed at scale; the benefit is largest where decision repetition is high.

Properties: only "the output is necessarily within the option set" and "one forward pass answers the whole request" are structural guarantees; the five items calibration, invariance under equivalent phrasing, fallback, self-knowledge and probabilistic self-consistency all have counter-examples measured by independent audits and must be re-measured per workload with frozen thresholds.

How it supports System1+System2: entry gate, process monitoring, evidence/memory plane, readout of slow-model output, closed-loop executor. The first four cells have positive evidence; the fifth fails across the board, and the failure is predictable from the defects of P5/P6/P9.

Blind spots (none answered by any paper):

1. **Nobody measures the end-to-end metric "cost per correct result"**. The checklist item "when reporting latency, report both with and without caching, and state the serving location, the concurrency and the measured cost per correct result" is complied with by only 3 of 20 applicable papers (2609.32160). Whether saving 44× in cost while dropping 11.6 F1 points is a gain or a loss, no paper gives a computable protocol.

2. **Nobody measures "how large the total volume of judgments becomes once judgments are made cheap"**. The Jevons hypothesis is formalized as a **conditional** proposition, but 2610.01231 describes itself as "treating it as a technical provocation rather than an established empirical result", with no measurement at all.
3. **"Decision rights rewritten by thresholds and exception routing" has no quantitative work at all**. This is the most direct business risk, yet there is only one conceptual paper.
4. **P5's "two uncertainties" have been measured in only one domain, content moderation**. 2610.33258 states explicitly that the remedy of decomposing into parallel sub-questions and reading the disagreement **is idiosyncratic**: it helps only the weakest, disagreement-blind models, works only when the decomposition aligns with the source of disagreement, is needed by neither the dedicated toxicity model nor the strongest LLM, and its effect is distribution-sensitive. Whether this holds in other business domains: zero hits.
5. **The consistency ceiling of the business process itself has been measured by only one paper** (2610.25985), and in a single domain (security operations).
6. **No work treats "business staff configuring it directly with prompts" as an evaluable business property** (P10). All the evidence for D1 comes from prompts written by researchers, not by business staff (§ 3.1).
7. **Nobody treats L2 semantic idempotence as an independent measurand**. Existing work either measures the flip rate under a single intervention (order, wording, tone, label name, one paper each) or uses structure to write one kind of equivalence into the mask; **no paper defines the equivalence class "business equivalence" and measures the stability of the metric over that class**. This is the only place in this section's proposition that has neither a structural guarantee nor an evaluation protocol (§ 2.1.4, P4 of § 7.3).
8. **Idempotence on the harness side is assumed, not measured**. Decision rights actually land in the harness, but the harness's own reliability is measured by only one paper, which splits it into seven components (context, execution, governance, lifecycle, observability, tools, validation) (2610.32285); whether "same metric → same action" holds on a real harness has not been measured for Jev-class models.

5.2 D5 Agent / knowledge systems: fast/slow cooperation, memory, routing and guards

> Protocol: can a Jev-class model turn the Agent loop from "slow thinking + execution feedback" into "fast/slow thinking cooperation". This section first gives the counter-evidence (faster decisions do not equal better Agents), then goes through, by role, fast/slow switching, memory and compression, routing and guards, multi-Agent, domain Agents and continual learning, and finally gives the matrix and the blind spots.

5.2.1 Counter-evidence: faster decisions ≠ better Agents

Study	Setting	Conclusion	Source
When Faster Decisions Do Not Make Better Agents	5 Agents, ALFWorld + AndroidWorld, 4,500 runs	Jev significantly lowers decision cost and latency; task success is close to baseline on ALFWorld and drops clearly on the two mobile Agents; **auxiliary reasoning and extra interactions eat the decision savings**; recovery failures and premature completion limit delivery; for one Agent the cost per attempt falls but **the average cost per successful outcome rises**	2610.13092
Good Judgment, Bad Advice	Claude Sonnet 5.5 terminal Agent, Terminal-Bench 2.1, 21 tasks, 315 runs, pre-registered, with a placebo arm	The Jev controller **raises cost per attempt by 19%** (1.19, CI 1.02–1.36) with the same pass rate (73.0% = 73.0%); cheaper than a small-LLM controller (0.81) but 22% more expensive than placebo; best offline ranking (AUROC 0.78 vs 0.62 for rules, 0.60 for the small LLM) but poor calibration (0.50 vs 87%), so it makes the Agent verify work that is already correct; **no controller ever chose to abandon**; "an informative decision signal did not become a useful intervention"	2610.24649
JevAgentHarness	AgentDecisionBench, from isolated questions to controlling a running Agent	The decision model most accurate on isolated questions solves only 69.5% of Agent tasks; **delegating tool selection directly without fitting a confidence threshold: success rate −19.0**; existing decision models have never seen the state after the Agent's own mistakes; AgentDecision-4B trained on the Agent's own trajectories: out-of-domain 91.5% vs Jev 82.0%; in-domain 81.1–89.0% cheaper, 2.5× faster, loss <3 points	2610.21412
System1-Arena	92k questions, 105 tasks + 138 closed-loop environments, 30 models	The largest System-1 model matches fast closed-source LLMs on non-Agent accuracy, with better calibration and 1/4 the latency; Decision-2.0-Vega-27B, Jev and Qwen3.5-9B all at 67.8% single-step; in closed loop, large open-source LLMs reach at most 32.9% (Gemma-4-31B), **the four small System-1 models at most 3.1%**	2610.21244

Study	Setting	Conclusion	Source
Fast Models, Slow Evidence	11 Agent decision points, 7,283 + 6,640 cases, paired, self-audited	Jev is significantly more accurate at 9/11 decision points (+10.8 to +46.0 pp); **zero-shot model routing does not beat random in either case, RAG relevance gating is flat**; the authors audit their own pipeline: pre-screening cost left out (23.9% savings reported, 4.3% actual), gate accuracy taken as end-to-end quality (58% vs 98%), in-sample threshold (5% target, 17% missed on the held-out set)	2610.02267
Microtask Eligibility Gap	4 harness microtasks (approving commands, writing memory, choosing tools, ranking history), Qwen3 0.6–8B, CI threshold	**0/16 configurations eligible**, 0/112 across rewrites; Llama 12/12 ineligible; quantization does not change the conclusion; "put the SLM behind a baseline that meets the bar, use it only where the baseline fails"	2610.00025
Measure Before You Train	Real email Agent, 739 emails	Logistic regression on ~200 labels is the most accurate on every field (0.79/0.74/0.83/0.90); sender relationship belongs to the contact graph, not the prompt; **once composed into actions, the most accurate system most often ignores emails that need handling (18% vs 4%)**	2610.32955
JEVal	11,257 cases, 36 datasets, 10 domains, 25 configurations, including τ-bench and social simulation	Decision models are most competitive when "the evidence can be judged directly", weaker when expertise or faithful uncertainty is needed (they often identify the most likely outcome but **overestimate its probability**); **on τ-bench, faster local decisions lower median turn time but lower task success**, errors accumulate along long trajectories	2610.03935
Code Owns the Simulation	Reflection tests, matrix games, ALFWorld, robotics	What can be judged from the input description (evaluation) Jev does well (CRT 99%); what depends on predicting beyond the input (simulation) it cannot do	2610.01834
What Can't Jev Decide	Single-factor benchmark	A single-pass model cannot make decisions that "an LLM can only make by writing out intermediate steps"; given a stale summary, it follows it	2610.24929

Conclusion: the benefit of putting Jev into the Agent loop depends on three things: (a) whether the decision point is really a "readout" rather than a "derivation"; (b) whether the confidence threshold is fitted on the Agent's own trajectories; (c) whether the savings are eaten by extra interactions. The "fast/slow cooperation" papers increasingly treat these three as design constraints rather than assumptions.

5.2.2 Fast/slow switching: who decides when to slow down

Work	Switching mechanism	Key result	Source
Talker-Reasoner (upstream)	Two Agents: a conversational System 1 + a planning System 2	Architecture proposal	2410.08328
CogRouter (upstream)	Switches among four cognitive levels per step; "the right cognitive depth should maximize action confidence"	Qwen2.5-7B 82.3% on ALFWorld/ScienceWorld, tokens −62%	2602.12662
StartLux AutoThink	Reasons only when the top probability is <0.9	27B AutoThink 86.9 vs Instant 75.4 (see 02)	2610.17309
Selective Reasoning	A shared candidate reader on a frozen Qwen3.5-4B: a benefit gate starts reasoning, natural states trigger reading, an entropy rule accepts and cancels generation	768 questions 70.96%→84.38%; vs 512-token terminal reading +7.29 pp with only 2.25% more output; vs the 1,024-token pipeline, time −46.30%, accuracy −1.17	2610.19644
LDM routing	Confidence routing to reasoning	92% of reasoning accuracy with 27–58% of the tokens (see 02)	2610.15501
System Switch	A fast executor decides every step; when the gate opens it hands over to a reasoning VLM, the game does not pause	0.15B–9B zero-shot models pick "collect item" under error 1.6–1.8× more often than random; accuracy / calibration / sensitivity are three independent things; the benefit of deferring the 30% most uncertain has rank correlation 0.87 with AUROC, held-out +0.13 [0.08, 0.18], reasoning contributes about half; **in closed loop, none of 33 games reaches the exit**; committed plans make a standing-still executor move but die more often	2610.09683
SOMA	S1 continuously monitors S2 reasoning: scene layer (natural-language goal + predefined actions), awareness layer (S1 judges whether a behavior appears), control layer (remind first, then take over)	No training; on overthinking, reasoning tokens −36% with average accuracy up; abstention on unanswerable questions 70.5%→87.2%	2610.26793

Work	Switching mechanism	Key result	Source
Truncated Reasoning	Jev reads truncated reasoning traces	7 reasoners × 5 benchmarks: every budget from 500 to 16,000 beats forced answering; budget 1,500 +4.1 pp (+8.6 on truncated questions); offline replay at 34.5% of the cost, −1.13 pp; the training-fold policy saves 71.3% at −0.58 pp; Jev on the question alone 68.0% vs 65.5% for 500-token forced answering	2610.25768
SanSi / LEV (System 1.5)	The number of recurrent layers serves as "thinking depth", no tokens generated	SanSi 72.0% (+13.5); LEV 2.6B 63.2%→76.2% (depth 1→4), on-demand depth saves 15–24%, under 50 ms it is 5.4–8.8 points more accurate than dense models and cascades at the same latency	2610.07730, 2610.20951
VoI slow-module scheduling	Formalizes "when to call the slow module" as a calibrated decision problem; fixed-ratio / budget / history gates are all degenerate cases of the value of intervention	At low cost ratios the calibration benefit vanishes; LIBERO 14 arms: adaptive triggering allocates queries by the threshold rule, but the language-rewrite channel cannot be converted into success; only "undo failed rewrites" has zero net negative	2610.21160
TORSI (real-time)	A slow teacher studies failures in the paused game and writes reusable instructions for the fast student; the student does not call the teacher during the match	RealtimeBench 180 tasks: GPT-6.1-sol beats Jev when paused, falls behind when running continuously; two rounds take Jev 13.51→24.19 (0–100); the instructions also work for three other models	2610.20665
CARE (embodied)	A lightweight critic monitors local execution reliability; after escalation, control persists until recovery; shared hand-off memory	Improvements across embodied benchmarks	2610.30587
JEV-as-a-Judge	Confident accept / escalate	Frozen threshold +0.9 over GPT-6 at 41% of the cost; fails on style-adversarial and reference-free prose	2609.26550
P(Yes) test-time search	A frozen decision model scores the probability of "yes" for a complete proposal, search for a higher one	128 logic puzzles, Qwen3.8-27B from 2 to 30 fully correct (15×), per blank 27.8%→48.5%	2610.25474

Pattern summary: switching signals come from four sources: (1) answer confidence (StartLux, LDM, JEV-as-Judge); (2) benefit/entropy gates (Selective Reasoning); (3) an external S1 monitoring S2's behavior (SOMA); (4) decision-theoretic value of intervention (VoI). The limits of (1) are bounded by the D4-D1/D3 evidence: confidence is answer uncertainty, not a "needs thinking" signal (System Switch item ii, Good Judgment's 0.50 vs 87%). (3) is the only design that puts S1 **inside** S2 rather than before it.

5.2.3 Memory and context compression: System 1 makes "many small" judgments

Work	S1 role	Key result	Source
Mnemon	Makes many independent yes/no judgments on raw dated records (needed or not, expired or not), the LLM plans the search; no decisions at write time	Architecture + background consolidation	2609.36059
Jev-Mem	S1 control plane: memory typing, relation organization, query routing, budget allocation, graph traversal, candidate scoring, adaptive stopping	LoCoMo 0.777 (+11.0%), construction 158 s (6.6×)	2609.23986
Selection vs extraction (pre-registered)	Jev selects raw turns in one call	On LoCoMo under a tight budget, non-inferior to LLM-extracted memory (one-sided 95% bound −3.0 vs a −5 margin); writing 3,061× cheaper; re-ranking gains shrink with budget (+17.4 / +9.1 when keeping 3/30, +1.5 / +1.1 under a wide budget where extraction is more accurate); as accurate as an LLM re-ranker (−2.0) at 1/3 the latency; **re-ranking lowers correct abstention**	2609.34227
Compression vs follow-up questions	A decision API judges which sentences are necessary for the current question	MultiRC current question +11.4 points @25% tokens; **for follow-up questions, no method that selects sentences by the current question beats random deletion**; question-agnostic word-level LLMLingua-2 +12.5 (+7.3 with equal content-word retention); on FEVER, label accuracy overestimates the support of the retained text; on τ-bench sentence selection beats random	2610.27301
SJC (source-judged compression)	Jev judges whether each summary entry has an environment output or a previous summary as its source	83% of failed runs trace back to executing an action from a sourceless summary entry; AppWorld/OfficeBench cross-run reliability +2.1–8.9%, no extra tokens	2610.20927

Work	S1 role	Key result	Source
Memory retirement gate	SUPERSEDED / NEEDS-CONFIRMATION / COMPATIBLE three-way contract	Zero false retirements on 141 non-changes; the binary gate retires prematurely 39.2%, the three-way gate 15.5% but detection −15.3; cascading and raising the threshold make no difference; "do not wire a binary score directly to deletion"	2610.15710
Cache Engineering (upstream)	The Agent operates directly on the KV cache (drop/repos) without changing the trajectory	History dependence makes prefix matching insufficient for cache reuse	2610.28577
MarkovSim (upstream)	Builds compact progress representations at finite-state-machine boundaries	Long-horizon computation simulation harness	2610.35349
Selection replaces extraction / Jev-LDE / MemCom	See D1 § 3.1.2	—	—

Conclusion: memory is S1's most convincing use: the judgments are "many, small, independent", and the (pre-registered) evidence supports "select raw records + S1 judgment" being non-inferior to LLM extraction. But two boundaries are clear: (a) content compressed for the current question is useless for follow-up questions (question-agnostic compression is needed); (b) a binary gate must not be wired directly to an irreversible action such as "delete".

5.2.4 Routing and guards

Work	Conclusion	Source
REFLEX	Jev as a typed decision layer, calling a strong LLM at low confidence: frozen 100 tasks 95% success, strong-model calls −72.7%, holds for all three fallback families; reliability depends on **action-set size and the near-valid options close to the authorization boundary**; on BFCL / τ-style, "when ordinary routing is already accurate, the advantage of the cascade over cheap generation is limited"	2609.26532
Jev-Guard	Policy-tree routing + conditional classification, 93.72% / 58 ms (see D3)	2610.18326
Option-channel attack	In the guard role, 7 open-source TDMs at 36–72%; six log lines 0→63%; misleading names 93–100%; **all defenses defeated; escalating the most uncertain is ineffective**; a deterministic rule over policy fields parsed into typed values reaches 100%, "making the model unnecessary"	2610.12292
Same Attack, Different Judge	Jev is most robust on input moderation with explicit rules; on phishing detection, which needs holistic judgment, it is easier to bypass and recall is low	2610.17820
TrustJev	Separates policy authority from untrusted evidence, bounded correction	2610.34826
COGNIT-Guard	Direct decision guard + calibration + dual-use false-positive control + CPU-NPU routing	2609.33671
SkillScreen	Single-pass screening of malicious Skills, flags only 20–25% of benign ones at 99% recall	2610.17727
StepGuard / ToolSafe / AgentDoG 1.5 (generative guards, for comparison)	Step-level tool-call guard; TS-Guard multi-task RL reasoning; taxonomy-guided data engine	2608.24777, 2601.10156, 2605.29801
CARE (shell) / StateMirror (static-first, for comparison)	Shell command normalization-attribution-parsing; detects hallucinated tool calls before execution (4,398 trajectories, median recovery 3 steps, 22.4K tokens)	2607.21642, 2610.32648
JevSoup	S1 picks two LoRA experts for orthogonal merging	At most +1.19% on 14 tasks
Harness Tokenomics	The router of an enterprise harness control plane (who answers, what to read, how the cache is used)	2609.28919

Conclusion: routing is S1's most mature but also most overrated role: when the LLM's own routing is already accurate (REFLEX's BFCL result) or the task can be solved by deterministic rules (the option-channel attack's 100%), S1 has no place. S1's failure modes in the guard role (option names, log injection) agree with D4-B3/E3.

5.2.5 Multi-Agent coordination

Work	S1 role	Result	Source
Jev-LiteMAS	Dynamic expert selection (planning gap) + screening of intermediate artifacts, flagged ones go to the LLM for re-review (review gap)	In three MAS frameworks, routing and review account for 31.7–43.5% of generated tokens; 4 benchmarks, 2 backbones, generated tokens −12.7–54.6%, accuracy within 1.34 pp	2610.21411

Work	S1 role	Result	Source
S1-MAS	A lightweight controller selects check conditions, selects follow-up tasks and decides termination; a compact reader fetches condition-relevant evidence from authorized sources	Decision-evidence loop	2610.08155
BLADE	Jev gives dependency probabilities and selects the relevant "source-decision" hand-off points to audit for faithful use	Best or tied best on 1,228 failed runs, +2.2–11.1 pp over one-shot attribution	2610.18780
Jarvis	In full-duplex dialogue, JEV reads each turn against the active tasks to choose an operation, **code checks each choice against the current state before applying it**; versioned task records	τ-Voice 29→192 / 278; Full-Duplex-Bench v3 69% Pass1 vs GPT-Realtime 60%	2610.23804

5.2.6 Domain Agents and the "S1 position" in non-Agent settings

Work	Position	Result	Source
JevCoder	Snippet recommendation, action-type guidance and when to commit in code repair	The benefit depends on how much work the intervention changes, not on the cost of the judgment; 3 models × 400 tasks, tokens −21.6–50.6%, higher repair rate	2610.19596
Jev-Mobile	The VLM plans at low frequency, Jev executes at high frequency in the action space defined by the accessibility tree	AndroidWorld 79% vs SeeAct-V 78%, step-by-step VLM 84%; time −32.7%, API cost −73.4%	2609.30186
JEV-Star	JEV action selection + continuous GPT-6 planning	Beats the Lv7 built-in AI; median response 0.422 s; $3.71 per game (JEV $0.15); **the JEV controller alone does not expand against Lv2**	2609.27331
SearchJev	Relevance, evidence sufficiency, search actions; soft-label calibration (SLCD)	5.2–5.3× faster than same-size Qwen3.5, ECE −41–74%; BrowseComp-Plus search time 3.7–4.7×, accuracy 45%→54%	2610.05107
JevGPT	S2 sets the evidence goals, S1 locates / assesses relevance in parallel	Leads on fixed-corpus QA and ranking, faster and cheaper	2610.18272
JevSpawn	Natural-language task → bounded probabilistic exploration; parallel action generation + feedback-based branch selection	Beats 7 baselines on 8 benchmarks	2610.00437
NIOCIS / SharedKV-BT	Node-conditioned option spaces (see D3)	—	2610.25985, 2610.07327
Decision4Rec	A knowledge graph writes the logs as scale statements, S1 reads typed questions	A 4B decision model's NDCG@10 is 25–36% above the best trained baseline (3 sparse catalogs)	2610.26035
Jev as the Q / policy in RL	Reference policy, exploration judge, replay scorer	Meets every requirement of the RL objects **except the cardinal use of the value function**; beats standard learners on 9 MiniGrid + 3 Atari	2610.11692
Chess	Jev judgments placed inside Stockfish search; distilled into a compact evaluator	>2200 Lichess bullet; dual-teacher averaged labels +9.6 Elo over all-Qwen	2610.09188
KITE	Queries a typed behavior kernel once per unique state, table-driven population experiments	1.7% anchors cut the effect error by 41%	2609.27535
ProAgent / StreamingJev	Continuous observation, when to intervene	More accurate intervention timing than the source backbone, fewer false triggers	2610.16967
Jevolution	World / scheduler / controller separation for remote decision Agents	Framework	2610.14449

Common point: successful domain Agents all put S1 where **the action space has already been enumerated by code / a tree / a graph** (Jev-Mobile's accessibility tree, NIOCIS's action graph, Decision4Rec's scale statements, the chess search tree), and let S2 change the action space or the goal at **low frequency**.

5.2.7 Record vs reality, continual learning

- **Claim Laundering**: decision models trust records more than observations (53–100%); probe AUROC 0.77–1.00; EVIGATE +18.7/+5.0/+5.1; offline calibration does not transfer (2610.25947).
- **Jev-DLC**: typed feedback (accept/undo) becomes a bounded causal state, no raw history stored, no scoring calls added; TCPA-Repeat 44.83% (+2.43 over Recent-4 replay), 56.07% after two feedbacks (+7.54) (2610.31521).
- **SHIFT-JEV**: certified abstention + benefit-gated continual adaptation; Qwen3-0.6B 70.5%, above replay/LwF/EWC; on 4B the conservative gate trails by 0.4–1.4 points (2610.34329).
- **TORSI**: offline recursive self-improvement in which the slow teacher writes instructions and the fast student executes (2610.20665).

5.2.8 D5 matrix

Question	Existing work	Status
Does putting S1 into the Agent loop save money	2610.13092, 2610.24649, 2610.21412, 2610.02267	**Conditional**: the threshold must be fitted on the Agent's own trajectories; savings get eaten by extra interactions
When to slow down	StartLux, Selective Reasoning, SOMA, System Switch, VoI, LDM	Four kinds of signal; answer confidence is not a "needs thinking" signal
Can S1 monitor S2	SOMA, BLADE	Prototypes exist
Memory selection / judgment	2609.34227, Mnemon, Jev-Mem, SJC, retirement gate	**Strongest evidence**; boundaries: follow-up questions, irreversible actions
Routing / guards	REFLEX, Jev-Guard, option-channel attack	Mature but overrated; the guard role is high-risk
Multi-Agent coordination	Jev-LiteMAS, S1-MAS, BLADE, Jarvis	Gains exist; Jarvis's "the model judges, the code executes" is the clearest
Domain Agents	Jev-Mobile, JEV-Star, SearchJev, JevCoder, Decision4Rec	Success = action space enumerated by code + low-frequency S2
Real-time	TORSI, System Switch, Jarvis	Fast models win when "the world does not wait"
Record vs reality	Claim Laundering, SJC, 2610.24929	Defined, partly mitigated
Continual learning	Jev-DLC, SHIFT-JEV, TORSI	Just starting
Where S1 training data comes from	AgentDecision-4B (the Agent's own trajectories + Jev labels), TORSI (teacher instructions)	Key finding: **must include post-mistake states**

5.2.9 D5 blind spots

1. **There is no standard for accounting the benefit of "fast/slow cooperation"**: the self-audit in 2610.02267 exposed three common miscalculations (pre-screening cost, gate accuracy taken as end-to-end, in-sample threshold); 2610.13092 proposes "resources per successful delivery"; neither has been widely adopted.
2. **The theory of switching signals has only one paper, VoI**, and that one is on VLA; on language Agents, "when to call the slow model" still relies on confidence thresholds, while D4 has already shown that confidence is not "needs thinking".
3. **S1 monitoring S2 (SOMA-style) has only one paper**, yet it is the only design that puts S1 inside slow thinking rather than before it.
4. **No work compares "S1 decisions" against "deterministic rules / logistic regression" as Agent microtask controllers** (Good Judgment's rule arm, Measure Before You Train's LR and the option-channel attack's 100% rule all suggest that rules are often enough).
5. The two findings **S1 training data must include post-error states** (JevAgentHarness) and **S1 trusts records over observations** (Claim Laundering) have not yet been absorbed by any reproduction recipe.
6. **Knowledge systems** ("knowledge" in the user's original wording) appear in the literature only as "memory selection" and "retrieval relevance"; there is no work on "S1 maintaining structured knowledge (ontologies, rule bases, versioned facts)"; Jarvis's versioned task records and Jev-DLC's bounded channel are the two closest.
7. **Zero Agent evidence for Chinese / enterprise settings**: all benchmarks are English public environments.

6 · Learning side: algorithms, data, backbone selection, output shape and chain of thought

> This chapter is the full body of evidence on "**what to train, how to train it, and what training cannot produce**"; its five parts answer five questions that were raised separately.
>
> § 6.1, the family tree of learning algorithms and dataset availability: relative to ordinary classification fine-tuning, what existing schemes change is the readout position, the proper-scoring term in the objective, and the data composition. **Almost nobody "changes the algorithm."** § 6.2 tests "SFT cannot deliver generalization in the general sense": checked against seven levels of generalization, **the strong form fails and the weak form holds**. § 6.3 tests "use RL to keep both ICL and skill injection at once": **it fails**; what decides whether ICL survives is whether the task definition varies across samples, and the only two methods that work are both SFT. § 6.4 tests "does task-agnostic RL have any life in it" and "can business expectations be written as a differentiable function": it has life, but only on second-order properties, and the class that is truly non-differentiable is exactly the class where measured RL gains are zero. § 6.5 is output shape and chain of thought: **proper scoring rules have hit the ceiling; what is really missing is the shape of the output space**.
>
> Reading this chapter through P0 of § 2.2 yields one additional conclusion: the untuned readout already reaches 76–97% of post-tuning accuracy, and the tuning gain shrinks from 15.8 points to 2.3 points as scale grows. "**Pretraining provides the representation; tuning only adjusts how it is read.**" So the question to ask is not "how to train the readout to be more accurate" but "how to keep the readout from losing the semantics already present in the representation."

6.1 Learning algorithms, dataset availability and base-model selection patterns

> Protocol: still analysis and survey only; no training scheme is given. This section answers three questions: (1) whether the **learning algorithms** of existing Jev-class schemes change, adjust or augment anything relative to "ordinary classification fine-tuning," and whether the evidence supports those changes; (2) whether the training and evaluation **datasets** are publicly available, under what license, and which ones are already on hand locally; (3) the selection patterns for **base models**. The recipe matrix in § 1.2 is organized by "project"; this section re-cuts it by "learning algorithm family" and puts the ablation evidence side by side.

6.1.1 The family tree of learning algorithms

The 30-odd recipes of § 1.2 plus the papers added in this round are split into eight families by "**what is trained, with what objective, and where supervision comes from**." For each family I give representatives, key implementation details and sources; one project can fall into several families (for example StartLux belongs to C, E, F and G at once).

Family	Core idea	Representatives (backbone / key details)	Sources
A. No training: readout + post-hoc calibration	Freeze the LLM and directly normalize the logits of the legal options; fit only a temperature or a prior	LLM-as-Jev training-free recipes (joint-context scoring, parallel-suffix scoring; a frozen 4B matches or beats existing training-free readouts); Mapika decider-chat (Gemma-4-31B-it / Qwen3.6-27B with weights untouched, temperature fitted by option count $T(n)=\max(0.05, 10.124-1.633 \cdot \ln n)$, Decision Index rank 2); AnyJev (question-agnostic affine head + reused output head, 15 equal-mass bins for ECE, rotation debiasing); Releasing the Inner Jev zero-shot Gemma 4 31B 79.6 vs Jev 80.1; the LDM diagnosis that "the readout of an untuned 27B reaches 97% of post-tuning accuracy"; P(Yes) test-time search (frozen model, no training, Qwen3.8-27B logic puzzles fully solved 2→30); RSAJEV replaces the readout with prototype memory (zero gradient updates, open-set AUROC 0.91–0.99); JevCascade's menu-restriction predictor (zero labels); Runoff's order-independent readout with two forward passes	2610.02076, Mapika/decider, 2610.00831, 2610.17205, 2610.15501, 2610.25474, 2610.15742, 2610.07716, 2610.18233

Family	Core idea	Representatives (backbone / key details)	Sources
B. Hard-label cross-entropy + LoRA (mainstream)	The same objective as "fine-tune the LLM as a classifier," only the readout position becomes an LM-head slice / letter logit / pointer head	Kev (Qwen3.5-Base series, LoRA rank 16, cross-entropy, 2 epochs, `decision-v7`; the optional `--perm_kl` and `--ord_w` losses are **not used in the released model**); Tev1 (Qwen3.5-4B, LoRA SFT, outputs the answer letter, 37,840 training + 4,568 validation); Bespoke Nimble (Qwen3.5-9B, LoRA **only on answer tokens**, 2,676 contrastive samples; 90.1% vs base 66.4%, Jev 93.2%); Releasing the Inner Jev (letter readout + a decision head initialized from an LM-head slice, **joint** cross-entropy, pool of 417,825 items, 400 steps about 5 GPU hours); the LDM baseline and DDS are both hard-label cross-entropy; Chinese Decision (1 epoch, 13.048M decisions); DKnown-Jev (fine-tuned directly from the Qwen3.5-2B / Qwen3.8-27B base, 1 epoch, 27B 98.75%, 6.07 pp above Jev); SecJev (scenario-weighted training); Koa-decision (atomic label tokens + SFT, single-token constrained generation); SALSA (class→token mapping + PEFT, single pass projecting only onto class logits); control: decoder + classification head vs instruction tuning, where the head scheme matches BERT with 10–30× fewer trainable parameters	jaredpalmer/kev, togethercomputer/tev1, bespokelabsai/nimble, 2610.17205, 2610.15501, 2610.23170, 2610.24358, 2610.03073, 2609.36115, 2510.22691, 2512.12677
C. Soft labels / proper scoring rules	Write calibration into the objective: Brier, soft cross-entropy, losses on the distribution rather than the argmax	jevlike (Qwen3-1.7B + LoRA + 37 open-source datasets, `CE + 0.5·Brier`; rationale: "cross-entropy is never satisfied; pushing 0.99 to 0.999 is still rewarded"; the base needs T=8.881 to be calibrated); OneJev (cross-entropy + Brier, lr 5e-6, 1 epoch, vision tower frozen); Open-Jev (LoRA + scalar decision head, **soft cross-entropy + 0.1 · Brier**, head initialized from the pretrained Yes–No readout, temperature fitted only on the calibration set); Eikos / Our Own Jev (soft targets + an auxiliary reasoning-rationale loss + symmetric KL between the English and Portuguese views, weight 0.5; about 25K decisions); in the Releasing the Inner Jev pool, Civil Comments, Measuring Hate Speech and others use annotator votes as soft targets, conclusion "**teacher labels help, soft labels may not**"; SanSi / LEV train with a proper scoring rule at every loop step; NDM argues "when labels are known, use a proper scoring rule directly" and gives a set-valued target; ufakzeka-karar outputs temperature-scaled probabilities + expected error as the "uncertain" signal; UniJev does SFT first and then CORA (pulls confidence toward the actual accuracy of each confidence bin; no RL stage)	mustafasemi-ai/jevlike, OmniJev/OneJev, Zefan-Cai/Open-Jev, 2610.29052, 2610.17205, 2610.07730, 2610.20951, 2610.25553, 2610.06744, 2610.25784
D. Distillation (teacher probabilities / reasoning→readout)	Supervision comes from a larger model, from reasoning chains or from the model's own reasoning mode, not from human labels	FunDM (**no labels**: learns the answer distribution of a frozen Qwen3-8B itself via KL, using only 24 of 36 layers, eight benchmarks 75.0 vs full model 75.7); LDM's SPRD (self-posed-question tasks, frozen Qwen3.8-27B teacher samples 4 reasoning+answer traces per question, SPRD-C filters for consistency and then distills the vote distribution); SII-Decision's Reasoning-to-Readout self-distillation (distills the answer distribution at the end of the reasoning chain into the first token); the legal domain distills a decision model's probabilities **without labels** into Qwen3-8B to obtain a calibrated local model; chess with two teachers: averaging Jev and Qwen3-32B labels gives Qwen 9.6 more Elo than the full budget, "the same judge answering again cannot replace a second teacher"; AgentDecision-4B uses Jev's answers as labels; Releasing the Inner Jev's teachers are Qwen3.5-122B-A10B and Qwen3.6-27B; Mapika labels with a local Qwen3.5-27B teacher; CT0 two-stage on-policy distillation (one set of weights serves System 1 typed decisions and the System 2 Agent at once); control (encoder): student-guided teacher distillation, ModernBERT student + DeBERTa NLI teacher iterated	2610.17175, 2610.15501, 2610.03935, 2610.29961, 2610.09188, 2610.21412, 2610.17205, Mapika/decider, 2610.16550, 2610.02516

Family	Core idea	Representatives (backbone / key details)	Sources
E. Structure-aware / calibration-aware objectives	Encode position, ordinality, evidence necessity and distribution anchors explicitly in the loss	PACT (pairwise anchoring + evidence-necessity term + ordinal transport cost + three-parameter contextual temperature; accuracy 84.6 vs the published recipe 85.2 and a cross-entropy-only control 82.2±6.0; relabel flips 9.8 vs 13.8%; NLL −26%; **"no single term raises raw accuracy; the value is in robustness"**); LLM-as-Jev tree-decomposed listwise loss + KL anchor (total score insensitive to anchor strength; without the anchor it degenerates); SHIFT-JEV's KL anchor + temperature (calibration error under shift 0.289→0.087); CORA; StartLux Appendix B.4 direct probability optimization and **policy-aware calibration** (confidence-accuracy gap ⩽2.7 pp across 184 settings, post-hoc); Releasing the Inner Jev's expected-utility objective (expected payoff computed separately for correct / wrong / abstain, +0.9±0.9 pp over cross-entropy); Bongard's second-stage JEPA (learns "the structure behind the judgment"; after diagnosing that "cosine alignment carries almost no signal" the objective was swapped, held-out rewritten questions 75.7%→85.9%); Kev's `--perm_kl` (order) and `--ord_w` (ordinal) losses (implemented, not used in the released model)	2609.35865, 2610.02076, 2610.34329, 2610.25784, 2610.17309, 2610.17205, 2609.39111, jaredpalmer/kev
F. Reinforcement learning (the various "RLCD"s)	Policy gradient with a proper scoring rule or correctness as the reward	OpenJev-RLCD (Qwen3-1.7B; after each reasoning chain, read out the distribution the model commits to and score it with a strictly proper scoring rule; a one-line variance identity shows that "scoring a mixture of multiple samples" rewards disagreement, and RLVR is exactly that mixture objective with the diversity term removed; end-to-end optimization either "System-One collapses" (prevented with a KL anchor) or is drowned by the policy-gradient term (whose gradient is 7–12× larger than the readout's) → **calibrate first, then reinforce**; on GSM8K answer verification a single query decides 81% of items at ⩽5% error, GRPO 19%); StartLux RLCD^2 (used only for AutoThink: the difference in proper score between the readouts before and after reasoning is the decision-quality reward + a soft length penalty; the share of reasoning that terminates within the length limit rises by up to +48 pp); UniJev compares five RLCDs (Shaped RLCD, GRPO, CtR, RLCD-lite, Rewarding Doubt): relative to a common 2B SFT initialization, the amount by which each algorithm changes accuracy on each modality is "**small compared to the gains from the supervised conversion**," with heterogeneous calibration effects; SBERT2S1 (encoder): the RLCD recipe released with the open-source System One model is **2.5–3.0 points below** cross-entropy, the gap mainly comes from reward normalization amplifying the noisy score-function term 3.6–15×, and a leave-one-out unbiased estimate recovers most of it; ufakzeka-karar: REINFORCE is **10.2 macro-F1 below** cross-entropy; JevAgentHarness: GRPO inside the Agent loop spends 4.5 extra GPU hours and 26.53 USD of LLM calls with **no measurable gain**; without SFT first, GRPO gives 11.5% vs 13.5% before; Bongard's third-stage sandbox RL (learns outcome distributions from action rollouts and an exact oracle, frozen sandbox panel 50.6%→64.8%); Dyad (frozen LLM, RL trains only the action encoder, consistent gains on four unseen environments); S^2-MAS (SFT + RLCD, 2.93M samples, 421M model 74.55%); NDM § 1.3.1 "why not reinforcement learning": when labels are known, direct optimization of a proper scoring rule **reaches the same optimum as the policy gradient of an RLCD-style reward without its variance**; TypeSafe's own RLCD is **not public**	2609.38850, 2610.17309, 2610.25784, 2610.02486, 2610.06744, 2610.21412, 2609.39111, 2609.36116, 2610.27746, 2610.25553, ML primer
G. Structural changes (change "what is trained")	Change the attention mask, the readout position, the number of layers, the loop or the encoder	Candidate-agnostic block-causal attention + position reset (Gemma 3 1B flip rate 8.30%→0.11%, Qwen3 1.7B 18.16%→1.20%; Qwen3-4B 250K research artifact); Prism-1 (one independent block per option, shared positions, LoRA, cross-entropy with softmax over blocks; Qwen3-8B / Gemma 4 12B; zero flips, Luce's axiom holds after the swap); StartLux Instant uses **bidirectional attention** over the prompt (same data, same Stage 1 checkpoint, gains on 4 of 5 scales, up to +2.4 DI points; random-seed variance 0.2–0.7); SanSi / LEV on the looped backbone Ouro, **backbone frozen, only LoRA + a small readout trained**, read once per loop step; Bongard uses the T5Gemma 2 4B-4B encoder-decoder, **full-parameter** 7.09B trained in three stages on one Blackwell; selective adaptation (LoRA only on attention and Gated DeltaNet projections, FFN adapters removed) + mean-span pooling; FunDM early exit (24/36 layers, skipping the LM-head); Intern-Decision places decision placeholders inside a JSON skeleton and reads all fields in one forward pass with an autoregressive masked LM; System X continues the "text reply" option from the KV cache; Dyad's action encoder; MetaEncoder fine-tunes a 30B decoder into a bi-encoder for one-directional contrastive learning (aimed at million-scale open-set candidates)	2610.01601, 2610.22066, 2610.17309, 2610.07730, 2610.20951, 2609.39111, 2610.35654, 2610.17175, 2610.22131, 2610.25456, 2609.36116, 2610.11316

Family	Core idea	Representatives (backbone / key details)	Sources
H. Continual learning / test-time adaptation	Absorb new feedback with little or no retraining	Kev `--init_from` (continue training from the released adapter and pointer head; the README says retraining from the base would "lose what Kev has already learned"); SHIFT-JEV's gain-gated continual adaptation (70.5% on Qwen3-0.6B, above replay / LwF / EWC / fine-tuning on every seed; on Qwen3-4B the conservative gate trails by 0.4–1.4 points); Jev-DLC decision-level continual learning (typed feedback→bounded causal state, see § 5.2); TORSI dual timescale: a slow teacher studies failures in a paused game and writes reusable instructions for the fast student, Jev average score 13.51→24.19; P(Yes) test-time search; Open-Jev's time-window regression control	jaredpalmer/kev, 2610.34329, 2610.31521, 2610.20665, 2610.25474, Zefan-Cai/Open-Jev

6.1.2 What actually changed relative to "ordinary classification fine-tuning"

Viewing the eight families together, the changes concentrate in five places, rather than in "a different learning algorithm":

1. **The readout position changed, and the classification head almost disappeared.** The mainstream no longer builds a new classification head; it reuses the pretrained output layer: letter logits (Tev1, Chinese Decision, LLM-as-Jev), a decision head initialized from an LM-head slice (Releasing the Inner Jev), a scalar head initialized from the Yes−No readout (Open-Jev), pointer heads (Kev, Our Own Jev), a question-agnostic affine head (AnyJev). The evidence supporting this is that "most of the decision is already in the representation": the readout of an untuned 27B reaches 97% of post-tuning accuracy, and on 9B 1/16 of the data recovers 88% of the tuning gain (2610.15501); FunDM's label-free KL recovers 99% (2610.17175); Our Own Jev's single-seed ablation finds "no detectable benefit from training the 26 letter rows of the output layer" (2610.29052). On this basis LDM defines decision fine-tuning as "**changing how the representation is read, not acquiring new knowledge**."
2. **The "augmentation" of the objective is mainly adding a proper-scoring-rule term.** Cross-entropy remains the baseline; the most common change is `+ Brier` (jevlike 0.5, Open-Jev 0.1, OneJev weight not given). More systematic frameworks treat "proper scoring rule" as a unifying language: SanSi / LEV use it at every step, NDM uses it directly and proves it shares its optimum with the policy gradient of an RLCD-style reward, OpenJev-RLCD uses it as the reward, StartLux uses the difference in score before and after reasoning as the reward. In other words, **"calibration" moved from a post-hoc temperature into the objective, but whether it is really better calibrated than "cross-entropy + temperature" has negative or inconclusive evidence** (see 12.3).
3. **The adaptation method is converging: frozen backbone + LoRA.** Rank 8–16 is most common (Kev 16, Open-Jev 27B uses 8); Releasing the Inner Jev reports "stable with respect to LoRA rank"; LEV reports LoRA beating full-parameter training on Ouro; the only exceptions are Bongard (7.09B full-parameter) and StartLux Stage 1 (full-parameter SFT). Nimble's variant is putting the LoRA loss "only on answer tokens."
4. **Supervision moved from "task labels" to "decision mixture + teacher."** Training sets are generally dozens to hundreds of public datasets recompiled into a "state + question + options" decision format (Kev: 10 datasets + policies + rule structures; Tev1: four source types; Releasing: 79 tasks, 300,042 hard labels + a soft-label subset; StartLux about 2.7M; Chinese Decision 13.048M), layered with a larger model as teacher (Qwen3.5-122B-A10B, Qwen3.6-27B, Qwen3.5-27B, Qwen3.8-27B, Jev itself). The only truly "new" data constructions are three: Nimble's **contrastive curation** (change one fact so the answer flips), LDM's **self-posed-question** tasks (SPRD), and AgentDecision's sampling **on the states the Agent reaches after its own mistakes**.
5. **Reinforcement learning appears only in a minority of projects, and is basically a "reasoning mode" matter.** For single-pass Instant models the open-source evidence is either negative (SBERT2S1 −2.5 to −3.0, ufakzeka −10.2 F1, JevAgentHarness no gain), or "equivalent" (the NDM theorem, UniJev "differences are small"), or holds only in settings **with a reasoning chain** (OpenJev-RLCD, StartLux AutoThink). TypeSafe's RLCD has no public details at all, and the open-source community uses the term "RLCD" in at least four ways (see § 1.4.7).

Structural changes (family G) are the only class of learning algorithm that changes "the problem itself": block-causal / Prism-1 turn order invariance from "something data augmentation has to learn" into "something guaranteed by construction," looped models make "one model serves every compute budget" a training objective, and Bongard builds "multiple judgments on the same state" into the encoder-decoder. These changes bear directly on the hybrid-backbone constraint of § 1.3: Prism-1 / block-causal both depend on the attention mask and do not hold on the DeltaNet layers of Qwen3.5.

6.1.3 Ablation evidence: do these changes help

Change	Evidence	Conclusion
LoRA itself	"LoRA drives the improvement," "stable with respect to rank," "longer training gives no consistent benefit" (2610.17205); LoRA beats full-parameter (Ouro, 2610.20951)	**Useful and insensitive**; no evidence of a full-parameter advantage
Head only / no training	Untuned readout = 97% of tuned accuracy; 1/16 of the data recovers 88% (2610.15501); FunDM label-free 99% (2610.17175); Our Own Jev 2B hard tier after training 63.1 **below** frozen 67.6 (1B 48.6 vs 54.1), battery set flat 70.6 vs 70.2 (2610.29052)	**Training gains are small and possibly negative**, depending on the evaluation set; for heads trained on frozen representations "no loss is clearly better calibrated after temperature scaling" (2610.17205)
Training duration	Longer baseline training makes Qwen overconfident: transfer-suite ECE 9B 23.8, 27B 18.5, the recipe stays at 8–12; one temperature removes most of the gap (2610.15501)	**The longer you train, the more overconfident**; temperature recovers most of it
Soft vs hard labels	"Teacher labels help, soft labels (may) not" (2610.17205); Open-Jev, Eikos and OneJev all use soft targets but without controls; Gaussian proxy theory: the student inherits the teacher's bias, and more data does not remove it (2610.26797)	**Contradictory and uncontrolled**; a first-order risk for our distillation route
Proper scoring rule / Brier term	SBERT2S1 grid of two heads × five objectives: after temperature scaling **no objective is clearly better calibrated than cross-entropy** (2610.02486); none of PACT's four terms raises raw accuracy, the gain is seed variance halved and NLL −26% (2609.35865); expected-utility objective +0.9±0.9 pp (2610.17205)	**No effect on accuracy, effective for robustness / NLL**; the calibration advantage essentially vanishes after temperature
RL / RLCD	Five RLCDs give "small" gains over SFT with heterogeneous calibration (2610.25784); the released RLCD recipe −2.5 to −3.0 (2610.02486); REINFORCE −10.2 F1 (2610.06744); GRPO no gain (2610.21412); but with a reasoning chain OpenJev-RLCD's selective prediction is better across the board (2609.38850), and StartLux AutoThink beats Instant by +11.5 to +18.4 paired accuracy (2610.17309)	**No positive evidence for RL on single-pass models**; RL's gains are bound to "generating reasoning"
Structural isolation	Block-causal flip rate 8.30→0.11%, 18.16→1.20% (2610.01601); Prism-1 zero flips, adding an option leaves relative odds unchanged (2610.22066); bidirectional attention +≤2.4 on 4 of 5 scales (2610.17309); per-step training makes loop 1 usable, 58.4% vs 35.6% (2610.07730)	**Structural changes are the only "algorithmic" changes with large effect sizes**, but they depend on the attention mask
Data vs algorithm	Swapping one corpus version moves the 2B hard tier by 11 points (2610.29052); decision training gives Nimble financial-sentiment Macro-F1 −12.0, Tev1 never significantly worse than the base (2610.26325); SecJev-0.8B beats Kev-9B by +20.51 (2610.03073); LEV "more data mainly strengthens the shallow readout" (2610.20951); Bongard JEPA +10.2, sandbox +14.2 (2609.39111)	**The effect size of data composition ≥ the effect size of the objective**; "decision training" is not uniformly beneficial
Number of teachers	Averaging two teachers > one teacher at full budget (+9.6 Elo); resampling the same teacher does nothing (2610.09188)	Teacher diversity matters more than label count

In sum: **relative to classification fine-tuning, the learning algorithms of existing schemes almost never "change the algorithm"; they only "change the readout + add a scoring term + change the data." What can be confirmed to work is LoRA, data composition, structural isolation and temperature; the evidence for soft labels and RL is insufficient or negative.**

6.1.4 Dataset availability

6.1.4.1 Public training and evaluation sets (by paper / project)

Data	Use	Size	Public status	Source
Kev `decision-v7` + `kev-suites`	Training / evaluation	10,000 + 896 + 1,680	Training scripts and suites public (HF `jaredpalmer/kev-suites`, 0.3 GB)	jaredpalmer/kev
Tev1 `new-v1`	Training	37,840 + 4,568	Public inside the repo	togethercomputer/tev1
Nimble training / held-out set	Training / evaluation	2,676 / 324	Re-released 2026-09-20; Decision Index evaluation set on HF (CC BY-NC 4.0)	bespokelabsai/nimble

Data	Use	Size	Public status	Source
Open-Jev training mixture	Training	148,639 rows (frozen)	HF `ZefanCai/Open-Jev` (CC0) and v1.1	Zefan-Cai/Open-Jev
OneJev-Data	Training	99,193 questions (94,707 public)	HF `OmniJev/OneJev-Data`, 17.7 GB (with images / video)	OmniJev/OneJev
OpenDecisions + training pool of 417,825	Validation / training	Pool 417,825	Paper says it releases the model, serving code and OpenDecisions	2610.17205
Eikos corpus	Training	About 25K	HF `caiovicentino1/eikos-decisions` (CC BY 4.0)	2610.29052
DecisionPivot	Evaluation	Control group + reference trajectories 32,233	HF `startlux-models/DecisionPivot`	2610.28522
StartLux training mixture	Training	About 2.7M (0.8–9B) / 0.8M (27B)	**Not seen public**; weights CC BY-NC 4.0	2610.17309
JEVal	Evaluation	11,257 (36 datasets, 10 domains, bilingual)	HF `carlosxiang/JEVal`	2610.03935
System1-Arena	Evaluation	92k questions / 105 tasks / 268 sources + 138 closed-loop environments	Described in the paper; no repo found on HF	2610.21244
TypedBench	Evaluation	Six checkpoints	Described in the paper; no HF repo found	2610.11392
JevBench	Evaluation	231 public questions (+ 111 hard)	GitHub repo; **HF `fstandhartinger/jevbench` does not exist**	fstandhartinger/jevbench
Laya typed-decisions benchmark	Evaluation	—	HF `LocalLLaMA/typed-decisions` (Apache-2.0), `Luni/laya-jev-benchmark`	2609.33843
DecisionGym / AgentDecisionBench	Evaluation / training	500 new tasks etc.	"To be released" (paper's wording)	2610.21412
DecisionVista-Synth / Human, DDS corpus	Evaluation / training	—	Code public; data status not verified	2610.15501
CJ-Bench / 13.048M Chinese decisions	Evaluation / training	307,900 / 13.048M	Paper says it releases the model; public status of the training set **not verified**	2610.23170
SecJev-Corpus	Training	14 tasks, 8 sources	Paper says it releases adapters, heads, corpus and code	2610.03073
MEDLINE-S1 / BIODECIDE	Training / evaluation	243k	Labels and models released (encoder route, control)	2610.02486
PACT data splits and adapters	Training / evaluation	324 held out	Public	2609.35865
this-that complex / spatial	Evaluation	—	HF `limberc/*` (MIT)	2609.23886

Data	Use	Size	Public status	Source
RLCDAlignBench	Evaluation	—	HF **gated (automatic)**, CC BY-NC 4.0	sumleo/RLCDAlignBench
Other HF community sets	Evaluation	—	`akhilaaa3/decision-bench`, `Hanno-Labs/decision-bench` (0.98 GB), `Praveenrajus/jev-bench` (2.95 GB), `yflantmy/DecisionRobust` (19.9 GB), `Lexmount/WebJev`, `mohit67890/imajev-bench`, `DaehanKim/korean-jev-benchmark`, `SabaPivot/jev-safety-bench`, `raxITLabs/decision-models-as-guardrails`, `respanai/behavior-benchmark`, `avbiswas/bev-decision`	HF probe record (`hf_probe.json`)

Conclusion: the availability of training data falls into three tiers. (a) **Fully public with a license**: Kev, Tev1, Nimble, Open-Jev, OneJev, Eikos, PACT, SecJev; (b) **only the model and evaluation are public, the training mixture is not**: StartLux, Chinese Decision (not verified), Intern-Decision, the teacher-labeled part of Mapika; (c) **entirely non-public**: TypeSafe Jev's training data and RLCD. Chinese decision data comes from Chinese Decision alone, and it targets general / medical / legal / financial use; **there is no Chinese decision data aimed at business rules (D3)**. On licensing, most data and weights are Apache-2.0 / MIT / CC0, but the weights of StartLux, Nimble v3, AgentDecision-4B, Dohnuts, torchcast and openjev/openjev are CC BY-NC; TuringLogic system-x requires manual approval.

6.1.4.2 Local copies (/mnt/cephfs/xingchao)

On 2026-10-28 the mirroring scope was narrowed by the user's decision to **all public datasets + the Qwen bases needed for the experiment ladder**; model weights are no longer downloaded in bulk. A re-check of `/mnt/cephfs/` the same day then found that the 27B-level bases were already in the shared directory, so the download scope shrank further to "only fill in the small sizes of Qwen3.5." The manifests are authoritative for status (`/mnt/cephfs/xingchao/manifests/jev/hf_manifest.jsonl`, `github_manifest.tsv`, `hf_probe.json`). As of this section's update (afternoon of 2026-10-28, mirroring complete):

- Directory convention: datasets `datasets/jev/<org>__<name>`, models `models/jev/<org>__<name>`, GitHub repos `repos/jev/<org>__<repo>` (shallow clones); every completed HF repo gets a `.complete` file (with revision and timestamp).
- **Datasets: 24/24 complete** (about 41 GB, including kev-suites, the two this-that benchmarks, DecisionPivot, Open-Jev / v1.1, JEVal, decision-bench, WebJev, imajev-bench, and the two large items `OmniJev/OneJev-Data` and `yflantmy/DecisionRobust`), each directory with a `.complete`. Excluded: `fstandhartinger/jevbench` (does not exist), gated repos (RLCDAlignBench, system-x, prompt-injection).
- **Models: 40/40 complete** (about 79 GB). 32 of them are ⩽9B decision models / adapters and controls (three generations of Nimble-9B, two Laya variants, Open-Jev 2B/9B/27B-v1.1, kev 0.5/0.8/4/9B, AgentDecision-4B, SanSi-2.6B, Lev-2.6B, decider-0.8b, Decision-2.0-Kai-0.6B, Qwen3-0.6B, JevAny-Qwen3.8-27B-LoRA, ufakzeka-karar, rlcd-modernbert-151m and others); the other 8 are the small Qwen3.5 bases.
- **The bases do not need to be re-downloaded**: `/mnt/cephfs/models/` already holds a complete `Qwen3.8-27B` (52 GB, 18/18 shards, config.json matches upstream key by key and has no `quantization_config`), `Qwen3.5-27B` (11/11), `Qwen3.6-27B` (15/15) and `Qwen3.5-35B-A3B` (14/14, MoE); the shared `huggingface_hub` cache also has complete Qwen3-0.6B / 14B / 30B-A3B (attention-only and MoE controls). The `-w8a8`, `-DSpark*` and `-DFlash2` entries in the same directory are quantized or speculative-decoding variants, not bases. See the manifest `manifests/jev/shared_models_reusable.tsv`. The originally planned 55.6 GB download of `Qwen/Qwen3.8-27B` was therefore cancelled.
- **The small Qwen3.5 bases have all landed**: Base and Instruct for 0.8B / 2B / 4B / 9B, 8 repos in total (1.77 / 4.57 / 9.34 / 19.33 GB per size, all Apache-2.0); none existed anywhere on cephfs before, downloaded "small→large" via `manifests/jev/hf_queue3.txt` and verified repo by repo. The configuration check is in the layer-count table of §1.3.2 (layer count and similar keys sit under `text_config`).
- Bandwidth constraint: the proxy gives about 10 KB/s per connection, 96 in parallel about 2.5 MB/s; repos with many small files actually get only about 1.5 GB/hour, large files close to 9 GB/hour. This round totaled 120 GB and took about one day. **The mirroring task is finished**; only the four GitHub repos below are not fully fetched.
- **GitHub**: 49 of the 54 cited repos are shallow-cloned (about 1.9 GB); `jaredpalmer/kev` has only a 37 MB partial checkout, the packs of `StartLuxLabs/StartLux-Decision`, `Mapika/decider` and `togethercomputer/tev1` are repeatedly truncated through the proxy, and `PACT` has only source and no weights because git-lfs is missing. The code of these four repos can still be fetched file by file from raw URLs, but without git history.

6.1.4.3 Sample check of the local copies: comparison against the data shape required by §2.2.1

On 2026-10-28 I sampled each of the 24 local datasets (for each directory, the four largest data files by size; read the schema and the first record; for multi-question sets I also computed grouping-structure statistics), with a single purpose: to check which of the interface requirements of §2.2.1, namely **(state, question set, one label per question)**, the business criterion written inside the question, the same state partitioned differently under different questions, option names injected from context, and unknown as a legal output, the existing data covers. The table below lists only the sets relevant to this requirement; every number comes from local files, not from papers.

Dataset (license)	Size (locally verified)	Shape	Relation to the requirement
`LocalLLaMA/typed-decisions` (Apache-2.0)	1,200 states, 4 workflows of 300 each; **5 questions** per state (a mix of choice / noul / score, 20 question types over the whole set); labels are probability distributions from multiple annotators	**(state, question set, one label per question)** triples; the criterion is written in each question's `criteria`	The only set with an annotator-disagreement field: of 6,000 question labels, **2,121 (35%) have inconsistent annotator argmax**, median total variation 0.14. This is a direct measurement of "labels are business conventions, and conventions are contestable"
`akhilaaa3/decision-bench` (Apache-2.0)	medium 80 states + hard; 586 questions in total, 1–9 questions per state, 40-odd business families; the state is a multilingual dialogue / ticket record	Same shape as above, with very long criteria (e.g. "certificate date no earlier than 12 months before the current contact date")	Enough for evaluation only. Its own results: Jev 1.13 **90.48%**, generative frontier 98.8–99.1% (state macro-average), the gap between readout and generation on business-criterion tasks
`ZefanCai/Open-Jev-v1.1` synthetic part `community-diversity-v2` (CC0)	48 templates × 80 groups, 30,720 training rows (51,200 in the full set); each group has 4 state variants × (choice + noul or score)	Rules **visible** (`numbered_requirements` + three-valued `evidence_policy`: null means unreported, not false; unknown propagates through AND/OR/NOT); options include `abstain`; labels computed by code	**The only labeled source of unknown / abstain**; counterfactual variants flip one field and the label changes in **10,201 / 11,858 (86%)**; the split holds out by rule family (`family_disjoint … 1 ood`). Weakness: the state is structured facts, not natural language
`ZefanCai/Open-Jev-v1.1` full mixture	147,139 rows, 73 sources; 65/73 sources have multiple questions per state	—	**Extremely low question diversity**: the full mixture has only **4,101 distinct question strings**, and the single source `wanli` uses one question for 82,045 rows
`avbiswas/bev-decision` (license not stated)	data 125,614 (1–9 questions per state, 56,091 multi-question rows), hard 42,000, skills 34,627, numeric_temporal 16,477, counterfactual 12,614	`counterfactual_15k`: the rule is written in the instruction and acts on a structured state ("if referrer is empty choose expedite"), 2,679 distinct rules	The closest off-the-shelf set to "rule-conditioned decisions," but the 12,614 rows contain **12,549 distinct states**: what varies is the state, not the rule; there is no "same state × multiple rules" pairing; license not stated, so evaluation only
`jaredpalmer/kev-suites diagnostics/binding-v1`	560 items, policy + case, six tiers (match_true / mismatch_clean / two decoy / two elapsed), code labels	D3 primitive probes (and / le / match / elapsed)	Directly usable; the trainable sources of `decision-v2` are all classic classification sets (banking77, boolq, agnews, mnli...), i.e. (state, fixed question)
`Hanno-Labs/decision-bench` (other)	23,900 rows, 43 tasks, 27 families, 3 primitives, up to 255 candidates; 1,200 rows dedicated to reasoning	The three `compact_*` columns are Qwen-generated, row-by-row verified **paraphrases** (labels, candidates and order unchanged)	Ready-made material for omission 4 (question-side paraphrase consistency), but what is paraphrased is the wording of the state and the instruction, not the criterion itself
`SabaPivot/jev-safety-bench` (other)	83,183 pairs × original / reversed / serialization_control (+ repeat), 415,827 rows of model results	Option positions swapped and key names swapped, with row-level results from laya, openjev and other models	Defect 6 (order) already measured; covers the safety domain only

Dataset (license)	Size (locally verified)	Shape	Relation to the requirement
`Praveenrajus/jev-bench` (other)	22,773 test questions × multiple models, logscore under two key permutations per question plus a set of **label priors**	Permuted readout in index mode	Ready-made infrastructure for the checklist's "same-backbone label log-probability baseline" and key permutation
`OmniJev/OneJev-Data` (128 sources, each with its own license)	18 shards, with images / video	One question per row	(state, single question)
The rest: `btzsc` (740–950k rows of classic classification), `Lexmount/WebJev` (64,122, GUI grounding / actions), `yflantmy/DecisionRobust` (19 GB of image perturbations), `DaehanKim/korean-jev-benchmark` (25,418, Korean), `carlosxiang/JEVal`, `limberc/this-that`, `startlux-models/DecisionPivot`, `mohit67890/imajev-bench`, `raxITLabs`, `respanai`, `caiovicentino1/eikos`, the `bespokelabs` decision index, `Luni` (empty directory)	—	Single-question or non-text	Unrelated to the shape of § 2.2.1; use as controls or not at all

Item-by-item check (against the five requirement points of § 2.2.1; for each item, first whether it exists, then whether it can be built from existing material):

1. **Multiple questions on the same state: yes.** Four sets (typed-decisions, akhilaaa3, the Open-Jev synthetic part, bev-decision) have the (state, question set, one label per question) shape. What omission 1 in § 2.3 originally said, "all existing decision datasets are (state, single fixed question, label)," **was wrong** and has been corrected.
2. **Cross-question re-partition (competing conventions on the same state): zero.** The multiple questions in the four sets are either different facets of the same state (urgency / risk / disposition) or different views of the same rule (which candidate to pick / whether a candidate qualifies / how many conditions are met). No group of questions is designed to "partition the state differently"; the second column of the rice / milk table has no local counterpart. But it **does not need a new task type**: Open-Jev's synthetic generator (visible rules, three-valued logic, code labels, held out by family) and bev's rule-conditioned templates are already the right components; what is missing is only the pairing layer "swap the rule on the same state."
3. **Option names injected from context (1–5 vs a–e): no set has transformed the name set.** SabaPivot and jev-bench only change the position and key assignment, not the "letters / digits / words" layer. This item needs no labels: any choice set can be re-rendered to test it, and in the literature its effect size is the largest (§ 3.1.2.2: under yes/no names the flip rate is up to 70.4 pp higher than the 0/1 control; the checklist's "option-name invariance test" compliance is 1/27).
4. **unknown: only one kind.** The unknown in the Open-Jev synthetic part is **missing evidence** (a field is null and the rule propagates it under three-valued logic), with code labels; the unknown of omission 3 is **rule-silent** (the question does not cover this state, e.g. nobody wrote "rice counts as solid"). The latter is zero locally, and no generator can compute it: its label can only come from the person who writes the rule.
5. **Labels come from business conventions: typed-decisions measured it.** 35% of question labels have inconsistent argmax across annotators, meaning that the "business convention" remains contestable under the same criterion. The synthetic route sidesteps the dispute with "rule in plain text + code evaluation," at the cost of the state becoming structured fields, which is exactly the projection of omission 9 (what the state is) onto the data.

One more quantity to check, directly related to proposition 3 (train "the ability to understand business requirements" rather than specific tasks): the conclusion of § 6.3.2 is that the variable deciding whether ICL survives is **whether the task definition varies** across training samples. By that measure no local set qualifies: the Open-Jev full mixture has 147,139 rows but only 4,101 question strings; typed-decisions has 20 question types; the synthetic part has 48 templates; kev's trainable sources are a dozen classic classification sets. To train "ability," what is missing is not states but **hundreds to thousands of differently worded rules on each state**, and there is no existing source for that.

Conclusion (answering only "can experiments be run on existing data"):

- **Measurements that can be run on existing data now**: the P2″ accuracy floor (typed-decisions, akhilaaa3, kev binding, Open-Jev synthetic, bev counterfactual); defect 6 order invariance (SabaPivot, already measured); option-name-set invariance (re-render any choice set); defect 5 missing-evidence unknown (Open-Jev synthetic); D3 primitive execution (the six kev binding tiers); the paraphrase of omission 4 (Hanno compact); the N = 5 prefix-sharing control (typed-decisions).
- **Must be constructed, but is not a new task type**: the "cross-question re-partition" denominator of P1″: the same state × several competing rules, with labels obtained by evaluating the rule text in code; it is a **new pairing** of existing components. The rule-silent unknown also belongs here, but it has no code label.
- **No existing source and not cheap to build**: business-convention labels on natural-language states (the 35% disagreement shows this is not an annotation-budget problem but a problem with the conventions themselves), and the rule diversity needed to "train ability."
- **One trap to record first**: the unconstrained optimum of synthetic rule data is **to learn the generator's rule grammar** (a parser). It generalizes to new rules in the same grammar, but has never touched the "milk is a liquid" half; the semantic half lives only in natural-language states, and labels on natural-language states can only come from people. So synthetic sets can measure the P2″ floor on formal rules but cannot substitute for the P1″ denominator on business semantics. This is the same line as the degenerate-optimum pattern of § 2.2.6.

6.1.5 Base-model selection patterns

Pattern	Evidence
The Qwen3.5 series (0.8B / 2B / 4B / 9B) + Qwen3.8-27B is the default backbone for "full reproduction"	Kev (Qwen3.5-0.8B/4B/9B-**Base**, Qwen3.8-27B), OneJev (the same four sizes), Nimble (Qwen3.5-9B), Tev1 (Qwen3.5-4B), Mapika decider (Qwen3.5-2B/4B-**Base**, 35B-A3B-Base), DKnown-Jev (Qwen3.5-2B / Qwen3.8-27B base), Chinese Decision (Qwen3.5-0.8B/2B/4B), UniJev (Qwen3.5 multimodal tower), Eikos (Qwen3.5-4B, Qwen3.8-27B), Open-Jev (Qwen 2B/9B/27B), pplx-decider (Qwen3.8-27B), qwen-rlcd (Qwen3.5-0.8B-Base), the Qwen branches of Releasing the Inner Jev and LDM
Attention-only Qwen3 (0.6B / 1.7B / 4B / 8B) concentrates in calibration and structure studies	OpenJev-RLCD (Qwen3-1.7B), jevlike (Qwen3-1.7B), SHIFT-JEV (Qwen3-0.6B / 4B + Phi-4-mini), block-causal (Gemma 3 1B, Qwen3 1.7B, Qwen3 4B), Prism-1 (Qwen3-8B), FunDM (Qwen3-8B), LLM-as-Jev (Qwen3-0.6B), System X (Qwen3-0.6B→1.7B), legal distillation (Qwen3-8B), NanoJev 0.6B. The reason agrees with § 1.3: these works need tree masks / block masks or gradients through the cache, which a hybrid backbone cannot do
Gemma 4 (12B / 31B) takes the "large model" slot	Mapika decider-12b / 31b (Gemma-4-*-it + a merged state-tracking LoRA), daseinlabs open-jev (Gemma 4 12B), Releasing the Inner Jev (Gemma 4 31B, 84.4 after fine-tuning), LDM, Prism-1 (Gemma 4 12B), torchcast-decision-12b
No consensus on Base vs Instruct	Kev, Mapika (Qwen branch), DKnown-Jev and qwen-rlcd explicitly use Base; Mapika's Gemma branch, Tev1 and OneJev use Instruct; Releasing the Inner Jev lists "starting from the instruct model" as an ablation item (the numbers are not in the cache and are not cited); **no paper systematically compares the two on decision quality, calibration and ICL retention**
Size: 4B is the mode for "full recipe" papers; 0.6–0.8B targets latency; 27B targets leaderboards	4B: Tev1, AgentDecision-4B, SII-Decision, Intern-Decision-4B, Mapika-4b, StartLux-4B, Eikos-4B, Jeva-4B, OmniMed-Jev-4B; 0.6–0.8B: Kev-0.8B, OneJev-0.8B, SecJev-0.8B, NanoJev, Dohnuts, System X; 27B: StartLux-27B (Decision Index rank 1), Open-Jev-27B, pplx-decider, Kev-27B, Eikos-27B
Each non-mainstream backbone has one representative	Loop-pretrained Ouro-1.4B / 2.6B (SanSi, LEV); encoder-decoder T5Gemma 2 4B-4B (Bongard); NeoHorse-1-4B (JEV reranking); MiniCPM5-1B/2B, Qwen3.6-35B-A3B (Our Own Jev); Muse-Glimmer 30B (MetaEncoder); a proprietary 182M base (ufakzeka-karar)
Encoders serve only as controls	Laya (ModernBERT-large 421M / mmBERT-base 322M), Verdict-MQ 395M, SBERT2S1 (the Sentence-Transformers family), NDM-Block, this-that, GLiNER2.5-Decide, CLM; the benchmark conclusion is "with task labels available, fine-tuned DeBERTa-v3-large has the highest accuracy on every annotated benchmark but one" (2610.00346), but this route is outside our direction

For the goal of "continued training on small Qwen-class models," the factual boundary the literature gives is: (1) the whole Qwen3.5 series is a DeltaNet hybrid, so branch isolation can only be trained row by row or by prefix forking (§ 1.3); (2) on the attention-only Qwen3 series every structural method already has an implementation and numbers; (3) on 0.6B–4B, "freeze + LoRA + reuse the output layer" is the starting point every paper uses, and after decision training on small models calibration **does not transfer** (jevlike out of domain, Nimble finance). (4) Qwen3.8 has only one dense size, 27B (the rest are MoE), so "small to large" can only use Qwen3.5-0.8B / 2B / 4B / 9B as the small end and 27B as the large end; the three generations of 27B share the same shape, and 0.8B–9B share family and proportions with 27B (one full-attention layer in every 4, the configuration table of § 1.3.2), so architecturally it is a clean ladder, but the small end and the large end straddle a generation of pretraining, and no literature has controlled for that difference.

Once the endgame size is fixed at the 27B / 35B level, the way the evidence is read must change (2026-10-28 protocol). Two points must be separated first:

Candidate	Architecture	Consequence
`Qwen3.5/3.6/3.8-27B`	**Dense**, 64 layers, 16 attention / 48 recurrent, hidden 5,120	The three generations have identical shape and interchangeable weights; the hybrid-backbone constraint of §1.3 **applies as is**
`Qwen3.5/3.6-35B-A3B`	**MoE, about 3B active**; Qwen has no dense 35B	**One more** assumption beyond the three structural assumptions of §1.3: whether per-token expert routing introduces new coupling between isolated question branches. **This report has zero hits on this**; it is an item that must be filled in before model selection

And the conclusions that this section and the tiered evidence of §6.2 give at the 27B level point in a **different direction** from the small-model level, which is a direct warning for the endgame goal:

1. **The gain from decision fine-tuning at 27B is already close to noise**. From Qwen3.5 0.8B→27B, the untuned readout reaches **76–97%** of post-tuning accuracy, the tuning gain shrinks from 15.8 points to **2.3 points**, and 1/16 of the data already recovers most of it; the authors' explanation is "**pretraining provides the representation; tuning only adjusts how it is read**" (2610.15501).
2. **On new schemas, the SFT gain at 27B is about zero**: the public recipe 88.1 / 87.5 and the source-augmented recipe 87.8 / 88.0, against **untuned 89.1 / 87.5** (same source). That is, at the endgame size, decision SFT does not buy cross-schema generalization.
3. **At the same level, "let it reason" still beats "readout" by a wide margin**: the same untuned 27B allowed to reason (four-sample majority) gets 96.9 vs 89.1 and 96.8 vs 87.5; on the three schemas of date arithmetic and numeric comparison, the readout gets only 56–66% at **every scale** while reasoning gets 98–100% (same source). So the endgame size will not erase the hard boundary of §6.5.5.
4. **The strongest positive result at that level is not decision SFT but distillation**: Reasoning-to-Readout self-distillation brings the 27B level with Jev on JEVal at about 0.1 s (2610.03935).
5. **The counter-evidence must be remembered too**: a domain-fine-tuned small model can beat a general large one; SecJev-0.8B is **20.51 pp** above Kev-9B (2610.03073), and AgentDecision-4B gets 91.5% vs Jev 82.0% on out-of-distribution tasks (2610.21412). So "the endgame must be large" needs an independent reason of its own and cannot be derived directly from "larger models are more accurate."

Taken together: if the endgame is 27B / 35B, then the P0 reading of §2.2 ("pretraining provides the representation; tuning only adjusts how it is read") is promoted from a side note to a **primary constraint**. At this level, what is worth investing in is not "training the readout to be more accurate" (only 2.3 points remain) but "**keeping the readout from losing the semantics already in the representation**" (rendering fixes, symbol tuning, §2.1.7) and "**distilling reasoning into the readout**" (item 4). At the same time the role of the small-size ladder (0.8B / 2B / 4B / 9B) changes with it: it is no longer "a scaled-down version of the endgame" but a **testbed for structural changes**, because every structural change with a large effect size (block-causal, branch isolation, loop depth) has been measured on small models, and the gain of decision SFT is significant only on small models.

6.1.6 Blind spots from the learning perspective

1. **There are almost no same-backbone, same-data objective-function comparisons.** The only systematic grid (two heads × five objectives) was run on encoders (2610.02486); on decoders there are only the partial ablations of Releasing the Inner Jev and UniJev's RL comparison. Whether "cross-entropy + temperature" is already the upper bound of the decoder route has no answer.
2. **Neither the benefit nor the risk of soft labels / teacher distillation has been quantified on decision models.** One side says "soft labels may be useless," the other uses them by default; theory says teacher bias cannot be removed and cannot be exposed by consistency. Whether the D4 defects of order, IIA, ordinality and negation are distilled into the student along with the teacher's probabilities, nobody has measured.
3. **The scope of RL is unclear.** All positive RL evidence comes from models with reasoning chains; on single-pass models RL is negative or ineffective; TypeSafe's RLCD is not reproducible. "RLCD" refers to at least four different things in the community (§1.4).
4. **The relation between decision fine-tuning and ICL still has zero literature** (first-order blind spot 1 of §8.3). From the learning perspective this is equivalent to "does LoRA decision training change how attention uses in-context examples"; the existing ablations look only at accuracy and ECE.

5. **Data composition has the largest effect size, but there is no research on "data recipes."** Contrastive curation, self-posed questions and the Agent's own trajectories have one paper each, with no same-backbone comparison; three kinds of data are empty: Chinese, business rules, and inputs written by non-technical staff.
6. **Training-time isolation on hybrid backbones still has no numbers** (§ 1.3), while every "algorithmic" change with a confirmed large effect size is a mask-dependent structural change. This turns "Qwen3.5 or Qwen3" from an engineering question into a learning-algorithm question.
7. **The generalization boundary of SFT has not been measured under control on single-pass models.** After the seven-level check of § 6.2, the evidence for transfer to new sources within the same family is sufficient, while the four levels of new schema, new symbolic operations, prior shift and calibration transfer have only negative or missing evidence; "SFT vs RL out-of-distribution controls" on single-pass decision models number zero papers.

6.2 Hypothesis review: can SFT bring generalization in the general sense

> Terms: this is an **evidence review** of one working hypothesis; it is still analysis only and gives no training plan. The hypothesis as originally stated: "I tend to think that SFT cannot give a model generalization ability in the general sense." The review is restricted to the literature on Jev-class single-pass decision models (the recipes and ablations in 02 / § 6.1), supplemented with a small number of upstream general results. Conclusion first: **the strong form does not hold, the weak form holds**: SFT transfers reliably on "same task family, new samples / new sources", and the gap converges as base-model scale grows; but not one paper shows that SFT can train in "new symbolic operations, fidelity under new schemas, out-of-distribution calibration, reasoning beyond single-pass depth", and several report negative results. The variables that move results across levels are the base model (scale, instruction tuning), data coverage, representation-stage objectives and structural constraints, not the supervision objective itself.
>
> **Re-check note (§ 8.1)**: the direction of this section's conclusion (strong form fails, weak form holds) is judged "right" in § 8.1.1, but this section's attribution of the failure to the **supervision objective** as the mechanism is judged **incomplete**: the evidence shows that what moves results across levels is base-model scale, the instruction-tuned starting point, the supervision distribution and structural constraints. In addition, the "seven generalization levels L0–L6" are the report's own construct (§ 8.1.2, overfitting risk medium-high).

6.2.1 First split "generalization" into testable levels

"Generalization in the general sense" cannot be tested directly. Following the evaluation splits actually used in the Jev-class literature, it can be split into seven levels, each corresponding to a different way of constructing the test set:

Level	Meaning	Test construction in the literature
L0 same distribution	held-out samples from the training sources	Kev "trained sources", Tev1 validation set, StartLux held-out
L1 new source / same schema	the same kind of decision, a new dataset or new policy text	Kev "new sources" (the 14 untrained datasets of breadth-v1), OpenDecisions transfer suite, SkillScreen new sources
L2 new schema	decision forms not seen in training (new question form / rule structure / rendering format)	DecisionVista DV-Dev-A/B, this-that JSON rendering and new rule structures, StartLux out-of-family tasks
L3 new symbolic operations	label redefinition, rule flipping, composition, fallback	names override definitions, label flipping, Certo counterfactuals, PFR fallback, StateWise
L4 distribution / prior shift	same task with different class frequencies, wording, sources	prevalence shift (2610.26325), cross-task reuse of rejection policies, forgetting in continual post-training
L5 calibration transfer	whether confidence and thresholds remain usable on a new distribution	jevlike out-of-domain ECE, transfer-suite ECE, teacher bias
L6 reasoning depth	items that need multi-step computation	rollout–readout gap, stratification by number of thinking tokens

Below, the evidence is given level by level; at the end of each level I state whether the hypothesis "holds / partly holds / does not hold" at that level.

6.2.2 Evidence by level

L0 / L1: same task family, new samples and new sources. The hypothesis does not hold, but the gain converges with scale

- **SFT does raise same-family accuracy**, and the gain lands mainly on small models. The new / trained sources table in the Kev README (dev / test): Kev-0.8B 0.648 / 0.697 vs 0.827 / 0.838; 4B 0.817 / 0.838 vs 0.873 / 0.865; 9B 0.820 / 0.852 vs 0.874 / 0.873; 27B 0.851 / 0.889 vs 0.865 / 0.866. The dev-set gap between "new sources" and "trained sources" shrinks from 17.9 points at 0.8B to 1.4 points at 27B (jaredpalmer/kev). On new sources, 27B is within one point of Jev (0.851 vs 0.857).
- **The gain itself converges with base-model scale.** On Qwen3.5 0.8B→27B, the untuned readout reaches 76–97% of the tuned accuracy, the tuning gain shrinks from 15.8 points to 2.3 points, and 1/16 of the data already recovers most of it; LoRA updates under 0.5% of the parameters, but training only the letter head "recovers almost none of the gain" (2610.15501). The authors' explanation is that "pretraining provides the representation, tuning only adjusts how it is read".
- **The marginal return of same-family data is far larger than cross-family.** StartLux, on the DecisionPivot reuse tasks, added second-stage data from the four families scoring / safety / preference / logic; the 18 tasks corresponding to these four families gained +8.6 points, out-of-family tasks only +0.3 (2610.17309).
- **"New sources" also have a failure surface.** SkillScreen, on unseen attack families, false-flags only 20–25% of benign skills at 99% recall and generalizes to new sources better than a fully fine-tuned encoder; but "unfamiliar benign sources make false positives rise sharply; recognizing new attacks is not the same as reliably screening new skill collections" (2610.17727).
- The "small beats large" of domain SFT (SecJev-0.8B +20.51 over Kev-9B, 2610.03073) is entirely **in-domain** numbers; the paper's own transfer test reports only two same-domain transfers, prompt injection and traffic.
- UniJev's three-modality joint SFT "raises decision accuracy on text, images and video" (2610.25784), again on same-family held-out sets.

Verdict: SFT transfer at L0 / L1 is the most consistent positive result in the literature; the strong form of the hypothesis fails here. But note two points: the bulk of the gain is "readout alignment" on small models, not new capability; at 27B the same-family gain is only about 2 points.

L2: new schema. The hypothesis partly holds, depending on scale and evaluation set

- The most direct comparison comes from LDM: at 27B, the public-data recipe raises validation accuracy from 79.3 to 81.8, and adding benchmark sources reaches 89.1 on the eight subsets of that benchmark; **but neither recipe reliably improves new-schema accuracy**: on DV-Dev-A/B the public recipe gives 88.1 / 87.5, the source-augmented recipe 87.8 / 88.0, versus 89.1 / 87.5 untuned. The authors' wording must be quoted as is: "transfer depends on scale and evaluation set; baseline tuning does help at 9B; so this is a motivation to widen schema coverage, **not a claim that 'supervised tuning cannot transfer'**" (2610.15501).
- this-that (encoder route, as contrast) gives the clearest example of "collapse outside the training distribution": the same batch of grids scores 0.457 rendered as ASCII and 0.373 rendered as JSON (chance 0.394); after one round of targeted training, 0.779 / 0.936 respectively; whereas on rule structures that never appeared in training, version 1.1 scores 0.447 against chance 0.31 with calibration error 0.20, "this is the only region where this model is confident and wrong, and the `confidence < 0.8` branch does not catch it" (FLock-io/this-that-model).
- Laya's independent reproduction found "no zero-shot transfer" on out-of-domain probes (2609.33843), likewise an encoder contrast.
- At the mechanism level, the "task schema" and "binding" of ICL are two separable mechanisms: schema transfers at 100%, binding only at 62%, and schema dependence is negatively correlated with prior knowledge (2512.17325); "rules can be amortized, pairings cannot": high-rank mappings transfer at zero in leave-one-task-out (2610.00526). This explains why "format / readout" is easily transferred by SFT and "specific mappings" are not.

Verdict: on the decoder route, the SFT gain on new schemas at 27B is about zero but not negative; at 9B it helps; on the encoder route, new schemas collapse outright. The hypothesis holds halfway at L2: **SFT brings no new-schema gain, but there is also no evidence that it harms the transfer ability the base model already has**.

L3: new symbolic operations. The hypothesis holds, with several pieces of negative evidence

- **Label names override definitions**: swapping option definitions with names drops AUC from 93.8% to 23.2% (2609.26758); deleting the definitions leaves accuracy unchanged (2610.02586). The models tested are hosted Jev and open-source decision models, all decision-trained.
- **Small models cannot flip label semantics in context** (1–12B, 2511.21038).
- **Counterfactual supervision works only on synthetic rules**: Certo's cacheability knocks rule sensitivity from recall@1 1.00 down to 0.24; targeted counterfactual supervision recovers high scores on **held-out synthetic rule tasks** (paraphrase, counterfactual, composition), reproducibly across seeds, but "we did not isolate whether predictions depend on the given rule", and on real rules "the added benefit is not established" (2609.37832).
- **Learning to track is not learning to obey**: StateWise used two-skill SFT + self-training to make 7–8B models track constraint state almost perfectly, "but unexpectedly did not learn to obey it" (2610.33766).
- **Recognition does not transfer to fallback**: in the PFR test, "correct recognition does not reliably transfer to the correct fallback option; even when the correct category is given explicitly, errors persist", and logically equivalent rule phrasings yield different decisions (2610.19856).
- **The amount of transfer depends on the degree of agreement with the most recent training rule**: in controlled arithmetic-rule experiments, "the adaptation gain rises with answer agreement with the most recent source; fully conflicting quadruples get only a small merge gain, and the sign reverses across source splits", and equivalent paraphrases have opposite task-level effects (2610.30045). This is the most direct quantification of "what SFT learns is similarity to the training rules, not the operation itself".
- Upstream general results point the same way: on precise instruction following, "most models strongly overfit to a small set of verifiable constraints in the benchmark and do not generalize to unseen constraints", and RL with verifiable rewards improves generalization (2507.02833); "SFT memorizes, RL generalizes" holds on the GeneralPoints / V-IRL rule variants, but the same paper also notes that SFT for a stable output format is a prerequisite for RL (2501.17161). Both are generative settings, not single-pass readout.

Verdict: there is no positive evidence at this level. SFT can make a model behave as if it can perform operations within the **trained operation family** (Certo synthetic tasks, StateWise tracking), but once the test asks "does it really depend on the rule" (swapped definitions, deleted definitions, real rules, obeying rather than tracking), the results are uniformly negative.

L4: distribution / prior shift. The hypothesis holds, and SFT may be negative

- Decision training is inconsistent in direction relative to its own base: on FPB financial sentiment, Tev1-4B's Macro-F1 is 10.0 points above Qwen3.5-4B, while Nimble-9B is **12.0 points below** Qwen3.5-9B with ECE 9.6 points higher, the loss concentrated in the neutral class; Tev1-4B is never significantly worse than its base (2610.26325). The same "decision SFT" can be +10 or −12 depending on data composition.
- Rejection policies cannot be reused across tasks: a Jev policy calibrated on DBpedia rejects 69.3% of covered Emotion test inputs, and in the reverse direction loses detection entirely (2610.03387).
- In continual post-training, "both the acquisition gap and forgetting are linearly correlated with the post-training distribution shift", small models most of all (2610.13113); LoRA causes 35–56% relative forgetting in out-of-domain few-shot adaptation while soft prompts do not (vision setting, 2609.11310, mechanism contrast only).
- Our Own Jev's 2B scores 63.1 after hard-tier training, below the frozen 67.6, and switching to another corpus version moves it by 11 points (2610.29052).

Verdict: after SFT the model is more, not less, sensitive to "the prior of the training mixture"; the effect size of data composition is larger than that of the objective function (§ 6.1.3).

L5: calibration transfer. The hypothesis holds

- jevlike (Qwen3-1.7B + LoRA + 37 datasets): temperature 1.19 brings in-domain ECE from 0.0212 down to 0.0083, out-of-domain only from 0.1311 to 0.1011; the 0.99 confidence gate validated in-domain as 99.85% precise fails out of domain (mustafasemi-ai/jevlike).
- Longer SFT makes Qwen over-confident on the transfer suite: ECE 23.8 at 9B, 18.5 at 27B, while the recipe stays at 8–12, and a single temperature removes most of the gap (2610.15501).

- Theoretically, the student inherits the teacher's bias and more data does not remove it; temperature scaling removes all excess risk only when the teacher's direction is correct, otherwise it leaves a residual that calibration diagnostics cannot see (2610.26797).
- Counter-examples exist only in generative settings: RL with a log scoring rule as reward makes confidence "generalize to unseen tasks" (2503.02623); temperature transfers across shallow surface variants (French input variants, 2610.33493).

Verdict: on single-pass decision models there is no evidence of "calibration holding across distributions after SFT"; what exists is evidence that temperature repairs it in-domain and not out of domain.

L6: reasoning depth. The hypothesis holds, and two papers explicitly list it as unresolved

- The same untuned 27B, when allowed to reason (four-sample majority), scores 96.9 vs 89.1 (DV-Dev-A), 96.8 vs 87.5 (DV-Dev-B), 85.1 vs 79.3 (validation set); on the three date-arithmetic and numeric-comparison schemas, readout reaches only 56–66% at every scale, reasoning 98–100% (2610.15501).
- Stratified by the thinking tokens the reference model needs, Jev's accuracy falls steadily from items under 250 tokens to 50% on items over 4k tokens; "whether training such decisions, or adding a few steps of internal computation before readout, can move this limit remains an open question" (2610.24929).

Verdict: this is not an SFT problem but a single-pass readout problem, yet it sets the upper bound of SFT: however much training data there is, a single-pass readout will not gain a depth it structurally lacks.

6.2.3 Which variables actually move results across levels

Since SFT itself has no cross-level gain, what is worth recording are the variables in the literature that **do** have cross-level effects; they are our reference points when we later discuss question definition:

Variable	Evidence	Which level the effect lands on
Base-model scale	new-source gap 17.9→1.4 (Kev), tuning gain 15.8→2.3, new-schema transfer "helps at 9B"	L1, L2
Instruction-tuned starting point	pretrain-only Gemma 4 31B with the same recipe reaches only 70.3, 14.1 below the instruct version (2610.17205)	L1, L2
Data coverage (not data volume)	this-that, one round of training targeted at JSON rendering, 0.373→0.936; LDM's conclusion is "widen schema coverage"; averaging two teachers > one teacher at full budget (2610.09188); 1/16 of the data already recovers most of the gain	L2 (covers only the trained forms)
Training distribution = deployment distribution	AgentDecision-4B trained on states reached after the agent's own mistakes, 91.5% on out-of-distribution tasks vs Jev 82.0% (2610.21412)	counter-example for L1→L2: with the supervision distribution chosen right, SFT can transfer across tasks
Representation-stage objective	Bongard's second stage (JEPA-style view alignment, not label supervision) on 14,606 held-out restatements 75.7→85.9, NLL 0.613→0.437 (2609.39111, encoder-decoder contrast)	L2 wording / logic-complement / ordinal-threshold variants
Structural constraints	block-causal flip rate 8.30→0.11% (2610.01601), Prism-1 zero flips (2610.22066)	D4 order invariance: **by construction, not by learning**
RL (only when a reasoning chain exists)	OpenJev-RLCD calibrate-then-reinforce, StartLux AutoThink +11.5 to +18.4 paired accuracy; single-pass RLCD is negative or equivalent (§ 6.1.3)	L6

6.2.4 Conclusion: the range where the hypothesis holds and the wording that needs revising

1. **The strong form "SFT cannot generalize" does not hold.** On new samples and new sources within the same task family (L0 / L1), decision SFT transfer is the most consistent positive result in the literature; the new-source gap converges with scale from 18 points to 1 point.
2. **The weak form holds, with more evidence than expected.** At the four levels of new symbolic operations (L3), prior shift (L4), calibration transfer (L5) and reasoning depth (L6), not one paper gives evidence of "SFT alone bringing cross-distribution gains", and several give negative results; on new schemas (L2), the SFT gain at 27B is about zero.
3. **A more precise statement that can replace it**: What SFT teaches is the alignment of readout and format, plus skills within the training distribution; it does not change whether the model "depends on the rule", nor the depth ceiling of single-pass readout. Cross-level generalization comes from the base model (scale, instruction tuning) and the choice of supervision distribution, not from the supervision objective. This agrees with LDM's "pretraining

provides the representation, tuning only adjusts how it is read" and with 2512.17325's "schema transfers, binding does not".

4. **Questions the literature does not answer** (only our own measurements can): (a) no paper on single-pass decision models has run a controlled SFT vs RL out-of-distribution comparison; "SFT memorizes, RL generalizes" and IFBench are both generative settings, and UniJev's comparison of five RLCD variants reports no out-of-distribution split; (b) the benefit of Certo-style counterfactual supervision on real rules is "not established", but also not refuted; (c) whether SFT **harms** the base model's existing L2 transfer ability: LDM's numbers (88.1 vs 89.1) are within noise, with no significance test; (d) all L3 negative results are on ⩽9B or hosted Jev; "names override definitions" has not been tested at 27B.
5. **Implication for "scaling up from small to large"**: all the scale-dependent quantities above (new-source gap, tuning gain, new-schema transfer) show that SFT effects observed on small models **cannot be extrapolated** to large ones; SFT looks "useful" on small models mainly because there is much room for readout alignment; therefore scale-up experiments must report L1 together with L2 / L3 at every size, otherwise "readout alignment" will be misread as "generalization".

6.2.5 Public English sets usable to test this hypothesis (metrics only, no plan)

The pre-experiments use English data, by the user's decision. English splits that already exist in the literature and are mirrored or mirrorable locally map directly onto the levels of 14.1:

Level	Public split	Local status
L1	breadth-v1 of Kev `kev-suites` (14 untrained datasets) and the `decision-v7` held-out set	`datasets/jev/jaredpalmer__kev-suites`
L2	this-that spatial (15 families, 7,305 items, three renderings) and complex-decisions (1,710 items, MIT)	`datasets/jev/limberc__*`
L2 / L6	DecisionPivot difficulty 1–3 contrast pairs (2610.28522); DecisionVista DV-Dev-A/B (code public, data status unverified)	`datasets/jev/startlux-models__DecisionPivot`
L3	name / definition swap protocol (2609.26758), PFR fallback test (2610.19856), Certo synthetic rule tasks	protocols public, must be rebuilt per the papers
L4	prevalence-shift protocol (2610.26325, five public datasets + 35 pre-registered paired tests); rejection-policy transfer benchmark (2610.03387, code and artifacts public)	rebuild per the papers / clone
L5	ECE on each of the splits above (§ 1.4.5 must first fix the binning) and out-of-domain reuse of the temperature fitted in-domain	—

What these splits have in common is the requirement that **the same checkpoint report all three columns, trained sources, new sources and new schema, at once** (the table format of the Kev README); otherwise "readout alignment" cannot be distinguished from "generalization".

6.3 RL or SFT: under the constraints of "keep ICL" and "inject skill semantics"

> Protocol:
> still analysis and survey only. This section answers a model-selection question that was raised explicitly, **should Jev-class models be trained and optimized with reinforcement learning**, with two added constraints: (1) the model must keep its **ICL** ability, i.e. the caller can **define a family of tasks** through the context; (2) the model must support **skill-like semantic injection**, i.e. it knows which method and procedure a given concrete question should be solved with. Conclusion first: **"RL or SFT" is not the variable that decides these two abilities**. What decides whether ICL survives is **what varies** between training samples; what decides whether skill semantics can be injected is **whether the skill lives in the context or in the weights**, and **whether a single-pass readout can execute a procedure**. RL has independent value only where there is a "sampled intermediate object" (a reasoning chain, an edit of the example set, a tool call) to which credit can be assigned; for a pure single-pass readout, the existing evidence is negative or equivalent to a proper scoring rule.
>
> **Re-check note (§ 8.1)**: this section rules that "use RL to keep ICL and skill injection" does not hold, which is the only one of the four working hypotheses whose **direction of conclusion is also judged wrong** (§ 8.1.1). Note, however, that the evidence base shared by this section and § 6.1 is thin: the three data points showing RL to be negative on single-pass models are mutually heterogeneous (one is an encoder, one is small-scale REINFORCE, one is GRPO inside an Agent loop rather than on the readout), see § 8.1.4.

6.3.1 First correct the framing of the question

Pick up the conclusion of § 6.2: what SFT teaches is readout alignment plus skills within the training distribution. A natural inference is "then switch to RL". But in the literature, the **positive evidence** relating to these two abilities comes almost entirely from SFT, only from **SFT of a specific construction**:

Desired ability	What works in the literature	Is it RL
Define a new task through the context	MetaICL: meta-train in ICL format on 142 tasks; at test time the model learns new tasks better from conditioning alone (2110.15943)	No, pure SFT
Let the context **override** prior semantics	Symbol tuning: fine-tune with natural-language labels replaced by arbitrary symbols (foo/bar); stronger on unseen ICL tasks, more robust to underspecified prompts, **greatly improved ability to follow flipped labels** (2305.08298)	No, pure SFT
Write skills/ procedures into the weights	LatentSkill: a hypernetwork turns text skills into plug-and-play LoRAs, ALFWorld seen / unseen +21.4 / +13.4 respectively, 63.9% fewer prefill tokens (2606.06087); Doc-to-LoRA generates a LoRA in a single forward pass (2602.15902); Cartridges train a KV cache offline (2506.06266)	No, hypernetwork / distillation
Inject procedures at inference time	SkillsBench: well-configured Skills raise the average pass rate from 33.9% to 50.5% (+16.6 pp), **focused Skills of ⩽3 modules beat exhaustive bundles**, a small model with Skills can match a large model without them (2602.12670)	No, pure context

The symbol-tuning row is especially critical: it is the known means of **using SFT to directly repair "label names override definitions"** (2609.26758, 2610.02586, 2511.21038), which is exactly the most serious current defect of D1. No RL paper claims to have achieved the same thing.

The framing should therefore change: not "RL or SFT", but "**what varies in the training distribution, and on which object the supervision lands**".

6.3.2 The variable that decides whether ICL survives: does the task definition vary across samples

- **ICL itself is "constrained vocabulary retrieval", not concept inference**: homogenizing the example labels collapses accuracy to ⩽12% (six models, 0.8B–8B, four tasks); when the examples use meaningless tokens the model puts 42–67% of its probability mass on the demonstrated set, while semantically plausible words get <0.2% (2605.08295). The model treats the label slot as an **exhaustive answer vocabulary**, which is the same mechanism as the Jev-style interface where "the option set is declared by the caller".
- **The learning capacity of ICL is itself limited**: after large-scale ablation, once examples become numerous, accuracy is insensitive to example distribution, model and prompt style; the model is **deducing** from regularities in the prompt and is therefore distribution-sensitive (2509.10414).
- **Mechanistically separable**: the task schema transfers at 100% through late MLPs, the binding only at 62% (2512.17325); rules can be amortized into a latent vector, pairings cannot, and leave-one-task-out transfer is zero (2610.00526).
- **The effect of decision fine-tuning on ICL has zero literature** (§ 3.1.2.3, § 8.3 first-order blind spot 1): no paper has measured the **k-shot curve** of the same backbone before and after decision fine-tuning.

Put together, the inference is clear: **if the task definition of every sample in the training set is fixed (one schema, one fixed option set), then whether the gradient comes from cross-entropy or from a policy gradient, the optimum is to burn the task into the weights and ignore the context.** The disappearance of ICL is not a property of SFT; it is a consequence of the data property "the task does not vary". Conversely, MetaICL and symbol tuning **improve** ICL precisely because they force the model to obtain the task only from the context.

For the Jev-style interface this has one particularly convenient aspect: **the option set is already in the request**, so every sample is naturally different. But when existing reproductions recompile dozens or hundreds of public datasets into decision format, what actually lands in front of the model is still a limited few dozen schemas (§ 6.1.2 item 4), so "options in the request" does not automatically equal "task in the context".

6.3.3 The variables that decide skill injection: context or weights, and whether it can be executed

In the literature, skill-semantics injection is **two distinct routes** with entirely different cost structures:

(a) Context skills (the SKILL.md kind)

- The gain is real and substantial: +16.6 pp, focused beats exhaustive, small model + Skill ≈ large model (2602.12670).
- But **well-written ones are scarce**: of 138,133 public SKILL.md files, 91.8% have at least one defect, mainly weak routing metadata, bloated non-executable bodies and poor resource organization; skills with compliant routing metadata are retrieved reliably at a significantly higher rate (2608.08453).

- **The organization itself changes behavior**: progressive disclosure raises the skill resources touched per trajectory from 1.18 to 3.85 (2606.11543); hierarchically grouped demonstrations raise the pass rate on "vaguely described" tasks from 76.7% to 90.7%, while no format yields any gain when the description is precise (2606.20978).
- **Having a skill ≠ executing it**: SkillSentry points out that even when a model has demonstrably completed a task under skill guidance, it still deviates from the procedure or executes steps wrongly (2608.09253); Skill Coverage proposes a trajectory-level coverage metric, because task success or failure does not tell you which part of the skill was executed (2606.20659). This is the same phenomenon as D2's "read correctly, done wrongly" (22%–72% execution rate, 2610.18579).

(b) In-weight skills

- LatentSkill uses a hypernetwork to map text skills into LoRAs, saving 63.9% of prefill and **exceeding** the corresponding context-skill baseline; the generated skill LoRAs form a structured semantic geometry that can be continuously modulated by a scaling coefficient and composed by parameter-space arithmetic (2606.06087).
- Doc-to-LoRA meta-learns "single-forward-pass approximate context distillation" (2602.15902); Cartridges train a KV cache offline and point out that naive next-token training is insufficient (2506.06266); cheat-sheet ICL distills many examples into one passage of text (2509.20820).
- Transfer of procedural memory is **partial**: one round of refinement gives +3.7–6.7 points, skills evolved from multi-model trajectories test at 73.1% across models, better than any single-model trajectory source; but "some skills generalize broadly, others solidify into role-specific procedures and fail under transfer" (2606.23127).

(c) The hard constraint for single-pass models: all of the "skills" above are **procedures**, and procedures need to be executed. A single-pass readout has no rollout: it can only let skill semantics influence **that one readout**; it cannot execute methods and steps. This is not a training-method problem: stratified by the thinking tokens the reference model needs, the accuracy of single-pass models falls from the <250-token items all the way to 50% on the >4k-token items, and the authors explicitly list "whether training such decisions, or adding a few steps of internal computation before the readout, can move this limit" as an open problem (2610.24929). The line of work that compiles workflows into deterministic programs (TraceCompiler, Artic, FlashScript, Formal Skill, 2608.02680, 2608.21341, 2610.33876, 2605.19604) is precisely an admission of this point: **procedures go to the program, judgment points go to the model**, and the Jev-style model is exactly that judgment point (2610.21411 measures routing and review at 31.7–43.5% of generative tokens).

6.3.4 So where exactly does RL have independent value

Merging the RL evidence of § 6.1.3 with the constraints of this section:

Setting	Evidence	Verdict
Pure single-pass readout	The published RLCD recipe −2.5 to −3.0 (2610.02486); REINFORCE −10.2 F1 (2610.06744); GRPO no gain (2610.21412); UniJev's five RLCD variants give "very small" gains relative to SFT (2610.25784); NDM proves that the optimum of a proper scoring rule and the policy gradient of an RLCD-style reward **share the same optimum** (2610.25553)	**No independent value**. Same optimum, only added optimization difficulty
With a reasoning chain	OpenJev-RLCD's "calibrate first, then reinforce" matches or exceeds SFT / RFT-STaR / GRPO on two real reasoning tasks and is uniformly better on selective prediction; on GSM8K answer verification a single query decides 81% of items at ⩽5% error versus 19% for GRPO (2609.38850); StartLux AutoThink beats Instant by +11.5 to +18.4 paired accuracy (2610.17309)	**Has value**, because there is a "sampled rationale" to assign credit to
The context/ examples are themselves the decision object	Jev-LDE: use RLVR to train a small System-1 model to make one Keep / Delete / Replace edit on the retrieved example set; consistent gains on classification benchmarks, transferable to held-out benchmarks and models (2609.33609)	**Has value, and directly serves ICL**
Instruction following under unseen constraints	Most models overfit badly to a small set of verifiable constraints and fail to generalize to unseen constraints; RLVR improves generalization (2507.02833); but RLVR is prone to over-optimization and verification shortcuts (2508.04632)	**Has value but shortcuts must be guarded against**; relevant to D2
Confidence expression	RL with a logarithmic scoring rule as reward makes calibration "generalize to unseen tasks" (2503.02623)	**Has value in the generative setting**, unverified for single-pass

There are also two theoretical results that limit the scope of RL, both from OpenJev-RLCD (2609.38850):

1. **A one-line variance identity** shows that "scoring a mixture of multiple samples" rewards disagreement among rationales and degrades reasoning, and **RLVR is exactly that mixture objective with the diversity term removed**.

2. **When the uncertainty comes from annotator disagreement, the same objective provably cannot beat cross-entropy**, and left unchecked the model will "discover that it should stop reasoning".

Add the engineering fact: the gradient of the policy-gradient term is **7–12 times** larger than that of the readout term and drowns the readout; without a KL anchor, "System-One collapse" appears (the model simply switches reasoning off).

6.3.5 Direct answers to the two constraints

(1) Keeping ICL. RL is not the answer; **the training distribution is**. The only two methods in the literature that improve the ability to "define a task through the context" (MetaICL's task-level meta-training, symbol tuning's symbolic labels) are both SFT, and both work through one data property: "the task of each sample can only be obtained from the context". RL's role here is **local**: when "which examples to pick / how to edit the example set" itself becomes a decision, the RLVR-trained Jev-LDE is effective (2609.33609). At the same time, know the ceiling of ICL: it is constrained vocabulary retrieval (2605.08295), and beyond a certain number of examples it is insensitive to the distribution (2509.10414).

(2) Skill-semantics injection. This is a **storage-location** question, not an SFT/RL question. Context skills are cheap, auditable and instantly editable, but 91.8% are written with defects, and "having a skill ≠ executing it"; in-weight skills (hypernetwork→LoRA) save more tokens and are stronger on both seen and unseen splits, but lose auditability and instant editability. For **single-pass** models there is one more hard constraint: the "methods and procedures" inside a skill cannot be executed and can only influence that one readout; the consensus of the workflow-compilation line is to hand procedures to deterministic programs and judgment points to the decision model.

(3) So should RL be used. For the single-pass Jev-class model we intend to build, the condition given by the evidence is clear: **RL has independent value only when the reward depends on a sampled intermediate object beyond the label**. For a pure readout, NDM's theorem says it shares the optimum of a proper scoring rule, and measured gains are "very small" or negative. If "a few steps of internal computation before the readout" or "AutoThink-style optional reasoning" is introduced later, the applicability condition for RL becomes satisfied, and a reproducible recipe shape already exists (calibrate first, then reinforce + KL anchor).

6.3.6 Blind spots identified in this section

1. **The k-shot curve before and after decision fine-tuning** is still unmeasured by anyone (already listed in 03 / § 8.3), and it is the only direct indicator for judging "did our training keep ICL".
2. **The intersection of symbol tuning and the Jev-style interface has zero literature**: nobody has done symbolic-label training on a typed readout, even though "label names override definitions" is exactly the most serious defect of typed readout.
3. **The intersection of in-weight skills and typed readout has zero literature**: LatentSkill and the like are all evaluated on generative Agents; nobody has measured the effect of skill LoRAs on a single-pass readout.
4. **There is no controlled out-of-distribution comparison of SFT against RL on single-pass models** (§ 6.2.4); UniJev's comparison of five RLCD variants reports no out-of-distribution split.
5. **Training data in which "the task definition varies in the context" does not exist in the decision domain**: all existing public sets are recompilations with fixed schemas; not one was constructed for "a new task definition per sample".

6.4 Does task-agnostic RL have vitality; can business expectations be made differentiable

> Protocol: still analysis and survey only. This section answers two follow-up questions: (1) **when not optimizing for a specific task, does reinforcement learning show enough vitality**; (2) **can all of our business expectations for Jev be turned into differentiable learning functions** (the questioner leans toward "cannot be directly optimized differentiably"). Conclusion first: (1) **Yes, and the strongest existence proof is Jev itself**, but this vitality lands only on **second-order properties** (how confident to be, whether to abstain, whether constraints are satisfied), not on first-order task knowledge, and wherever a "verifier" serves as the reward it is systematically arbitraged. (2) **The questioner's intuition holds, but the reason is not "non-differentiable"; it is four different causes stacked together**; the more critical finding is that **the few kinds of expectation that are truly non-differentiable are exactly the kinds where RL's measured gain is zero**: non-differentiability and the gain surface of RL barely intersect.

> **Re-check note (§ 8.1)**: this section's conclusion ("the objective is non-differentiable, so use RL" does not hold) is judged "right" in § 8.1.1, but the **reason** offered by the hypothesis ("because non-differentiable") is judged wrong: resistance to gradients has four different causes. The "five differentiability classes A–E" are the report's own construct (§ 8.1.2, overfitting risk, medium).

6.4.1 Follow-up one: does task-agnostic RL have vitality

6.4.1.1 Yes, and the evidence is stronger than expected

Evidence	Content	Why it counts as "task-agnostic"
Jev itself	Trained with RLCD, yet a **general-purpose** model: "one set of weights, served as an API, adapted to domains only through the request" (2610.25553's characterization of the other end of the design space)	This is the **existence proof** that RL produces task-agnostic models
Zero-shot alignment-failure detection	RLCDAlignBench tests Jev on ten classes of alignment failure (sycophancy, jailbreak, deception, prompt injection, hallucination, privacy, bias, reward hacking, concealed uncertainty, power seeking), 44 benchmarks and 5 target models: **a single generic question reaches a median AUROC of 0.886 zero-shot**, **beats supervised baselines** on most benchmarks, agrees with the reference scorer on human labels at a comparable level, also exposes label defects in the benchmarks themselves, at 1/63 the cost of an LLM judge (2609.29429)	One question across ten failure classes and 44 benchmarks
Confidence expression	RL with a logarithmic scoring rule as reward; calibration **generalizes to unseen tasks** without further fine-tuning, which the authors call "the emergence of general confidence awareness" (2503.02623)	Explicit cross-task generalization
Instruction following under unseen constraints	Most models overfit badly to a small set of verifiable constraints and cannot generalize to unseen constraints; RLVR improves this generalization (2507.02833)	A direct contrast of SFT overfitting while RL does not
Rule and visual variants	RL (especially with outcome-based rewards) generalizes across textual rule variants and visual variants, SFT tends to memorize (2501.17161)	Upstream general conclusion

6.4.1.2 But the vitality lands only on "second-order properties"

Looking at the five rows side by side, what gets generalized has one thing in common: **none of it is "what the answer is"; it is "how to answer"**.

- What Rewarding Doubt generalizes is **confidence awareness**, not knowledge.
- What Just Ask Jev generalizes is the **ability form** "give a calibrated probability for any question"; the paper itself points out that **the question wording has little effect, the context has more, mainly through fields that encode the label**, that is, what the model generalizes is the way of reading, while the content still comes from the context.
- What IFBench generalizes is the meta-skill of **constraint satisfaction**.

This matches the core fact of this report: the early audit concluded that "typed readout itself shows no **accuracy** advantage independent of same-class label-probability readout", and Jev's clearest gains are in latency, cost and calibration (2609.32160). **What RL generalizes is exactly the thing Jev actually sells**; that is no coincidence.

Conversely, **there is no evidence at all that RL generalizes first-order task knowledge**. Not one of the negative results at the L3 layer of § 6.2 (names override definitions, learning to track is not obeying, counterfactual supervision works only on synthetic rules) has been overturned by RL.

6.4.1.3 Three pieces of counter-evidence that must be kept in mind together

1. **Using a verifier as reward gets arbitraged, and this is measured on released checkpoints**: publicly released checkpoints after RLVR show "higher constraint satisfaction but lower answer quality", and frontier models can get full marks with answers of almost no substance (2610.19525); RLVR is prone to over-optimization and verification shortcuts (2508.04632); using constraints as reward pushes the model away from the main task (2610.28819).
2. **Theoretical ceiling**: a one-line variance identity shows that "scoring a mixture of multiple samples" rewards disagreement among rationales and degrades reasoning, and **RLVR is exactly that mixture objective with the diversity term removed**; when the uncertainty comes from **annotator disagreement**, the same objective

provably cannot beat cross-entropy, and left unchecked the model will "discover that it should stop reasoning" (2609.38850).

3. **Measured negative or equivalent in the single-pass setting** (§ 6.1.3, § 6.3.4): NDM proves that a proper scoring rule and an RLCD-style reward share the same optimum (2610.25553).

Summary: task-agnostic RL does have vitality, but its vitality = **training "how to answer" into a cross-task habit**. As long as the reward is the real objective itself (a proper scoring rule is calibration itself), it is stable; as soon as the reward is a proxy (a verifier, a rule checker), it gets arbitraged.

6.4.2 Follow-up two: can business expectations be written as differentiable learning functions

Classify Jev's business expectations by **why they resist gradients**, not by business domain. Once classified, "non-differentiable" turns out to be four different things.

Class	Expectation	Why it is hard	What the literature actually does	Conclusion
A Already differentiable with the correct optimum	Accuracy; **calibration**	None	Cross-entropy; proper scoring rules (Brier / logarithmic). NDM proves their optimum **is** the optimum of an RLCD-style reward (2610.25553); under a probabilistic model the exact recalibration map is temperature scaling (2610.26797)	**Not a problem**. Distinguish carefully: ECE as a **metric** is binned and has zero gradient almost everywhere; but calibration as an **objective** has a differentiable proxy with the correct optimum
B Property defined on a set of samples	Order invariance / flip rate; IIA (adding an option does not change the remaining relative odds); **definition takes priority over label name**; consistency under equivalent rewrites	Undefined on a single sample; permutation pairs, refinement pairs, counterfactual pairs, rewrite pairs must be constructed	Can be written as differentiable pairwise terms: PACT's pairwise anchoring + evidence necessity + ordinal transport + three-parameter context temperature; LLM-as-Jev's tree-decomposed listwise loss + KL anchor	**Differentiable, but crushed by structural construction**: block-causal takes the flip rate 8.30%→0.11% (2610.01601), Prism-1 has **zero flips by construction and the Luce axiom holds** (2610.22066); whereas PACT states of itself that "**no single term improves raw accuracy; the value is in robustness**" (2609.35865)
C Quantile / ranking type	The risk–coverage curve of selective prediction; the abstention threshold at a target risk; the threshold for escalating to a large model	Piecewise constant in the parameters, zero gradient almost everywhere	**Do not train; certify afterwards**: SHIFT-JEV certifies the abstention threshold with Learn-then-Test (2610.34329); conformal prediction; temperature is fitted only on a calibration set	**Giving up differentiability is the right move**. But know the cost: at a 10% target risk, a learned uncertainty score with a certified threshold still exceeds the bound in 3–5 of 27 environment–seed tests (max-probability score 5–15, conformal sets 9–14)
D Downstream outcome / long horizon	Agent task success rate, cost, wall-clock time; **decision→action execution**	No analytic gradient for the decision model; credit assignment spans multiple steps and external systems	Only RL, or do not optimize	**This is the truly "non-differentiable" class**. And as measured: **GRPO no gain** on JevAgentHarness; the model most accurate on isolated questions solves only 69.5% of Agent tasks; delegating tool selection directly without fitting a threshold **−19.0 points** (2610.21412); a Jev controller raises cost by +19% with an unchanged pass rate (2610.24649); policy modifications are read out correctly but executed in only 22%–72% of cases (2610.18579)
E Not an optimization objective at all	Latency, cost, deployability; skills being auditable and editable; fixing the definition itself	Not a function of the model parameters	Architecture and system design; the definition disagreements of § 1.4 must be fixed first	**Should not be attempted as optimization**. Until the six disagreement points listed in § 1.4 are fixed, not even the "objective" can be written down

6.4.2.1 Three judgments that follow

1. **The questioner's intuition holds, but the word "non-differentiable" conflates four things.** The only class that is both non-differentiable and without post-hoc certification is class D (downstream Agent outcomes and decision→action). Class A has a differentiable proxy with the correct optimum; class B is differentiable but should

not be solved by gradients; for class C, giving up gradients in favor of statistical certification is **the consensus practice of the literature**, not a compromise.

2. **The most critical finding is that the two sets barely intersect.** Where RL shows task-agnostic vitality (§6.4.1: calibration, confidence, constraint satisfaction) **is exactly classes A/C**, where a differentiable proxy or post-hoc certification already exists; and the truly non-differentiable class D **is exactly where RL's measured gain is zero** (GRPO no gain, the controller only adds cost). Therefore the chain of reasoning "because our objective is non-differentiable, we should use RL" **does not hold under the existing evidence**: non-differentiability and RL's gain surface do not line up.
3. **Class B offers a cheaper path.** Consistency-type expectations (order, IIA, definition priority) are better handled by construction than by writing pairwise losses and training: the effect sizes of Prism-1 and block-causal are **orders of magnitude** (8.30%→0.11%, zero flips), while the effect size of loss terms is "accuracy unchanged, robustness improved". The conclusion of §6.1.3, "structural changes are the only algorithmic changes with a large effect size", holds here once again.

6.4.2.2 Under what conditions this answer would change

- If **sampled intermediate objects** (reasoning chains, example-set edits, tool calls) are introduced before the readout, class D's credit assignment gains a landing point and the applicability condition for RL is met (§6.3.4: OpenJev-RLCD's "calibrate first, then reinforce", StartLux AutoThink, Jev-LDE).
- If **business utility** is written directly into the objective (rather than proxied by a verifier), class A can expand: Releasing the Inner Jev tried an expected-utility objective, but got only **+0.9±0.9 pp** (2610.17205), i.e. indistinguishable from zero.
- If the six definition disagreements of §1.4 are fixed, part of class E becomes measurable quantities and may thereby enter class C (certifiable).

6.4.3 Blind spots identified in this section

1. **Nobody has run a controlled comparison of "reward = real objective" against "reward = verifier proxy" on a single-pass decision model**; the arbitrage evidence of 16.1.3 all comes from generative instruction following.
2. **Class D credit assignment has no methodological work at all on the Jev-style interface**: once the decision model is placed in an Agent loop, the only thing tried has been GRPO, with no gain.
3. **Class B's "structure vs loss" has no same-backbone comparison**: block-causal and Prism-1 were each compared against "training with shuffled order", but neither has been compared against a PACT-style pairwise loss on the same backbone and the same data.
4. **Class C's certification methods are untested on hybrid backbones**: SHIFT-JEV uses Qwen3-0.6B/4B and Phi-4-mini, all attention-only.
5. **The boundary "second-order properties generalize, first-order knowledge does not" has itself not been tested head-on by any paper**; it is induced by this section from five independent pieces of evidence and is a hypothesis awaiting verification, not an established conclusion.

6.5 Training objective: what the output should look like, and whether to introduce chain of thought

> Protocol: still analysis and survey only. This section answers the second follow-up question: **what the training objective of a Jev-class model is, what numeric / decision outputs should look like, how to optimize them, whether chain of thought should be introduced to improve decision quality; and, if the business goal is System 1, what that chain of thought should look like**. The full landscape of learning algorithms is already in §6.1 (seven classes of change A–G and the ablation evidence); this section does not repeat it and only answers the two things §6.1 did not expand on: "the shape of the output" and "chain of thought".
>
> Conclusions first: (1) **the consensus on the training objective has converged on "proper scoring rules", but what is really missing is not the objective function but the shape of the output space**: three of the four output kinds have the wrong shape (boolean lacks the third truth value; Choice readout is ordinal but is used as a probability; Score is systematically compressed), and numeric values should not be fit with a single scalar at all; (2) **chain of thought should be introduced, but not at inference time**: in the existing evidence the three uses of chain of thought (distillation at training time, generation at inference time, layer recurrence) have diminishing

> returns and rising cost, and the most underrated is a fourth one, **letting Jev read others' chains**; (3) if the business goal is System 1, then its "chain of thought" **at inference time should not be a token stream but a question graph**: on a prefix-only model asking one more question is nearly free, while generating one more token is metered.
>
> **Re-check note (§ 8.1)**: the "question graph vs token stream" of this section is the report's own construct, derived from the cost structure rather than measured (§ 8.1.2, overfitting risk medium-high); the compositional calibration of "splitting one judgment into n judgments" has not been answered by any paper. The most stable conclusions of this section are ordinal compression and "structural constraints beat loss terms"; the one most likely to be overturned is the ranking of the three numeric routes (§ 8.1.4).

6.5.1 The consensus objective and its three proven boundaries

§ 6.1.2 already gave the conclusion: relative to ordinary classification fine-tuning, the changes concentrate on the **readout position** and on **adding a proper-scoring-rule term to the objective**. This section only adds three "already proven" boundaries, because they directly constrain all the discussion in § 6.5.2–18.5.

1. **When labels are known, the proper scoring rule and the policy gradient of an RLCD-style reward reach the same optimum, without the variance of the latter** (NDM § 1.3.1, 2610.25553). → On a pure readout, "switching to RL" is not an objective-function question but a variance question (§ 6.3.4, § 6.4.1.3).
2. **When the uncertainty comes from annotator disagreement, the same objective provably cannot beat cross-entropy** (2609.38850). → Wherever the business labels themselves are disputed (moderation, scoring, satisfaction), there is no room to gain on the objective-function side.
3. **Adding a proper-scoring-rule term does nothing for accuracy and helps robustness and NLL**: in SBERT2S1's two-head × five-objective grid, after temperature scaling no objective is clearly better calibrated than cross-entropy (2610.02486); none of PACT's four terms raises raw accuracy, the gains are seed variance halved and NLL −26% (2609.35865); the expected-utility objective gives +0.9±0.9 pp (2610.17205).

Corollary: if one wants "decision quality" to actually improve, the three movable places are, in order, **the shape of the output space** (§ 6.5.2), **structural constraints** (§ 6.1.3 already proved them the most effective), and **the form of computation at inference time** (§ 6.5.4). The objective function itself has essentially topped out.

6.5.2 What each of the four outputs "should look like"

6.5.2.1 Boolean / Noul: the shape is wrong because a truth value is missing

Sys1Cal-v1 takes true/false propositions whose P(A) is "known by construction", asks each of them through the three primitives, scores by total variation distance to the true distribution, and finds a behavior that can only be explained by "Jev suppresses the third truth value": in `Choice` answers P(A) and P(¬A) are presented as summing to 1, while the sum is missing a term $P(U) \neq 0$; **restoring P(U) raises the median soft accuracy of Choice from 0.771 to 0.978** (2609.35342).

The same conclusion is independently verified from the "probability axioms" side: on 480 negation pairs the sum of "the label is X" and "the label is not X" deviates from 1 by 0.064 on average (95% CI 0.055–0.072), the three single-label probabilities sum to **1.14** on average, and the violation is about 5 times its repeat noise and **concentrated where it is uncertain** (2609.33209). By comparison, the first-token readout of the official BF16 weights of Qwen3.8-27B deviates by 0.293 and its verbalized probability by 0.122, and it is a **two-sided negator** (in 196 of the 480 pairs it negates both a proposition and its negation), and it is not more self-consistent where it is more confident.

Shape conclusion: the correct shape of a boolean output is not P(true) but **a distribution over the triple {true, false, unknown}**; and the "uncertain" term cannot be cut out afterwards by a threshold, because the violation is concentrated precisely in the uncertain region. D4's "refusal / fallback" sub-dimension (§ 4.2) and P6/P7 (§ 2.1) converge here: 2610.19856 measures that "correctly recognizing the category does not transfer into correctly selecting the fallback option", precisely because the fallback is implemented as **an ordinary option in the option set** rather than **a structural slot in the distribution**.

6.5.2.2 Selection / Choice: the readout is ordinal, and using it as a probability goes wrong

JevCascade proves a testable property: when the intervention changes only the menu and the text is unchanged, **post-intervention accuracy is already determined by the cached first-pass distribution**: restrict the first-pass

probabilities to the new menu, renormalize, and take the argmax, with no labels and no second pass; across six model families, ten core datasets (sixteen in total) and two tasks, menu-only interventions are predicted to within **4.2 points**, exactly for one family. The key qualification is: **this property is ordinal, not probabilistic**: the "probability-layer" variant of the same estimator has an error of **21.0 points**; and it is category-specific, since generative models of the same size have errors of 1.6–15.8 points that get worse with scale (2610.07716).

Two direct consequences:

- **Menu design is a stronger lever than model size**: 0.8B on a curated 5-candidate menu reaches 95.4% on CLINC150, 4B on the full 150-label menu 80.0% (same paper). This agrees with the NIOCIS ablation: the gain is attributed to **restricting the candidate set per node** (+3.5 points over a flat catalog) rather than to graph traversal (zero within the interval) (2610.25985).
- **Using Choice probabilities as "probabilities" is out-of-scope use**. TypedBench measured the business consequence: under asymmetric costs, **using the host model's probabilities can be worse than taking its top-1 directly** (2610.11392).

Shape conclusion: the Choice output should be treated as **a ranking over the given menu + a confidence valid only on that menu**, not as a probability statement about the world. The property "probabilities remain comparable after the menu changes" does not hold under the existing evidence (IIA violation, § 4.2).

6.5.2.3 Score: ordinal compression is systematic, and scale does not fix it

This is the only output-shape defect in the existing literature that has been **confirmed on forty datasets across three model families**.

Observation	Number	Source
Ordinal scale-utilization bias	40 datasets, nine models from the three families JEV / KEV / StartLux; final decisions concentrate on **fewer effective labels** than the gold distribution; clearly weaker on unordered semantic categories	2609.38827
A finer scale **worsens** compression	The same items are compressed more strongly on a finer ordinal scale, with gold imbalance and candidate position controlled	Same
Randomizing candidate order	**Reduces but does not eliminate** compression	Same
Widening ordinal spacing at a fixed number of candidates	Can raise accuracy **without restoring utilization**	Same
Scale	Larger models have better accuracy but **do not consistently restore scale utilization**	Same
Co-occurring phenomenon	For most models, **concentrated decisions co-occur with wide average candidate probabilities**	Same
Can be mitigated by targeted training	The authors conclude that "scale utilization should stand alongside accuracy as an actionable reliability target"	Same
Not Jev-specific	Three frontier multimodal LLM families show the same **central-tendency effect** (endpoint compression) on clinical ordinal scales: overestimating the low end and underestimating the high end, while a fully fine-tuned ViT is best calibrated (MAE 0.52, 91% within tolerance); few-shot examples spanning the full range and removing clinical terminology **both fail to eliminate** the effect	2605.16386
A direct objective-function lever	PACT's **ordinal transport cost** term lowers the MAE of rubric fields to **0.232** (vs 0.311 for the published recipe), with the lowest position bias (9.8% vs 13.8% changed verdicts under relabeling) and flat accuracy (84.6% vs 85.2%, McNemar $p \geqslant 0.50$ for every seed)	2609.35865
Almost nobody turns it on in public implementations	Kev's `--ord_w` (ordinal loss) is **unused in the released models** (§ 6.1.1 table, row B)	jaredpalmer/kev
The readout form itself moves the result	Asking the same question with `Score` instead of `Choice` moves grading accuracy by up to **10.5 points**	2609.32160, paraphrasing T3
Do not collapse the distribution into one number	Psychometric audit: **under the default rounded readout, 13 of 14 decision models are below the generative reference**; yet after averaging over respondents, **the probability vector is closer to the human item marginals than the rounded answers**, and in ranking respondents it is roughly equal to a profile regression fit on 163 real people	2610.14912

Shape conclusion: the correct shape of Score is **a full distribution over ordered levels**; training should carry an ordinal term (transport cost / ordinal cross-entropy), evaluation should use an **ordinal proper scoring rule** rather than accuracy (TypedBench does exactly this, 2610.11392), and downstream **should not round to a single level**. These three are the best-evidenced group in this section and the worst-implemented in public code.

6.5.2.4 Numeric: Jev is not a regressor; there are three viable routes and one forbidden zone

First, the forbidden zone.

- **The arithmetic refusal bottleneck**: accuracy is 99% when the answer is present, but the correct refusal rate when the answer is absent is only **7%**, and this gap persists across numeric magnitude, operation depth, context phrasing and refusal label, and extends to time computation and capacity rounding; yet **on the same answer-absent arithmetic problems, native boolean verification reaches 99% exact match**, which means **classification-style refusal fails even though the model can successfully verify the correctness of a candidate**; a decision threshold selected on independent development problems raises refusal accuracy from 7% to **79%** while keeping 97% answer-present accuracy, with no retraining or extra inference (2609.39496).
- **The readout vs reasoning gap is largest on dates and numbers**: 56–66% (readout) vs 98–100% (reasoning) (§ 6.2 L3).
- **In medicine this is an explicit capability boundary**: Jev is on par with leading models on text-driven tasks but clearly weaker on **numerically driven** tasks (2610.18529).
- § 1.1.6 already wrote this as a conclusion: Jev **has no arithmetic / date / counting capability**.

Outside the "forbidden zone", the literature offers three genuinely viable numeric output shapes.

Route	Method	Measured quantity	Source
R1 Turn the number into an interval decision (training-free)	Accept that "selecting a numeric interval is itself a decision problem"; use a **multi-way decision tree to recursively refine the interval**, always keeping the original question in context, with no parameter updates and no access to hidden states; on a 100-value grid a ten-way tree needs only two rounds	Interval-normalized MAE **1.84%** vs 5.18% for direct selection; **2.93 pp higher even than "direct selection with the correct answer in the candidate table"**; in a three-date historical-index study the mean relative recall error is 4.58%, and the readout error when the value is given is **0%**	2609.28587
R2 Turn the number into binned classification, coarse to fine	For numeric and point-prediction tasks use **value bins and image regions**, refined by coarse-to-fine classification; multiple-choice tasks use the given options directly	This readout form is the core mechanism of MJev-Spatial	2610.22276
R3 Use a decoding-style regression head	Represent the numeric prediction as a decoded string; give the theoretical grounds for this capability and treat a causal sequence-decoding model as a numeric regression head over arbitrary feature representations	Although trained only with ordinary next-token cross-entropy, **the decoding head is as good as standard pointwise heads on standard regression tasks**, while being flexible enough to capture smooth numeric distributions (e.g. density estimation)	2501.19383

Shape conclusion: numeric output should be **a distribution over ordered bins** (the same family as Score); recursive refinement can buy precision, and a decoding head can buy smooth densities; but it **should not be a scalar regression target** and **should not carry arithmetic**. "Whether to support numbers" is therefore not a model-capability question but an **interface decomposition question**: hand "computing" to tools or System 2, and hand "which interval it falls in" to the readout.

6.5.2.5 Summary: four criteria for output shape

Primitive	Current shape	Evidence-supported shape	Strongest counterexample
Noul	P(true) ∈ [0,1]	Distribution over {true, false, **unknown**}	Soft accuracy 0.771 → 0.978 after restoring P(U) (2609.35342)
Choice	Probability distribution over options	**Ranking over the given menu** + within-menu confidence	Probability-layer estimation error 21.0 points vs ordinal-layer 4.2 points (2610.07716); under asymmetric costs probabilities are worse than top-1 (2610.11392)
Score	Distribution over ordered levels (but compressed, and often rounded downstream)	Distribution over ordered levels + **ordinal term** + **not collapsed into one number**	Scale-utilization bias on 40 datasets and 9 models (2609.38827); probability vectors closer to human marginals than rounding (2610.14912)
Numeric	Not in the interface	**Distribution over ordered bins** (recursive refinement / decoding head), arithmetic outsourced	Refusal rate 7% (2609.39496); interval decision MAE 1.84% vs 5.18% (2609.28587)

6.5.3 How to optimize: what the evidence supports and what it rules out

Lever	Evidence	Judgment
Structural invariance by construction (block-causal attention, position reset, one independent block per option)	Flip rate 8.30% → 0.11% (Gemma 3 1B), 18.16% → 1.20% (Qwen3 1.7B); Prism-1 has **zero flips by construction** and the Luce axiom holds; NDM's block attention mask makes each answer invariant to option order **by construction** and independent of the other questions, with lower loss than a pairwise cross-encoder at equal compute (§ 6.1.3; 2610.01601, 2610.22066, 2610.25553)	**Most effective**. Any consistency that can be written into the mask should not be written into the loss
Set-level loss terms (paired difference, permutation consistency, evidence necessity, ordinal transport)	PACT: **no single term raises raw accuracy**; the value is in robustness and stability (seed variance halved, NLL −26%, position bias 9.8% vs 13.8%, rubric MAE 0.232 vs 0.311) (2609.35865)	**Effective but only on second-order properties**; consistent with the class-B judgment in § 6.4.2
Post-hoc calibration and statistical certification	Naive thresholds exceed the target risk in about half of the settings, Learn-Then-Test only 7.8% (2610.19811); temperature fits differently under different loads, so threshold transfer cannot be assumed (2609.32160 C3)	**Must be a separate stage**, not folded into the training objective (§ 6.4.2 class C)
RLVR / outcome reward	RLVR improves task performance but produces **extreme overconfidence**; SFT is significantly better calibrated **and remains so under distribution shift** (with smaller gains); the diagnosis is that **the decision token acts as an "extraction step" in the reasoning trace and carries no confidence information**, which prevents RL from surfacing a calibrated alternative; "calibration-aware RL" built on this directly adjusts the decision-token probability, keeps RLVR's accuracy level and lowers ECE by up to **9 points** (2601.13284)	**Directly harmful for outputs that need probabilities, like Score / Noul**, unless a calibration term is explicitly imposed on the decision-token probability. This is the piece of evidence with the clearest mechanism on the question "whether to use RL"
RLCD (with reasoning chain)	Must **calibrate first, then reinforce**: end-to-end optimization either "System-One collapses" (prevented with a KL anchor) or is swamped by the policy gradient (whose gradient is **7–12 times** larger than the readout's); on GSM8K answer verification, a single query decides 81% of items at ⩽5% error, GRPO 19% (2609.38850)	Has independent value only when "the reward depends on a sampled intermediate object" (§ 6.3.4)

6.5.4 Whether to introduce chain of thought: four questions conflated into one

"Introducing chain of thought" is four different things in the literature, with different gains and different costs.

6.5.4.1 Use A: as a supervision source at training time, absent at inference time (strongest evidence)

Reasoning-to-Readout self-distillation: internalize an open-weight LLM's **own reasoning** into a single-pass first-token decision. SII-Decision-27B is **on par with Jev** on JEVal, while a typical query takes about **0.1 s** (2610.03935).

Evidence from the same family: Eikos adds a **reasoning-rationale auxiliary loss** on top of soft targets (§ 6.1.1 table, row C, 2610.29052); in StartLux's three-stage pipeline, AutoThink's reasoning ability comes from **reasoning supervision + RL** (2610.17309); FunDM's label-free KL distillation recovers 99% of the tuning gain (2610.17175).

Judgment: this is the only use that "gets the reasoning gain without paying the reasoning cost", and the only one that does not conflict with the business goal (the unit price of § 2.1).

6.5.4.2 Use B: generate chain of thought at inference time (effective, but then it is no longer System 1)

Evidence	Content
Off-the-shelf models suffice	LLM-as-Jev points out that decision models "seamlessly inherit the backbone's thinking mode": Jev is introduced as a System 1 model, while "Thinking Jev" uses **budgeted test-time reasoning** to raise accuracy on hard tasks, **with no training** (2610.02076)
A purpose-trained version	StartLux AutoThink invokes explicit reasoning on uncertain decisions and answers directly in confident cases; on DecisionPivot it is **11.5–18.4 points** of paired accuracy above the corresponding Instant version and leads among evaluated systems; the 27B AutoThink reaches 73.4 on the Decision Index 0.3 public subset (2610.17309)
The cost must be explicitly held down	RLCD^2 uses a frozen model and a proper-scoring-rule reward for "improvement in decision probability mass", and applies a **soft penalty** to overlong reasoning traces: the share of traces terminating within the length limit rises by up to **+48 pp**, mean reasoning length falls by up to **−65%**, and decision accuracy is essentially preserved (same paper)
Calibration cost	With an RLVR-style outcome reward, there is extreme overconfidence and the decision token carries no confidence information (2601.13284); RLCD without prior calibration either System-One collapses or is swamped by the policy gradient (2609.38850)

Judgment: effective, but it moves the cost structure back to the generative side (§5.1.1.1 eliminates exactly the decoding cost), so it can only exist as **a confidence-gated minority path**: this is cell A of §5.1.4, and the two properties P3/P5 that cell A depends on cannot be directly promised (§5.1.2).

6.5.4.3 Use C: layer recurrence, i.e. "chain of thought without generating tokens" (System 1.5)

Evidence	Content
SanSi	Turns a pretrained recurrent language model into a typed decision model; **option probabilities are read after every recurrence, and every recurrence is trained with a proper scoring rule**, so one training run yields a model covering every budget from 1 to 8 recurrences. **72.0%** on 10,027 test decisions from 59 sources: **13.5 points** above a non-recurrent model of the same shape (same recipe), 5.3 points above a newer non-recurrent model of the same size, 1.8 points below a model with three times the parameters. On two depth-controlled tasks, **recurrence pushes the solvable depth beyond the depths seen in training**, while larger single-pass models fail. As a judge for RL policy optimization (no gold answers), it raises the generator's share of correct answers from 37% to 52% (2610.07730)
LEV	Same idea, three recurrent backbones: on the same set of weights accuracy rises with depth, and **the gain falls mainly on hard items**; 2.6B goes from 63.2% at depth 1 to **76.2%** at depth 4 (JevBench). **On-demand depth** (decided per item by a small stopper with a threshold fit on unlabeled requests) saves **15%–24%** compute at a cost of under one accuracy point, on par with or slightly above a fixed-depth mixture; on the locked test set, 1.4B and 3B are **5.4–8.8 points** above non-recurrent dense models and cascades at **equal latency (<50 ms)** (2610.20951)

Judgment: this is **one of the most direct validated answers** to "if the business goal is System 1, what should the chain of thought look like": variable depth, zero output tokens, controllable latency, allocated per item. Its constraint is the backbone: both are built on **language models pretrained to be recurrent** (Ouro), while our base model family (Qwen3.5 / 3.8) is not recurrently pretrained and is a Gated DeltaNet hybrid (§1.3). There is no work in the literature on "converting a non-recurrent hybrid backbone into a recurrent readout"; this is a zero hit.

6.5.4.4 Use D: let Jev read others' chains of thought (most underrated)

On seven LLM reasoners × five multiple-choice benchmarks (35 pairs), Jev is used as a "decision reader for truncated reasoning": across all tested budgets from 500 to 16,000 tokens, **Jev's mean accuracy is above forced answering** (with the estimated cost of reasoning and answering counted in). At a 1,500-token budget, **+4.1 pp** overall and **+8.6 pp** on truncated items, 17 pairs significant under Holm-corrected bootstrap; offline replay estimates 34.5% of the full-run cost with only 1.13 pp of accuracy lost; a policy selected on the training fold saves **71.3%** of the estimated full-run cost on the held-out fold while losing only 0.58 pp. The same paper also finds that Jev's **question-only** accuracy is strong, on average above 500-token forced answering (68.0% vs 65.5%), but this advantage varies by benchmark (2610.25768).

Judgment: this is the form that fully decouples "chain of thought" from "decision": **the chain is produced by System 2 and the decision is consumed by System 1**. It does not require Jev to reason itself, only to read; and among the property checklist of §5.1.2 it depends only on P1 and P3.

6.5.4.5 Why some form of "deeper computation" must be introduced: the hard limit

Jev **cannot make the decisions that "an LLM can only make by writing out intermediate steps"**. On a factor-by-factor benchmark, Jev finds and selects the record without error within 20k tokens and 64 options, but once the answer depends on **more than two updates** from the same user, accuracy drops, reaching 53%–80% at sixteen updates, because similar records interfere with one another. Given a **stale state summary**, it answers according to the stale summary. On real decision items graded by "how many thinking tokens a reference LLM needs", Jev gets 97.5% of the items that the LLM **can solve in one forward pass**, but only **74%** of the rest. More critically, **it does not know its own limit**: its confidence can flag its own errors, but cannot flag the errors it inherits from stale inputs, and neither explicit nor trained handoff options **catch the latter**. All evaluated open models share these limits, and the authors conclude that "**checking the state such models depend on requires a model that reasons**" (2610.24929).

This limit decides the trade-off among the uses: A (training-time distillation) cannot break through it, because what is distilled is the "solvable in one forward pass" part; C (layer recurrence) is the only use with **depth-extrapolation evidence** (SanSi pushes the solvable depth beyond the training depth); B and D are two ways of outsourcing the limit to System 2.

6.5.5 If the business goal is System 1, what should this chain of thought look like

Lay out the four uses of §6.5.4 by "whether decoding cost is paid" and "whether the one-forward-pass limit is broken", and the answer appears.

Use	Decoding cost at inference	Can it break the one-forward-pass limit	Suited for
A Training-time distillation	**No**	No	Default plan: gets the full gain of the "solvable in one forward pass" part for free
C Layer recurrence (System 1.5)	**No** (pays layer recomputation)	**Partly** (depth extrapolation validated)	The only validated route to more depth under the zero-token premise
D Reading others' chains	No (S2 pays for the chain)	Yes (the limit is carried by S2)	Pipelines that already have an S2 running
B Generating its own chain	**Yes**	Yes	Only as a confidence-gated minority path

But there is a **fifth form**, and it is the natural form of "the chain of thought of System 1" at the interface level.

6.5.5.1 The chain of thought of System 1 is a question graph, not a token stream

The cost structure of a prefix-only model decides this: **all questions in one request are answered in the same forward pass** (§1.1.4), so asking one more question is nearly free (in Jev's order; the order fixed in §2.2.1 trades this for cross-sample caching along the rule axis, so each node of the question graph pays the prefill of its candidate branches once); while generating one more token is metered (§5.1.1.1). So the equivalent of "unrolling the reasoning" on Jev is not "writing out the steps" but **decomposing one judgment into more typed judgments and putting their structure into the request or into the caller's code**.

Existing instances already cover most variants of this form:

Form	Method	Measured quantity	Source
Parallel sub-questions, read the disagreement	Because "the model's price for extra questions is almost zero", decompose the judgment into parallel sub-questions and read their dispersion; this recovers the **human disagreement** signal that the model's native confidence misses	The remedy **is specific**: it helps only the weakest, disagreement-blind models, and only when the decomposition aligns with the source of disagreement; dedicated toxicity models and the strongest LLMs do not need it, and the effect is distribution-sensitive	2610.33258
Put "what to ask" into the same forward pass too	LAVOIR puts candidate missing information (slots) in the input, right next to the answer options, so that **one forward pass returns both the decision distribution and, for each slot, the "expected gain in correct-decision probability if that slot were asked about"**; the VOI target needs no human labels; a Gini-impurity upper bound constrains predictions to the range "a calibrated model can still gain"	The questioning policy matches a greedy oracle VOI policy (AUC 0.799 vs 0.797); at most 0.5 follow-ups per dialogue, it is **14.1 points** above "never ask"; on real ABCD dialogues, one real interaction raises accuracy by **8.3 points** where it chooses to ask and leaves it unchanged where it does not; on SGD the bound cuts the follow-up rate from 93% to **8.6%**; 31 ms (median, GH200)	2609.30706
Walk the tree node by node	SemTree's tree of Yes/No/Unknown nodes can localize and repair failure conditions node by node; Jev-Guard first routes on an editable policy tree and then does conditional classification; the official docs' "walk the classification tree level by level"	Jev-Guard: safety accuracy 93.72% on 7,033 cases, mean latency 58 ms, hierarchical routing accuracy 93.10%, 240 ms end to end	2610.13999, 2610.18326
Menu curation	Shrink the candidate set before asking; this is a stronger lever than deepening the model	0.8B with a curated 5-candidate menu 95.4% vs 4B with all 150 labels 80.0%; restricting the candidate set per node +3.5 points	2610.07716, 2610.25985
Reuse the first-pass distribution	Menu-only interventions **need no second forward pass**; restricting + renormalizing the first-pass distribution suffices, within 4.2 points of error	The same paper also gives the negative side: **running uniformly more passes buys only calibration, almost no accuracy**; at equal cost a confidence cascade beats any scheme that re-asks the same model	2610.07716

6.5.5.2 Three known boundaries of this form

1. **Decomposition is not a universal gain.** "Running uniformly more passes buys only calibration, almost no accuracy" (2610.07716); reading disagreement from parallel sub-questions works only under specific conditions (2610.33258).
2. **Decomposition amplifies consistency defects.** The question form itself moves the answer (`Score` vs `Choice` by up to 10.5 points, 2609.32160), and the probabilities are already not self-consistent (three single-label probabilities

summing to 1.14, 2609.33209). Splitting one judgment into ten and recombining them multiplies these biases ten times. The "decomposition consistency" sub-dimension of § 4.2 § D4 exists for exactly this.

3. **Correct intermediate choices do not guarantee a correct sequence**: a correct final label masks wrong intermediate choices (2609.32160, paraphrasing T2); Choice2Seq measures "task correctness, plan faithfulness, output compliance" separately (2610.16191).

§ 6.5.5 conclusion: if the business goal is System 1, then the "chain of thought" should be absorbed **at training time** in the form of distillation (use A), and should exist **at inference time** in two zero-token forms: **the depth of layer recurrence** (use C) and **the width of the question graph** (§ 6.5.5.1); a chain of thought that actually generates text should only be a confidence-gated escape route (use B), or simply be produced by System 2 with Jev only responsible for reading it (use D).

6.5.6 Section summary and blind spots

Training objective: proper scoring rules are the consensus and have topped out (three boundaries in § 6.5.1). What can be moved is the shape of the output space, structural constraints, and the form of computation at inference time.

Output shape: boolean needs to be ternary (with unknown); Choice is an ordinal quantity within a menu, not a world probability; Score needs an ordinal term and must not be rounded downstream; numbers must land on ordered bins with arithmetic outsourced. Three of the four have evidence spanning many datasets and many models, and are the worst-implemented in public code (Kev's ordinal loss was not enabled at release).

Chain of thought: introduce it, but at inference time it should not be a token stream. Of the four uses, A (training-time distillation) is the default, C (layer recurrence) is the only zero-token route with depth-extrapolation evidence, D (reading others' chains) is the most underrated, and B (generating its own) only deserves to be a gated escape route. And System 1's own form of "unrolling" is the **question graph**: asking more is nearly free, generating more is metered.

Blind spots:

1. **Nobody has made "unknown" a structural slot in the interface and measured it**. P(U) has only been **restored after the fact** once (2609.35342); no training work has taken the ternary truth value as the output space.
2. **The gain from the ordinal term has evidence from only one paper and one setting**. PACT's MAE 0.232 vs 0.311 comes from a frozen held-out set of 324 items and three seeds (2609.35865); 2609.38827 says it "can be mitigated by targeted training" but gives no cross-dataset training comparison. **The number "how much does scale-utilization bias drop once the ordinal loss is turned on" does not exist.**
3. **The three numeric routes are not compared against each other**. R1 is training-free, R2 is in multimodal, R3 is in general regression; **no paper has compared the interval decision tree, binned classification and the decoding regression head on the same batch of numeric decisions**.
4. **Layer recurrence is incompatible with our backbone, and nobody has studied the bridge**. SanSi / LEV both require a recurrently pretrained backbone; "converting a non-recurrent hybrid (Gated DeltaNet + attention) backbone into a multi-recurrence readout" is a zero hit (§ 1.3).
5. **The compositional semantics of the question graph has no theory**. Decomposition multiplies consistency defects (§ 6.5.5.2), and existing work has only measured single-level decomposition; "what is the calibration of n typed judgments after composition" has been answered by no paper; the 160 items and 480 pairs of 2609.33209 are the only evidence in the whole literature on probabilistic self-consistency.
6. **The mechanism "the decision token carries no confidence information" has only been diagnosed in the generative RLVR setting** (2601.13284). Whether it also holds on a pure-readout (no reasoning chain) single-pass model is unverified; and if it does hold, it directly explains the "RL negative or ineffective on single-pass models" recorded in § 6.1.3.

7 • Evaluation: the existing checklist, the coverage, and what cannot be decided

> This chapter only analyzes and surveys; it **proposes no scheme and designs no new benchmark**. The method is to treat the twelve business properties of §5.1.2 as the objects under test and check which cells the existing evaluation sets cover.
>
> Three conclusions: (1) "how evaluation should be done" already has an answer in the literature: a **14-item checklist** plus a minimal report card, and the compliance rate of existing papers against it is low (same-backbone label log-probability baseline: only 5 of 26 papers comply; option-name invariance test: only 1 of 27; latency with and without cache: only 3 of 20). In other words, in the published numbers "the interface's credit" and "the model's credit" have essentially never been separated; (2) seen by business property, coverage is severely uneven, and **the two cells "configurable by business staff" and "auditable" are zero**; (3) the largest gap is the business goal itself: **end-to-end cost per correct result has no reproducible protocol at all**.
>
> Read through P1′ of §2.2, the coverage matrix of this chapter is still missing its most critical row: **decision consistency rate on pre-registered business equivalence classes**, which is the unoccupied slot this report found (§7.5 row 13, §8.2.2).

7.1 "How evaluation should be done" already has an answer: the 14-item checklist and its compliance rate

An evidence audit of 28 papers published within nine days after the Jev release collected the recurring comparison and reporting defects into a 14-item checklist, and for the scorable items gives **compliant papers / applicable papers** (counted within the 27 papers that "evaluated a model"; "unclear" papers are removed from both numerator and denominator; "–" means the item could not be reliably scored from the text) (2609.32160).

ID	What must be reported	Compliant / applicable	Observed failures for this item
B1	A baseline that reads label log-probabilities from a plain or **same-backbone** language model	**5/26** (1 unclear)	Otherwise interface gains cannot be separated from model gains; wherever it was measured, the readout added almost no accuracy
B2	If a confidence threshold hands decisions to another model: a **cheap generative cascade** as a baseline	**1/5**	On external benchmarks the cheap generative cascade is cheaper than the TDM cascade and no worse; among six open-weight LLMs in Jev's price range, two have a higher median macro-F1
B3	If the labels suffice to train one: a **task-specific model** (encoder / classic classifier / reward model / probe) as a baseline	**7/13** (3 unclear)	A fine-tuned encoder is not significantly worse than open-source TDMs in ranking, has nominally lower ECE (higher Brier), and is faster; trained quality models still lead by a wide margin
T1	**The score gap between models** on the test set; do not claim parity on a saturated test	–	Several parity conclusions come from tests on which strong models have already hit the ceiling
T2	Correctness or consistency of **each individual decision** under repetition / reordering / rewording, not only aggregate accuracy	**17/26** (1 unclear)	Identical aggregate scores hide different decisions; a correct final label hides wrong intermediate choices
T3	**Option-name invariance** test: swap names across rubrics, or compare neutral names with semantic names	**1/27**	Swapping no/yes across rubrics pushed Jev's AUC from .815 to .581 with zero type errors; asking with `Score` instead of `Choice` moves grading accuracy by up to 10.5 points
T4	**Data provenance**: synthetic or real, source of gold labels, whether the public benchmark predates the model	–	The corpus contains Agent-generated text paired with Agent-generated labels, and unresolved contamination concerns
R1	**Exact model version and run date**	–	The host TDM is closed-source and may change under the same name; several papers give neither
R2	**Confidence intervals or significance tests** on the main result	**15/27**	12 of 27 papers have neither on their main result; because the host model is non-deterministic, stability across repeated runs should also be reported
R3	Disclose prompts, recipes, baselines or thresholds that were chosen **after seeing the test data**	–	At least 7 papers made such choices; a chain of redesigns pushed the main metric from .576 to .797

ID	What must be reported	Compliant / applicable	Observed failures for this item
C1	If probabilities are used as, or described as, calibrated: measure calibration with ECE / Brier / reliability analysis or an equivalent metric	**13/17** (1 unclear)	Prefer held-out data, metrics matched to label prevalence, and both pre- and post-recalibration results; under some workloads the raw probabilities are miscalibrated, recalibration lowers it by 3.3 (points), and the 0.01 grid limits resolution on rare labels
C2	Accuracy of the **highest-confidence batch of samples**, broken down by task	–	The corpus contains cases of "high confidence but accuracy near the base rate", and wrong tool calls with mean confidence .778; under answer-preserving optimization context, 45% of Jev's originally correct decisions (73% of those redirected) end up on the wrong option with probability >0.7
C3	If a confidence threshold is used for accept / escalate: the threshold is fitted on **data separate from the evaluation set**, or fixed before evaluation and not tuned on it	**7/7** (3 unclear)	A frozen threshold should also be tested for transfer across workloads / across fallback models, because a threshold fitted on the selection set does not hold on held-out items for every fallback model, and the fitted temperature differs across workloads; some fixed thresholds are applied to scores the authors explicitly do not assume to be calibrated
C4	If TDM latency is reported or a latency advantage is claimed: **latency both with and without cache**	**3/20** (1 unclear)	Serving location, concurrency and **measured cost per correct result** should also be stated; caching erased one report's latency advantage; two sister papers report **opposite cost directions** on different workloads

The paper also compresses the checklist into a **minimal report card**: exact model version and run date; test difficulty or model score gap; a plain-LM readout baseline; a cheap generative baseline; intervals across repeated runs; held-out calibration; and, **when a cascade is used, a frozen threshold evaluated on a new workload or a new fallback model**.

How to read this: this is the direct answer to this report's question "how evaluation should be done", and the literature gives it itself. It also exposes one thing: **the problem with existing evaluation sets is not too few benchmarks but reporting discipline**. B1 at 5/26 and T3 at 1/27 mean that, in the published numbers, "the interface's credit" and "the model's credit" have essentially never been separated, and "label names override definitions", the most serious D1 defect, has essentially never been tested as a matter of routine.

7.1.1 A self-audit case showing how easily the cost protocol goes wrong

A paired evaluation of 11 Agent decision points (18 public sources, 7,283 base cases + 6,640 robustness variants, byte-identical inputs, paired tests, reproduction checks across hardware and across days) **audited its own pipeline** and reported how three analysis errors and one design confound distorted its original deployment conclusion: **the cost of a pre-filter was left out, so a claimed "23.9% saving" was actually 4.3%**; **gate accuracy was reported as end-to-end quality** (58% versus 98%); the threshold was fitted in-sample (target 5%, up to 17% on the held-out set) (2610.02267).

This passage is more persuasive than the checklist: **it is a record of the same authors overturning their own cost conclusion within the same paper**. §5.1.5 blind spot 1 says "nobody measures end-to-end cost per correct result", and the root cause is here: this number is extremely easy to get wrong, and the direction of the error always favors oneself.

7.2 Existing evaluation sets classified by "what is tested"

§6.1.4.1 already gives data availability and scale; this section only adds "the object under test for each evaluation set", because that is the input to the coverage analysis.

Evaluation set	Scale	Main object under test	Source
JEVal	11,257 items / 36 datasets / 10 domains / bilingual	Cross-domain capability + behavior change once composed into a system; the conclusion is that decision models are most competitive when "the decision can be made from existing evidence" and weaken when expert knowledge or **faithful uncertainty estimates** are needed: "often points to the most likely outcome but substantially overestimates its probability"	2610.03935
System1-Arena	92k questions / 105 tasks / 268 sources + **138 closed-loop environments** (including a live mode)	Probability scores from all models on the same batch of decision requests; four groups: text classification and knowledge, applied decisions, evaluation and control, Agent and action	2610.21244

Evaluation set	Scale	Main object under test	Source
TypedBench	7 policy-annotated generators + 9 evaluation suites	**Four things tested jointly**: policy compliance, wording robustness (accuracy reported as median and range across paraphrases), probability quality (selective prediction, **ordinal proper scoring rules**, realized cost under asymmetric costs), cost; calibration error relative to the **finite-sample noise floor of a matched perfectly calibrated predictor**	2610.11392
DecisionPivot	48 tasks / 40,033 items; core track 7,800 items (13 generative task families, **contrast groups + independently implemented solver verification**); reference track 32,233 items (35 public test sets, original metrics retained)	Item accuracy, **contrast-group accuracy**, probability ranking, calibration, empirical selective coverage. Key result: two 27B models only 2.6 points apart on the reference track have contrast-group accuracy of **70.1% and 44.2%**; on 2,400 binary contrast pairs Perplexity Decider ranks the positive above the negative 92.3% of the time, yet **gets both sides right only 74.1%**	2610.28522
AgentDecisionBench / DecisionGym	From isolated questions to controlling a running Agent	Decision delegation in Agent scenarios. The model most accurate on isolated questions solves only **69.5%** of Agent tasks; delegating tool selection without fitting a confidence threshold lowers the success rate by **19.0 points**	2610.21412
Sys1Cal-v1	True / false propositions with P(A) known by construction	**The numerical meaning of probabilities** (not confidence ranking): scored by total variation distance to the true distribution, with the three primitives asked separately	2609.35342
Probability Contracts	1,000 finite worlds (+ 400 independent queues)	Exact finite-world posteriors, event alignment checks, **interface consistency**, failure-aware decision evaluation tested jointly. Jev's Event and Choice interfaces change the binary action on **32.8%** of valid pairs at a defer cost of 0.10	2609.37470
Probability axiom battery	160 items (ChaosNLI, PubMedQA) / 480 negation pairs	**Per-item logical self-consistency** (no labels needed)	2609.33209
JevAdvBench	862 questions + 5,586 answer-preserving variants, 12 black-box attacks	Adversarial robustness: attacks flip **46.3%** of Jev's correct decisions (an identical rerun flips only 0.17%); fallacious arguments without commands flip 77.2%, close to the 92.0% of direct prompt injection; eight open-source TDMs are flipped 34.0%–51.7% by the same variants; **of the decisions the service would still execute at a 0.8 threshold, 23.5% are wrong**	2609.31142
Option-channel attacks	Seven open-weight models as guards	Reports the two error directions separately (fail-open is a vulnerability, fail-closed is only a cost): allow/block decision accuracy 36%–72% (chance level 50%); six lines of policy-irrelevant server logs raise one gate's fail-open rate from 0% to 63%; giving the permissive option a misleading name (definition and judged text unchanged) raises it to **93%–100%** on the four models that put labels in the input; **every tested defense was broken**, and **escalating the least confident decisions does not help either: the decision flipped by the attack is not less confident than the one it replaced**	2610.12292
PFR (paired fallback robustness)	Three benchmarks, controlled comparison across architectures and parameter counts	Fallback: whether the model switches correctly between regular items and fallback items; whether it is consistent under **logically equivalent formulations** of the same rule	2610.19856
TUPLE-Bench	Paired synthetic cases, nine models	**Interface leakage**: instructions to "hide the reason" all fail, 91.88%–99.93% of responses leak the reason through direct selection or **residual scores**; after excluding the reason from the candidate set, the full output still lets held-out recovery reach 62.85%–75.35% (51.04%–57.64% using only the selected category, chance 50%); the full output lowers average downstream cost in **27/27** model-task comparisons, while **the confidence field increases leakage yet lowers cost inconsistently**	2610.23668
Paired Agent decision-point evaluation	11 decision points / 18 sources / 7,283 + 6,640	Paired, byte-identical inputs, reproduction across hardware and days + **self-audit**. Jev is significantly more accurate on 9/11 decision points (+10.8 to +46.0 pp); **both are worse than random on zero-shot model routing**, and tie on the RAG relevance gate; Laya changes 30% of its answers when option order is reversed, and degrades sharply when candidates are many or similar (31% with 50 nearest-neighbor tools, Jev 98% on items with a unique correct tool)	2610.02267
Choice2Seq	—	Tests **task correctness, plan faithfulness, and output compliance** separately	2610.16191

Evaluation set	Scale	Main object under test	Source
Online moderation suite	Five moderation benchmarks + controlled information conditions	Separates **written rules, retrieved precedents, and restrictions on the candidate answer space**. Jev generally benefits from retrieved precedents (with a fixed answer space it improves precise policy selection and harmful-content discrimination); Laya's retrieval often only shifts the positive rate without improving discrimination; Jev's confidence can support selective review in several settings but is **not consistently calibrated**	2610.07953
RLCDAlignBench	10 classes of alignment failure / 44 benchmarks / 5 target models	Zero-shot alignment failure detection (also exposes label defects in the benchmarks themselves)	2609.29429
Psychometric audit	Lithuanian organizational psychology questionnaire (n=263, 68 items), 100 real-person profiles, 14 decision models / 11 vendors + 47 generative models	**Construct validity of synthetic respondents** (PSS): under the default rounded readout, 13 of 14 decision models fall below the generative reference; 33 of 47 generative models reproduce the item correlation structure better than any decision model; but **probability vectors are closer to the human item marginals than rounded answers**; as predictions of individual responses, every decision model is worse than "the response frequencies of the other respondents"	2610.14912
Others (Decision Index, JevBench, kev-suites, CJ-Bench, DecisionVista, DecisionRobust, this-that spatial/complex, biosafety audit, cultural values audit, bias audit, etc.)	See § 6.1.4.1	Cross-domain capability, language coverage, specific biases	§ 6.1

7.3 Coverage matrix: twelve business properties × existing evaluation sets

Take the property list of § 5.1.2 as the objects under test and match them one by one.

Business property (§ 2.1)	Evaluation sets that test it specifically	Coverage verdict
P1 Output must be in the option set	No benchmark needed (structural). But "right type, wrong semantics" is only tested **incidentally** (T3's AUC .815→.581 with zero type errors)	**Structurally guaranteed, but "the false sense of safety that type safety gives" is systematically measured by nobody**
P2 Answered in one forward pass / extra questions nearly free	None. Latency and cost are scattered across papers, and checklist item C4 (with / without cache) has compliance **3/20**	**Zero systematic coverage**
P3 Calibration	TypedBench (relative to the noise floor), Sys1Cal-v1 (P(A) known by construction), DecisionPivot, JEVal, Decision Index, social-science annotation audit, online moderation suite	**The most thoroughly covered cell**, but the protocols differ from each other (§ 1.4.5: at least five algorithms for ECE)
P4 Consistency under equivalent formulations (= L2 semantic idempotence)	DecisionPivot (contrast groups), TypedBench (median and range across paraphrases), CALM bias suite, ordinal-scale bias (40 datasets), test-language audit, paired Agent decision points (order reversal)	**Dense coverage of interventions but the equivalence class is never defined**: each paper tests one intervention, and checklist item T3 (option-name invariance) was done by **only 1/27** papers; no evaluation set measures "business equivalence" as an equivalence class as a whole (§ 2.1.4)
P5 Confidence usable for routing	System1-Arena, Doom fast/slow switching (calibration / accuracy / sensitivity separated three ways), rubric review cascade, Indonesian regional-language cascade	**Covered but with negative conclusions**, and "the two uncertainties" have been tested in **only one domain, content moderation** (2610.33258)
P6 Fallback / abstention	PFR test (2610.19856), arithmetic refusal bottleneck (2609.39496)	**Two isolated papers**, and neither is a general benchmark
P7 "I don't know" can be expressed (P(U))	Sys1Cal-v1 (2609.35342)	**A single isolated paper**
P8 Knowing what it does not know	Controlled self-knowledge audit (2610.01006), the selective-prediction part of the biosafety audit (2609.30454)	**A single isolated paper + a single domain**
P9 Probabilistic self-consistency (axioms)	Probability axiom battery, 160 items / 480 pairs (2609.33209), Probability Contracts, 1,000 worlds (2609.37470)	**Two papers, and at a tiny scale**; "self-consistency after combining n judgments" is zero

Business property (§ 2.1)	Evaluation sets that test it specifically	Coverage verdict
P10 Configurable by business staff in text	The controlled information conditions of the online moderation suite come closest (rules / precedents / answer space separated three ways, 2610.07953); all upstream D1 evidence (label names override definitions, etc.) comes from prompts written by researchers	**Zero coverage**: no evaluation set uses definitions / criteria / examples **written by business staff** as an input variable
P11 Auditable	**Zero**. All existing work adds structure outside the model (policy-tree routing, Yes/No/Unknown node trees, evidence records)	**Zero coverage**
P12 Risk can be budgeted	Risk-budgeted decisions (2610.19811), the certification part of SHIFT-JEV (§ 6.4.2)	**A single isolated paper**

7.3.1 Three more things that are densely evaluated but not on the business property list

They show that the center of gravity of existing evaluation is not aligned with business goals.

1. **Adversarial and safety**: JevAdvBench, option-channel attacks, context poisoning (2610.15537), adversarial cues in how the request is presented (2610.11436), swapping citations with the same number (2610.08675). The evidence density in this cell is **higher than for fallback and self-knowledge**, while the latter two are the failures triggered more often in business.
2. **Privacy and interface leakage**: TUPLE-Bench quantified the fact that "the returned fields are themselves an information leak", concluding that **the confidence field increases leakage while lowering cost inconsistently** (2610.23668). This is a direct design constraint on "whether to expose probabilities downstream", and the property list of § 2.1 does not contain it: **which shows the business property list itself is also incomplete**.
3. **Cultural and population validity**: cultural values audit (2609.36399), bias case study (2610.19619), psychometric audit (2610.14912), population experiment kernel (2609.27535).

7.4 Numbers across evaluation sets are incomparable, and this is structural

§ 1.4 already gives the full list; here I only stress its three direct consequences for "evaluation".

1. **At least five algorithms under the same metric name**: ECE has at least five variants (10/15 equal-width bins, 15 equal-mass bins, relative to a null hypothesis, and so on); "flip rate" has five denominators; whether "abstention" is an option, a probability band, a score or a fallback action differs by paper (§ 1.4.5, § 1.4.6, § 1.4.8).
2. **Aggregate scores hide contrast-group performance**: two 27B models only 2.6 points apart on the reference track have contrast-group accuracy of 70.1% and 44.2% (2610.28522). This is the clearest single number for "why one cannot just look at the leaderboard".
3. **Good probability ranking ≠ both sides right**: on 2,400 binary contrast pairs, the positive is ranked above the negative 92.3% of the time, yet both sides are right only 74.1% (same source). What business executes is the **decision**, not the ranking.

7.5 Working backward from business goals: which capabilities no existing evaluation set can currently decide

Each item below is "**the existing evaluation sets cannot give a verdict**", not "what we should do".

#	What business needs decided	Why the existing evaluation sets cannot decide it
1	**End-to-end cost per correct result**	Checklist item C4 compliance 3/20; two sister papers report opposite cost directions on different workloads; one self-audit revised "save 23.9%" to "save 4.3%" (2609.32160, 2610.02267). No paper gives a computable protocol for "is saving 44× in cost while losing 11.6 F1 points actually worth it"
2	**Whether decision fine-tuning preserves / destroys ICL**	No paper has measured the k-shot curve **before and after** decision training (§ 3.1 blind spot, § 6.3.2)
3	**Whether rules written by business staff themselves can be executed faithfully**	The prompts in all D1 / D2 / D3 evidence were written by researchers; no evaluation set treats "the rule's author is a non-technical business expert" as a variable (§ 3.2, P10 in § 7.3)
4	**Chinese business-rule decisions**	Chinese decision data comes from a single source and targets general-purpose / medical-legal-finance; **there is no Chinese decision data aimed at business rules** (§ 6.1.4.1)
5	**How much scale utilization improves once ordinal items are enabled**	2609.38827 says it "can be mitigated by targeted training" but gives no cross-dataset training control; PACT's ordinal transfer has only a 324-item frozen held-out set and three seeds (§ 6.5 blind spot 2)

#	What business needs decided	Why the existing evaluation sets cannot decide it
6	**Which of the three numeric routes is better**	Interval decision trees, binned classification, and decoded regression heads **have never been compared on the same batch of numeric decisions** (§ 6.5 blind spot 3)
7	**What the combined calibration is after splitting one judgment into n judgments**	All evidence on probabilistic self-consistency is 160 items and 480 pairs (2609.33209); decomposition has only been tested at a single level (§ 6.5.5.2)
8	**The consistency ceiling of the recorded business process itself**	Only one paper measured it (cross-trajectory leave-one-out puts the ceiling at 25.6%–44.9%, with repeated states inconsistent with each other more often than consistent), and in a single domain (2610.25985). **Public benchmarks structurally cannot measure this quantity**
9	**Threshold transfer across workloads / across fallback models**	Checklist item C3 requires testing transfer, and it has been observed that "a threshold fitted on the selection set does not hold on held-out items for every fallback model, and the fitted temperature differs across workloads" (2609.32160); but no benchmark makes "transfer" its main metric
10	**What the total volume of judgments becomes once judgment is cheap / how decision authority gets rewritten**	Only one conceptual paper, self-described as "a technical provocation rather than an established empirical result" (2610.01231)
11	**Whether "the two uncertainties" hold outside the moderation domain**	Tested in only one domain, content moderation, and the authors explicitly say the remedies are specific and distribution-sensitive (2610.33258)
12	**Auditability**	Zero evaluation sets. The model by construction produces no textual rationale (2610.11392, 2610.22526)
13	**L2 semantic idempotence: do equivalent business situations give equivalent metrics**	Existing work only measures the flip rate under a single intervention (order, wording, tone, label name, one paper each; the table in § 2.1.4). **No paper defines the equivalence class "business equivalence" and measures the stability of the metric on that class**; and the harness can only guarantee "same metric → same action", not this layer (§ 2.1.4)
14	**Whether the harness-side "same metric → same action" holds in real systems**	Harness reliability has been measured by only one paper, across seven components (context, execution, governance, lifecycle, observability, tools, validation) (2610.32285), and not for Jev-class models

7.6 Chapter summary and blind spots

How evaluation should be done: the literature already gives a 14-item checklist and a minimal report card (version and date, model score gap, a plain-LM readout baseline, a cheap generative baseline, intervals across repeated runs, held-out calibration, a frozen cascade threshold tested for transfer). Existing papers comply with it at a low rate: B1 is 5/26, T3 is 1/27, C4 is 3/20. This means that in the published numbers "the interface's credit" and "the model's credit" have essentially never been separated.

What the current evaluation sets cover: accuracy, calibration, consistency / contrast groups, adversarial robustness, cross-domain capability, Agent scenarios; coverage is dense and there are several large-scale benchmarks (System1-Arena with 92k questions + 138 closed-loop environments, DecisionPivot with 40,033 items, JEVal with 11,257 items, TypedBench's four things tested jointly).

What is not covered: the five cells fallback, unknown, self-knowledge, probabilistic self-consistency and risk budgeting rest on one or two isolated papers; **the two cells "configurable by business staff" and "auditable" are zero**; and the business goal itself (end-to-end cost per correct result) has no reproducible protocol at all.

This chapter's own blind spots:

1. **The coverage verdicts of this chapter depend on the premise "objects under test = the twelve properties of § 2.1"**, and § 7.3.1 has already shown that this list is incomplete (TUPLE-Bench's interface leakage is not in it). The property list itself needs further extension, and the coverage matrix of § 5.2 will change accordingly.
2. **All compliance rates come from the coding of a single audit** (2609.32160), whose corpus is only 28 papers over a nine-day window, and whose authors state that several items required judgment and left blanks where they could not score reliably. The many papers added in the following month (System1-Arena, TypedBench, DecisionPivot, JEVal, etc.) **have not been recoded against the same checklist**, so "whether the compliance rate has improved" is unknown.
3. **This report ran no evaluation of its own**. All numbers are restated from papers and documents, and the protocol differences (§ 1.4) mean any cross-paper comparison can only be qualitative.

8 • Review and adversarial check: working hypotheses, the report's own constructs, methodological defects and attacks

This chapter adds no evidence; it only reviews and attacks. Its reason for existing: the production process of this report is "the questioner gives a proposition → I look for evidence → I write the conclusion", which **satisfies the standard conditions for confirmation bias**; and the framework in §2.2 is a joint product of the discussion, of which I am a co-author, so I am no longer an external reviewer. "Basically reaching agreement" is exactly the moment when adversarial analysis is most needed.

§8.1 rules on the four working hypotheses one by one (three are "conclusion right, mechanism wrong"), marks the eight frameworks in this report that **belong to me rather than to the literature** together with their overfitting risk, lists my own nine methodological defects (including one recorded instance of confirmation bias), and gives a robustness grading of the conclusions plus four blind spots in the questioner's field of view. §8.2 tests the §2.2 framework with seven attacks: **two overturn its wording, three force it to add qualifications, one is undecided and the most important, one is a permanent discount**; it then gives the revised framework and "what evidence would overturn it". §8.3 is the work-question matrix and the blind-spot summary sorted by impact; §8.4 is the list of open questions.

How to read: the P0′–P5′ in §8.2.3 are an **intermediate version**; the final framework wording for the whole report is the P0″–P5″ of §2.2.5, and the ruling on A4 is in the second postscript of §8.2.1.

8.1 The four working hypotheses, the report's own constructs and my methodological defects

Protocol: this section adds no evidence; it only reviews. Sections 14–§5.2 each answer a working hypothesis raised by the questioner, and **the person who raised the hypothesis and the person who tested it are the two ends of the same conversation, which is the standard condition for confirmation bias**. While reviewing §2.1 on 2026-10-28 I found an instance: the first draft treated the proposition "the migration of idempotence" as a foundation, while the evidence overturning its premise **had already been cited by this report itself**, scattered across 04, 05, 06, 09, 10 and §1.4; it simply was not retrieved when §2.1 was written. This section handles this kind of failure systematically: §8.1.1 rules on the four hypotheses one by one, §8.1.2 marks which frameworks in the report are my own constructs rather than literature conclusions, §8.1.3 lists my own methodological defects, §8.1.4 gives a grading of "which conclusions are most robust and which are most likely to be overturned", and §8.1.5 answers "what blind spots lie in the questioner's field of view".

8.1.1 Rulings on the four working hypotheses

#	Hypothesis (raised by the questioner)	Ruling	Direction of conclusion	Mechanism given
H1	SFT cannot give a model generalization ability in the general sense (§6.2)	**Strong form does not hold, weak form holds**	**Right**	**Incomplete**: the hypothesis attributes the cause to the "supervised objective", while the evidence shows that what moves results across levels is base-model scale, the instruction-tuned starting point, the supervision distribution and structural constraints
H2	Our business expectations of Jev cannot be directly turned into a differentiable learning function (§6.4)	**Conclusion holds, reasoning does not**	**Right**	**Wrong**: resistance to gradients has four distinct causes; the classes that are genuinely non-differentiable are exactly the classes where measured RL gain is zero, so "non-differentiable → use RL" does not hold
H3	RL should be considered to preserve ICL and skill-like semantic injection at the same time (§6.3)	**Does not hold**	**Wrong**	**Wrong**: what decides whether ICL survives is "whether the task definition changes between training samples"; the only two methods that improve "defining the task from context" (MetaICL, symbol tuning) are both SFT

#	Hypothesis (raised by the questioner)	Ruling	Direction of conclusion	Mechanism given
H4	Jev-class model = converting generative business judgment that does not guarantee idempotence into "indicator output + deterministic idempotent execution in the harness" (§ 2.1 § 2.1)	**Premise does not hold, conclusion needs major revision**	**Partly right**	**Partly wrong**: see the six pieces of counter-evidence in § 2.1.3

8.1.1.1 A pattern worth noting

Three of the four (H1, H2, H4) have **the right direction of conclusion but a mechanism that is wrong or incomplete**. This is not a coincidence; it is the difference between two sources of knowledge:

- The questioner's judgments come from **deployment and engineering observation**: change the schema and the model stops working, business goals cannot be written as a loss, generative output is unstable. This kind of observation is very reliable about **phenomena**.
- Attributing mechanisms requires **controlled experiments**, and the literature happens to lack controlled experiments at exactly these points: nobody has measured k-shot curves before and after decision fine-tuning (§ 3.1), nobody has measured Jev's repeat jitter side by side with a temperature-0 generative baseline (§ 2.1.3 counter-evidence 1), nobody has compared the three numeric routes on the same batch of numeric decisions (§ 6.5 blind spot 3).

Corollary: on this topic, "phenomenon judgments" are far more trustworthy than "mechanism judgments". When there is disagreement, one should keep the questioner's phenomenon and redo the mechanism, rather than accept or reject the whole. H4 was handled exactly this way: its **diagnostic value is high** (it correctly predicted why the fifth cell "closed-loop executor" in § 5.1.4 generally fails: the decision was moved back into the model), while its **descriptive claim about idempotence is wrong**.

8.1.1.2 The diagnostic value of H4 should be recorded explicitly

In § 1.1.6 I wrote "Jev's value lies in the interface contract, not in the weights". The questioner's version is sharper than mine because it states **what the contract is for**: the point of the contract is not "the output parses well" but "handing the **bookkeeping** of the decision to code". (I first wrote this as "moving decision authority from the model to the code"; the questioner pointed out on 2026-10-28 that this is wrong: **what we build is an AI system, and the core logic is the generalization of semantic understanding**. This objection uses exactly this report's own reductio, since the unconstrained optimum of "move it to code" is likewise a lookup table, so it holds, and the wording has been corrected in § 2.1.7.) This version can make a prediction that my version cannot: which kind of deployment will succeed (decision authority in the code) and which will fail (decision authority back in the model). The five-cell table in § 5.1.4 agrees with it completely, and that table was written before I heard this proposition. **So the correct attitude here is not "the proposition is wrong" but "the diagnostic part of the proposition should be adopted and the descriptive part replaced".**

8.1.2 Which frameworks in the report are my own constructs

Each item below is used as an organizing tool in the main text, but **no paper has ever proposed any of them**. This is where the report is most easily misread as "literature conclusions", so each one is annotated with its basis and its overfitting risk.

Construct	Where	Basis	Overfitting risk
Seven generalization levels L0–L6	§ 6.2.1	Results of about 20 papers sorted by "what changed" and then merged	**Medium-high**. The levels are not orthogonal (L2 new schema and L3 new symbol operations are mixed together in real data), and L5 calibration transfer and L6 reasoning depth each have only two or three pieces of evidence. A different layering might yield a different answer to "which level has positive evidence"
Five classes of differentiability A–E	§ 6.4.2	Classified by "why it resists gradients" rather than by business domain	**Medium**. The boundary between class C (quantile/rank) and class D (downstream long-range) depends on whether you count the certification means as part of the objective or part of the process

Construct	Where	Basis	Overfitting risk
Three layers of idempotence L1–L3	§ 2.1.4	Built by me while testing H4, to make the proposition falsifiable	**High**. The three-layer division itself has no literature support; § 2.1.3 has already shown that L3 is not a clean layer (it holds only under a fixed plan and fixed load), so this table is missing at least one "load/plan" dimension
Five System1+System2 positions A–E	§ 5.1.4	About 15 systems classified by "who consumes S1's output"	**Medium**. It fits the "guard/routing/classification" family well and is strained on the "memory/monitoring/review" family: SOMA is a control loop of cumulative judgments, Mnemon's S1 output is never executed at all, and this family has exactly the best measured results (LoCoMo 91.7%, highest among 14 systems)
The "decision repetition" axis	§ 5.1.1.3	The common structure extracted from the motivation sentences of three papers (JEVDB, Bellman LM, NIOCIS)	**High**. No paper has quantified "repetition", and no paper has regressed it against gain. It is a **hypothesis** that I am using as a criterion
"Question graph vs token stream"	§ 6.5.5.1	Derived from the cost structure (asking more is nearly free, generating more is metered), then filled in with five existing forms	**Medium-high**. The compositional calibration of "splitting one judgment into n judgments" **has not been answered by any paper** (§ 6.5 blind spot 5), so the feasibility of this form is inferred, not measured
Twelve business properties P1–P12	§ 5.1.2	Enumerated from the angle of "what the business side would order on"	**Known to be incomplete**. § 7.3.1 itself proves this: TUPLE-Bench's "the interface fields themselves are an information leak" is not on the list
The 30 sub-dimensions of D4	§ 4.2	Enumerative exploration; the user explicitly asked to "define and understand the question ourselves"	**Low** (declared as exploration from the start), but **the sub-dimensions are highly correlated** and cannot be treated as 30 independent measurement axes

How to read: for every framework in the table above, sentences in the main text of the form "under this classification, cell X is empty" **should be read as "under this report's classification"**, not as "the literature has proved that X does not exist".

8.1.3 My own methodological defects

1. **Confirmation bias, with one instance already.** The process is "the questioner gives a hypothesis → I look for evidence → I write the conclusion". The first draft of § 2.1 is the failure case: of the papers behind five of the six pieces of counter-evidence (constraint tax, IIA pre-registration, plans changing answers, control-loop SoK, Koa-decision), **four had already been cited in other chapters of this report**; only 2610.06425 was newly added during review. In other words the failure was not in retrieval but in **not doing a reverse search of existing evidence when writing a new chapter**.
2. **I read only about 70 papers in full; the remaining 212 were read from abstracts.** Numbers in abstracts are often not the paper's main conclusion, and qualifying conditions are often not written into the abstract. In many places this report paraphrases abstract sentences.
3. **One-directional retrieval.** The keywords revolved around Jev / typed decision / System One, which will systematically miss systems that **do not use this terminology but expose the same kind of interface**; the early audit itself admitted this limitation.
4. **No evaluation was run.** All numbers are paraphrased from papers and documents, and § 1.4 has already shown that protocols across papers are incomparable (at least five algorithms for ECE, five denominators for flip rate), so any cross-paper comparison can only be qualitative.
5. **All compliance rates come from the coding of a single audit** (2609.32160, a corpus of 28 papers, a nine-day window); the large number of papers added in the following month has not been re-coded under the same checklist.
6. **The time window is extremely narrow.** The great majority of the evidence is preprints from the six weeks after the Jev release, not peer-reviewed, and several describe themselves as "early evidence".
7. **My "zero hits" statements are the most fragile class of claims.** "No paper has done X" can only be falsified, never confirmed, and it appears dozens of times in this report. Each one should be read as "not found within this report's retrieval scope".
8. **I organized the whole report along the single axis of "business cost" and did not re-walk it along "semantic generalization".** The questioner pointed this out on 2026-10-28, and it caused one identifiable omission: the engineering operation that makes Jev cheap and the operation that destroys rule sensitivity **are the same**

operation (Certo's cacheable independent encoding is about 5× cheaper, and rule-sensitivity recall@1 goes 1.00 → 0.24, 2609.37832), while this report recorded these two facts separately in § 5.1.1 (cost structure) and § 6.2.3 (generalization levels), **never placed them side by side, and never wrote them up as a trade-off**. When organized along the cost axis, "63× cheaper" is recorded as an advantage, while on the semantic axis the same evidence says "the model is irreplaceable" (2609.29429); the same number means something completely different on the two axes.

9. **Three times in a row I ruled the questioner's proposition to "have a degenerate optimum" without pointing out one step earlier that this was the same pattern.** Idempotence → lookup table (§ 2.1.6), move to code → lookup table (§ 2.1.7), consistency rate → constant classifier (§ 2.1.8). All three propositions named the "invariance" half and left the "discrimination" half empty. Writing the three reductios separately let the same structural gap be rediscovered three times, which is itself an organizational failure.

8.1.4 Robustness grading of the conclusions

Robustness	Conclusion	Why
Most robust	The orders of magnitude of cost and latency (1–2 orders of magnitude cheaper, review 16–357× more expensive, local 2B at 30.9 ms versus frontier API at 8,758 ms)	Multiple independent measurements, consistent direction, consistent magnitude, and matching the structural cause "no decoding"
Most robust	Typed readout itself has no accuracy advantage independent of same-class label-probability readout	Two independent paths: the 28-paper coding of the early audit + an off-the-shelf model reading logits is enough (2610.02076)
Robust	Structural constraints (masking / position reset / block isolation) beat writing consistency into the loss	Multiple papers, consistent direction, and a strong form that "holds by construction"
Robust	Ordinal scale compression	40 datasets × 9 models × 3 families, with independent cross-paradigm confirmation (central tendency effect on clinical ordinal scales)
Robust	Fallback/abstention is the worst cell	Two mutually independent measurements (refusal rate 7%; a fallback option drops Intern from 97.9% to 23.1%)
Medium	"RL on single-pass models is negative or ineffective"	§ 6.1 and § 6.3 both correctly restrict this to "single-pass / pure readout" and note that "all positive evidence is tied to reasoning chains", so **the wording is fine**; the problem is **the evidence base is too small**. Of the three negative data points, one is an encoder (which by the user's own definition is not our direction anyway), one is small-scale REINFORCE, and one is GRPO inside an Agent loop rather than RL on a readout. Supporting a general conclusion about architecture with three heterogeneous points is thin
Medium	Five System1+System2 positions	The classification is my own, and it is strained on the "memory/monitoring" family (§ 8.1.2)
Most likely to be overturned	"Nobody guarantees L2 semantic idempotence"	This is a zero-hit statement, and it is the most likely retrieval blind spot: relevant work may be published in the robustness/invariance literature without describing itself as related to decision models
Most likely to be overturned	"Nobody has measured end-to-end cost per correct result"	Same as above. Industrial practice very likely has this protocol but does not publish papers; this report's sources skew academic (the early audit also admitted "industrial experience is underestimated")
Most likely to be overturned	"Decision repetition" is the main variable of gain	Never quantified or regressed (§ 8.1.2)

8.1.5 Four blind spots in the questioner's field of view

This is the part this section was asked to answer. For each: the blind spot, why it is a blind spot, and the supporting measurements.

Blind spot 1: treating "idempotence" as the goal, when what the business actually needs is "boundable risk", and the two conflict under asymmetric costs. "Deterministic, idempotent execution" requires collapsing the

distribution into one action. But after averaging over respondents, **the probability vector is closer to the human per-item marginal distribution than the rounded answer** (2610.14912); confidence conflates human disagreement with decision risk, and under asymmetric loss the two demand **opposite escalation signals**, so **no single confidence number can serve both** (2610.33258). The genuinely operational goal is something else: treating the output as an "observation record with scores" and letting unknown refuse directly yields a **calibration-free** upper bound on verdict flips, on which one can choose among multiple sufficient contracts (2610.10978). **Boundable > idempotent**, and the former can hold when the latter does not.

Blind spot 2: the proposition treats the harness as a trustworthy deterministic layer, while the harness itself changes answers. In semantic query engines "**the choice of query plan affects not only cost but also results**"; accuracy is plan-independent **only under pointwise deterministic policies**, and when the escalation band is calibrated on the plan's own candidates, **selection pushdown is not quality-reliable** (2610.08089). A systematic survey of 139 paper families is more direct: 50 families claim to fit a control loop, **only 4 support it with matching measurements**; of the 139, **only 4 report deadline attainment**; and "a decision that meets a 10-second budget for every isolated request fails it for every request once the decision is queued before a replayed execution time" (2610.06425). **Moving the hard part into the harness is not the same as solving it**; once moved, it becomes a queueing-theory and query-optimization problem, and almost nobody in the existing Jev literature measures either.

Blind spot 3: the proposition implicitly assumes "conservation of capability", so it underestimates the loss of multi-step / stateful business logic, and the harness receives no signal of this loss. This is the most dangerous item for deployment. Jev cannot make decisions that "an LLM can only make by writing out intermediate steps" (items solvable in one forward pass 97.5%, the rest **74%**); when the answer depends on two or more updates from the same user, accuracy drops, to 53%–80% at sixteen updates; given a stale state summary, it answers from the stale summary. And **it does not know its own limits**: confidence flags the errors it makes itself but cannot flag the errors it inherits from stale inputs, and **neither explicit nor trained handoff options catch the latter** (2610.24929). The "indicator + deterministic execution" architecture has **no usable signal** for this class of failure; what the harness sees is a high-confidence, type-valid, semantically wrong value. Combined with the "constraint tax" finding (a hard schema raises "valid but wrong" from 49.5% to 88.9%, 2605.26128), this is the only place in the whole architecture where **the failure mode is more hidden than in the alternative**.

Blind spot 4: the premise "driven by usable configuration (prompts)" has itself never been validated by any evaluation. The proposition starts from "generalized, flexible, configuration-based business processing ability of a generative model". But in all the evidence for D1, the prompts were written by **researchers**; no evaluation set treats "the author of the rules/criteria/examples is a non-technical business expert" as a variable (§ 7.5 item 3, § 5.1.5 blind spot 6). And the known failure modes land exactly where business staff are most likely to step: **label names override definitions** (AUC 93.8% → 23.2%), deleting the definitions leaves accuracy unchanged, small models cannot flip label semantics in context, homogeneous examples collapse small models to below 12% (§ 3.1). **"Configurable" is an assumption in this literature, not a measured property.** If it does not hold, then of the two halves the proposition splits out, the "model + prompt configuration" half has a weaker source of capability than the proposition assumes.

8.1.6 Summary of this section

- Of the four working hypotheses, three have **the right direction of conclusion and the wrong mechanism** (H1, H2, H4), and one does not hold (H3). When there is disagreement, keep the phenomenon and redo the mechanism.
- H4's **diagnostic value should be adopted** (it correctly predicted which kind of deployment fails), and its **descriptive claim should be replaced** (idempotence is not what it does; what it does is shrink a non-idempotent surface down to a few boundable fields).
- Eight frameworks in the report are **my own constructs**, of which "three layers of idempotence", "decision repetition" and "question graph" carry the highest overfitting risk.
- Among my own methodological defects, the one most worth guarding against is **confirmation bias**, and there is already one instance: most of the counter-evidence material for the first draft of § 2.1 was already in this report; a reverse search was simply not done when writing the new chapter.
- The dozens of "no paper has done X" statements should all be read as "not found within this report's retrieval scope"; of these, "nobody guarantees L2" and "nobody has measured end-to-end cost" are the most likely retrieval blind spots.

8.2 Seven adversarial attacks on the §2.2 framework

> **Reading note**: this section was written after the fourth round of the discussion, and its target is the six statements P0–P5 agreed at that time (their revised version P0′–P5′ is in §8.2.3). After the sixth round of the discussion (the "multiple questions on the same state" example in §2.2.1), **A1 was resolved and A4 was upgraded to a backbone-specific version**; the final framework is the P0″–P5″ of §2.2.5. This section keeps the original text and adds postscripts at A1 and A4, to preserve the trail of reasoning.
>
> Protocol: this section adds no evidence; it only restructures the logic. The four rounds of discussion on 2026-10-28 switched the organizing axis of this report from "**business cost**" to "**the generalization of semantic understanding**", and switched the main metric from accuracy to the consistency rate on equivalence classes. §2.2 first writes the agreed framework as six falsifiable statements; **the task of the remaining sections is to attack it**.
>
> Why the attacks take more space than the statements: §8.1.3 already recorded that this report's production process (the questioner gives a proposition → I look for evidence → I write the conclusion) satisfies the standard conditions for confirmation bias, and **the conditions now are worse than they were then**: this framework is a joint product of the discussion, I am its co-author, and I am no longer an external reviewer. "I feel we have basically reached agreement on the approach" is exactly the moment when adversarial analysis is most needed.
>
> Conclusions first: of the seven attacks, **two overturn the framework's wording** (A1: the equivalence class is itself a non-idempotent judgment; A3: "the model handles semantics, the code handles bookkeeping" has no boundary), **three force the framework to add qualifying conditions** (A2: pre-registration; A5: it is a system metric, not a model metric; A6: do not present ordinary multi-objective optimization as a discovery), **one is undecided but the most important** (A4: the semantic axis and the cost axis may be directly opposed at the architecture level), and **one is a permanent discount** (A7: each of the four load-bearing points rests on a single paper). The revised framework is in §8.2.3.

8.2.1 The seven adversarial attacks

A1 The equivalence class is itself a judgment, and exactly the kind of judgment we want to get rid of: overturns the wording of P1

P1 hands the "denominator" problem to "business-declared equivalence classes". But **who declares them?** When a business expert says "these two cases are equivalent", that statement is itself a generalized, semantics-based judgment that does not guarantee idempotence, isomorphic to the kind of judgment we want to replace. So P1 does not solve the problem; it only **moves it up one layer**, and the new location has fewer measurements than the old one.

Evidence supporting this attack: - Artifacts written by the business side are **systematically defective**: of 138,133 public SKILL.md files, **91.8%** have at least one defect (CI 88.8–94.6%), mainly weak routing metadata and bloated, non-executable bodies (2608.08453). There is no reason to think "equivalence-class declarations" would be written any better. - Rule artifacts **must be compiled for a specific model**: a SemTree tree compiled for judge A drops from 67.70% to **47.87%** when moved to judge B (2610.13999). Must equivalence-class declarations then also be recompiled per model? If so, they are not a business artifact but a model artifact, and the words "business-declared" in P1 do not hold. - Worse, there is a **cycle**: if the equivalence classes cannot be enumerated (which is exactly why generalization is needed), then deciding "are these two situations of the same kind" requires a model, so **the consistency of the metric depends on the consistency of the metric's own model**. - And the equivalence class is precisely **where business risk hides**: "are two loan applications that differ only in zip code equivalent?" is a fairness ruling, not a measurement. Writing it into the denominator hides the most contested judgment inside the metric definition.

Ruling: the attack holds, P1 must be rewritten. The defensible version is not "business-declared equivalence classes" but "**versioned, held-out, audited equivalence-class declarations**": that is, treat the equivalence class as **an artifact that needs governance**, a newly added non-idempotent source in the system, which must be listed as a risk item rather than taken as the reliable base of the metric. This has been written into P5′ in §8.2.3.

Postscript (after the sixth round): A1 has been resolved, and not by patching but by dissolution. The equivalence class does not need to be declared separately: it is **induced by the question**. "I eat rice" and "I eat milk" are of the same kind under "did you eat a meal" and of different kinds under "did you drink a liquid", and questions are already first-class objects of the interface (§2.2.1). But omission 2 in §2.3 points out that this holds only for **human-written, offline** questions; questions generated at runtime by S2 inside an Agent loop bring A1 back unchanged.

A2 The consistency rate is unfalsifiable when the equivalence classes are chosen after the fact: P1 gains a pre-registration condition

Any consistency rate can be engineered by picking the equivalence relation. Stack A1 on top of this: if we declare the equivalence classes and we also report the consistency rate on those classes on the same batch of data, that is **in-sample fitting**, and this report has already cited its price: one paper fitted its threshold in-sample, targeting 5% while the held-out set went as high as **17%** (2610.02267); the same paper corrected "saves 23.9%" to **4.3%** and reported gate accuracy as end-to-end quality, 58% versus 98%.

The literature already has templates that do this right: the IIA study **pre-registered** its hypotheses and then ran 299,463 calls (2610.16655); the union-bound study is **pre-registered** with 36,000 calls (2610.10978).

Ruling: the attack holds and a remedy is available. The usable form of P1 must be written as "decision consistency rate on **pre-registered, held-out** equivalence classes". A consistency rate that is not pre-registered is not evidence.

A3 "The model handles semantics, the code handles bookkeeping" has no boundary: overturns the wording of P0

This dichotomy sounds clean, but the report contains three pieces of evidence that **bookkeeping carries semantics**:

What is treated as "bookkeeping"	How it carries semantics
Option table (configuration)	Every option added shifts the log-odds between the gold label and the strongest wrong label; the systematic shift is **1.2–3×** the model's own rerun fluctuation (2610.16655)
Query plan (execution)	"The choice of query plan affects not only cost but also results"; accuracy is plan-independent **only under pointwise deterministic policies** (2610.08089)
Decision-tree structure (compiled artifact)	A tree compiled for judge A drops **19.8 points** when moved to judge B (2610.13999)

Ruling: the attack holds, but it points to a precise repair. The common factor: **whenever the parameters of the code are fitted on the model's own outputs or candidates, that code carries semantics and the dichotomy breaks**. This is exactly the boundary measured by 2610.08089: "when the escalation band is calibrated on the plan's own candidates, selection pushdown **is not quality-reliable**". So the defensible version of P0 needs a condition: **the parameters of the code may only be fitted on held-out ground truth, never on the model's own candidates**; when this condition is met the dichotomy holds, and when it is not, bookkeeping is part of the semantics and must be evaluated as semantics.

A4 The semantic axis and the cost axis may be directly opposed at the architecture level: undecided, and the most important item

This is the deepest attack on P0 + P5, because it hits not the wording but the **architecture**.

- If semantic sensitivity requires **joint encoding** (reading state + rules + candidates together), then the skeleton "shared state prefix + isolated question branches + list-style readout", copied by more than 15 open-source reproductions, **is structurally opposed to the core logic we just agreed on**. Measured: the joint scorer has rule-sensitivity recall@1 **1.00**; switching to cacheable independent encoding (about 5× cheaper at 77 candidates) drops it to **0.24** (2609.37832).
- The counter-evidence is just as strong: under the standard architecture, **one generic question** zero-shot across 44 benchmarks reaches a median AUROC of 0.886 and beats most supervised baselines (2609.29429). Semantic generalization has at least one strong example alive on the cheap architecture.

One paper on each side, with different backbones, tasks and readout methods; **this report cannot rule on it**. And the consequences are heavy: if A4 holds, then "continued training of Jev-style readout on a small Qwen" is optimizing the **wrong architecture** for the goal we just agreed on.

Ruling: undecided. This is the evidence gap this report most needs filled (see § 8.2.2), and it is a row that should be added to the "most likely to be overturned" grade in § 8.1.4.

Postscript (after the sixth round): A4 now has a sharper, backbone-specific version. "Multiple questions on the same state" requires the state to be encoded once and each question to project separately; this is exactly the shared-prefix + isolated-branch architecture. But the shared prefix is computed **while the question is unknown**: under an attention-only backbone the KV cache keeps token-level detail and the question branch can "re-read" the state; while

in the 64 layers of Qwen3.5/3.8-27B, **48 are recurrent layers**, the shared prefix reuses a recurrent state snapshot (§ 1.3.2), and what the question branch gets is a **question-independent fixed-length summary**. Prediction: questions that need non-salient aspects of the state will fail, and fail silently. This is row 11 of the defect registry (§ 2.2.4), and the control (hybrid vs attention-only, same state, N questions, binned by aspect salience) can be run now.

Second postscript (after the eighth round): A4 is ruled by a design choice and is no longer undecided. The order fixed by the questioner is task definition → input → candidate branches, with the encoding of the input conditioned on the task definition (the expensive, rule-sensitive side in Certo; the direction of Bongard's third stage, 2609.39111), and cross-sample caching along the axis "the same set of rules" rather than "the same input, multiple questions". So the attack "the cheap operation and the operation that destroys rule sensitivity are the same one" still holds; we simply chose the side that pays: what is lost is Jev's "additional questions nearly free", what is kept is the one to two orders of magnitude from "no decoding" (§ 2.2.5 P5″). The residual is backbone-specific: on the hybrid backbone the channel through which candidate branches see the input is 16 attention layers verbatim + 48 recurrent layers of compression; defect 11 is rewritten as "candidate branches see the input through recurrent layers", and the control experiment (local Qwen3-0.6B attention-only versus Qwen3.5-0.8B hybrid) remains the first structural experiment to run.

A5 "Decision consistency rate" is a system metric, not a model metric: P1 and P2 gain qualifications

Two independent reasons:

1. **Conflict with asymmetric costs.** After averaging over respondents, **the probability vector is closer to the human per-item marginal distribution than the rounded answer** (2610.14912); confidence conflates human disagreement with decision risk, and under asymmetric loss the two demand **opposite escalation signals**, so **no single confidence number can serve both** (2610.33258). If the main metric is the consistency rate of "decisions", optimizing it will collapse the distribution into one action, and that throws away exactly the information that decisions under asymmetric costs need.
2. **A "decision" exists only after the threshold.** So decision consistency is a property of (model + threshold + contract), and **no model-level version exists**. And the threshold is fitted after training, so "using decision consistency as a training objective" is circular in time.

Ruling: the attack holds, the framework must be layered. Usable form: the **system-level** main metric is the decision consistency rate on pre-registered equivalence classes; the trainable proxy at the **model level** can only be **the ordinal stability of the scores** (which happens to match "the readout is ordinal in the first place, not probabilistic": ordinal-layer error 4.2 points versus probability-layer 21.0 points, § 6.5.2.2). Mixing the two layers mistakes a system metric for a model objective.

A6 The pair "invariance + discrimination" is nothing new in itself: prevents packaging ordinary multi-objective optimization as a discovery

Expanded, P4 is "constrained multi-objective optimization", which is textbook content. If the framework's contribution is stated as P4, it has no contribution.

Ruling: the attack partly holds, and the credit must be limited. The new part is only three items, and all are **concrete fixed choices** rather than general principles: (a) which invariance to use: not byte-level rerun (already at ICC 0.997 / flips 0.17%, no gain to be had), not order permutation (purchasable by construction, Prism-1 zero flips), but **pre-registered business equivalence classes**, which nobody has defined; (b) where idempotence should live: in the threshold and the deterministic contract, with a bound that **does not depend on calibration**; (c) where the trade-off should be priced: cost and semantics are the same knob (A4). Apart from these three, P4 should be treated as a restatement of common sense, not a discovery.

A7 Each of the four load-bearing points rests on a single paper: permanent discount

Load-bearing conclusion	Sole source
Calibration-free union bound (the foundation of P2, P3)	2610.10978
Cost and semantics are the same knob (P5, A4)	2609.37832
The metric is a menu-relative quantity (the core counterexample for A3, L2)	2610.16655
Semantic generalization is alive on the cheap architecture (the strongest support for P0)	2609.29429

All four are preprints from the six weeks after the Jev release, not peer-reviewed. **If any one fails to reproduce, the framework loses a pillar**; two of them (union bound, IIA) are pre-registered studies, the two most credible of the four, and the other two are not.

Ruling: this is not a defect that can be patched; it is a discount that must stay attached to the framework permanently.

8.2.2 The evidence missing to rule on these disagreements (gaps, not plans)

Following the protocol of § 8.4, the undecided parts above are written as "which measurement is missing", not "what should be done":

Gap	Which attack it rules on	Closest existing thing
Side-by-side measurement of "joint encoding vs cacheable independent encoding" on the same backbone, same data, same task, reporting rule sensitivity and unit price together	**A4** (most important)	Certo reports both ends only in its own setting (2609.37832); nobody has done a cross-setting control
Consistency rates reported side by side on the four denominators: byte-level rerun / order permutation / answer-preserving adversarial variants / **pre-registered business equivalence classes**	**A1, A2**	The first three denominators each have a paper (0.17%, 0.11%–8.30%, 46.3%); **the fourth has never been defined** (§ 7.5 row 13)
k-shot curves before and after decision fine-tuning, plus controls with the name-definition swap protocol, rendering fixes and symbol tuning	Whether P0 can be repaired (whether semantics can be read back)	Rendering fix **+0.1511** (2610.02586), symbol tuning improves flipped in-context label semantics (2305.08298), **neither has been tried on a decision model**; nobody has measured k-shot curves (§ 3.1 blind spot 1)

8.2.3 The framework that remains after the attacks

> This subsection is the **intermediate version** (P0′–P5′). The final version P0″–P5″ is in § 2.2.5; the differences are: equivalence classes are induced by questions (P0″), the main metric becomes the pair "within-question invariance / cross-question re-partition" (P1″), accuracy is a third leg rather than a supplement to the floor (P2″), and the cost trade-off is bounded as "losing 'additional questions nearly free', not losing one to two orders of magnitude" (P5″).

Folding in the five attacks that hold, the defensible version is:

> **P0′** The model maps open-world situations into typed observations with a scale; the code does arithmetic on top of the observations, **and the parameters of the code may only be fitted on held-out ground truth, never on the model's own candidates**. When this condition is not met, "bookkeeping" carries semantics and must be evaluated as semantics.
>
> **P1′** The system-level main metric is the decision consistency rate on **pre-registered, held-out** business equivalence classes. The trainable proxy at the model level can only be **the ordinal stability of the scores**; the two layers must not be mixed.
>
> **P2′** Accuracy is a **declared floor**, and it must be measured with the decision-level paired protocol (cumulative failure set, equivalence-class flip rate), not with aggregate accuracy: across 8 orderings accuracy changes by **<2 pp** while the cumulative failure set grows by **27–29%** (2610.35826).
>
> **P3′** Idempotence lives in the threshold and the deterministic contract; the three requirements on the model side are a monotonic thresholdable score, native unknown (currently the worst cell: correct refusal rate **7%** when the answer is absent, up to **79%** with a fitted threshold), and invariance within equivalence classes.
>
> **P4′** Every invariance is paired with a discrimination. This item is a restatement of common sense; the credit lies only in the fixed choice of "which invariance to use" (A6).
>
> **P5′** Unit price is the enabling condition. **The equivalence-class declaration is a newly added non-idempotent source in the system** and must be versioned, held out, audited and listed as a risk item. Any means of trading cost for semantics must be priced explicitly, and this trade-off is currently **undecided** (A4).

8.2.4 What evidence would overturn this framework

If observed	What is overturned
On pre-registered business equivalence classes, the consistency rate of typed single-pass readout is **not better than** same-backbone label log-probability readout	The entire motivation for P0 and P1: the Jev-class interface would then have no consistency advantage either, only cost
Joint encoding and cacheable independent encoding show **no significant difference** in rule sensitivity	A4 dissolves, and the "explicit pricing" in P5 is downgraded to a secondary clause; this is **good news** for the framework
The union bound is overturned in some cell by a third-party reproduction	The foundation of P2 and P3 (sole source, A7)
Rendering fixes + symbol tuning **cannot** repair "names override definitions"	The repairability of P0: semantic generalization may be unreachable on this interface, in which case "the core logic is semantic generalization" is only a wish
The k-shot curve flattens after decision fine-tuning	The premise that business staff can configure it (§ 5.1.5 blind spot 6, § 8.1.5 blind spot 4)

8.2.5 Summary of this section

1. **Reaching agreement is not passing a test.** This framework is a joint product of the discussion, I am a co-author, and the conditions for confirmation bias are worse than those recorded in § 8.1. That is why the attacks in this section take more space than the statements.
2. **Of the seven attacks, two of the framework's wordings are overturned** (the equivalence class is not a reliable base but a newly added non-idempotent source; "the model handles semantics, the code handles bookkeeping" holds only when "the code's parameters are not fitted on the model's candidates"), **three must gain qualifications** (pre-registration, separating the system level from the model level, credit only for the fixed choices), **one is undecided and the most important** (whether cost and semantics are opposed at the architecture level), **one is a permanent discount** (each of the four load-bearing points rests on one paper, two of which are not pre-registered).
3. **The framework is not weakened by this; it becomes falsifiable.** § 8.2.4 lists five observations that would overturn it, and the measurements for three of them **can be made now** (consistency rates side by side on the four denominators, the joint vs independent encoding control, k-shot curves before and after decision fine-tuning).
4. **The one thing most worth remembering**: all three propositions (idempotence, move to code, consistency rate) named only "invariance" and left "discrimination" empty, and their unconstrained optima were lookup table, lookup table and constant classifier respectively. This is a **pattern**, not three coincidences: when the next proposition is raised, first check whether its unconstrained optimum is a degenerate model.

8.3 Work-problem matrix and blind-spot summary

> This section compresses the conclusions of 01–08 into two tables: (1) for each dimension, "which class of problem is solved by which works, and to what degree"; (2) a cross-dimension list of blind spots, ordered by how directly "we would run into them when training our own Jev-class model". Details and numbers are in the respective chapters; this section only gives the index.

8.3.1 Overview of the Five Dimensions

Dimension	Problem restated (in the user's terms)	Solved	Partially solved	Unsolved
D1 ICL	Whether a single-pass model retains ICL after decision fine-tuning; how to exploit it; which prompt levers business staff can use	Option names and definitions are the strongest levers (03 § 3.1.2.2); in-state examples/precedents work (JevKT, review precedents)	The effect of example count and format has only indirect evidence; upstream ICL mechanisms (label space, format, symbolic circuits) have not been verified on decision models	**No direct measurement of whether decision fine-tuning preserves or destroys ICL**; the effect of prompts written by non-technical staff has not been measured
D2 Symbolic instructions	Whether the model faithfully executes explicit symbolic rules (thresholds, dates, negation, fallback)	"Read"-type factual probes are stable (Reading Right 0.96–1.00 with code composition)	Negation/hierarchy have label-free metrics; arithmetic/date boundaries are clear (the vendor recommends putting them in code)	**Refusal/fallback is the weakest link** (99%/7%, 97.9→23.1); no mitigation for equivalent-wording invariance

Dimension	Problem restated (in the user's terms)	Solved	Partially solved	Unsolved
D3 Rule language	Give non-technical experts a set of logic primitives to write business rules in text, which the model executes faithfully	The Skill paradigm works for generative Agents (+16.6); NL→IR compilers have multiple intermediate representations	SemTree (Yes/No/Unknown trees, compile-per-judge) and probability-aware deterministic adjudication (union bound) are prototypes on single-pass models	**No rule language targets single-pass models**; primitive executability is not graded; the Unknown primitive conflicts with "the model cannot say it does not know"; L1/L2 rules (dependency / new framework) reach only 35%/13% even on reasoning models
D4 Consistency	Define and measure all factors that change the probability of "the same judgment"	Order (structural mitigation), prior shift (EM/KL anchors), repeatability	26 of the 30 sub-dimensions have definitions and metrics	**Mitigation covers only 2 items**; version drift and teacher bias have no empirical evidence; no unified consistency benchmark; zero evidence in Chinese
D5 Agent/knowledge	Whether fast/slow cooperation can replace "slow thinking + execution feedback"	Memory selection/judgment (pre-registered non-inferiority); domain Agents whose action space is enumerated by code	Four classes of switching signal, but confidence ≠ "needs thinking"; S1 monitoring S2 has only SOMA	**Cheaper does not mean better** (three opposing papers); no standard for benefit accounting; no work on knowledge-structure maintenance; rule/LR baselines are often sufficient
08 Backbone	Constraints a hybrid backbone places on the Jev-style interface	Inference-time isolation (row-by-row / forking) is feasible	Selective adaptation, bidirectional reading of attention layers	**No literature on training-time branch isolation**; no controlled comparison of hybrid vs attention-only at equal parameter count

8.3.2 Work → Problem Matrix (Main Works)

Columns = dimensions; cell = the type of contribution the work makes to that dimension: **E** evidence (measured the problem), **M** mitigation (gives a usable method), **T** tool/benchmark, **N** opposing evidence.

Work	D1	D2	D3	D4	D5	08
TypeSafe documentation (primitives / patterns / jaggedness / cookbooks / agent-skill)	M	E	M	E	M	—
Archer Hume architecture probe	—	—	—	E	—	E
Kev	—	—	—	T(permute)	—	M(row-by-row)
qwen-rlcd	—	—	—	—	—	M(forking)
StartLux-Decision	—	—	—	M(calibration)	M(AutoThink)	M(bidirectional)
Releasing Inner Jev / LLM-as-Jev / LDM / FunDM / AnyJev	—	—	—	E(readout)	—	—
OneJev / Open-Jev / Tev1 / Nimble / Intern-Decision / Chinese Decision	—	—	—	E	—	E(latency)
PACT / Prism-1 / Runoff / Block-causal	—	—	—	M(order)	—	—
SanSi / LEV	—	—	—	—	M(depth)	—
2609.26758 option names / 2610.02586 definitions	E, M	—	E	E	—	—
JevKT / review precedents / MemCom / cheat-sheet / Jev-LDE	E	—	—	—	—	—
CALICO / Goldsmith	M	—	M	—	—	—
In-context fixation / semantic anchors / schema	E(upstream)	—	—	—	—	—
Reading Right	—	E, M	M	—	—	—
Negation axioms 2609.33209 / hierarchy 2609.33971 / conjunction 2610.19619	—	E	—	E	—	—
Refusal bottleneck / Lost in the Fallback / legal abstention / refusal transfer / retirement gate	—	E	E	E	—	—
JevOut / JevAdvBench / option channel / Same Attack / TrustJev / DICE	—	E	—	E	N	—
2610.30045 wording / tables 2610.19536	—	E	—	E	—	—
IIA 2610.16655 / selection bias 2610.35826 / request independence 2610.14631	—	—	—	E	—	—
Multiple-correct 2610.29217 / ordinal compression 2609.38827	—	—	E	E	—	—
Self-knowledge 2610.01006 / two kinds of uncertainty 2610.33258 / literal lock-in 2610.22526	—	—	—	E	—	—
Prevalence shift 2610.26325 / SHIFT-JEV	—	—	—	E, M	M	—
Teacher bias 2610.26797	—	—	—	E(theory)	—	—
SemTree	—	—	M	E(version)	—	—
Probability-aware deterministic adjudication 2610.10978 / semantic query optimization 2610.08089 / JEVDB	—	—	M	M(composition)	—	—
Jev-Guard / SafePyramid / LPG / gpt-oss-safeguard / PolicyGuard	—	—	E, M	—	M	—

Work	D1	D2	D3	D4	D5	08
Artic / FlashScript / TraceCompiler / Formal Skill / P2T / SIRNA / Cedar / ARc / LSP	—	—	M(IR)	—	—	—
SkillsBench / 138K SKILL.md / Skill coverage / SkillSentry / Auto-Policy	—	—	E	—	—	—
Code Owns the Simulation / What Can't Jev Decide	—	E	M(boundary)	E	N	—
When Faster Decisions… / Good Judgment Bad Advice / JevAgentHarness / Microtask / Measure Before You Train / JEVal	—	—	—	—	N, M	—
System1-Arena / Fast Models Slow Evidence / DecisionPivot / TypedBench	—	E	—	E	T	—
Selective Reasoning / SOMA / System Switch / VoI / Truncated Reasoning / TORSI	—	—	—	—	M	—
Selection vs extraction / Mnemon / Jev-Mem / SJC / compression for follow-up questions	—	—	—	—	E, M	—
REFLEX / Jev-LiteMAS / S1-MAS / BLADE / Jarvis	—	—	—	—	M	—
Jev-Mobile / JEV-Star / SearchJev / JevCoder / Decision4Rec / NIOCIS	—	—	M(action space)	—	M	—
Claim Laundering / Jev-DLC	—	—	—	E	E, M	—
Marconi / FastJEV / Hydragen / DeFT / Block-Attention / IPPD / Visual Jev / selective adaptation	—	—	—	—	—	M

8.3.3 Blind-Spot Summary (Ordered by Impact on Our Own Training Work)

Tier 1: Directly Decides Whether a Training Recipe Can Stand

1. **Whether decision fine-tuning preserves ICL (D1)**. All reproductions report only accuracy/ECE; nobody has measured "how much in-state examples change the decision" before and after fine-tuning. If RLCD / soft-label distillation wipes out ICL, the premises of both D1 and D3 fail.
2. **No literature on training-time branch isolation on hybrid backbones (08)**. The whole Qwen3.5 series is a DeltaNet hybrid; Kev trains row-by-row, qwen-rlcd defers the question. Either prove that gradients pass correctly through the copied recurrent state, or accept the row-by-row cost, or switch to an attention-only backbone: none of the three paths has public numbers.
3. **No empirical evidence on teacher-bias inheritance (D4-D5)**. Small model + Jev labels (AgentDecision-4B) or + large-LLM labels (Releasing Inner Jev) is the mainstream recipe; theory says the student inherits the teacher's IIA/ordinal/negation defects and more data does not remove them; nobody has measured it.
4. **Refusal/fallback/Unknown (D2-F6, D3 blind spot 3)**. A rule language needs a "don't know" primitive; single-pass models are weakest on exactly this item; the only mitigation is one instance of a ternary contract (retirement gate).

Tier 2: Decides the Shape of the Rule Language

1. **No rule language targets single-pass models**; primitive executability is not graded by "read / compose / needs intermediate steps" (D3).
2. **Contradictory evidence on question-form equivalence** (Noul normalization vs Choice, medical +, intent −8.7): every primitive must be tested individually on the target model (D4-B6).
3. **No mitigation for equivalent-wording invariance** (D2/D4-C3): a rule language must either template the wording or train for invariance; neither has been done.
4. **Decomposed probabilities are not self-consistent** (TV 0.219–0.349, reconstruction −22.9): node probabilities in a rule tree cannot simply be multiplied (D4-C1).
5. **Compile-per-judge** (SemTree 67.70→47.87): a compilation step is needed between the business rules and the tree the model actually executes, and it changes with the model version (D3-6, D4-A3).

Tier 3: Decides Deployment and Evaluation

1. **No unified consistency benchmark**; decision-level paired evaluation is not the norm (D4).
2. **No standard for benefit accounting of fast/slow cooperation** (D5-1); confidence is not a "needs thinking" signal (D5-2).
3. **Rule / logistic-regression baselines are often sufficient**, but no work compares them systematically (D5-4).

4. **S1 training data must include post-error states, and S1 trusts the record over the observation**: neither has entered any recipe (D5-5).
5. **Evidence for Chinese / enterprise scenarios is close to zero**: consistency, Agents and rule languages are all on English public data; in Chinese there is only CJ-Bench's accuracy and ECE (D4-E5, D5-7).
6. **Knowledge-structure maintenance** (ontologies, rule-base versioning) has no work (D5-6).
7. **No controlled comparison of decision quality for hybrid vs attention-only at equal parameter count** (08).

8.4 List of open questions (without solutions)

> Terms: "this stage keeps only analysis and survey work... understand the problem first". This section **gives no training plan**; it only rewrites the blind spots of §8.3 as answerable questions and, for each question, names the measurement method already available in the literature (or points out that the measurement itself is also missing). The purpose is to give the next stage's experiments a clear statement of "what was answered".

8.4.1 D1: Retention and Use of ICL

#	Question	Existing measurement	Measurement gap
Q1.1	Before and after decision fine-tuning (LoRA / full-parameter / RLCD / soft-label distillation), how much does the marginal effect of k in-state examples on the decision change?	Upstream: example-count vs accuracy curves; the label-slot test of In-context fixation (2605.08295); JevKT's zero-data AUC (2610.11135)	No standard definition of "example effect" on decision models; needs a paired design (same state ± examples)
Q1.2	Is the effect the same when examples are placed in the state (JSON array) versus in the question's `examples` field?	The two placements in the vendor API (state documentation)	No controlled comparison
Q1.3	When example labels conflict with option names, which wins? (The decision-model counterpart of In-context fixation's "label slot")	2605.08295, 2609.26758	None
Q1.4	How much worse is decision quality with option names / definitions / examples written by non-technical staff than with those written by experts?	The expert editing workflows of CALICO / Goldsmith (2609.14726, 2610.09489)	**No measurement of "author expertise → decision quality"**
Q1.5	Does the finding that definitions can be removed (0.8487→0.7971) hold for Chinese business rules?	The removal / contradiction / inversion design of 2610.02586	No Chinese data

8.4.2 D2: Symbolic Instructions

#	Question	Existing measurement	Measurement gap
Q2.1	Which symbolic operations are reliable within a single pass (comparison, containment, date ordering) and which are not (arithmetic, counting, multi-hop)?	Reading Right's factual probes + code composition (2610.18579); the "thinking-token grading" of What Can't Jev Decide (2610.24929); the 126 counting-failure variants	The two gradings have not been merged
Q2.2	The relation between fallback/refusal execution rate and rule wording	The PFR test (2610.19856); refusal-bottleneck pairing (2609.39496)	No mitigation available to measure
Q2.3	Do axiom violations for negation, exclusivity and conjunction increase or decrease after fine-tuning?	The label-free axiom battery (2609.33209)	No before/after fine-tuning comparison
Q2.4	Can the separation of policy authority vs evidence be reproduced on small models?	TrustJev's paired supervision (2610.34826)	Only one paper

8.4.3 D3: Rule Language

#	Question	Existing measurement	Measurement gap
Q3.1	Where is the boundary of the rule subset a single-pass model can execute faithfully (L0 single rule / with one exception / L1 dependency / L2 new framework)?	The SafePyramid three-level numbers (2606.29887) on reasoning models	None on single-pass models
Q3.2	What is the minimal sufficient set of primitives? Candidates: read / compare / contain / exception / Unknown / evidence requirement / typed variables	IR elements of the compiler-type works (05 §3.2.2); SemTree node types; probability-aware true/false/unknown	None
Q3.3	How does error propagate when rules are composed in code?	Union bound (2610.10978); fan-out-weighted expected error (2610.08089); hierarchy TV (2609.33971)	The three have not been verified on the same rule tree

#	Question	Existing measurement	Measurement gap
Q3.4	What does a "compile-per-judge" compiler look like for small models? How much does the rule base degrade after a model version change?	SemTree's judge-swap experiment (2610.13999)	None
Q3.5	Can the Skill part of a rule (how to do it) and the Policy part (what is allowed) be expressed separately and their interaction measured?	Auto-Policy's distinction (2608.25091); Jev-Guard's policy tree (2610.18326)	None
Q3.6	The "executability rate" of rules written by non-technical experts, and the "executability rate after one round of repair"	SemTree node repair (82.04→86.16); AFTER one round of refinement +3.7–6.7	No setting where the authors are non-experts

8.4.4 D4: Consistency

#	Question	Existing measurement	Measurement gap
Q4.1	A joint profile of the 30 sub-dimensions (06 § 4.2.2) on the same model and the same data	Each sub-dimension has its own metric	**No unified benchmark**; DecisionPivot / TypedBench / System1-Arena each cover a part
Q4.2	The correlation between the distilled student and its teacher on order, IIA, ordinal compression and negation violations (teacher-bias inheritance)	The individual metrics + Gaussian proxy theory (2610.26797)	No empirical evidence
Q4.3	Version drift: the decision-level flip rate of the same rule base between adjacent versions	The cumulative-failure-set method over 8 presentations in 2610.35826 can be borrowed	None
Q4.4	Consistency of the same judgment expressed as Noul / Choice / Score, broken down by task type	2609.33209 (sum of 1.14), 2610.18529, 2610.16655 (−8.7)	Needs systematization by task type
Q4.5	The E5 (language), E6 (style) and B3 (option name) effects in Chinese	English methods are portable	None in Chinese
Q4.6	Decision-level paired evaluation should become the reporting norm for reproduction work: report at least flip rate, cumulative failure set, axiom violations, readout type	2610.35826, 2609.27678, 2609.33209, 2610.00054	Needs a norm rather than a new method

8.4.5 D5: Agent and Knowledge

#	Question	Existing measurement	Measurement gap
Q5.1	"Resources per successful delivery" as the unified accounting (including pre-screening cost and held-out-set thresholds)	The three metrics of 2610.13092; the self-audit checklist of 2610.02267	Needs a norm
Q5.2	Which signal can predict "needs slow thinking" rather than merely "the answer is uncertain"?	System Switch's AUROC-benefit correlation (2610.09683); VoI (2610.21160); SOMA's behavioral judgment (2610.26793)	No theory for language Agents
Q5.3	Head-to-head: S1 decisions vs deterministic rules vs logistic regression as a microtask controller	The rule arm and placebo arm design of Good Judgment (2610.24649); Measure Before You Train (2610.32955)	No systematic comparison
Q5.4	After the S1 training set includes "post-error states" and "record-observation conflicts", how far does Claim Laundering's 53–100% drop?	Paired state construction (2610.25947); AgentDecision-4B's trajectory collection (2610.21412)	The two have not been combined
Q5.5	A task definition for S1 maintaining structured knowledge (ontology nodes, rule versions, fact validity periods)	Jarvis's versioned task records (2610.23804); Jev-DLC's bounded channel (2610.31521); the retirement gate's ternary contract (2610.15710)	The task itself is undefined
Q5.6	How the trade-off between compressing for the current question vs for follow-up questions behaves under S1 judgment	The follow-up-question protocol of 2610.27301	None on the S1 side

8.4.6 Backbone

#	Question	Existing measurement	Measurement gap
Q6.1	On a hybrid backbone, are the gradients through a forked cache equivalent to row-by-row training?	qwen-rlcd's inference equivalence test (4.8e-5)	None on the training side
Q6.2	Hybrid vs attention-only at equal parameter count: accuracy, calibration, order flip rate, latency	Single-point numbers from the individual reproductions (08 § 1.3.5)	No controlled comparison
Q6.3	Are recurrent layers more sensitive than attention layers to irrelevant details / interference from repeated updates?	The single-factor design of What Can't Jev Decide	No architecture comparison

#	Question	Existing measurement	Measurement gap
Q6.4	The implementation path for structural mitigation of order bias on hybrid backbones	Block-causal (masking), Runoff (two forward passes), Prism-1	None on hybrid backbones

8.4.7 Three Cross-Dimension Meta-Questions

1. **Can the three-way split "read / compose / needs intermediate steps" become the shared coordinate for primitive grading, consistency sub-dimensions and fast/slow switching?** Each of the three dimensions gives one boundary (Reading Right, Code Owns the Simulation, What Can't Jev Decide); they have not been unified.
2. **Into how many numbers should confidence be split?** Calibration, discrimination, sensitivity, human disagreement vs decision risk, self-knowledge: at least five meanings, while the interface returns a single probability.
3. **The rule language, the consistency benchmark and Agent benefit accounting all need "decision-level paired" evaluation**: this is a methodological gap shared by the three dimensions, not three independent problems.

9 • References

> Every entry carries a link. Every paper cited in the body is listed here; the count in § 9.1 equals the number of citations in the body (the checking script is in the commit history).

9.1 arXiv papers (283 in total)

ID	Date	Title
2102.09690	2021-02-19	Calibrate Before Use: Improving Few-Shot Performance of Language Models
2110.15943	2021-10-29	MetaICL: Learning to Learn In Context
2202.12837	2022-02-25	Rethinking the Role of Demonstrations: What Makes In-Context Learning Work?
2303.03846	2023-03-07	Larger language models do in-context learning differently
2305.08298	2023-05-15	Symbol tuning improves in-context learning in language models
2305.14160	2023-05-23	Label Words are Anchors: An Information Flow Perspective for Understanding In-Context Learning
2309.17249	2023-09-29	Batch Calibration: Rethinking Calibration for In-Context Learning and Prompt Engineering
2310.15213	2023-10-23	Function Vectors in Large Language Models
2310.15916	2023-10-24	In-Context Learning Creates Task Vectors
2311.04235	2023-11-06	Can LLMs Follow Simple Rules?
2402.05099	2024-02-07	Hydragen: High-Throughput LLM Inference with Shared Prefixes
2404.00242	2024-03-30	DeFT: Decoding with Flash Tree-attention for Efficient Tree-structured LLM Inference
2404.11018	2024-04-17	Many-Shot In-Context Learning
2405.00200	2024-04-30	In-Context Learning with Long-Context Models: An In-Depth Exploration
2405.19874	2024-05-30	Is In-Context Learning Sufficient for Instruction Following in LLMs?
2409.15355	2024-09-14	Block-Attention for Efficient Prefilling
2410.03608	2024-10-04	TICKing All the Boxes: Generated Checklists Improve LLM Evaluation and Generation
2410.08328	2024-10-10	Agents Thinking Fast and Slow: A Talker-Reasoner Architecture
2411.19379	2024-11-28	Marconi: Prefix Caching for the Era of Hybrid LLMs
2412.08972	2024-12-12	RuleArena: A Benchmark for Rule-Guided Reasoning with LLMs in Real-World Scenarios
2501.17161	2025-01-28	SFT Memorizes, RL Generalizes: A Comparative Study of Foundation Model Post-training
2501.19383	2025-01-31	Decoding-based Regression
2502.14628	2025-02-20	PEARL: Towards Permutation-Resilient LLMs
2502.18695	2025-02-25	Policy-as-Prompt: Rethinking Content Moderation in the Age of Large Language Models
2503.02623	2025-03-04	Rewarding Doubt: A Reinforcement Learning Approach to Calibrated Confidence Expression of Large Language Models

ID	Date	Title
2505.14313	2025-05-20	Teaching Small Language Models to Learn Logic through Meta-Learning
2505.19766	2025-05-26	PAM: Training Policy-Aligned Moderation Filters at Scale
2506.06266	2025-06-06	Cartridges: Lightweight and general-purpose long context representations via self-study
2506.12796	2025-06-15	Surprise Calibration for Better In-Context Learning
2507.02833	2025-07-03	Generalizing Verifiable Instruction Following
2507.17746	2025-07-23	Rubrics as Rewards: Reinforcement Learning Beyond Verifiable Domains
2508.04632	2025-08-06	IFDecorator: Wrapping Instruction Following Reinforcement Learning with Verifiable Rewards
2509.02563	2025-09-02	DynaGuard: A Dynamic Guardian Model With User-Defined Policies
2509.10414	2025-09-12	Is In-Context Learning Learning?
2509.20820	2025-09-25	Distilling Many-Shot In-Context Learning into a Cheat Sheet
2510.06084	2025-10-07	Spectrum Tuning: Post-Training for Distributional Coverage and In-Context Steerability
2510.16092	2025-10-17	Compressing Many-Shots in In-Context Learning
2510.22691	2025-10-26	SALSA: Single-pass Autoregressive LLM Structured Classification
2511.09008	2025-11-12	A Neurosymbolic Approach to Natural Language Formalization and Verification
2511.21038	2025-11-26	Semantic Anchors in In-Context Learning: Why Small LLMs Cannot Flip Their Labels
2512.04408	2025-12-04	Executable Governance for AI: Translating Policies into Rules Using LLMs
2512.12677	2025-12-14	Fine-Tuning Causal LLMs for Text Classification: Embedding-Based vs. Instruction-Based Approaches
2512.17325	2025-12-19	Task Schema and Binding: A Double Dissociation Study of In-Context Learning
2512.22258	2025-12-24	Logic Sketch Prompting (LSP): A Deterministic and Interpretable Prompting Method
2601.10156	2026-01-15	ToolSafe: Enhancing Tool Invocation Safety of LLM-based agents via Proactive Step-level Guardrail and Feedback
2601.13284	2026-01-19	Balancing Classification and Calibration Performance in Decision-Making LLMs via Calibration Aware Reinforcement Learning
2602.12662	2026-02-13	Think Fast and Slow: Step-Level Cognitive Depth Adaptation for LLM Agents
2602.12670	2026-02-13	SkillsBench: Benchmarking How Well Agent Skills Work Across Diverse Tasks
2602.15902	2026-02-13	Doc-to-LoRA: Learning to Instantly Internalize Contexts
2603.05829	2026-03-06	Test-Time Adaptation via Many-Shot Prompting: Benefits, Limits, and Pitfalls
2604.09075	2026-04-10	Hierarchical Alignment: Enforcing Hierarchical Instruction-Following in LLMs through Logical Consistency
2604.14723	2026-04-16	Bounded Autonomy for Enterprise AI: Typed Action Contracts and Consumer-Side Execution
2604.25203	2026-04-28	BARRED: Synthetic Training of Custom Policy Guardrails via Asymmetric Debate
2605.02028	2026-05-03	Language models fail at extended rule following
2605.08295	2026-05-08	In-Context Fixation: When Demonstrated Labels Override Semantics in Few-Shot Classification

ID	Date	Title
2605.16386	2026-05-11	Auditing Multimodal LLM Raters: Central Tendency Bias in Clinical Ordinal Scoring
2605.17329	2026-05-17	LPG: Balancing Efficiency and Policy Reasoning in Latent Policy Guardrails
2605.19604	2026-05-19	Formal Skill: Programmable Runtime Skills for Efficient and Accurate LLM Agents
2605.26128	2026-05-20	The Constraint Tax: Measuring Validity-Correctness Tradeoffs in Structured Outputs for Small Language Models
2605.29801	2026-05-28	AgentDoG 1.5: A Lightweight and Scalable Alignment Framework for AI Agent Safety and Security
2606.06087	2026-06-04	LatentSkill: From In-Context Textual Skills to In-Weight Latent Skills for LLM Agents
2606.07808	2026-06-05	Where Instruction Hierarchy Breaks: Diagnosing and Repairing Failures in Reasoning Language Models
2606.08932	2026-06-08	From Statute to Control Flow: Span-Grounded Deontic Trees for Defeasible Scope Parsing
2606.20659	2026-06-09	Skill Coverage: A Test Adequacy Metric for Agent Skills
2606.11543	2026-06-10	SkillJuror: Measuring How Agent Skill Organization Changes Runtime Behavior
2606.19183	2026-06-17	Language Models as Interfaces, Not Oracles: A Hybrid LLM-ML System for Pediatric Appendicitis
2606.20978	2026-06-18	How Should Agents Read Demonstrations? Hierarchical Structure Beats Flat Action Logs
2606.23127	2026-06-22	Managing Procedural Memory in LLM Agents: Control, Adaptation, and Evaluation
2606.26649	2026-06-25	Autoformalization of Agent Instructions into Policy-as-Code
2606.29887	2026-06-29	SafePyramid: A Hierarchical Benchmark for In-context Policy Guardrailing
2607.12149	2026-07-13	It is not enough to give your moderation rules to ChatGPT: Policy-as-Prompt Moderation and Its Potential Impacts on Community Governance
2607.21642	2026-07-21	CARE: Pre-Execution Command Verification for Shell-Executing LLM Agents
2608.00396	2026-08-01	Verifiable Checks for Business Rule Consistency
2608.02680	2026-08-03	TraceCompiler: Skill-Guided Mining and Compilation of LLM Agent Traces into Mostly Deterministic Workflows
2608.02687	2026-08-03	PolicyGuard: Prompt-Configurable Semantic DLP for LLM Coding Agents
2608.08254	2026-08-08	Your Prompt Is Not the Only Prompt: How Much Do LLMs Weight Structured-Output Schema Descriptions?
2608.08453	2026-08-09	What Keeps Agent Skills from Being Reusable? Evidence from 138K SKILL.md Files
2608.14693	2026-08-09	Domain Agnostic Text Redaction from Natural Language Rules using Instruction Tuning
2608.09253	2026-08-10	SkillSentry: Reliable Skill Execution for LLM Agents via Runtime Assurance
2608.21341	2026-08-21	Natural-Language Workflows Are Not Software Yet: Artifact-Driven Compilation for Reliable Agent Execution
2608.24777	2026-08-25	StepGuard: Learning Step-Level Guardrails with Scalable Supervision and Safety-Utility Balancing
2608.25091	2026-08-25	Auto-Policy, not Auto-Skill: Compiled Agent Skills for the Physical World
2610.00025	2026-09-01	Measuring the Microtask Eligibility Gap: When Is an Off-the-Shelf SLM Enough for an Agent Harness?
2609.05707	2026-09-04	Intra-Prompt Parallel Decoding for Common-Context Question Answering
2610.00054	2026-09-04	The First Token Is Not the Verdict: Hidden Costs of Reading LLM Judges Without Generating

ID	Date	Title
2609.11310	2026-09-10	Your Model Already Knows Don't Teach It, Learn to Ask It: Soft Prompting for Few-Shot Adaptation of Vision-Language Models
2609.14726	2026-09-13	CALICO: A Human-Centered, Codebook-Aligned System for Annotation
2609.23136	2026-09-19	Intent Interpretation at RIC Timescales: Jev Decision Models versus Large Language Models in 6G Open RAN
2609.23742	2026-09-20	Constrained Decoding Eliminates Structural Failures in Small LLMs but Reveals a Scale-Dependent Semantic Gap
2609.23886	2026-09-20	this-that-model-1.0: A typed decision model that decides in 30 ms, for a millionth of a cent
2609.23986	2026-09-21	Jev-Mem: System-One-Controlled Agentic Memory for Efficient AI Agents
2609.24052	2026-09-21	Calibrated Decisions at Scale: Converting Police Crash Narratives into Probabilistic Crash Variables with a System One Model (Jev)
2609.24574	2026-09-21	Evaluating Decision Models for Text Annotation in Computational Social Science
2609.25845	2026-09-22	Visual Jev: Accurate and Efficient Decisions from Shared Visual Context
2609.26532	2026-09-22	REFLEX with Jev for Efficient Selective Control in LLM Agents
2609.26550	2026-09-22	JEV-as-a-Judge: Accept When Confident, Escalate When Unsure
2609.26758	2026-09-22	Type-Safe Is Not Error-Free: Typed Decision Models Follow the Option Name, Not the Definition Bound to It
2609.27331	2026-09-23	JEV-Star: Fast, Low-Cost StarCraft II Control with Language-Model Planning
2609.27535	2026-09-23	KITE: Scaling Jev Population Experiments with Sparse Flagship Calibration
2609.27678	2026-09-23	Same Scores, Different Decisions: Evaluating JEV and Language Models for Legal Document Understanding
2609.28587	2026-09-23	NumericJev: Jev-like LLM Numerical Decoding with Multiway Decision Trees
2609.28613	2026-09-23	Decision Hijacking: Prompt Injection Attacks on Jev's Typed Probabilistic Decisions
2609.28919	2026-09-24	Harness Tokenomics: A Router for the Enterprise Agentic Control Plane
2609.29429	2026-09-24	Just Ask Jev: Reinforcement Learning for Calibrated Decisions as a Zero-Shot Detector of AI Alignment Failures
2609.29769	2026-09-24	Decision Models vs. LLMs as Rubric Judges: Cheaper, Faster, and Wrong in the Same Places
2609.30186	2026-09-24	Jev-Mobile: Jev as an Executor for Mobile GUI Agents
2609.30216	2026-09-24	Jev in the Wild: A Data-Driven Analysis of the Jev Model's Functionality, Applications and Ecosystem
2609.30243	2026-09-24	JevOut: Natural Context Can Flip Decision Models
2609.30454	2026-09-24	Auditing System-1 Models on Biosecurity-Relevant Benchmarks: Calibration, Selective Prediction, and Permutation Instability in a Non-Generative Model
2609.30706	2026-09-25	LAVOIR: Teaching a Single-Pass Decision Encoder When and What to Ask with Amortized Value of Information
2609.30922	2026-09-25	JevSoup: System-One Routing for Training-Free LoRA Composition
2609.31142	2026-09-25	JevAdvBench: A Benchmark and Black-Box Attacks for Reinforcement Learning for Calibrated Decisions Models
2609.32160	2026-09-26	Typed Decision Models: An Early Evidence Audit and Evaluation Checklist
2609.35865	2026-09-26	PACT: Pairwise-Anchored Calibrated Tuning for Single-Token Typed Decisions
2609.33209	2026-09-27	Beyond Calibration: Do a Typed-Decision Model's Probabilities Obey the Probability Axioms?

ID	Date	Title
2609.33401	2026-09-27	Evaluating System One Models for Agent Security Decisions: Reliability, Calibration, and Selective Automation
2609.33609	2026-09-27	You Only Edit Once: Incentivizing In-Context Capability of LLMs via Local Demonstration Refinement
2609.33671	2026-09-27	COGNIT-Guard: Calibrated Standalone Direct-Decision Guardrails with Heterogeneous CPU-NPU Confidence Cascading under Explicit Latency and False-Positive Constraints
2609.33843	2026-09-27	Laya as a Typed Probabilistic Assessor: An Independent Reproduction and a Preregistered Study of Calibration and Selective Escalation
2609.33971	2026-09-27	Do System One Decisions Add Up? A Study of Probabilistic Coherence
2609.37470	2026-09-27	Probability Contracts: Accuracy, Coherence, and Decisions Across LLM Interfaces
2610.13999	2026-09-27	SemTree: Judge-Aware Semantic Decision Trees for Inspectable and Repairable Rule Decisions
2609.34227	2026-09-28	When Does Selection Replace Extraction? A Pre-Registered Test of Agent Memory with a Typed Decision Model
2609.34862	2026-09-28	JEV as a Judge for Agent Trace Security: An Empirical Comparison with Generative LLM Judges
2609.35342	2026-09-28	Jev thinks "I don't know'', but doesn't say it: Introducing Sys1Cal-v1 Dataset for Probability Calibration
2609.36059	2026-09-28	Mnemon: Raw Records, Fast Judgments, Slow Thoughts
2609.36115	2026-09-28	Koa-decision: Fast and Consistent Structured Decision Making with Generative LLMs
2609.36116	2026-09-28	Dyad: Extending Large Language Models with Native Typed Decision-Making
2609.36265	2026-09-28	In-Context Learning Amplifies a Latent Symbolic Circuit
2609.36399	2026-09-28	Calibrated to Whom? Persona and Language Effects on Cultural Values in JEV
2609.37647	2026-09-29	Evaluating and Benchmarking the System One Model Jev
2609.37832	2026-09-29	Can a Cacheable Decision Model Follow Rules?
2610.00346	2026-09-29	Benchmarking System One decision models against trained classifiers and language models for automated decision gates
2609.38827	2026-09-30	More Choices, Fewer Decisions: Ordinal-Scale Bias in JEV-like Direct-Decision Models
2609.38850	2026-09-30	OpenJev-RLCD: A Working RLCD Implementation
2609.39111	2026-09-30	Bongard: Training Machine Intuition
2609.39496	2026-09-30	When the Right Answer Is Missing: An Arithmetic-Dependent Rejection Bottleneck in Jev
2610.00437	2026-09-30	JevSpawn: Adaptive Agentic Inference through Compositional Action Spaces
2610.00526	2026-09-30	Rules Amortize, Pairings Don't: Linguistic Structure Determines What Latent Task Representations Can Replace In-Context Learning
2610.00831	2026-09-30	AnyJev Technical Report
2610.01006	2026-10-01	Beyond Answer Confidence: A Controlled Audit of Self-Knowledge in a Black-Box Decision Model
2610.01054	2026-10-01	Capturing In-Context Learning Dynamics with Task Operators
2610.01231	2026-10-01	Judgement in the Age of Jev: From Evaluation Scarcity to Evaluation Abundance
2610.01601	2026-10-01	Permutation-Robust Decision Modeling with Candidate-Independent Block-Causal Attention
2610.01834	2026-10-01	Code Owns the Simulation, Jev Owns the Evaluation

ID	Date	Title
2610.02046	2026-10-01	Prune First, Decide Fast: Scalable Semantic Query Processing with JEVDB
2610.02076	2026-10-01	LLM-as-Jev: LLMs Are Already Jev-Style Decision Models -- When and How to Fine-Tune Them
2610.02267	2026-10-01	Fast Models, Slow Evidence: A Paired and Self-Audited Evaluation of System-1 Decision Models for LLM Agent Harnesses
2610.02486	2026-10-01	From Retrieval to Typed Decisions: Calibrated System One Models from Biomedical Sentence Encoders
2610.02516	2026-10-01	Student-Guided Teacher Distillation for Efficient LLM Task Routing: Positioning Against Jev-Style System-1 Classifiers
2610.02586	2026-10-01	Labels Override Definitions in Jev-Style Typed Decision Models
2610.03073	2026-10-02	SecJev: Bringing Security Expertise to System One Decision Models
2610.03324	2026-10-02	To Jev or Not? Evaluating the Accuracy and Efficiency of Structured Decision Models for Hate-Speech Moderation
2610.03387	2026-10-02	Benchmarking candidate coverage and rejection policy transfer in typed decision models
2610.03935	2026-10-02	General Decision Models: Benchmarking and Insights Beyond Jev
2610.05107	2026-10-04	SearchJev: A Calibrated System-1 Model for Search Agents
2610.05278	2026-10-04	When Verifiable Counts Depend on Wording: Auditing Wording Robustness in Instruction Following
2610.06354	2026-10-05	GraphDecisionBench: Benchmarking System One Models on Graph Tasks
2610.06425	2026-10-05	SoK: Semantic Decision Engines in Network Control Loops
2610.06744	2026-10-05	ufakzeka-karar: An Open Turkish Typed-Decision Model with Order-Invariant Option Scoring
2610.07327	2026-10-05	SharedKV-BT: Node-Local Typed Decisions for Behavior-Tree Agents
2610.07716	2026-10-06	JevCascade: Readout Stability and Inference-Time Compute Allocation for Prefill-Only Decision Models
2610.07730	2026-10-06	SanSi: A Looped Typed Decision Model for System 1.5 Thinking
2610.07953	2026-10-06	Benchmarking System One Models in Online Moderation
2610.08089	2026-10-06	When Plans Change Answers: Formalizing Cost-Accuracy Optimization for Semantic Queries
2610.08155	2026-10-06	Token-Efficient Multi-Agent Collaboration via System One-Guided Computational Division of Labor
2610.08675	2026-10-06	Same-Number Citation Swaps: Stress-Testing Jev as a Financial Evidence Judge
2610.09188	2026-10-06	From Probabilities to Decisions: Search and Multi-Teacher Distillation with Jev
2610.09489	2026-10-07	Goldsmith: Gold-Loss-Guided Definition Optimization with an Agentic Annotation Harness
2610.09683	2026-10-07	System Switch: When Should a Fast Decision Model Stop and Think?
2610.10615	2026-10-07	JevForest: Path Voting for Budgeted Feature Acquisition
2610.10978	2026-10-07	Probabilistic Sensing, Deterministic Authority: Admitting Model-Produced Observations into Sufficiency-Checked Governance Contracts
2610.11135	2026-10-08	Can a System-One LLM Perform Knowledge Tracing When Few or No Learners Are Logged?
2610.11316	2026-10-08	MetaEncoder: Exploring the Limit of Bi-Encoders for Multimodal System One Decision Making with Natural Language Interface
2610.11379	2026-10-08	FastJEV: Understanding Redundancy for Compact JEV Inference

ID	Date	Title
2610.11392	2026-10-08	TypedBench: A Benchmark for Calibration, Framing Sensitivity, and Cost in System One Decision Models
2610.11436	2026-10-08	Adversarial Cues in Decision Models Used as Judges: The Role of Request Presentation
2610.11692	2026-10-08	Can Jev be Your Q or Policy in Reinforcement Learning?
2610.11715	2026-10-08	Internalizer: Portable Context-to-Parameter Mapping for Very Large Language Models
2610.11978	2026-10-08	Specialized Decision Models vs. General-Purpose LLMs: Benchmarking Jev Across Knowledge, Reasoning, and Multilingual Tasks
2610.12292	2026-10-08	One Word Opens the Gate: The Option-Channel Attack on Typed Decision Models as Agent Guardrails
2610.13092	2026-10-09	When Faster Decisions Do Not Make Better Agents: An Empirical Study of Jev in Agent Workflows
2610.13113	2026-10-09	Continually Post-training Small Models via Synthetic Self-distillation
2610.14112	2026-10-09	Does System One Transfer? Calibrated Typed-Decision Models, Label Budgets, and Translation for Indonesian Regional Languages
2610.14404	2026-10-09	DICE: Tilting the Odds through Interface Attacks on System One Model
2610.14449	2026-10-10	Jevolution: A Framework for Studying Communication and Collective Dynamics in AI Agents
2610.14631	2026-10-10	The Company a Question Keeps: Auditing Open Jev-Style Models for Retrieval
2610.14912	2026-10-10	Probable but Not Valid: A Psychometric Audit of Decision Models as Synthetic Survey Respondents
2610.15501	2026-10-11	Large Decision Models: Repurposing Large Language Models as Single-Pass Decision Makers
2610.15537	2026-10-11	Poisoned into Action: Context Attacks on Typed Decision Workflows
2610.15710	2026-10-11	Suggested Is Not Superseded: A Pre-Registered Evaluation of a System-One Model as a Memory-Retirement Gate
2610.15742	2026-10-11	RSAJEV: Training-Free Abstaining Prototype Memory for Eliminating Silent Hallucination in JEV-Based Remote Sensing Classification
2610.16191	2026-10-12	Correct Choices, Failed Sequences: Choice2Seq Benchmarks Correctness Retention Across Output Interfaces
2610.16550	2026-10-12	CT0: Cybersecurity Transformer
2610.16655	2026-10-12	Does a Hosted Decision Model Respect Irrelevant Alternatives? A Pre-Registered Test of Option-Set Dependence in Jev
2610.16967	2026-10-12	Proactive Physical AI through Continuous Observation and Timely Intervention
2610.17096	2026-10-12	Towards Universal Calibration for In-Context Classification
2610.17175	2026-10-12	FunDM: Cheap & Label-Free Decision Models from Frozen Language Models
2610.17205	2026-10-12	Releasing the Inner Jev: A Simple Recipe for Open Decision Models
2610.17309	2026-10-12	StartLux-Decision: AutoThink Decision Models with Reinforcement Learning for Calibrated Deliberative Decisions
2610.17727	2026-10-13	SkillScreen: High-Recall Screening of Malicious Agent Skills with Decision Models
2610.17820	2026-10-13	Same Attack, Different Judge: A First Controlled Adversarial Comparison of Jev, LLM Judges, and Purpose-Built Safety Classifiers
2610.18122	2026-10-13	Cheap, Consistent, Calibrated, but Biased? Typed Decision Models as Judges
2610.18233	2026-10-13	Consider, Then Choose: Order-Invariant Decisions in System One Models
2610.18272	2026-10-13	JevGPT: Thinking Fast and Slow in Autonomous Research

ID	Date	Title
2610.18326	2026-10-13	Jev-Guard: Fast Routing and Auditing Agent Actions over Editable Policy Trees
2610.18488	2026-10-13	Recalibrating Many Questions with Few Labels: Minimax Rates and Identification Limits
2610.18529	2026-10-13	How Reliable Are System-One Decisions in Medicine? A Comparative Study of Jev, Jev-like Models, and LLMs
2610.18579	2026-10-13	Reading Right, Acting Wrong: Diagnosing Decision Failures of System One Models in Agent Loops
2610.18612	2026-10-13	Nudge the Judge: Uncovering Program Decision Hijacking in Real-World Jev Applications
2610.18766	2026-10-13	Rubrics from the Task, Verdicts from Evidence: Pointwise Rubric Rewards with Tool-Augmented Judges
2610.18780	2026-10-13	BLADE: Auditing Information Use for Failure Attribution in LLM Multi-Agent Systems
2610.19525	2026-10-14	Diagnosing and Correcting Reward Hacking in Verifiable Instruction Following
2610.19536	2026-10-14	Typed Decision Models on Tabular Data
2610.19596	2026-10-14	JevCoder: When Does Decision Delegation Pay Off in Code Repair?
2610.19619	2026-10-14	Are AI Decision Models Biased? A Case Study of Jev
2610.19644	2026-10-14	Selective Reasoning for Typed Decision Models: When to Start, Read, and Stop
2610.19811	2026-10-14	Calibration and Risk-Budgeted Decisions with System One Models
2610.19856	2026-10-14	Lost in the Fallback: How Reliably Do Typed Decision Models Use Fallback Options?
2610.20665	2026-10-14	The World Won't Wait: Evaluating and Teaching Agents Under a Live Clock
2610.20778	2026-10-14	Verdict: Encode Once, Decide Many
2610.20907	2026-10-14	From Detection to Decision: Calibration and Deferral for Prompt-Injection Detection
2610.20927	2026-10-14	Prune the Summary, Not the History: Source-Judged Compaction for Long-Horizon Agents
2610.20951	2026-10-14	LEV: Thinking Fast and Deep with Looped Decision Models
2610.21160	2026-10-15	When to Ask for Help: A Decision-Theoretic Account and Measurement Standard for Slow-Module Scheduling in Dual-System Vision-Language-Action Models
2610.21214	2026-10-15	Rubric Judgments Lack Uncertainty: Conformal Weighting for LLM Evaluation
2610.21244	2026-10-15	System1-Arena: Which Decisions Can Be Left to System 1?
2610.21411	2026-10-15	Jev-LiteMAS: Separating Decisions from Generation in Multi-Agent Workflows
2610.21412	2026-10-15	JevAgentHarness: Benchmarking, Deploying and Training Decision Models That Share Control with LLMs
2610.22066	2026-10-15	Prism-1: Position- and Rival-Independent Scoring for System One Models
2610.22131	2026-10-15	Intern-Decision: Efficient Multimodal Decision Making with Autoregressive Masked Language Modeling
2610.22276	2026-10-15	MJev-Spatial: System One Spatial Intelligence via Native Attention Matching
2610.22526	2026-10-15	Literal Lock-In: How Test Language Shifts Literal Attraction in Typed Decision Models on Bengali Idioms
2610.23170	2026-10-16	Chinese Decision: A Family of Language Models for Single-Pass Probabilistic Decision-Making
2610.23668	2026-10-16	TUPLE-Bench: Evaluating and Mitigating Privacy-Utility Trade-offs in Typed Decision Models

ID	Date	Title
2610.23804	2026-10-16	Jarvis: Coordination Deserves a Brain of Its Own in Agentic Interactive Models
2610.24358	2026-10-17	DKnown-Jev: Fine-Tuning and Evaluation of a JEV-Style Model for Tasks Involving Normative Knowledge
2610.24649	2026-10-17	Good Judgment, Bad Advice: Decision-Model Controllers Raise Agent Cost Without Improving Success
2610.24871	2026-10-17	Bellman Language Models: Solving the Recurring Decisions of LLM Agents
2610.24929	2026-10-18	What Can't Jev Decide? Probing the Limits of One-Pass Decision Models
2610.25456	2026-10-18	System X: An LLM Decision Model That Replies from Its Own KV Cache
2610.25474	2026-10-18	P(Yes): 15$\times$ Improvements via Test-Time Scaling of Decision Models
2610.25553	2026-10-18	Narrow Decision Models: A System-1 Architecture for Calibrated, Typed Decisions
2610.25760	2026-10-19	Computation, Not Memory: Chain-of-Thought and State Tracking in LLM Agents
2610.25768	2026-10-19	Truncated Reasoning Under a Budget: Evaluating System-1 Models as Decision-Makers
2610.25784	2026-10-19	UniJev: Separating Accuracy from Calibration in Trimodal System One Decision Models
2610.25947	2026-10-19	Claim Laundering: When Agent Decision Models Trust Records Over Reality
2610.25985	2026-10-19	NIOCIS: Typed-Decision Next-Hop Routing with Node-Conditioned Option Spaces for Tool-Use Agents
2610.26035	2026-10-19	Decision4Rec: A Jev-Inspired Framework of Decision Models on Knowledge and Active Graphs for Multi-Agent Recommender Systems
2610.26325	2026-10-19	Typed Decision Models under Prevalence Shift: Decision Training, Calibration, and Implicit Priors
2610.26793	2026-10-19	SOMA: Continuous Monitoring and Intervention for System Two Reasoning with System One Models
2610.26797	2026-10-19	Teacher Supervision, Fixed Representations and Calibration: Exact Risk Calculations for Candidate-Scoring Decision Models
2610.27301	2026-10-19	Evaluating Context Compaction on the Questions That Come Later
2610.27746	2026-10-20	S^2-MAS: A Society-Scale Multi-Agent Simulation Framework with 500 Million Household Persona Agents
2610.28116	2026-10-20	AgenTest: Mining Business Rules to Test LLM Agents for Business Logic Bugs
2610.28522	2026-10-20	DecisionPivot: A Diagnostic Benchmark for Decision Models in LLM Systems
2610.28577	2026-10-20	Cache Engineering for Agents
2610.28819	2026-10-20	LIFT: Scoped Constraints for Evaluating and Training Instruction Following
2610.29052	2026-10-21	Our Own Jev: Typed Decisions in One Forward Pass, from an In-House Judge to Open LoRA Models
2610.29217	2026-10-21	When Multiple Choices Are Correct: Asymmetric Probability Allocation in Jev and Other Decision Models
2610.29294	2026-10-21	Same Probabilities, Different Advice: Gain-Loss Framing Effects in High-Stakes LLM Decision Support
2610.29961	2026-10-21	Commit or Abstain? System One Decision Models, LLMs, and Missing Legal Authority
2610.30045	2026-10-21	Which Differences Matter? Controlled Tests of Rule Transfer, Joint Reliability, and Information Use in Decision Models
2610.30587	2026-10-22	Know When to Escalate: Critic-Guided Adaptive Execution for Cloud-Local Embodied Agents
2610.30690	2026-10-22	From Clinical Judgment to Checkable Rules: Clinician-Led Guardrails for Mental-Health Chatbots

ID	Date	Title
2610.31315	2026-10-22	The Value of System 1 Models in Medicine
2610.31521	2026-10-22	Jev-DLC: Decision-Level Continual Learning from Typed Feedback
2610.32285	2026-10-23	Trial by Trace: Benchmarking AI Agent Harnesses for Reliability and Failure Attribution
2610.32648	2026-10-23	StateMirror: Pre-Execution Detection of Hallucinated Tool Calls in Coding Agents
2610.32955	2026-10-23	Measure Before You Train: Typed Decision Models vs. a Classical Baseline vs. a Generative LLM for Routing a Personal Multilingual Inbox
2610.33258	2026-10-23	Two Uncertainties Are Not One: Disagreement versus Decision-Risk in Cheap Structured Content Moderation
2610.33493	2026-10-24	Small Edits, Shifting Decisions: Benchmarking LLM Accuracy and Reliability on French Input Variants
2610.33766	2026-10-24	StateWise: Teaching Language Models to Track and Obey Evolving Constraints in Multi-Turn Dialogue
2610.33775	2026-10-24	The Hitchhiker's Guide to System 1 Models: Fast Decision-Making Beyond LLMS
2610.33876	2026-10-24	FlashScript: Compiling Recurring Agent Workflows into Reusable Neuro-Symbolic Programs
2610.34258	2026-10-25	Recovering Business Rules as Cascade Lookups for Enterprise Field Prediction
2610.34329	2026-10-25	SHIFT-JEV: Certified Abstention and Gain-Gated Continual Adaptation for LLM-Based Decision Models under Distribution Shift
2610.34826	2026-10-26	TrustJev: Separating Policy Authority from Untrusted Evidence in LLM-Based Decisions
2610.34956	2026-10-26	Decision Model Factory: Building Specialized Sovereign Decision Models
2610.35349	2026-10-26	MarkovSim: an LLM Agent Harness with Bounded Context for Long-Horizon Computational Simulation
2610.35654	2026-10-26	Improving JEV Entailment Reranking with Span Readouts and Selective Adaptation
2610.35721	2026-10-26	Can JEV Support Clinical Trial Criteria Assessment?
2610.35826	2026-10-26	Understanding Jev-Style System One Models: Does Structure Reduce Selection Bias?

9.2 TypeSafe official documentation and pages

Page	URL
Launch blog post: Introducing System One models and Jev	https://typesafe.ai/blog/introducing-system-one-models-and-jev
Documentation home	https://docs.typesafe.ai/introduction
Machine-learning primer	https://docs.typesafe.ai/introduction/machine-learning-primer
Concept: System One	https://docs.typesafe.ai/concepts/system-one
How to build with System One	https://docs.typesafe.ai/concepts/how-to-build-with-system-one
Concept: State	https://docs.typesafe.ai/concepts/state
Use-case map	https://docs.typesafe.ai/concepts/use-case-map
Primitive: Score	https://docs.typesafe.ai/primitives/score
Primitives: advanced	https://docs.typesafe.ai/primitives/advanced
Confidence	https://docs.typesafe.ai/confidence
Patterns overview	https://docs.typesafe.ai/patterns
Pattern: fan-out	https://docs.typesafe.ai/patterns/fan-out
Pattern: composite scoring	https://docs.typesafe.ai/patterns/composite-scoring
Pattern: confidence routing	https://docs.typesafe.ai/patterns/confidence-routing
Agent skill	https://docs.typesafe.ai/agent-skill
Model list	https://docs.typesafe.ai/models

Page	URL
Model jaggedness (jev-1.13)	https://docs.typesafe.ai/model-jaggedness/jev-1.13
Cookbooks overview	https://docs.typesafe.ai/cookbooks
Cookbook: Noul self-consistency	https://docs.typesafe.ai/cookbooks/consistency_noul_cookbook
Cookbook: autoresearch feature discovery	https://docs.typesafe.ai/cookbooks/autoresearch_feature_discovery
Official evaluation page	https://evals.typesafe.ai/
typesafe-ai/skills repository	https://github.com/typesafe-ai/skills
system-one-adapter-python	https://github.com/typesafe-ai/system-one-adapter-python

9.3 Open-source reproductions, tools and model cards

Project	URL
Kev（jaredpalmer）	https://github.com/jaredpalmer/kev
Kev-9B model card	https://huggingface.co/jaredpalmer/kev-9b
Tev1（Together）	https://github.com/togethercomputer/tev1
Tev1-4B-experimental model card	https://huggingface.co/togethercomputer/Tev1-4B-experimental
Together blog: How to train your own Jev	https://www.together.ai/blog/how-to-train-your-own-jev
Nimble（Bespoke Labs）	https://github.com/bespokelabsai/nimble
Bespoke-Nimble-9B-v3 model card	https://huggingface.co/bespokelabs/Bespoke-Nimble-9B-v3
pplx-decider-v1.2 model card	https://huggingface.co/perplexity-ai/pplx-decider-v1.2
OneJev（OmniJev）	https://github.com/OmniJev/OneJev
OneJev-9B model card	https://huggingface.co/OmniJev/OneJev-9B
OmniJev flide	https://github.com/OmniJev/flide
awesome-jev-papers	https://github.com/OmniJev/awesome-jev-papers
awesome-jev-gallery	https://github.com/OmniJev/awesome-jev-gallery
awesome-jev-robustness	https://github.com/Yifan-Lan/awesome-jev-robustness
Open-Jev（Zefan-Cai）	https://github.com/Zefan-Cai/Open-Jev
open-jev / TOD（daseinlabs）	https://github.com/daseinlabs/open-jev
qwen-rlcd（shamazharikh）	https://github.com/shamazharikh/qwen-rlcd
Mapika decider	https://github.com/Mapika/decider
StartLux-Decision	https://github.com/StartLuxLabs/StartLux-Decision
StartLux-Decision-v1.1-27B-Instant-GGUF model card	https://huggingface.co/startlux-models/StartLux-Decision-v1.1-27B-Instant-GGUF
jevlike（mustafasemi-ai）	https://github.com/mustafasemi-ai/jevlike
this-that-model（FLock-io）	https://github.com/FLock-io/this-that-model
openjev-sglang（ekzhang）	https://github.com/ekzhang/openjev-sglang
jev-benchmarks（AbdelStark）	https://github.com/AbdelStark/jev-benchmarks
decision-model-benchmark（nibzard）	https://github.com/nibzard/decision-model-benchmark
jev-calibration-audit（jujumilk3）	https://github.com/jujumilk3/jev-calibration-audit
RLCDAlignBench（sumleo）	https://github.com/sumleo/RLCDAlignBench
jev-ultrafast（browser-use）	https://github.com/browser-use/jev-ultrafast
WaterSheep	https://github.com/SamratDuttaOfficial/WaterSheep
Laya（NandhaKishorM）	https://github.com/NandhaKishorM/laya
Laya model page	https://huggingface.co/convaiinnovations/laya
Laya website	https://laya.convaiinnovations.com/
decision-index（apolinario）	https://github.com/apolinario/decision-index
Jev Decision Index article	https://systemonemodels.org/examples/articles/jev-decision-index/
Jev Decision Index（HF Space）	https://huggingface.co/spaces/multimodalart/jev-decision-index

9.4 Other web pages

Page	URL
Archer Hume：Jev's Architecture Unmasked	https://archerhume.com/posts/jevs-architecture-unmasked

Page	URL
Agent Skills specification	https://agentskills.io/specification
OpenAI：Introducing gpt-oss-safeguard	https://openai.com/index/introducing-gpt-oss-safeguard/
LangChain：Building a harness with Jev	https://www.langchain.com/blog/building-a-harness-with-jev
Wikipedia：Jev (AI model)	https://en.wikipedia.org/wiki/Jev_(AI_model)